Prospectus

Filed Pursuant to Rule 424(b)(4)
Registration No. 333-288549

5,000,000 Shares

shoulderinnovationsinc.jpg

Common Stock

This is an initial public offering of shares of common stock of Shoulder Innovations, Inc.

We are offering 5,000,000 shares of our common stock. Prior to this offering, there has been no public market for our common stock. The public offering price is $15.00 per share. We have been approved to list our common stock on the New York Stock Exchange under the trading symbol "SI."

We are an "emerging growth company" and a "smaller reporting company" as defined under the U.S. federal securities laws and, as such, have elected to comply with certain reduced public company reporting requirements in this prospectus and may elect to do so in future filings. See the section titled "Prospectus Summary—Implications of Being an Emerging Growth Company and a Smaller Reporting Company."

Investing in our common stock involves a high degree of risk. See the section titled "Risk Factors" beginning on page 20.

	Per Share	Total
Initial public offering price	$15.00	$75,000,000
Underwriting discounts and commissions[(1)]	$1.05	$5,250,000
Proceeds, before expenses, to us	$13.95	$69,750,000

(1) See the section titled "Underwriting" for additional disclosure regarding the estimated underwriting discounts and commissions and estimated offering expenses.

We have granted the underwriters an option for a period of 30 days to purchase up to an additional 750,000 shares of our common stock solely to cover over-allotments, if any.

At our request, the underwriters have reserved up to 6% of the shares of our common stock offered by this prospectus for sale at the initial public offering price through a directed share program to certain of our directors, officers and employees and others. See the section titled "Underwriting—Directed Share Program" for additional information.

The underwriters expect to deliver the shares of common stock to purchasers on August 1, 2025.

Neither the Securities and Exchange Commission nor any state securities commission has approved or disapproved of these securities or determined if this prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

Morgan Stanley **Goldman Sachs & Co. LLC** **Piper Sandler**

Jefferies

BTIG

The date of this prospectus is July 30, 2025.

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We have not, and the underwriters have not, authorized anyone to provide you any information or to make any representations other than those contained in this prospectus or in any free writing prospectus prepared by or on behalf of us or to which we have referred you. Neither we nor the underwriters take responsibility for, or provide any assurance as to the reliability of, any other information others may give you. This prospectus is an offer to sell only the shares offered hereby, and only under circumstances and in jurisdictions where it is lawful to do so. We are not, and the underwriters are not, making an offer to sell these securities in any jurisdiction where the offer or sale is not permitted. The information contained in this prospectus is accurate only as of the date of this prospectus, regardless of the time of delivery of this prospectus or any sale of the shares of our common stock. Our business, financial condition, and results of operations may have changed since that date.

For investors outside the United States: We have not, and the underwriters have not, done anything that would permit this offering or the possession or distribution of this prospectus or any free writing prospectus in connection with this offering in any jurisdiction where action for that purpose is required, other than in the United States. Persons outside the United States who come into possession of this prospectus must inform themselves about, and observe any restrictions relating to, the offering of the shares of common stock and the distribution of this prospectus outside the United States.

BASIS OF PRESENTATION

As used in this prospectus, unless the context otherwise requires, references to "Shoulder Innovations," the "company," "we," "us," and "our" refer to Shoulder Innovations, Inc.

The financial statements include the accounts of Shoulder Innovations, Inc. Our financial statements have been prepared in accordance with generally accepted accounting principles in the United States ("GAAP"). Our fiscal year ends on December 31 of each year. References to 2024 and 2023 refer to the year ended December 31, 2024 and December 31, 2023, respectively.

Certain monetary amounts, percentages and other figures included in this prospectus have been subject to rounding adjustments. Percentage amounts included in this prospectus have not in all cases been calculated on the basis of such rounded figures, but on the basis of such amounts prior to rounding. For this reason, percentage amounts in this prospectus may vary from those obtained by performing the same calculations using the figures in our financial statements included elsewhere in this prospectus. Certain other amounts that appear in this prospectus may not sum due to rounding.

TRADEMARKS AND TRADENAMES

"Shoulder Innovations," the Shoulder Innovations logos, and other trade names, trademarks, or service marks of Shoulder Innovations appearing in this prospectus are the property of Shoulder Innovations. Other trade names, trademarks, or service marks appearing in this prospectus are the property of their respective holders. We do not intend our use or display of other companies' trade names, trademarks, or service marks to imply a relationship with, or endorsement or sponsorship of us, by these other companies. Solely for convenience, trade names, trademarks, and service marks referred to in this prospectus appear without the ®, ™, and SM symbols, but those references are not intended to indicate, in any way, that we will not assert, to the fullest extent under applicable law, our rights or that the applicable owner will not assert its rights, to these trade names, trademarks, and service marks.

INDUSTRY AND OTHER MARKET DATA

This prospectus contains estimates, projections, and other information concerning our industry and our business, as well as data regarding market research, estimates, and forecasts prepared by our management or third parties. Information that is based on estimates, forecasts, projections, market research, or similar methodologies is inherently subject to uncertainties, and actual events or circumstances may differ materially from events and circumstances that are assumed in this information. While we believe the market and industry data included in this prospectus are reliable and are based on reasonable assumptions, these data and the industry in which we operate is subject to a high degree of uncertainty and risk due to a variety of factors, including those described in the section titled "Risk Factors." These and other factors could cause results to differ materially from those expressed in these estimates, publications, and reports made by third parties or us.

Unless otherwise expressly stated, we obtained such industry, business, market, and other data from reports, research surveys, studies, and similar data prepared by market research firms and other third parties, industry and general publications, government data, and similar sources. Our estimated number of physician visits annually for shoulder conditions is based on data from the National Ambulatory Medical Care Survey 2015-2016 conducted by the National Center for Health Statistics of the Centers for Disease Control and Prevention and the Physical Medicine & Rehabilitation Journal, Characteristics of Non-Spine Musculoskeletal Ambulatory Care Visits in the United States, 2009-2016, May 2021. In addition, certain procedure and market data are based on independent reports from third parties, including a leading healthcare market research provider and Transparency Market Research. Our estimated the annual growth in the number of shoulder arthroplasty procedures in the United States through 2029 is based on data from a leading healthcare market research provider regarding the number of procedures performed historically and an assumed growth rate in line with the recent annual growth rate in the number of procedures in the United States. In some cases, we do not expressly refer to the sources from which this data is derived. In that regard, when we refer to one or more sources of this type of data in any paragraph, you should assume that other data of this type appearing in the same paragraph is derived from the same sources, unless otherwise expressly stated or the context otherwise requires.

The content of these third-party sources, except to the extent specifically set forth in this prospectus, does not constitute a portion of this prospectus and is not incorporated herein.

Forecasts and other forward-looking information with respect to industry, business, market, and other data are subject to the same qualifications and additional uncertainties regarding the other forward-looking statements in this prospectus. See the section titled "Special Note Regarding Forward-Looking Statements."

PROSPECTUS SUMMARY

This summary highlights selected information contained in greater detail elsewhere in this prospectus. This summary is not complete and does not contain all of the information you should consider in making your investment decision. Before investing in our common stock, you should carefully read this entire prospectus. You should carefully consider, among other things, the sections titled "Risk Factors," "Special Note Regarding Forward-Looking Statements," "Business," and "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our financial statements and the related notes included elsewhere in this prospectus.

Overview

We are a commercial-stage medical technology company exclusively focused on transforming the shoulder surgical care market. We currently offer advanced implant systems for shoulder arthroplasty. These systems are a core element of our ecosystem, which we designed to improve core components of shoulder surgical care – preoperative planning, implant design and procedural efficiency – to benefit each stakeholder in the care chain. Our ecosystem is also comprised of enabling technologies, efficient instrument systems, specialized support and surgeon-to-surgeon collaboration. Together, these elements seek to address the long-standing clinical and operational challenges in the shoulder surgical care market by delivering predictable outcomes, procedural simplicity, and efficiency across all sites of care. We believe our exclusive focus on shoulder surgical care, combined with a highly specialized commercial organization and strong clinical data, positions us well to capture significant share in this large, growing market.

Shoulder pain is highly prevalent, often chronic, and can significantly reduce quality of life. The primary conditions that can result in shoulder pain and reduced functionality include osteoarthritis, rheumatoid arthritis, rotator cuff tears and shoulder fractures. Shoulder conditions are widespread, often debilitating, are commonly experienced concurrently as interrelated musculoskeletal disorders, and are estimated to result in more than eight million physician visits annually in the United States, based on data from the National Ambulatory Medical Care Survey 2015-2016 conducted by the National Center for Health Statistics of the Centers for Disease Control and Prevention ("CDC"). Despite this prevalence, we believe there has been a historical underutilization of surgical treatments for shoulder care due to several factors including patient hesitation to pursue surgical intervention, insufficient technology to appropriately treat shoulder conditions, complex shoulder anatomy, perceived difficulty of surgical intervention and barriers to patient access of care.

We believe the shoulder surgical care market today presents a significant market opportunity. Our initial focus within this broader market is on shoulder arthroplasty. Shoulder arthroplasty is an established surgical procedure involving the reconstruction of the shoulder joint with prosthetic implants through one of two main approaches: anatomic total shoulder arthroplasty ("aTSA") and reverse total shoulder arthroplasty ("rTSA"). Both approaches can be performed in inpatient hospital settings and in outpatient settings, including ambulatory surgery centers ("ASCs"). A key competitive advantage of ours has been the emergence of ASCs as a cost-efficient site of care with positive outcomes relative to hospital-based care. We expect that future growth in the shoulder surgical care market will be significantly driven by ASCs as hospitals face capacity constraints and are more limited in their ability to meet increasing demand.

We believe traditional implants used in shoulder arthroplasty procedures are hindered by several limitations, including poor biomechanical fit, suboptimal kinematics, difficult replacement and conversion procedures (aTSA to rTSA or stemless to stem), imprecise implant positioning due to limited surgical planning, inefficient and burdensome workflow designs and non-specialized case support. These limitations can result in continued pain, lack of mobility, postoperative complications, low rates of implant survivorship, necessity of revision surgeries and costly and inefficient procedures for healthcare providers.

We developed our ecosystem with an approach to innovation that prioritizes ease of use, flexibility, predictability of outcomes and site of care efficiency, attributes we believe are critical to win in our market. This ecosystem is comprised of our advanced implant systems, ProVoyance preoperative planning technology, efficient instrument system, specialized support and surgeon-to-surgeon collaboration. Our advanced implants are comprised of our aTSA and rTSA systems, which are designed to address the unique needs of patients and surgeons, and

include various, specifically designed components capable of a wide array of system configurations to facilitate different modes of operation (anatomic or reverse). Our InSet Glenoid technology serves as the foundation for our advanced implant systems and includes a novel "InSet" design that aims to reduce mechanical stress at the bone implant interface, improve fixation mechanics, enhance stability and reduce micromotion. Our implant systems leverage consistent surgical techniques and the same efficient, two tray instrumentation system.

In addition to our advanced implant systems, we offer a leading preoperative surgical planning technology: ProVoyance. We believe that surgeon-level engagement in preoperative planning provides for better care for patients, and that bespoke surgical plans can help facilitate consistent positioning of implants. ProVoyance integrates artificial intelligence ("AI") and machine learning ("ML") to transform planar CT imaging into 3D renderings of patient-specific anatomy ahead of surgery, and is cleared by the U.S. Food and Drug Administration (the "FDA") for preoperative shoulder planning. ProVoyance received 510(k) clearance in 2021 and is classified by the FDA as a Class II device. ProVoyance is listed on the FDA's AI/ML-enabled medical devices list, which is a resource maintained, published, and periodically updated by the FDA to identify AI/ML-enabled devices that have been authorized for marketing in the United States through any of the standard paths to market for medical devices, although it is not intended to be a comprehensive list of all such devices that incorporate AI/ML. We believe the differentiation and value proposition of ProVoyance is validated by high utilization rates across procedures using our advanced implant systems. For example, for the three months ended June 30, 2025, we estimate an implied utilization rate for our ProVoyance technology of approximately 98%, based on 1,478 surgical plans created using ProVoyance technology and 1,503 implant systems sold during such period. This implied utilization rate is based on real-world data from our customers during such period, as ProVoyance technology tracks and reports each surgical plan that our customers create, and the actual number of implant systems sold during the respective period.

A key component of shoulder arthroplasty procedures are instrument trays, equipped with the specific instruments, supplies, and equipment needed for the surgery. We have developed a proprietary two-tray instrument system designed to enable interoperability between our aTSA and rTSA systems and a range of humeral stem options. We believe our efficient, two tray instrument system can enable surgeons and staff to reduce operating room footprint, procedural setup time, sterilization time and expense, and procedural complexity.

To best support our surgeon customers, we have built our commercial organization around their unique needs. Our commercial organization is comprised of a dedicated commercial leadership team that drives our internal commercial efforts with an exclusive focus on shoulder care, a Customer Experience and Medical Education ("CEME") team that enhances surgeon engagement and training and a network of independent distributors. These three key components of our commercial organization work in tandem to form a commercial flywheel that is designed to build and reinforce relationships with surgeons and other stakeholders in the shoulder surgical care market, accelerate adoption, and enhance long-term retention.

We leverage our team's decades of experience developing and launching novel shoulder surgical care technologies to identify the unmet needs of patients and surgeons and develop solutions to address those unmet needs. With respect to our advanced implant systems, we commenced development efforts with our InSet Glenoid in 2009 and received 510(k) clearance in 2011. We commercially launched an initial aTSA system with our InSet Glenoid in 2016. Since this initial launch in 2016, we have successfully launched a wide range of new technologies to enhance our ecosystem and provide surgeons with the tools and support needed to deliver quality outcomes for patients requiring shoulder surgical care. For example, we commenced development efforts for our InSet PLUS Augmented Glenoid in 2019 and received 510(k) clearance in 2020. We commenced development efforts for our rTSA system in 2019 and received 510(k) clearance in 2021. We commenced development efforts for our short stem, stemless and I-Series humeral stem system options for our aTSA and rTSA systems in 2017, 2019 and 2021, respectively, and received our primary 510(k) clearances in 2018 and 2022, with an additional 510(k) clearance in 2024 for use of our primary I-Series humeral stem for use with anatomic fractures. Each of

these devices is classified by the FDA as a Class II device. We have a robust pipeline of new technologies in various stages of development and evaluation, including the anticipated expansion of our humeral stem line, indication expansions into fracture and revision, and implants tailored for metal-sensitive patients. For example, we commenced development of InSet 70, InSet 135 and InSet 185 stems to expand our I-Series humeral stem line in 2024, and we anticipate pursuing FDA clearance of these stems, as needed, over the next twelve months. We are also evaluating

expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets.

We have contributed to numerous publications that we believe evidence and strengthen our position as a leader in shoulder surgical care. There is a significant body of clinical evidence that supports the safety, efficacy, and durability of our implants in shoulder arthroplasty, including our InSet Glenoid technology. For example, a retrospective long-term follow-up analysis of patients who received our InSet Glenoid was published in the *Journal of Shoulder and Elbow Surgery* in 2019, which demonstrated a 72-point improvement in the mean American Shoulder and Elbow Surgeons ("ASES") outcome score, statistically significant improvements in pain scores and range of motion, with no surgical complications, no cases of glenoid loosening and no revision surgeries performed at a mean follow-up time of 8.7 years. We are committed to continued investment in obtaining further clinical evidence with the support of surgeons who are recognized as thought leaders in shoulder surgical care. We believe these efforts will continue to generate a substantial body of clinical evidence that will drive increased awareness and adoption of our products.

We have experienced significant growth in recent years, primarily driven by growth in our net revenue from the sale of our advanced implant systems. We generated net revenue of $31.6 million for the year ended December 31, 2024, compared to net revenue of $19.3 million for the year ended December 31, 2023, representing 64.0% year-over-year growth. We recognized a gross margin of 77.0% and net loss of $15.6 million for the year ended December 31, 2024, compared to a gross margin of 79.2% and net loss of $12.7 million for the year ended December 31, 2023. We generated net revenue of $10.1 million for the three months ended March 31, 2025, compared to net revenue of $7.2 million for the three months ended March 31, 2024, representing an increase of 41.0%. We recognized a gross margin of 76.9% and net loss of $4.7 million for the three months ended March 31, 2025, compared to a gross margin of 76.8% and net loss of $3.6 million for the three months ended March 31, 2024.

Market Overview

Market Opportunity

Shoulder pain is highly prevalent and can significantly impact and reduce quality of life as well as result in chronic pain. The primary conditions that can result in shoulder pain and reduced functionality include: osteoarthritis, rheumatoid arthritis, shoulder fractures, and rotator cuff tears. Shoulder conditions are widespread, often debilitating, are commonly experienced concurrently as interrelated musculoskeletal disorders, and are estimated to result in more than eight million physician visits annually in the United States, based on data from the CDC. These conditions can require surgical intervention.

We believe the shoulder surgical care market today presents a significant market opportunity. Our initial focus within this broader market is on shoulder arthroplasty, which represents a large, immediately addressable and rapidly growing market. We estimate that approximately 250,000 shoulder arthroplasty procedures will be performed in the United States in 2025, which we believe represents an approximately $1.7 billion market opportunity based on our average sales price. Based on data from a healthcare market research provider regarding the number of shoulder arthroplasty procedures performed, our internal estimates and analysis of such data, as well as our knowledge of our industry, we expect the number of annual procedures performed to grow by approximately 11% annually through 2029. While our current commercial focus is on the United States, we plan to pursue market access initiatives in other attractive, high-growth international markets. We believe a significant opportunity exists outside of the United States and, based on a market research report issued by Transparency Market Research in 2025, we estimate that the total international shoulder arthroplasty market is approximately $1.0 billion in 2025. Together, we believe these markets represent a global annual shoulder arthroplasty market of approximately $2.8 billion. These market opportunities represent the total overall revenue opportunity that we believe is available for our systems if 100% market share is achieved by us, and are not a representation that we will achieve any such market share. The market share we achieve is subject to a number of assumptions, risks

and uncertainties, including the adoption of our implant systems and our ability to execute on our commercial strategy. For more information, please see the section titled "Risk Factors—Risks Related to Our Business and Industry—Our business plan relies on certain assumptions about the market for our implant systems, however, the size and expected growth of our

addressable market has not been established with precision and may be smaller than we estimate, and even if the addressable market is as large as we have estimated, we may not be able to capture additional market share."

We also plan to continue investing in our robust product pipeline to expand our capabilities in shoulder arthroplasty and serve additional, adjacent segments of the shoulder surgical care market that we do not currently address. Our near-term development efforts include a revision solution, a fracture-specific system and a line of humeral head and glenoid technologies for patients who test positively for a metal hypersensitivity. In addition to these near-term development efforts, we are also evaluating expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets.

Limitations of Existing Product Offerings in Shoulder Arthroplasty

Despite their frequent use and wide adoption, we believe traditional implants used in shoulder arthroplasty procedures are hindered by several limitations, including poor biomechanical fit, suboptimal kinematics, difficult replacement and conversion procedures (aTSA to rTSA or stemless to stem), imprecise implant positioning due to limited surgical planning, inefficient and burdensome workflow designs and non-specialized case support. These shortcomings can impact both patients and surgeons and result in continued pain or discomfort, lack of mobility, postoperative complications, low rates of implant survivorship, necessity of revision surgeries and costly and inefficient procedures for healthcare providers.

We believe traditional shoulder arthroplasty products present several limitations, including:

- Fixation of the Glenoid Component in aTSA
- Poor Product Design Resulting in Suboptimal Kinematics in rTSA
- Difficult to Replace and Convert
- Imprecise Implant Positioning Due to Limited Surgical Planning
- Inefficient and Burdensome Surgical Workflow Design
- Non-Specialized Case Support

These limitations can result in the following:

- Failure to Reduce Pain and Improve Shoulder Function
- Frequent Post-Operative Complications
- Low Rates of Implant Survivorship
- Necessity of Revision Surgeries
- Costly and Inefficient for Healthcare Providers

Our Solutions

We developed our ecosystem with an approach to innovation that prioritizes ease of use, flexibility, predictability of outcomes and site of care efficiency, attributes we believe are critical to win in our market.

Key Elements of Our Ecosystem

Our ecosystem is comprised of the following key elements:

- **Advanced Implant Systems**: Our advanced implants include a diverse range of interchangeable InSet aTSA and rTSA systems that leverage our novel, InSet Glenoid and InSet humeral stem technologies.

- **ProVoyance Preoperative Planning Technology**: This preoperative planning technology integrates AI and ML to transform planar CT imaging into 3D renderings, allowing surgeons to create bespoke surgical plans considering patient-specific anatomy ahead of surgery.
- **Efficient Instrument System**: Our efficient instrument system supports both aTSA and rTSA procedures from start to finish with just two convenient trays.
- **Specialized Support**: Our team of dedicated shoulder specialists enable us to deliver a highly tailored experience to surgeons operating in a complex and technically demanding procedure category.
- **Surgeon-to-Surgeon Collaboration**: Our CEME team fosters a collaborative network of expert surgeon educators and promotes surgeon-to-surgeon training and peer education.

The image below depicts the key elements of our ecosystem:

Key Benefits of Our Ecosystem

Our ecosystem offers notable benefits that differentiate it within the shoulder surgical care market. These key benefits include**:**

- **Improved Fixation and Stability**: Our foundational InSet Glenoid technology is biomechanically designed to specifically address what we believe is the primary problem in legacy aTSA implants: glenoid loosening.
- **Restores Full Functionality**: We engineered our three humeral stem options to provide consistent, optimized biomechanics in both aTSA and rTSA.
- **Exceptional Longevity**: Our advanced implant systems are designed for longevity. In one published study, our InSet Glenoid demonstrated no surgical complications, cases of glenoid loosening or revision surgeries at a mean follow-up time of 8.7 years in a particularly challenging patient population.

- **Improved Implant Selection and Positioning**: ProVoyance preoperative planning technology empowers surgeons to create bespoke surgical plans that facilitate consistent, effective positioning of our implants.
- **Optimized Procedural Workflow and Efficiency**: Our efficient instrument system supports our implant portfolio across both aTSA and rTSA. This can provide significant workflow advantages for our customers

by minimizing the operating room footprint, reducing procedural setup time, lowering sterilization requirements and costs and reducing the risk of errors - key advantages for the ASC setting.

- **Access to Expert Advice**: Our commercial organization creates deep relationships with surgeons and allows us to support their practice with specialized customer service and case support before, during and after surgery. In addition to direct support, our commercial team connects surgeons to the broader shoulder surgical community in various settings with a goal to improve connectivity across the shoulder surgical community and contribute to better patient outcomes.

Our Success Factors

We attribute our success to a combination of the following factors. We believe these attributes are central to our business outcomes and will be significant factors in our continued success and growth.

- ***Disruptive Ecosystem to Address Existing Limitations Within Shoulder Surgical Care***
- ***Strong Clinical Results and Positive Outcomes for Patients and Surgeons***
- ***Well Positioned as Shoulder Surgical Care Market Grows in Outpatient Settings***
- ***Proven and Experienced Management Team***
- ***Unique Commercial Organization Dedicated to the Shoulder Surgical Care Market***
- ***AI-Enabled Business and Clinical Intelligence Technologies***

Our Growth Strategies

Our goal is to leverage our purpose-built ecosystem to become the leader for shoulder surgical care. The key elements of our growth strategy include:

- ***Increase Awareness of our Purpose-Built, Innovative Shoulder Surgical Care Ecosystem to Continue Taking Share Across Care Settings***
- ***Expand our Commercial Flywheel of Shoulder Specialists, Surgeon-to-Surgeon Collaboration, and Network of Independent Distributors***
- ***Capitalize on Our Unique Advantages to Capture Outsized ASC Growth***
- ***Increase our Addressable Market Through our Commitment to Continuous Innovation and Advancing Shoulder Surgical Care***
- ***Continue Building and Driving Marketing of our Technology Solutions***
- ***Pursue Expansion in International Markets***

Recent Developments

2025 Convertible Notes

On July 18, 2025, we issued convertible promissory notes to certain investors in an aggregate principal amount of $40.0 million (the “2025 Convertible Notes”). The 2025 Convertible Notes mature on September 1, 2028 and accrue interest at a rate of 5.0% per annum through June 30, 2026 and 10.0% per annum after June 30, 2026. Upon the completion of this offering, the 2025 Convertible Notes and any accrued interest will automatically convert into shares of our common stock at the applicable conversion price. The conversion price

is the lower of (a) 80% of the initial public offering price per share in this offering and (b) $280.0 million divided by the number of fully diluted shares of capital stock (on an as-converted basis) outstanding immediately prior to this offering but excluding the 2025 Convertible Notes (the "Fully Diluted Capitalization"), provided, that in no event shall such conversion price be less than the quotient obtained by dividing $210.0 million by the Fully Diluted Capitalization. Based on the initial

public offering price of $15.00 per share and without giving effect to accrued interest, the 2025 Convertible Notes will automatically convert into an aggregate of 2,805,049 shares of our common stock upon the completion of this offering.

Preliminary Estimated Selected Financial Results for the Three Months and Six Months Ended June 30, 2025

Our financial results for the three months and six months ended June 30, 2025 are not yet complete and will not be available until after the completion of this offering. Accordingly, set forth below are certain preliminary estimated selected unaudited financial results for the three months and six months ended June 30, 2025 and the corresponding periods of the prior fiscal year. We have provided ranges, rather than specific amounts, for the three months and six months ended June 30, 2025, because our closing procedures for the quarter-end financial closing process are not yet complete, these results are preliminary and subject to change, and there is a possibility that our actual results may differ materially from these preliminary estimates. These ranges are based on the information available to us as of the date of this prospectus. These preliminary estimated results for the three months and six months ended June 30, 2025 are derived from our preliminary internal financial records and are subject to revisions based on our procedures and controls associated with the completion of our financial reporting, including all the customary reviews and approvals, and completion by our independent registered public accounting firm of its review of such financial statements for the quarter ended June 30, 2025. These preliminary estimated results should not be viewed as a substitute for financial statements prepared in accordance with U.S. GAAP. Our independent registered public accounting firm, Deloitte & Touche LLP, has not audited, reviewed, compiled or performed any procedures with respect to this preliminary financial information and, accordingly, Deloitte & Touche LLP does not express an opinion or any other form of assurance with respect thereto. It is possible that we or our independent registered public accounting firm may identify items that would require us to make adjustments to the preliminary estimates set forth below as we complete our financial statements and that our actual results may differ materially from these preliminary estimates. Accordingly, undue reliance should not be placed on these preliminary estimates. These preliminary estimates are not necessarily indicative of any future period and should be read together with "Risk Factors," "Special Note Regarding Forward-Looking Statements," "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our financial statements and related notes included elsewhere in this prospectus.

	Three Months Ended June 30,			Six Months Ended June 30		
	2025 (estimated low)	2025 (estimated high)	2024 (actual)	2025 (estimated low)	2025 (estimated high)	2024 (actual)
	(unaudited) (in thousands)					
Preliminary estimated financial results:						
Net revenue	$ 10,813	$ 11,213	$ 8,260	$ 20,945	$ 21,345	$ 15,444
Cost of goods sold	2,570	2,770	1,906	4,911	5,111	3,572
Operating expenses:						
Selling, general and administrative expenses[1]	13,445	13,945	9,171	23,947	24,447	16,875
Research and development expenses	1,306	1,506	1,162	2,889	3,089	2,232
Total operating expenses	14,751	15,451	10,333	26,836	27,536	19,107
Net loss	(21,142)	(19,142)	(4,171)	(25,804)	(23,804)	(7,772)
Adjusted EBITDA[2]	(20,020)	(18,020)	(3,194)	(23,520)	(21,520)	(5,811)

(1) Includes expected stock-based compensation expense of between $139 thousand and $239 thousand and between $266 thousand and $366 thousand for the three months ended and six months ended June 30, 2025, respectively, and stock-based compensation expense of

$170 thousand and $338 thousand for the three months ended and six months ended June 30, 2024, respectively.

(2) Adjusted EBITDA is a non-GAAP financial measure. See below for a reconciliation of Adjusted EBITDA to the most comparable GAAP financial measure, net loss, for each of the periods presented above.

For the three months ended June 30, 2025, we expect net revenue to be between $10.8 million and $11.2 million, compared to $8.3 million for the three months ended June 30, 2024. The expected increase in net revenue is primarily due to an increase in demand and in the number of our systems sold, as well as an increase in the number of our customers. For the six months ended June 30, 2025, we expect net revenue to be between $20.9 million and $21.3 million, compared to $15.4 million for the six months ended June 30, 2024. The expected increase in net revenue is primarily due to an increase in demand and in the number of our systems sold, as well as an increase in the number of our customers.

For the three months ended June 30, 2025, we expect cost of goods sold to be between $2.6 million and $2.8 million, compared to $1.9 million for the three months ended June 30, 2024. The expected increase in cost of goods sold is primarily due to an increase in the number of our systems sold. For the six months ended June 30, 2025, we expect cost of goods sold to be between $4.9 million and $5.1 million, compared to $3.6 million for the six months ended June 30, 2024. The expected increase in cost of goods sold is primarily due to an increase in the number of our systems sold.

For the three months ended June 30, 2025, we expect selling, general and administrative expenses to be between $13.4 million and $13.9 million, compared to $9.2 million for the three months ended June 30, 2024. The expected increase in selling, general and administrative expenses is primarily due to an increase in personnel related expenses as a result of increased headcount, an increase in legal costs, an increase in commissions due to higher sales of our systems and an increase in depreciation of surgical instruments. Selling, general and administrative expenses also includes $1.1 million of costs related to the proposed public offering. For the six months ended June 30, 2025, we expect selling, general and administrative expenses to be between $23.9 million and $24.4 million, compared to $16.9 million for the six months ended June 30, 2024. The expected increase in selling, general and administrative expenses is primarily due to an increase in personnel related expenses as a result of increased headcount, an increase in legal costs, an increase in commissions due to higher sales of our systems and an increase in depreciation of surgical instruments. Selling, general and administrative expenses also includes $1.2 million of costs related to the proposed public offering.

For the three months ended June 30, 2025, we expect research and development expenses to be between $1.3 million and $1.5 million, compared to $1.2 million for the three months ended June 30, 2024. The expected increase in research and development expenses is primarily due to an increase in research and development expenses due to our investment in new product development efforts, including an increase in external consulting fees related to such efforts. For the six months ended June 30, 2025, we expect research and development expenses to be between $2.9 million and $3.1 million, compared to $2.2 million for the six months ended June 30, 2024. The expected increase in research and development expenses is primarily due to an increase in research and development expenses due to our investment in new product development efforts, including an increase in external consulting fees related to such efforts.

For the three months ended June 30, 2025, we expect net loss to be between $19.1 million and $21.1 million, compared to $4.2 million for the three months ended June 30, 2024. The expected increase in net loss is primarily due to expected non-cash charges of between $12.2 million and $14.2 million related to changes in the fair value of our preferred stock warrant liability and Series E purchase option, as well as expected increases in operating loss and interest expense. We exercised the Series E purchase option in June 2025 in connection with the closing of the second tranche of our Series E convertible preferred stock financing. Following this offering, the warrants will be reclassified to stockholders' equity and will no longer be subject to remeasurement. For the six months ended June 30, 2025, we expect net loss to be between $23.8 million and $25.8 million, compared to $7.8 million for the six months ended June 30, 2024. The expected increase in net loss is primarily due to expected non-cash charges related to changes in the fair value of our preferred stock warrant liability and Series E purchase option and an increase in operating loss.

For the three months ended June 30, 2025, we expect Adjusted EBITDA loss to be between $18.0 million and $20.0 million, compared to $3.2 million for the three months ended June 30, 2024. The expected increase in

Adjusted EBITDA loss is primarily due to an increase in net loss, including due to non-cash charges of between $12.2 million and $14.2 million related to changes in the fair value of our preferred stock warrant liability and Series E purchase option, and an increase in depreciation and amortization expense. For the six months ended June 30,

2025, we expect Adjusted EBITDA loss to be between $21.5 million and $23.5 million, compared to $5.8 million for the six months ended June 30, 2024. The expected increase in Adjusted EBITDA loss is primarily due to an increase in net loss, including due to non-cash charges related to changes in the fair value of our preferred stock warrant liability and Series E purchase option, and an increase in depreciation and amortization expense.

We define Adjusted EBITDA as net loss before interest expense, net, income tax expense, depreciation and amortization, stock-based compensation expense and loss on extinguishment of convertible promissory notes. For more information about why we consider such measure useful and a discussion of the material risks and limitations of such measure, please see "Management's Discussion and Analysis of Financial Condition and Results of Operations—Non-GAAP Financial Measures." The following table presents a reconciliation of Adjusted EBITDA to the most comparable GAAP financial measure, net loss, for each of the periods presented above:

	Three Months Ended June 30,			Six Months Ended June 30,		
	2025 (estimated low)	2025 (estimated high)	2024 (actual)	2025 (estimated low)	2025 (estimated high)	2024 (actual)
	(unaudited) (in thousands)					
Net loss	$ (21,142)	$ (19,142)	$ (4,171)	$ (25,804)	$ (23,804)	$ (7,771)
Interest expense, net	166	266	292	533	633	609
Income tax expense	—	—	—	—	—	—
Depreciation and amortization expense	617	817	515	1,285	1,485	1,013
Stock-based compensation expense	139	239	170	266	366	338
Loss on extinguishment of convertible promissory notes	—	—	—	—	—	—
Adjusted EBITDA	(20,020)	(18,020)	(3,194)	(23,520)	(21,520)	(5,811)

As of June 30, 2025, our cash, cash equivalents and marketable securities balance is expected to be $39.6 million as compared to $24.8 million as of June 30, 2024.

Summary of Risks Associated with Our Business

Our business is subject to a number of risks of which you should be aware before making a decision to invest in our common stock. These risks are more fully described in the section titled "Risk Factors" immediately following this prospectus summary. These risks include, among others, the following:

- We are an early-stage company with a history of significant net losses, we expect to incur operating losses in the future and we may not be able to achieve or sustain profitability.
- If we fail to manage our growth effectively, our business could be materially and adversely affected.
- We have a significant amount of debt, which may affect our ability to operate our business and secure additional financing in the future.
- We may require substantial additional funding, which may not be available to us on acceptable terms, or at all, and, if not available, may require us to delay, scale back, or cease our innovation efforts or operations.
- We operate in a very competitive business environment, and if we are unable to compete successfully against our existing or potential competitors, our business, financial condition and results of operations

may be adversely affected.

- If we fail to develop and retain an effective dedicated commercial leadership team, or if we are unable to successfully expand dedicated commercial leadership team, it could negatively impact our sales, and we may not generate sufficient net revenue to sustain profitability.

- Our business plan relies on certain assumptions about the market for our implant systems, however, the size and expected growth of our addressable market has not been established with precision and may be smaller than we estimate, and even if the addressable market is as large as we have estimated, we may not be able to capture additional market share.
- Our business is dependent upon the adoption of our implant systems by hospitals, ASCs, surgeons and patients.
- Our long-term growth depends on our ability to enhance our implant systems, expand our indications and develop and commercialize additional products in a timely manner. If we cannot innovate, we may not be able to develop or exploit new products in time to remain competitive.
- We face the risk of product liability claims that could be expensive, divert management's attention and harm our reputation and business. We may not be able to maintain adequate product liability insurance.
- Industry trends have resulted in increased downward pricing pressure on medical services and products, which may affect our ability to sell our products at prices necessary to support our current business strategy.
- If hospitals, ASCs and other health care facilities do not approve the use of our implant systems, our sales may not increase.
- We depend on third-party contract manufacturers and suppliers, some of which are single source, to produce and package all elements comprising our shoulder implant systems, and if these suppliers and manufacturers fail to supply us, our products or their components or subcomponents in sufficient quantities or at all, or in accordance with applicable regulatory requirements and our specifications, it will have a material adverse effect on our business, financial condition, and results of operations.
- Our results of operations will be materially harmed if we are unable to accurately forecast demand for our implant systems and manage our inventory.
- We may be unable to continue to successfully demonstrate to shoulder specialists or key opinion leaders the merits of our implant systems and technologies compared to those of our competitors, which may make it difficult to achieve market acceptance.
- The loss of any member on our executive management team or our inability to attract and retain highly skilled members of our dedicated commercial leadership and marketing teams as well as certain third-party engineers could have a material adverse effect on our business, financial condition and results of operations.
- If we are unable to obtain and maintain significant patent or other intellectual property protection for our products, or if the scope of our patents and other intellectual property rights do not adequately protect our products, our competitors could develop and commercialize products similar or identical to ours and we may be unable to gain significant market share and be unable to operate our business profitably.
- Our devices and operations are subject to extensive government regulation and oversight in the United States, and our failure to comply with applicable requirements could harm our business.
- Our relationships with customers, physicians and third-party payors are subject to federal and state health care fraud and abuse laws, false claims laws, physician payment transparency laws and other health care laws and regulations. If we or our employees, independent contractors, consultants, commercial partners, or vendors violate these laws we could face substantial penalties.

Corporate and Other Information

We were initially formed on July 1, 2009 as Shoulder Innovations, LLC, a Delaware limited liability company. On February 10, 2017, we completed a corporate conversion where Shoulder Innovations, LLC was converted into a

C-corporation under the laws of the State of Delaware and changed our name to Shoulder Innovations, Inc. Our principal executive offices are located at 1535 Steele Avenue SW, Suite B, Grand Rapids, MI 49507, and our telephone number is (616) 294-1026. Our corporate website address is www.shoulderinnovations.com. Information contained on, or accessible through, our website shall not be deemed incorporated into and is not a part of this prospectus or the registration statement of which it forms a part. We have included our website in this prospectus solely as an inactive textual reference.

Implications of Being an Emerging Growth Company and a Smaller Reporting Company

We are an emerging growth company as defined in the Jumpstart Our Business Startups Act of 2012 (the "JOBS Act"). We will remain an emerging growth company until the earliest of: (i) the last day of the fiscal year following the fifth anniversary of the consummation of this offering; (ii) the last day of the fiscal year in which we have total annual gross revenue of at least $1.235 billion; (iii) the last day of the fiscal year in which we are deemed to be a "large accelerated filer" as defined in Rule 12b-2 under the Securities Exchange Act of 1934, as amended (the "Exchange Act"), which would occur if the market value of our common stock held by non-affiliates exceeded $700.0 million as of the last business day of the second fiscal quarter of such year; or (iv) the date on which we have issued more than $1.0 billion in non-convertible debt securities during the prior three-year period. An emerging growth company may take advantage of specified reduced reporting requirements and is relieved of certain other significant requirements that are otherwise generally applicable to public companies. As a result, the information that we provide to our stockholders may be different than you might receive from other public reporting companies in which you hold equity interests.

As an emerging growth company, we have elected to take advantage of certain reduced disclosure obligations in the registration statement that this prospectus is a part of, and may elect to take advantage of other reduced reporting requirements in future filings. In particular:

- we will present in this prospectus only two years of audited financial statements, plus any required unaudited financial statements, and related section titled "Management's Discussion and Analysis of Financial Condition and Results of Operations";
- we will avail ourselves of the exemption from the requirement to obtain an attestation and report from our independent registered public accounting firm on the assessment of our internal control over financial reporting pursuant to the Sarbanes-Oxley Act of 2002, as amended (the "Sarbanes-Oxley Act");
- we will avail ourselves of relief from compliance with the requirements of the Public Company Accounting Oversight Board regarding the communication of critical audit matters in the auditor's report on the financial statements;
- we will provide less extensive disclosure about our executive compensation arrangements; and
- we will not be required to hold stockholder non-binding advisory votes on executive compensation or golden parachute arrangements.

In addition, the JOBS Act provides that an emerging growth company can take advantage of an extended transition period for complying with new or revised accounting standards. This provision allows an emerging growth company to delay the adoption of some accounting standards until those standards would otherwise apply to private companies. We have elected to use the extended transition period for any other new or revised accounting standards during the period in which we remain an emerging growth company; however, we may adopt certain new or revised accounting standards early.

We are also a "smaller reporting company" as defined in the Exchange Act. We may continue to be a smaller reporting company even after we are no longer an emerging growth company. We may take advantage of certain

of the scaled disclosures available to smaller reporting companies and will be able to take advantage of these scaled disclosures for so long as our voting and non-voting common stock held by non-affiliates is less than $250.0 million measured on the last business day of our second fiscal quarter, or our annual revenue is less than $100.0 million

during the most recently completed fiscal year and our voting and non-voting common stock held by non-affiliates is less than $700.0 million measured on the last business day of our second fiscal quarter.

The Offering

Common stock offered by us	5,000,000 shares.
Underwriters' over-allotment option of common stock offered by us	750,000 shares.
Common stock to be outstanding immediately after this offering	20,228,510 shares (or 20,978,510 shares if the underwriters exercise their over-allotment option in full).
Use of proceeds	We estimate that the net proceeds from this offering will be approximately $64.8 million (or approximately $75.2 million if the underwriters exercise their over-allotment option in full), based on the initial public offering price of $15.00 per share and after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us. We currently intend to use the net proceeds from this offering (i) to scale up our commercial organization through the hiring of additional sales representatives and expansion of our commercial leadership team, as well as to invest in additional instrument sets, (ii) to fund the research and development of continued general innovation in our implant systems and (iii) for working capital and other general corporate purposes. We will have broad discretion in the way that we use the net proceeds from this offering. See the section titled "Use of Proceeds" for additional information.
Directed share program	At our request, the underwriters have reserved up to 6% of the shares of common stock offered hereby, at the initial public offering price, to offer to certain of our directors, officers and employees and others. The sales will be made at our direction by Morgan Stanley & Co. LLC and its affiliates through a directed share program. The number of shares of common stock available for sale to the general public will be reduced to the extent these individuals purchase such reserved shares. Any reserved shares that are not so purchased will be offered by the underwriters to the general public on the same terms as the other shares offered by this prospectus. Except for any shares acquired by our directors and officers, shares purchased pursuant to the directed share program will not be subject to lock-up agreements with the underwriters. See the section titled "Underwriting—Directed Share Program" for additional information.
Risk factors	Investing in our common stock involves a high degree of risk. See the section titled "Risk Factors" for a discussion of risks you should consider carefully, together with all the other information included in this prospectus, before deciding to invest in our common stock.
New York Stock Exchange trading symbol	"SI"

The number of shares of our common stock to be outstanding after this offering is based on 15,228,510 shares of our common stock outstanding as of March 31, 2025, after giving effect to (i) the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) into shares of our common stock and (ii) the issuance of the 2025 Convertible Notes and the related Notes Conversion (as defined below), and excludes:

- 1,310,108 shares of our common stock issuable upon the exercise of outstanding stock options under our Stock Option Plan as of March 31, 2025, with a weighted-average exercise price of $2.08 per share;

- 461,236 shares of our common stock issuable upon the exercise of outstanding stock options granted subsequent to March 31, 2025 under our Stock Option Plan, with a weighted-average exercise price of $2.86 per share;
- 17,827 shares of our common stock issuable upon the exercise of a warrant to purchase shares of our common stock outstanding as of March 31, 2025, with an exercise price of $2.10 per share;
- 45,859 shares of our common stock issuable upon the exercise of warrants to purchase 875,000 shares of our Series B convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.372 per share of Series B convertible preferred stock.
- 130,736 shares of our common stock issuable upon the exercise of a warrant to purchase 2,494,457 shares of our Series D convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.54120 per share of Series D convertible preferred stock, which will convert into a warrant to purchase shares of our common stock immediately prior to the completion of this offering; and
- 2,743,112 shares of our common stock reserved for future issuance under our equity compensation plans, consisting of:
    - 2,493,739 shares of our common stock reserved for future issuance under our 2025 Incentive Award Plan (the "2025 Plan"), which became effective upon the effectiveness of the registration statement of which this prospectus forms a part (which number includes 46,000 shares of our common stock subject to restricted stock unit awards granted to certain of our directors pursuant to our 2025 Plan in connection with this offering; and
    - 249,373 shares of our common stock reserved for future issuance under our 2025 Employee Stock Purchase Plan (the "ESPP"), which became effective upon the effectiveness of the registration statement of which this prospectus forms a part.

Unless otherwise indicated, all information contained in this prospectus, including the number of shares of common stock that will be outstanding after this offering, assumes or gives effect to:

- the filing and effectiveness of our amended and restated certificate of incorporation, which will occur immediately prior to the completion of this offering;
- the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) into an aggregate of 12,320,676 shares of our common stock immediately prior to the completion of this offering;
- the automatic conversion of a warrant to purchase shares of our Series D convertible preferred stock outstanding as of March 31, 2025 into a warrant to purchase 130,736 shares of our common stock immediately prior to the completion of this offering (the "Warrant Conversion");
- the issuance of 2,805,049 shares of our common stock upon the conversion of our 2025 Convertible Notes, based on the initial public offering price of $15.00 per share, without giving effect to accrued interest, which conversion will occur upon the completion of this offering (the "Notes Conversion").;
- a 1-for-19.08 reverse stock split of our common stock, which we effected on July 23, 2025;
- no exercise of the outstanding options and warrants referred to above;
- no exercise by the underwriters of their over-allotment option; and

- no purchase of shares of our common stock by our directors, officers and employees through the directed share program described in the section titled "Underwriting—Directed Share Program."

Summary Financial Data

The following tables summarize our financial data for the periods and as of the dates indicated. The following summary statements of operations and comprehensive loss data for the years ended December 31, 2024 and 2023 and balance sheet data as of December 31, 2024, except for pro forma and pro forma as adjusted amounts, have been derived from our audited financial statements and the related notes included elsewhere in this prospectus. The summary statement of operations and comprehensive loss data for the three months ended March 31, 2025 and 2024 and the balance sheet data as of March 31, 2025 have been derived from our unaudited interim condensed financial statements that are included elsewhere in this prospectus. Our historical results are not necessarily indicative of the results that may be expected for any period in the future. You should read the following summary financial data together with our audited financial statements and the related notes included elsewhere in this prospectus and the section titled "Management's Discussion and Analysis of Financial Condition and Results of Operations." The summary financial data included in this section are not intended to replace the financial statements and the related notes included elsewhere in this prospectus.

Statements of Operations and Comprehensive Loss Data

	Three Months Ended March 31,		Year Ended December 31,	
	2025	2024	2024	2023
	(unaudited)			
	(in thousands, except for share and per share data)			
Net revenue	$ 10,132	$ 7,184	$ 31,623	$ 19,274
Cost of goods sold	2,341	1,666	7,282	4,004
Gross profit	7,791	5,518	24,341	15,270
Operating expenses:				
Selling, general and administrative expenses[1]	10,502	7,704	34,505	23,113
Research and development expenses	1,583	1,070	4,489	3,021
Total operating expenses	12,085	8,774	38,994	26,134
Operating loss	(4,294)	(3,256)	(14,653)	(10,864)
Other (income) expense:				
Interest expense, net	367	317	1,316	761
Loss on extinguishment of convertible promissory notes	—	—	—	1,127
Other (income) expense, net	1	27	(350)	(97)
Total other expense	368	344	966	1,791
Loss before income tax expense	(4,662)	(3,600)	(15,619)	(12,655)
Income tax expense	—	—	—	—
Net loss	$ (4,662)	$ (3,600)	$ (15,619)	$ (12,655)
Other comprehensive (loss) income, net:				
Unrealized (loss) gain on marketable securities	(116)	(42)	(159)	356
Total other comprehensive (loss) income, net	(116)	(42)	(159)	356
Comprehensive loss	$ (4,778)	$ (3,642)	$ (15,778)	$ (12,299)
Net loss per share attributed to common stock – basic and diluted[2]	$ (52.13)	$ (62.77)	$ (242.04)	$ (220.65)
Weighted-average shares of common stock outstanding – basic and diluted[2]	89,438	57,354	64,530	57,354

	Three Months Ended March 31,		Year Ended December 31,	
	2025	2024	2024	2023
	(unaudited)			
	(in thousands, except for share and per share data)			
Pro forma net loss per share of common stock, basic and diluted (unaudited)[3]	$ (0.48)		$ (1.69)	
Pro forma weighted-average shares of common stock outstanding, basic and diluted (unaudited)[3]	9,734,510		9,246,405	

(1) Includes stock-based compensation expense of $127 thousand and $168 thousand for the three months ended March 31, 2025 and 2024, respectively, and $0.8 million and $0.5 million for the years ended December 31, 2024 and 2023, respectively.

(2) Please see Note 11 to our audited financial statements and Note 6 to our unaudited financial statements included elsewhere in this prospectus for details on the calculation of basic and diluted net loss per share attributed to common stock.

(3) Pro forma net loss per share attributed to common stock, basic and diluted, is calculated giving effect to the conversion of (i) all outstanding shares of our convertible preferred stock into shares of our common stock; and (ii) all outstanding warrants into warrants to purchase shares of our common stock. Pro forma net loss per share attributed to common stock does not include the shares expected to be sold and related proceeds to be received in this offering or the issuance of the 2025 Convertible Notes and the related Notes Conversion.

Balance Sheet Data

	As of March 31, 2025		
	Actual	Pro Forma[1]	Pro Forma As Adjusted[2]
	(unaudited) (in thousands, except for share data)		
Balance Sheet Data:			
Cash, cash equivalents and marketable securities	$ 27,564	$ 87,564	$ 152,314
Working capital[3]	$ 42,622	$ 102,622	$ 167,372
Total assets	$ 57,524	$ 117,524	$ 182,274
Convertible preferred stock warrant liability	$ 970	$ —	$ —
Convertible preferred stock	$ 94,147	$ —	$ —
Accumulated deficit	$ (61,703)	$ (61,703)	$ (61,703)
Total stockholders' (deficit) equity	$ (59,314)	$ 95,803	$ 160,553

(1) Pro forma amounts give effect to (i) the receipt of approximately $20.0 million in aggregate gross proceeds from the issuance and sale of our Series E convertible preferred stock in June 2025; (ii) the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) into an aggregate of 12,320,676 shares of our common stock; (iii) the Warrant Conversion and the related reclassification of warrant liability to stockholders' equity; (iv) the issuance of the 2025 Convertible Notes and the related Notes Conversion; and (v) the filing and effectiveness of our amended and restated certificate of incorporation, which will become effective immediately prior to the completion of this offering.

(2) Pro forma as adjusted amounts give effect to (i) the pro forma adjustments set forth in footnote (1) above and (ii) the issuance and sale of 5,000,000 shares of common stock in this offering, at the initial public offering price of $15.00 per share, after deducting the estimated underwriting discounts and commissions and estimated offering expenses payable by us.

(3) We define working capital as current assets less current liabilities. See our financial statements and the related notes included elsewhere in this prospectus for further details regarding our current assets and current liabilities.

Key Business Metrics

The following table sets forth the number of implant systems sold in each of the three-month periods indicated:

	Three Months Ended									
	Mar. 31, 2023	June 30, 2023	Sept. 30, 2023	Dec. 31, 2023	Mar. 31, 2024	June 30, 2024	Sept. 30, 2024	Dec. 31, 2024	Mar. 31, 2025	June 30, 2025
Implant systems sold	559	564	618	943	971	1,121	1,037	1,220	1,443	1,503

Non-GAAP Financial Measures

In addition to our results and measures of performance determined in accordance with U.S. GAAP, we believe that non-GAAP financial measures can be useful in evaluating and comparing our financial and operational performance over multiple periods, identifying trends affecting our business, formulating business plans and making strategic decisions. We utilize and present Adjusted EBITDA for these purposes. We define Adjusted EBITDA as net loss before interest expense, net, income tax expense, depreciation and amortization, stock-based compensation expense and loss on extinguishment of convertible promissory notes. The following table presents Adjusted EBITDA for each of the periods indicated:

	Three Months Ended March 31,		Year Ended December 31,	
	2025	2024	2024	2023
	(in thousands)			
Adjusted EBITDA[1]	$ (3,500)	$ (2,617)	$ (11,353)	$ (8,675)

(1) Adjusted EBITDA is a non-GAAP financial measure. For a reconciliation of Adjusted EBITDA to the most directly comparable U.S. GAAP financial measure, information about why we consider such measure useful and a discussion of the material risks and limitations of such measure, please see "Management's Discussion and Analysis of Financial Condition and Results of Operations—Non-GAAP Financial Measures."

RISK FACTORS

Investing in our common stock involves a high degree of risk. You should carefully consider the risks described below, as well as the other information in this prospectus, including our financial statements and the related notes included elsewhere in this prospectus and the section titled "Management's Discussion and Analysis of Financial Condition and Results of Operations," before deciding whether to invest in our common stock. The occurrence of any of the events or developments described below could have a material adverse effect on our business, financial condition, results of operations, and prospects. In such an event, the market price of our common stock could decline, and you may lose all or part of your investment. Please also see the section titled "Special Note Regarding Forward-Looking Statements." Additional risks and uncertainties not presently known to us or that we currently believe are not material may also impair our business, financial condition, results of operations and prospects.

Risks Related to Our Financial Condition and Capital Requirements

We are an early-stage company with a history of significant net losses, we expect to incur operating losses in the future and we may not be able to achieve or sustain profitability.

We have incurred annual net losses since our inception, and we expect to incur additional substantial losses in the foreseeable future. For the years ended December 31, 2023 and 2024, we had a net loss of $12.7 million and $15.6 million, respectively. For the three months ended March 31, 2025 and 2024, we had a net loss of $4.7 million and $3.6 million, respectively. As of March 31, 2025, we had an accumulated deficit of $61.7 million. To date, we have financed our operations primarily through equity and debt financings and from sales of our shoulder replacement implants. The losses and accumulated deficit have primarily been due to the substantial investments we have made to develop our implant systems, costs related to our sales and marketing efforts and general research and development expenses, including costs related to regulatory initiatives to obtain marketing approval. These investments may be more costly than we expect, and if we do not achieve the benefits anticipated from these investments, or if the realization of these benefits is delayed or not enough to offset our higher operating expenses, such investments may not result in increased revenue or growth in our business, which will lead to increased net losses. In addition, as a public company, we will incur significant legal, accounting and other expenses that we did not incur as a private company. Accordingly, we expect to continue to incur operating losses for the foreseeable future. Our failure to achieve and sustain profitability in the future will make it more difficult to finance our business and accomplish our strategic objectives, which would have a material adverse effect on our business, financial condition and results of operations and cause the market price of our common stock to decline. In addition, failure of our solutions to significantly penetrate the target markets would negatively affect our business, financial condition and results of operations.

If we fail to manage our growth effectively, our business could be materially and adversely affected.

We have experienced recent rapid growth and anticipate further growth in the future. For example, for the year ended December 31, 2023, our net revenue was $19.3 million compared to our net revenue of $31.6 million for the year ended December 31, 2024, an increase of 64% year-over-year. For the three months ended March 31, 2025 our net revenue was $10.1 million compared to our net revenue of $7.2 million for the three months ended March 31, 2024, representing an increase of 41.0%. This growth has placed significant demands on our management, financial, operational, technological and other resources, and we expect that our future growth will continue to place significant demands on our management and other resources and will require us to continue developing and improving our operational, financial and other internal controls. In particular, continued growth increases the challenges involved in a number of areas, including managing our supplier relationships and ensuring adequate inventory is available, recruiting and retaining sufficient skilled personnel for our dedicated commercial leadership team, contracting with independent distributors to market our implant systems, providing adequate training and supervision to maintain our high-quality standards and preserving our culture and values. We may not be able to address these challenges in a cost-effective manner, or at all. To achieve our revenue goals, we must also successfully increase our supply of implants and other products, components and instruments from third party manufacturers to meet expected customer demand. In the future, we may experience difficulties with quality control, component supply and shortages of qualified personnel, among other problems. These problems could result in delays in product availability and increases in expenses. Any such delay or increased expense could adversely affect

our business, financial condition and results of operations. In addition, rapid and significant growth will place a strain on our administrative and operational infrastructure. In order to manage our operations and growth, we will need to continue to improve our operational and management controls, hiring process, reporting and information technology systems and financial internal control procedures. If we do not effectively manage our growth, we may not be able to execute on our business plan, respond to competitive pressures, take advantage of market opportunities, satisfy customer requirements or maintain high-quality product offerings, which could have a material adverse effect on our business, financial condition and results of operations.

We have a significant amount of debt, which may affect our ability to operate our business and secure additional financing in the future.

As of March 31, 2025, we had an aggregate of approximately $15.0 million in principal outstanding under our term loan facility (the "Trinity Loan Agreement") with Trinity Capital Inc. ("Trinity Capital"). In July 2025, we issued $40.0 million aggregate principal amount of our 2025 Convertible Notes. The Trinity Loan Agreement, which we entered into in August 2023, as amended by the First Amendment to the Loan and Security Agreement on April 23, 2024 and the Second Amendment to the Loan and Security Agreement on July 21, 2025, consists of term loans of up to $45.0 million and bears interest at an annual rate equal to the greater of the prime rate plus 3.50% and 11.00%. Under the Trinity Loan Agreement, the prime rate is equal to the greater of 8.0% per year and the prime rate as reported in The Wall Street Journal. We must make interest payments under the Trinity Loan Agreement, which has diverted and will continue to divert resources from other activities. For the three months ended March 31, 2025, we incurred interest expense, net of $0.4 million, which included payments made under the Trinity Loan Agreement. Our obligations under the Trinity Loan Agreement are collateralized by substantially all of our assets, including intellectual property, and we are subject to customary affirmative and negative covenants limiting our ability to, among other things, relocate or dispose of assets, undergo a change in control, merge or consolidate, enter into certain transactions with affiliates, make acquisitions, incur debt, pay dividends, grant liens, store certain amounts of inventory or equipment with third parties and make investments, in each case subject to certain exceptions.

The covenants related to the Trinity Loan Agreement, as well as any future financing agreements into which we may enter, may restrict our ability to finance our operations and engage in, expand or otherwise pursue our business activities and strategies.

While we have not previously breached and are not currently in breach of these or any other covenants contained in our Trinity Loan Agreement or other debt arrangements, there can be no guarantee that we will not breach these covenants in the future. Our ability to comply with these covenants may be affected by events beyond our control, and future breaches of any of these covenants could result in a default under the Trinity Loan Agreement. If not waived, future defaults could cause all of the outstanding indebtedness under the Trinity Loan Agreement to become immediately due and payable and terminate commitments to extend further credit and foreclose on the collateral granted to it to collateralize such indebtedness. If we do not have or are unable to generate sufficient cash available to repay our debt obligations when they become due and payable, either upon maturity or in the event of a default, our assets could be foreclosed upon and we may not be able to obtain additional debt or equity financing on favorable terms, if at all, which may negatively impact our ability to operate and continue our business as a going concern.

If we default under the Trinity Loan Agreement, the Trinity Capital on behalf of the lenders will be able to declare all obligations immediately due and payable and take control of our pledged assets, potentially requiring us to renegotiate our agreement on terms less favorable to us or to immediately cease operations. Further, if we are liquidated, the lender's rights to repayment would be senior to the rights of the holders of our common shares to receive any proceeds from the liquidation. The lenders could declare a default under the Trinity Loan Agreement upon the occurrence of specific events such as our failure to pay or our failure to comply with specified covenants, thereby requiring us to repay the loan immediately. Any declaration by the lenders of an event of default could significantly harm our business and prospects and could cause the price of our common shares to decline.

In order to service this indebtedness and any additional indebtedness we may incur in the future, we need to generate cash from our operating activities. Our ability to generate cash is subject, in part, to our ability to

successfully execute our business strategy, as well as general economic, financial, competitive, regulatory and other factors beyond our control. We cannot assure you that our business will be able to generate sufficient cash flow from operations or that future borrowings or other financings will be available to us in an amount sufficient to enable us to service our indebtedness and fund our other liquidity needs. To the extent we are required to use cash from operations or the proceeds of any future financing to service our indebtedness instead of funding working capital, capital expenditures or other general corporate purposes, we will be less able to plan for, or react to, changes in our business, industry and in the economy generally. This may place us at a competitive disadvantage compared to our competitors that have less indebtedness.

Also, the anticipated benefit of any strategic alliance, joint venture or acquisition may not materialize or such strategic alliance, joint venture or acquisition may be prohibited. The Trinity Loan Agreement restricts our ability to pursue certain mergers, acquisitions, amalgamations or consolidations that we may believe to be in our best interest. Additionally, future acquisitions or dispositions could result in potentially dilutive issuances of our equity securities, the incurrence of debt, contingent liabilities or amortization expenses or write-offs of goodwill, any of which could harm our financial condition. We cannot predict the number, timing or size of future joint ventures or acquisitions, or the effect that any such transactions might have on our operating results.

We may acquire other complementary companies, products or technologies, through the licensing of products or technologies from third parties or other strategic alliances, and the failure to manage acquisitions, investments, licenses or other strategic alliances, or the failure to integrate them with our existing business could fail to result in a commercial product or net sales, divert our management's attention, result in additional dilution to our stockholders, cause us to incur significant expense and otherwise disrupt our business.

Our success depends on our ability to continually enhance and broaden our system offerings in response to changing surgeon and patient needs, competitive technologies and market pressures. Accordingly, from time to time we may consider opportunities to acquire, make investments in or license other technologies, products and businesses that may enhance our capabilities, complement our existing products and technologies or expand the breadth of our markets or customer base. Although we currently have no agreements or commitments to complete any such transactions, we may in the future seek to acquire or invest in businesses, applications or technologies that we believe could complement or expand our portfolio, enhance our technical capabilities or otherwise offer growth opportunities. However, we cannot assure you that we would be able to successfully complete any acquisition we choose to pursue on favorable terms, if at all, or that we would be able to successfully integrate any acquired business, product or technology in a cost-effective and non-disruptive manner. Potential and completed acquisitions, strategic investments, licenses and other alliances involve numerous risks, including:

- difficulty assimilating or integrating acquired or licensed technologies, products, employees or business operations;
- issues maintaining uniform standards, procedures, controls and policies;
- unanticipated costs associated with acquisitions or strategic alliances, including the assumption of unknown or contingent liabilities and the incurrence of debt or future write-offs of intangible assets or goodwill;
- diversion of management's attention from our core business and disruption of ongoing operations;
- adverse effects on existing business relationships with suppliers, independent distributors, health care facilities, shoulder specialists and other health care providers;
- risks associated with entering new markets in which we have limited or no experience;
- potential losses related to investments in other companies;
- potential loss of key employees of acquired businesses;
- unanticipated or undisclosed liabilities of any target; and
- increased legal and accounting costs relating to the acquisitions or compliance with regulatory matters.

If we do acquire, make investments in or license other technologies, products and businesses, we may not ultimately obtain the expected benefits of such acquisition or investment, strengthen our competitive position or achieve our goals, and any acquisitions we complete could be viewed negatively by our customers, investors and industry analysts.

To date, the growth of our operations has been wholly organic, and we have limited experience in acquiring other businesses or technologies. We do not know if we will be able to identify acquisitions or strategic relationships we deem suitable, whether we will be able to successfully complete any such transactions on favorable terms, if at all, or whether we will be able to successfully integrate any acquired personnel, operations and technologies, or effectively manage the combined business following an acquisition. Our ability to successfully grow through strategic transactions depends upon our ability to identify, negotiate, complete and integrate suitable target businesses, technologies or products and to obtain any necessary financing. Acquisitions could also result in dilutive issuances of equity securities, the use of our available cash, or the incurrence of debt, which could harm our operating results. To finance any acquisitions, investments or strategic alliances, we may choose to issue shares of our common stock as consideration, which could dilute the ownership of our stockholders. If the price of our common stock is low or volatile, we may be unable to consummate any acquisitions, investments or strategic alliances using our common stock as consideration. Additional funds may not be available on terms that are favorable to us, or at all. In addition, if an acquired business fails to meet our expectations, our business, financial condition and results of operations may be negatively affected. These efforts could be expensive and time-consuming and may disrupt our ongoing business and prevent management from focusing on our operations.

We may require substantial additional funding, which may not be available to us on acceptable terms, or at all, and, if not available, may require us to delay, scale back, or cease our innovation efforts or operations.

Even if this offering is successful, we may require additional capital in the future to maintain and expand our operations. Our operations are capital-intensive and are expected to increase as we expand our dedicated commercial leadership team, research and development efforts and product offerings, among other growth initiatives. Changing circumstances could result in lower revenues or cause us to consume capital significantly faster than we currently anticipate, and we may need to raise capital sooner or in greater amounts than currently expected because of circumstances beyond our control. If we raise additional funds through the issuance of equity, equity-linked or debt securities, those securities may have rights, preferences or privileges senior to those of our common stock, and our existing stockholders may experience dilution. Any debt financing secured by us in the future could require that a substantial portion of our operating cash flow be devoted to the payment of interest and principal on such indebtedness, which may decrease available funds for other business activities, and could involve restrictive covenants relating to our capital-raising activities and other financial and operational matters, which may make it more difficult for us to obtain additional capital and to pursue business opportunities. Additionally, any future collaborations we enter into with third parties may provide capital in the near term but limit our potential cash flow and revenue in the future. If we raise additional funds through strategic partnerships and alliances and licensing arrangements with third parties, we may have to relinquish valuable rights to our technologies, intellectual property or products, or grant licenses on terms unfavorable to us. We cannot be certain that we will be able to obtain additional financing on favorable terms, if at all. If we cannot raise funds on acceptable terms, if and when needed, we may not be able to continue as a going concern or we may not be able to grow our business or respond to competitive pressures or unanticipated requirements, which could seriously harm our business.

Risks Related to Our Business and Industry

We operate in a very competitive business environment, and if we are unable to compete successfully against our existing or potential competitors, our business, financial condition and results of operations may be adversely affected.

Our existing implant systems are, and any new products or procedures we develop and commercialize will be, subject to intense competition. There are a number of competing products that can be used in shoulder arthroplasty procedures. The industry in which we operate is competitive, subject to change and sensitive to the introduction of new products, procedures or other market activities of industry participants. Our ability to compete successfully will depend on our ability to continue to train surgeons on the use of our implant systems and other technologies for

shoulder surgical care and gain their acceptance of our systems or ecosystem, develop additional products to improve shoulder surgical care and expand our offerings that reach the market in a timely manner, receive adequate coverage and reimbursement from third-party payors and provide systems, technologies and instrumentation that are easier to use, safer, less invasive and perceived more effective than the products of our competitors. In addition, our ability to increase our customer base and achieve broader market acceptance of our products will depend to a significant extent on our ability to expand our marketing efforts. We plan to dedicate significant resources to our marketing and surgeon-to-surgeon collaboration programs, but if our marketing efforts and expenditures do not generate a corresponding increase in net revenue, it will negatively affect our business, financial condition and results of operations. In addition, we believe that developing and maintaining broad awareness of our systems in a cost-effective manner is critical to achieving broad acceptance and expanding our business in our target markets. Promotion activities may not generate sufficient or any awareness or increase net revenue, and even if they do, any increase in net revenue may not offset the costs and expenses we incur in building our brand and conducting such promotion activities. If we fail to successfully promote, maintain and protect our brand, we may fail to attract or retain the physician acceptance necessary to realize a sufficient return on our brand building efforts, or to achieve a level of brand awareness that supports broad adoption of our systems.

We compete with large, multinational companies, as well as with smaller orthopedic companies that offer innovations and improvements on traditional shoulder implants and technologies. For example, we compete with Arthrex, Enovis, Exectech, Johnson & Johnson, Smith & Nephew, Stryker, and Zimmer Biomet. Furthermore, conditions in our market could change rapidly and significantly as a result of technological advancements, partnerships, or acquisitions by competitors. We have seen and continue to see consolidation amongst our competitors. If our competitors have greater resources and access to funding than us, they may be able to finance the development of new technologies and products before we are able to do so, which may allow them to gain market share or enter new markets before us or provide lower-priced or better-quality offerings.

At any time, these competitors and other potential market entrants may develop new products, procedures or treatment alternatives that could render our products obsolete, uncompetitive or economically unviable. In addition, one or more of such competitors may gain a market advantage by developing and patenting competitive products, procedures or treatment alternatives earlier than we can, obtaining regulatory clearances or approvals more rapidly than we can or selling competitive products at prices lower than ours. If medical research were to lead to the discovery of alternative therapies or technologies that improve or cure certain shoulder diseases, including through the use of pharmaceuticals or breakthrough bio-technological innovations or therapies, our profitability could suffer through a reduction in sales or a loss in market share to a competitor. The discovery of methods of prevention or the development of other alternatives to our systems, or novel methods of conducting shoulder arthroplasty or other procedures, could result in decreased demand for shoulder arthroplasty procedures and our systems, and, accordingly, could have a material adverse effect on our business, financial condition and results of operations. Many of our current and potential competitors have substantially greater sales and financial resources than we do. These competitors may also have more established distribution networks, a broader offering of products, entrenched relationships with surgeons and distributors or greater experience in launching, marketing, distributing and selling products or treatment alternatives.

We also compete with our competitors to engage the services of independent distributors, both those presently working with us and those with whom we hope to work with as we expand. These independent distributors may also sell shoulder care products that our competitors offer. In addition, we compete with our competitors in acquiring technologies and technology licenses complementary to our products or procedures or advantageous to our business. If we are unable to compete successfully against our existing or potential competitors, our business, financial condition and results of operations may be adversely affected, and we may not be able to grow at our expected rate, if at all.

If we fail to develop and retain an effective commercial organization, or if we are unable to successfully expand our organization, it could negatively impact our sales, and we may not generate sufficient net revenue to sustain profitability.

Our net revenue and profitability is directly dependent upon the sales and marketing efforts of our commercial organization. Our commercial organization is comprised of a team of dedicated commercial leaders, our CEME team

and a network of independent distributors. In order to expand our business, we plan to expand our dedicated commercial leadership team with individuals who have strong technical backgrounds specializing in sales and marketing of products for shoulder arthroplasty procedures and within the shoulder surgical care market. As we increase our marketing efforts, we will need to retain, develop and grow the number of team members that we employ. We intend to invest in recruiting and training dedicated commercial leadership team members and clinical representatives as we expand our business. There is significant competition for personnel experienced in relevant medical device sales and in shoulder surgical care in particular. Once hired, the training process can be lengthy because it requires significant education for new dedicate commercial leadership team members and clinical specialists to achieve the level of clinical competency and expertise that is expected by surgeons that use our systems. In addition, team members typically require at least one year if not longer in the field to grow their network of accounts and achieve the productivity levels we expect them to reach in any individual territory. Our future success will depend largely on our ability to continue to hire, train, retain and motivate skilled members of our commercial organization with significant technical knowledge. If we are unable to attract, motivate, develop and retain a sufficient number of such team members, or if these team members do not achieve the productivity levels we expect them to reach, our net revenue will not grow at the rate we expect and our financial performance will suffer. Also, to the extent we hire personnel from our competitors, we may have to wait until applicable non-competition provisions have expired before deploying such personnel in restricted territories or incur costs to relocate personnel outside of such territories, and we have been in the past, and may be subject to future allegations that these new hires have been improperly solicited, and that they have divulged to us proprietary or other confidential information of their former employers. Additionally, because the market for our target team members is competitive, our competitors may try to hire our team members from us. If successful, we would be required to dedicate resources to recruiting, filling and training those vacant positions. Any of these risks may adversely affect our business.

We rely in part on independent distributors to sell our products to our customers, and if we are unable to maintain and expand our network of independent distributors, we may be unable to generate anticipated sales.

Within our commercial organization and as part of our commercial approach, we use independent distributors to sell our implant systems to hospitals, ASCs and other end users and to assist us in promoting market acceptance of, and creating demand for, our systems. Our distributor network is currently comprised of over 41 independent groups and 150 agents across the United States. If we are unable to come to commercially reasonable terms with a distributor and/or its agents, we may not generate the expected level of sales and may need to spend more of our capital resources to hire sales personnel as employees or to engage one or more alternative distributors in a particular region or market. In addition, our independent distributors that we contract with are not all exclusive to us and there is a risk that an independent distributor will give higher priority to the products of other medical device companies, including products directly competitive with our systems or may be required by larger medical devices companies to stop offering our systems. Though we have established initiatives to further focus our independent distributor channel on our systems, these initiatives may not translate to an increase in sales or penetration or may underperform relative to our expectations. In addition, we may not have insight into competitive lines at our distributors, including other commercial arrangements between such distributors and our competitors to sell our competitor's products. As such, there can be no assurance that an agent will devote the resources necessary to provide effective sales and promotional support to our systems. In addition, if an agent terminates its relationship with us and is retained by one of our competitors, we may be unable to prevent them from helping competitors solicit business from our existing customers and surgeons, which could adversely affect our sales and growth. In addition to expanding our internal commercial team members, we expect to continue to utilize and expand our relationships with independent distributors and agents to meet our growing business needs and accomplish our business strategies.

Our business plan relies on certain assumptions about the market for our implant systems, however, the size and expected growth of our addressable market has not been established with precision and may be smaller than we estimate, and even if the addressable market is as large as we have estimated, we may not be able to capture additional market share.

Our estimates of our addressable market are based on publicly-available information, a number of internal and third-party estimates and assumptions, including the prevalence of aTSA and rTSA procedures and management's

knowledge and experience in the shoulder surgical care market. For example, we believe that the aging of the general population and increasingly active lifestyles will continue and that these trends will increase the demand for effective devices and implant systems to be used in connection with shoulder arthroplasty procedures. Furthermore, we currently only operate in the United States, which is the largest market for shoulder arthroplasties, and any contraction in the this market or incorrect assumptions about the current and future development of the market would have a material impact on our results of operations and financial condition. While we believe our assumptions and the data underlying our estimates are reasonable, these assumptions and our estimates may not be correct. The projected growth of annual shoulder arthroplasty procedures, including in ASCs, and demand for our systems could materially differ from actual demand if our assumptions regarding these trends and acceptance of our systems by the medical community prove to be incorrect or do not materialize, or if non-surgical treatments or other surgical techniques gain more widespread acceptance as a viable alternative to our systems or products. In addition, even if the number of people who undergo shoulder arthroplasty procedures increases as we expect, technological or medical advances could provide alternatives to address shoulder conditions and reduce demand for shoulder arthroplasty procedures or for our particular systems. As a result, our estimates of the addressable market for our current or future products may prove to be incorrect. Even if the total addressable market for our current and future products is as large as we have estimated, we may not be able to penetrate this market to capture additional market share for the reasons discussed in this "Risk Factors" section. The estimated market opportunities included in this prospectus represent the total overall revenue opportunity that we believe is available for our systems if 100% market share is achieved by us, and are not a representation that we will achieve any such market share. If the actual number of people who suffer from shoulder conditions who would benefit from shoulder arthroplasty procedures with our systems, the price at which we can sell our current and future products or the addressable market for our systems and future products is smaller than we estimate, or if the total addressable market is as large as we have estimated but we are unable to capture additional market share, it could have a material adverse effect on our business, financial condition and results of operations.

Our business is dependent upon the adoption of our implant systems by hospitals, ASCs, surgeons and patients.

Our future growth and profitability depend on our ability to increase physician awareness of our implant systems or ecosystem and on the willingness of surgeons and hospitals and ASCs to adopt our implant systems. Surgeons may not adopt our technologies unless they are able to determine, based on experience, clinical data, medical society and peer-surgeon recommendations and other analyses, that our implant systems provide a safe and effective treatment for shoulder care. Even if we can raise awareness among surgeons, such surgeons, hospitals and ASCs may be slow in changing their medical treatment practices and may be hesitant to select our systems for a variety of reasons, including:

- lack of experience with us and concerns that we are relatively new to market;
- long-standing relationships with companies and distributors that sell other products;
- lack of availability of adequate third-party payor coverage or reimbursement;
- competitive response and negative selling efforts from providers of alternative treatments;
- perception regarding the time commitment and skill development that may be required to gain familiarity and proficiency with our systems, technologies and instrumentation;
- perceived liability risk generally associated with the use of new products and treatment options in both hospitals as well as ASCs that lack ambulatory services and may expose patients to greater risk and us and care providers to increased potential liability;
- lack or perceived lack of sufficient clinical evidence, including long-term data, supporting clinical benefits or the cost-effectiveness of our systems over existing treatments; and
- the failure of key opinion leaders to provide recommendations regarding our systems, or to assure surgeons, patients and healthcare payors of the benefits of our systems as an attractive alternative to other treatment options.

To effectively market and sell our systems and other future products, we will need to continue to educate the medical community about the safety, efficacy, necessity and efficiency of our systems and about the patient population that would potentially benefit from using our systems. We will also need to educate the medical community, including at hospitals and ASCs, as to the broader benefits and efficiencies of our systems and our ecosystem. We cannot assure you that we will achieve broad education or market acceptance. In addition, some surgeons may choose to utilize our systems on only a subset of their total patient population or may not adopt our offerings at all. If we are not able to effectively demonstrate that our offerings are beneficial for a broad range of patients, adoption will be limited and may not occur as rapidly as we anticipate or at all, which would have a material adverse effect on our business, financial condition and results of operations. We cannot assure you that our implant systems will achieve broad market acceptance among hospitals, ASCs, surgeons or patients. Any such failure to satisfy demand or to achieve meaningful market acceptance and penetration will harm our future prospects and have a material adverse effect on our business, financial condition and results of operations.

The seasonality of our business creates variance in our quarterly revenue, which makes it difficult to compare or forecast our financial results.

Our revenue fluctuates on a seasonal basis, which affects the comparability of our results between periods. In particular, we have experienced and expect to continue to experience seasonality in our business. For example, the orthopedic industry traditionally experiences lower sales volumes in the months in and surrounding the third calendar quarter than throughout the rest of the year as elective procedures generally decline during the summer months due to warmer weather and its corresponding impact on individual lifestyles. Medical device companies historically experience a decline in the number of orthopedic implant surgeries in the summer months, and we may experience similar seasonality in the future. We expect these seasonal factors to become more pronounced in the future as our business grows. These seasonal variations are difficult to predict accurately, may vary amongst different markets and at times may be entirely unpredictable, which introduces additional risk into our business as we rely upon forecasts of customer demand to build inventory in advance of anticipated sales. In addition, we believe our limited history commercializing our products combined with our rapid growth has, in part, made our seasonal patterns more difficult to discern, making it more difficult to predict future seasonal patterns.

Our long-term growth depends on our ability to enhance our implant systems, expand our indications and develop and commercialize additional products in a timely manner. If we cannot innovate, we may not be able to develop or exploit new products in time to remain competitive.

The market for our systems is highly competitive, dynamic, and marked by rapid and substantial technological development and product innovation. New entrants or existing competitors could attempt to develop products that compete directly with ours. Our success depends on our ability to continually enhance and broaden our product offerings in response to changing surgeon and patient needs, competitive technologies and market pressures. For us to remain competitive, it is important to develop and bring to market new products. Any promotional activities for new products may not generate patient or physician awareness or increase revenue, and even if they do, any increase in revenue may not offset the costs and expenses we incur in such product development. Demand for our systems and other future products could be diminished by equivalent or superior products and technologies offered by competitors. If we are unable to innovate successfully, our products could become obsolete and our revenue would decline as our customers purchase our competitors' products. Developing products is expensive and time-consuming and could divert management's attention away from our core business. The success of any new product offering or product enhancements to our existing offerings will depend on several factors, including our ability to:

- assemble sufficient resources to acquire or discover additional products;
- properly identify and anticipate surgeon and patient needs;
- develop and introduce new products and product enhancements in a timely manner;
- avoid infringing upon the intellectual property rights of third-parties;
- demonstrate, if required, the safety and efficacy of new products;

- obtain the necessary regulatory clearances, approvals or certifications for new products or product modifications;
- be fully compliant with the FDA regulations and, to the extent we expand internationally, be fully compliant with applicable foreign regulations marketing of new devices or modified products;
- produce new products in commercial quantities at an acceptable cost;
- provide adequate training to potential users of our products;
- receive adequate coverage and reimbursement for procedures performed with our products; and
- develop an effective and dedicated commercial leadership and marketing team.

If we are unable to develop or improve products, applications or features due to constraints, such as insufficient cash resources, high employee turnover, inability to hire and retain personnel with sufficient technical skills or a lack of other research and development resources, we may not be able to maintain our competitive position compared to other companies. Furthermore, many of our competitors have the capability to devote a considerably greater amount of funds to their research and development programs than we do, and those that do not may be acquired by larger companies that could allocate greater resources to research and development programs. Our failure or inability to devote adequate research and development resources or compete effectively with the research and development programs of our competitors could harm our business.

Further, our newer products, such as stemless products and I-95 stems, as well as ProVoyance, may not gain a substantial degree of market acceptance among surgeons, patients or healthcare providers. Our failure to gain widespread of acceptance of our new products could negatively affect our business, financial condition and results of operations.

We may expend our limited resources to pursue particular improvements or enhancements or new products and fail to capitalize on development opportunities that may be more profitable or for which there is a greater likelihood of success.

Because we have limited financial and managerial resources, we may forego or delay pursuit of opportunities with developments or initiatives that later prove to have greater commercial potential. Our resource allocation decisions may cause us to fail to timely capitalize on viable commercial products or profitable market opportunities. Furthermore, our spending on current and future research and development programs and products for specific indications may not yield any commercially viable products. If we do not accurately evaluate the commercial potential or target market for a particular product, we may relinquish valuable rights to that product through collaboration, licensing or other royalty arrangements in cases in which it would have been more advantageous for us to retain sole development and commercialization rights to such product.

We face the risk of product liability claims that could be expensive, divert management's attention and harm our reputation and business. We may not be able to maintain adequate product liability insurance.

Our business exposes us to the risk of product liability claims that are inherent in the testing, manufacturing and marketing of medical devices. This risk exists even if a device is cleared or approved for commercial sale by the FDA or approved and manufactured in facilities licensed and regulated by the FDA. Our products are designed to affect, and any future enhancements to our products will be designed to affect, important bodily functions and processes. Any side effects, manufacturing defects, misuse or abuse associated with our systems could result in patient injury or death. The medical device industry has historically been subject to extensive litigation over product liability claims, and we may face product liability suits in the future. We may be subject to product liability claims if our systems cause, or merely appear to have caused, patient injury or death. In addition, an injury that is caused by the activities of our suppliers, such as those who provide us with components and raw materials, may be the basis for a claim against us. Product liability claims may be brought against us by patients, healthcare providers or others selling or otherwise coming into contact with our systems and other future products, among others. If we cannot successfully defend ourselves against product liability claims, we will incur substantial liabilities and reputational

harm. If any of our products become the subject of a product liability claim, legal defenses are costly, regardless of the outcome. Thus, we may experience increased legal expenses as we defend any such matter, and we could incur liabilities associated with adverse outcomes that exceed our insurance coverage. In addition, regardless of merit or eventual outcome, product liability claims may result in:

- distraction of management's attention from our primary business;
- the inability to commercialize our products and develop enhancements to our products;
- decreased demand for our products;
- damage to our business reputation and brand;
- product recalls or withdrawals from the market;
- withdrawal of clinical trial participants;
- substantial monetary awards to patients or other claimants; or
- loss of sales.

Additionally, we could experience a material design or manufacturing failure in our products, a quality system failure, other safety issues or heightened regulatory scrutiny that would warrant a recall of some of our products. While we may attempt to manage our product liability exposure by proactively recalling or withdrawing from the market any defective products, any recall or market withdrawal of our products may delay the supply of those products to our customers and may impact our reputation. We may not be successful in initiating appropriate market recall or market withdrawal efforts that may be required in the future and these efforts may not have the intended effect of preventing product malfunctions and the accompanying product liability that may result. Such recalls and withdrawals may also be used by our competitors to harm our reputation for safety or be perceived by patients as a safety risk when considering the use of our products. Product liability lawsuits and claims, safety alerts and product recalls and withdrawals, regardless of their ultimate outcome, could have a material adverse effect on our business, financial condition and results of operations.

Although we have product liability insurance that we believe is appropriate, this insurance is subject to deductibles and coverage limitations. Our current product liability insurance may not continue to be available to us on acceptable terms, if at all, and, if available, coverage may not be adequate to protect us against any future product liability claims. If we are unable to obtain insurance at an acceptable cost or on acceptable terms or otherwise protect against potential product liability claims, we could be exposed to significant liabilities. While we have not been materially impacted by product liability claims, lawsuits and recalls to date, we may be materially impacted by such events in the future, and any such claim, lawsuit, recall or other claim with respect to uninsured liabilities or for amounts in excess of insured liabilities could have a material adverse effect on our business, financial condition and results of operations. Further, such product liability matters may negatively impact our ability to obtain insurance coverage or cost-effective insurance coverage in future periods.

Industry trends have resulted in increased downward pricing pressure on medical services and products, which may affect our ability to sell our products at prices necessary to support our current business strategy.

The trend toward health care cost containment through aggregating purchasing decisions and industry consolidation, along with the growth of managed care organizations, is placing increased emphasis on the delivery of more cost-effective medical therapies. For example:

- There has been consolidation among health care facilities and purchasers of medical devices, particularly in the United States. One of the results of such consolidation is that group purchasing organizations ("GPOs"), integrated delivery networks and large single accounts use their market power to consolidate purchasing decisions, which intensifies competition to provide products and services to health care providers and other industry participants, resulting in greater pricing pressures and the exclusion of certain suppliers from important market segments.

- Surgeons increasingly have moved from independent, outpatient practice settings toward employment by or affiliation with hospitals and other larger health care organizations, which aligns surgeons' product choices with the institutional providers' price sensitivities and adds to pricing pressures. Hospitals and health care facilities have introduced and may continue to introduce new pricing structures into their contracts to contain health care costs, including fixed price formulas and capitated and construct pricing. Such pricing structures may create incentive programs for surgeons that is less advantageous to our business model, including capitation models where surgeons are paid a fixed amount per patient, per period.

- Certain hospitals provide financial incentives to doctors for reducing hospital costs (known as gainsharing), rewarding physician efficiency (known as physician profiling) and encouraging partnerships with health care service and goods providers to reduce costs.

- Existing and proposed laws, regulations and industry policies, in both domestic and international markets, regulate or seek to increase regulation of sales and marketing practices and the pricing and profitability of companies in the health care industry.

More broadly, provisions of the Affordable Care Act ("ACA") could meaningfully change the way health care is developed and delivered in the United States, and may adversely affect our business and results of operations. For further discussion of these challenges, see the section titled " —Risks Related to Regulatory Matters—Legislative or regulatory reforms may have a material adverse effect on us." We cannot predict accurately what health care programs and regulations will ultimately be implemented at the federal or state level, or the effect of any future legislation or regulation in the United States or elsewhere. However, any changes that have the effect of reducing reimbursement by government health care programs and other third-party payors for procedures using our systems or reducing medical procedure volumes could have a material and adverse effect on our business, financial condition and results of operations. Any decline in the amount that payors reimburse our customers for our systems could make it difficult for customers to continue using, or to adopt, our systems and could create additional pricing pressure for us. If we are forced to lower the price we charge for our systems, or if we add more components to our systems, our gross margins will decrease, which will adversely affect our ability to invest in and grow our business. If we are unable to maintain our prices, or if our costs increase and we are unable to offset such increase with an increase in our prices, our margins could erode. While we have not experienced a material impact from any such industry trends to date, there is no assurance that our net revenue will not be subject to these headwinds in the future.

In addition, the largest medical device companies with multiple product franchises have increased their effort to leverage and contract broadly with customers across franchises by providing volume discounts and multi-year arrangements that could prevent our access to these customers or make it difficult, or impossible, to compete on price.

Our employees and independent contractors, including independent distributors and any other consultants, any future service providers and other vendors, may engage in misconduct or other improper activities, including noncompliance with regulatory standards and requirements, which could have an adverse effect on our results of operations.

We are exposed to the risk that our employees and independent contractors, including independent distributors and any other consultants, any future commercial collaborators, and other vendors may engage in misconduct or other illegal activity. Misconduct by these parties could include intentional, reckless and/or negligent conduct or other unauthorized activities that violate: federal, state or local laws and regulations, as well as the laws, regulations and rules of regulatory bodies such as the FDA; manufacturing standards; U.S. federal and state health care fraud and abuse laws and regulations, data privacy laws and other similar non-U.S. laws; or laws that require the true, complete and accurate reporting of financial information or data. For example, we are aware of limited instances in which our dedicated commercial leadership team and independent distributors have recommended using our systems for off label uses, including using certain aTSA products for rTSA procedures and vice versa. It is not always possible to identify and deter misconduct by employees and other third parties, and the precautions we take to detect and prevent this activity may not be effective in controlling unknown or unmanaged risks or losses or in protecting us from governmental investigations or other actions or lawsuits stemming from a failure to comply with such laws or regulations. In addition, we are subject to the risk that a person or government could allege such fraud or other

misconduct, even if none occurred. If any such actions are instituted against us, and we are not successful in defending ourselves or asserting our rights, those actions could have a significant impact on our business and financial results, including, without limitation, the imposition of significant civil, criminal and administrative penalties, damages, monetary fines, disgorgements, possible exclusion from participation in Medicare, Medicaid and other U.S. health care programs, other sanctions, imprisonment, contractual damages, reputational harm, diminished profits and future earnings and curtailment of our operations, any of which could adversely affect our ability to operate our business and our results of operations.

If rising prices or availability of raw materials continues to persist, our business and results of operations may be adversely affected.

Volatility in raw material prices and availability can impact our business and finances due to numerous factors beyond our control, including general, domestic, and international economic conditions, labor costs, production levels, competition, consumer demand, import duties, inflation and currency exchange rates. This volatility can significantly affect the availability and cost of raw materials that our suppliers purchase and are ultimately used in our systems, and may therefore have a material adverse effect on our business, results of operations, and financial condition.

In addition, the current U.S. administration has expressed strong concerns about imports from countries that it perceives as engaging in unfair trade practices, and has imposed tariffs or other restrictions on products, components or raw materials sourced from those countries. Moreover, these new tariffs, or other changes in U.S. trade policy, have triggered and may in the future trigger retaliatory actions by affected countries. For example, there have been and continue to be further indications that there may be an increase in tariff rates on various types of goods imported from Canada and Europe that could apply to the raw materials we require in our products, including certain types of metal powders used for coating our products. In the event that any such possible tariff increases remain in place or become enacted in the future, they could significantly increase the cost of materials and components that our suppliers import and use in our systems, which in turn could increase our supply costs. At this time, there can be no assurance that we will be able to pass any portion of such increases on to customers. We currently do not hedge against our exposure to changing raw material prices and are not aware as to whether our suppliers hedge. As a result, fluctuations in raw material prices could have a material adverse effect on our business, results of operations, and financial condition. Supply shortages or changes in availability for any particular type of raw material can delay supplier volume and production capabilities or cause increases in the cost of manufacturing our products. We may be negatively affected by changes in availability and pricing of raw materials, which could negatively impact our results of operations.

Risks Related to Administrative, Organizational and Commercial Operations and Growth

If hospitals, ASCs, and other health care facilities do not approve the use of our implant systems, our sales may not increase.

In order for surgeons to use our systems at hospitals, ASCs and other health care facilities, we are often required to obtain approval from those hospitals, ASCs and health care facilities. Typically, hospitals, ASCs and health care facilities review the comparative effectiveness and cost of products used in the facility. The makeup and evaluation processes for health care facilities vary considerably, and it can be a lengthy, costly and time-consuming effort to obtain approval by the relevant health care facilities. Additionally, hospitals, ASCs, other health care facilities, which manage purchasing for multiple facilities, may also require us to enter into a purchase agreement and satisfy numerous elements of their administrative procurement process, which can also be a lengthy, costly and time-consuming effort. If we do not obtain access to hospitals, ASCs, GPOs and other health care facilities in a timely manner, or at all, via their approvals or purchase contract processes, or otherwise, or if we are unable to obtain approvals or secure contracts in a timely manner or on terms that are economically viable to our business, or at all, our operating costs will increase, our sales may decrease and our operating results may be adversely affected. Furthermore, we may expend significant efforts on these costly and time-consuming processes but may not be able to obtain necessary approvals or secure a purchase contract from such hospitals, ASCs or health care facilities.

Performance issues, service interruptions or price increases by shipping carriers could adversely affect our business and harm our reputation and ability to provide our systems on a timely basis.

Expedited, reliable shipping is essential to our operations. We rely heavily on providers of transport services for reliable and secure point-to-point transport of our systems to our customers and for tracking of these shipments, including our InSet tray systems that are delivered directly to hospitals and ASCs, employees or independent distributors. Should a carrier encounter delivery performance issues such as loss, damage or destruction of our systems, it would be costly to replace our systems in a timely manner, could cause surgeries using our systems to be delayed or canceled and such occurrences may damage our reputation and lead to decreased demand for our systems and increased cost and expense to our business. Once a product leaves our facilities, we are no longer in control of the care of such shipped products and any damage that may occur during transit may not be readily detectable by the customer, including any penetrations of sterile barriers, temperature swings that may cause package seal rupturing or exposure to fine particulates or other debris that is not easily detectable. In addition, any significant increase in shipping rates could adversely affect our operating margins and results of operations. Similarly, strikes, severe weather, natural disasters, including fires and hurricanes, or other service interruptions affecting delivery services we use would adversely affect our ability to process orders for our systems on a timely basis.

If coverage or adequate levels of reimbursement from third-party payors for procedures using our implant systems, or any future products we may seek to commercialize, are not obtained or maintained, shoulder specialists and patients may be reluctant to use our systems and our business will suffer.

In the United States, health care providers such as hospitals and ASCs who purchase our implant systems generally rely on third-party payors, principally federally-funded Medicare, state-funded Medicaid and private health insurance plans, to pay for all or a portion of the cost of shoulder procedures and products utilized in those procedures. We may be unable to sell our implant systems, or any future products we may seek to commercialize, on a profitable basis if third-party payors deny coverage or reduce their current levels of reimbursement for procedures using our systems. Our sales depend largely on governmental health care programs and private health insurers reimbursing providers for procedures using our implant systems. Significant changes to operations at, funding of, or restructuring of such governmental authorities, including but not limited to a government shutdown, decreases in staff who are able to provide reimbursement services, reductions or other changes in funding provided to such governmental authorities, and changes in policy and enforcement priorities, may adversely affect our business. Hospitals, ASCs and other health care providers may not purchase our products if they do not receive adequate reimbursement from third-party payors for procedures using our products. Payors continue to review their coverage policies for existing and new therapies and may deny coverage for treatments that include the use of our systems or any future products we may seek to commercialize. Third-party payors, whether governmental or commercial, are developing increasingly sophisticated methods of controlling healthcare costs. In addition, no uniform policy of coverage and reimbursement for procedures using our solution exists among third-party payors. Therefore, coverage and reimbursement for procedures using our products can differ from payor to payor.

In addition, some health care providers in the United States have adopted or are considering bundled payment methodologies and/or managed care systems in which providers contract to provide comprehensive health care for a fixed cost per person. Health care providers may attempt to control costs by authorizing fewer elective surgical procedures, including shoulder surgeries, or by requiring the use of the least expensive procedure available. In addition, third-party payors increasingly are requiring evidence that medical devices are cost-effective, and if we are unable to meet this requirement, the third-party payor may not cover procedures using our products, which could reduce sales of our products to health care providers who depend upon third-party payor reimbursement for payment. Changes in coverage policies or health care cost containment initiatives that limit or restrict reimbursement for procedures using our systems may have an adverse effect on our business. For example, the Centers for Medicare and Medicaid Services ("CMS") only recently approved total shoulder arthroplasty for reimbursement when performed in an ASC on January 1, 2024, and any change or reversal in this decision would adversely affect our results of operations and financial condition.

We depend on third-party contract manufacturers and suppliers, some of which are single source, to produce and package all elements comprising our shoulder implant systems, and if these suppliers and manufacturers fail to supply us, our products or their components or subcomponents in sufficient quantities or at all, or in accordance with applicable regulatory requirements and our specifications, it will have a material adverse effect on our business, financial condition, and results of operations.

We utilize qualified medical device contract manufacturers and suppliers, the majority of which are single sources, to produce, package and sterilize the majority of the elements comprising our systems. For us to be successful, our suppliers must be able to provide us with products and components in substantial quantities, in compliance with regulatory requirements, in accordance with agreed upon specifications, at acceptable costs and quality, and on a timely basis. An interruption in our commercial operations could occur if we encounter delays or difficulties in securing these components, and if we cannot then obtain an acceptable substitute. Our significant single source suppliers include Avalign Technologies, Inc. ("Avalign Technologies"), which manufactures our humeral stems, Micropulse, Inc. ("Micropulse"), which manufactures and supplies our InSet Glenoid products, Trifecta Medical Group, which manufactures our instrument cases and trays and Revelation Medical Devices ("RMD"), which manufactures the surgical instruments used during procedures involving our systems. While we estimate replacing these single source suppliers could take up to between six and twelve months, in some of these examples alternative second source suppliers may not be readily available. Furthermore, while we have two sterilization partners, we utilize a single supplier to sterilize all of our polyethylene implants. We also do not have long-term supply agreements, and we typically purchase our products on a purchase order basis, which means that aside from any binding purchase orders we have from time to time, our supplier could cease supplying to us or change the terms on which it is willing to continue supplying to us at any time. Although we have identified alternate third parties who could provide this manufacturing service and expect to reduce our reliance on single source manufacturers and third parties we use for sterilization in the foreseeable future, we cannot guarantee you that we would be able to contract with such alternate third parties within a reasonable amount of time or at all, or upon similar pricing and volume terms, nor can we be assured that any such third party would be capable of producing products in sufficient volume and quality. For example, a replacement supplier may discover challenges with our products or we may not perfectly specify our product designs for such new suppliers' manufacturing systems, which may require us to refine our products and/or incur additional research and development expense, which could delay our sales and have an adverse effect on our financial condition and results of operations. Furthermore, if we are required to change the manufacturer of a critical component of any of our systems, we will be required to verify that the new manufacturer maintains facilities, procedures and operations that comply with our quality standards and applicable regulatory requirements, which could further impede the manufacture our systems in a timely manner or at all. A change in manufacturer could trigger the requirement to submit and obtain a new 510(k) clearance from the FDA, or similar international regulatory authorization before we implement the change, which could cause substantial delays. Any event, including those listed above, other circumstances that result in a prolonged business disruption or shutdown to one or more of single source suppliers' facilities, or the facilities of our other vendors and service providers, or a deterioration in our relationship with any of our single source suppliers, or any of our other vendors or service providers, in each case, could create conditions that prevent, or significantly and adversely affect, our sales, increase our expenses, create potential liabilities or damage our reputation, any of which could have an adverse effect on our business, financial condition and results of operations. Our marketing applications and other regulatory submissions may identify the facilities used by our third-party manufacturers for the manufacture of our systems, and these facilities are subject to inspections. We do not control the manufacturing process of, and are completely dependent on, third-party manufacturers for compliance with current good manufacturing practice ("cGMP") requirements for manufacture of our systems. If these third-party manufacturers cannot successfully manufacture material that conforms to our specifications and the strict regulatory requirements of the FDA or any comparable foreign regulatory authority, we will not be able to secure and/or maintain marketing authorization or certification of our products manufactured in their manufacturing facilities.

We seek to strategically maintain sufficient levels of inventory to help mitigate supply disruption, usually holding sufficient inventory to allow for manufacturing from approximately 90 days following the loss of a supplier, to accommodate varying demand mix and to achieve more efficient volume-based pricing on our components; however, we may not be accurate in our estimates which could result in insufficient inventory to meet demand or excess inventory and the risk of inventory obsolescence and expiration. Further, while we have entered into several

supply agreements in an effort to reinforce our supply chain, there is no guarantee the counter-party suppliers will adhere to the terms of these agreements.

We do not have complete control over all aspects of the manufacturing process of, and are dependent on, our manufacturing partners for compliance with current cGMP regulations applicable to our systems. Third-party manufacturers may not be able, or fail, to comply with cGMP regulations. If our third-party manufacturers cannot successfully manufacture products that conform to our specifications and the strict regulatory requirements of the FDA, they will not be able to secure and/or maintain registration for their manufacturing facilities. In addition, we do not have complete control over the ability of our third-party manufacturers to maintain adequate quality control, quality assurance and qualified personnel. The failure of our manufacturing partners to maintain acceptable quality requirements could result in quality issues, including recalls. If one of our manufacturing partners fails to maintain acceptable quality requirements or perform as agreed, we may have to identify and qualify a new supplier or develop in-house manufacturing capabilities, which could require significant capital investments.

Our suppliers may encounter problems during manufacturing for a variety of reasons, including, for example, failure to follow specific protocols and procedures, failure to comply with applicable legal and regulatory requirements, equipment malfunction and environmental factors, including damages to their facilities as a result of fires, storms or hurricanes, failure to properly conduct their own business affairs, and infringement of third-party intellectual property rights, any of which could delay or impede their ability to meet our requirements. Our suppliers may also not prioritize the production of our systems compared to the suppliers' larger customers so we may experience longer delays in receiving our requested orders. Our reliance on a third-party manufacturer and third-party suppliers also subjects us to other risks that could harm our business that we would not be subject to if we manufactured ourselves, including, among others:

- we may not be a major customer of many of our suppliers, and these suppliers may therefore give other customers' needs higher priority than ours, including some of our competitors who use the same manufacturing partners;
- third parties may threaten or enforce their intellectual property rights against our suppliers, which may cause disruptions or delays in shipment, or may force our suppliers to cease conducting business with us;
- we may not be able to obtain an adequate supply of components in a timely manner or on commercially reasonable terms;
- our suppliers, especially new suppliers, may make errors in manufacturing that could negatively affect the efficacy or safety of our systems or cause delays in shipment;
- we may have difficulty locating and qualifying alternative suppliers and our existing suppliers could be acquired by companies that have limiting or exclusive relationships with our competitors;
- switching components or suppliers may require product redesign and possibly submission to FDA, or notified bodies, which could significantly impede or delay our commercial activities;
- one or more of our single-source suppliers may be unwilling or unable to supply components of our systems;
- other customers, including our competitors, may use fair or unfair negotiation tactics or pressures to impede our use of the suppliers;
- the occurrence of a fire, natural disaster or other catastrophe, or the occurrence of geopolitical conflicts, as well as any sanctions or other actions resulting therefrom impacting one or more of our suppliers may affect their ability to deliver products to us in a timely manner;
- the occurrence of pandemics, epidemics and other public health emergencies may impact a manufacturing facility by limiting operating capacity or sideline critical employees involved in the manufacturing processes thereby affecting their ability to deliver products to us in a timely manner;

- our suppliers may encounter financial or other business hardships unrelated to our demand, which could inhibit their ability to fulfill our orders and meet our requirements;
- our suppliers may not maintain the confidentiality of our proprietary information or may mislabel raw materials resulting in inaccurate certifications; and
- higher manufacturing and product costs than more vertically integrated companies.

Any of these factors could cause delay or suspension of commercialization and marketing, regulatory submissions or required approvals, clearances or certifications, or cause us to incur higher costs. Furthermore, if our contract manufacturers fail to deliver the required commercial quantities of finished products on a timely basis and at commercially reasonable prices and we are unable to find one or more replacement manufacturers capable of production at a substantially equivalent cost, in substantially equivalent volumes and quality, and on a timely basis, we would likely be unable to meet demand for our systems and we would lose potential revenue. Any difficulties in locating and hiring third-party manufacturers, or in the ability of third-party manufacturers to supply quantities of our products at the times and in the quantities we need, could have a material adverse effect on our business. It may take a significant amount of time and resources (including costs) to establish an alternative source of supply for our products and to have any such new source authorized by the FDA or other bodies. Given our reliance on certain single-source suppliers, we are especially susceptible to supply shortages because we do not have alternate suppliers currently available that we could contract with in reasonable amount of time or at all, or upon similar pricing and volume terms, nor can we be assured that any such third party would be capable of producing in sufficient volume and of sufficient quality.

Moreover, some third parties are located in markets subject to infrastructure problems and exposed to significant natural disasters, such as hurricanes and wildfires. Failure of third parties to meet their contractual, regulatory, and other obligations may have a material adverse effect on our business, financial condition, and results of operations. Any of these matters could materially and adversely affect our business, financial condition, and results of operations.

Our results of operations will be materially harmed if we are unable to accurately forecast demand for our implant systems and manage our inventory.

In order to market and sell effectively, we must maintain significant levels of inventory of systems and surgical instrumentation. As a result, a significant amount of our cash used in operations has been associated with maintaining these levels of inventory. To ensure adequate inventory supply, we must forecast inventory needs and manufacturing orders based on our estimates of future demand. Our ability to accurately forecast demand for our systems could be negatively affected by many factors, including our failure to accurately manage our expansion strategy, product introductions by competitors, an increase or decrease in customer demand for our systems or for products of our competitors, our failure to accurately forecast customer acceptance of new products, unanticipated changes in general market conditions or regulatory matters and weakening of economic conditions or consumer confidence in future economic conditions. Inventory levels in excess of customer demand may result in inventory write-downs or write-offs, which would cause our gross margin to be adversely affected and could impair the strength of our brand. Furthermore, as we continue to grow and expand, we require increasing levels of inventory which may subject us to greater financial risk in the event of a recall where such inventory is no longer commercially viable. Conversely, if we underestimate customer demand for our systems, our manufacturers and suppliers may not be able to deliver systems to meet our requirements, which could result in damage to our reputation and customer relationships. In addition, if we experience a significant increase in demand, additional supplies of raw materials or additional manufacturing capacity may not be available to or at our existing manufacturers and suppliers when required on terms that are acceptable to us, or at all, which will negatively affect our business, financial condition and results of operations.

We seek to maintain sufficient levels of inventory in order to protect ourselves from supply interruptions. As a result, we are subject to the risk that a portion of our inventory will become obsolete or expire, which could have a material adverse effect on our earnings and cash flows due to the resulting costs associated with the inventory obsolescence charges and costs required to replace such inventory.

We may enter into collaborations, in-licensing arrangements, joint ventures, strategic alliances or partnerships with third parties that may not result in the development of commercially viable products, product improvements or the generation of significant future revenues.

In the ordinary course of our business, we may enter into collaborations, in-licensing arrangements, joint ventures, strategic alliances, partnerships or other arrangements to develop new products or product improvements and to pursue new markets. Proposing, negotiating and implementing collaborations, in-licensing arrangements, joint ventures, strategic alliances or partnerships may be a lengthy and complex process. Other companies, including those with substantially greater financial, marketing, sales, technology or other business resources, may compete with us for these opportunities or arrangements. We have entered into a second amended and restated software license agreement with Genesis Software Innovations, LLC ("Genesis Software"), pursuant to which we have exclusive rights to use certain preoperative surgery planning platform software. For a more complete description of this agreement, see the section titled "Business—Intellectual Property—Software License Agreement with Genesis Software."

We may not identify, secure, or complete any such transactions or arrangements in a timely manner, on a cost-effective basis, on acceptable terms or at all. We have limited institutional knowledge and experience with respect to these business development activities, and we may also not realize the anticipated benefits of any such transaction or arrangement. In particular, these collaborations may not result in the development of products that achieve commercial success or viable product improvements or result in significant revenues and could be terminated prior to developing any products. For example, while our license agreement with Genesis Software provides us with a preoperative software planning platform that is used by surgeons preparing for procedures, we do not generate any separate revenue specifically from ProVoyance.

Additionally, we may not be in a position to exercise sole decision-making authority regarding the transaction or arrangement, which could create the potential risk of creating impasses on decisions, and our future collaborators may have economic or business interests or goals that are, or that may become, inconsistent with our business interests or goals. It is possible that conflicts may arise with our collaborators, such as conflicts concerning the achievement of performance milestones, or the interpretation of significant terms under any agreement, such as those related to financial obligations or the ownership or control of intellectual property developed during the collaboration. If any conflicts arise with any future collaborators, they may act in their self-interest, which may be adverse to our best interest, and they may breach their obligations to us. For example, we have and will continue to enter into agreements with third parties that are owned or controlled by certain of our officers and directors. Robert Ball, our Chief Executive Officer and Executive Chairman, and is a co-founder and director of Genesis Innovation Group, Inc. ("Genesis Innovation") and Genesis Software. See the section titled "—Risks Related to Administrative, Organizational and Commercial Operations and Growth—Certain of our officers and directors also serve as directors of, and are affiliates of, Genesis Innovation, Genesis Software and cultivate(MD), which may give rise to a perceived or actual conflict of interest which may adversely impact your interests." In addition, we may have limited control over the amount and timing of resources that any future collaborators devote to our or their future products.

Disputes between us and our collaborators may result in litigation or arbitration which would increase our expenses and divert the attention of our management. Further, these transactions and arrangements will be contractual in nature and will generally be terminable under the terms of the applicable agreements and, in such event, we may not continue to have rights to the products relating to such transaction or arrangement or may need to purchase such rights at a premium. If we enter into in-bound intellectual property license agreements, we may not be able to fully protect the licensed intellectual property rights or maintain those licenses. Future licensors could retain the right to prosecute and defend the intellectual property rights licensed to us, in which case we would depend on the ability of our licensors to obtain, maintain and enforce intellectual property protection for the licensed intellectual property. These licensors may determine not to pursue litigation against other companies or may pursue such litigation less aggressively than we would. Further, entering into such license agreements could impose various diligence, commercialization, royalty or other obligations on us. Future licensors may allege that we have breached our license agreement with them, and accordingly seek to terminate our license, which could adversely affect our competitive business position and harm our business prospects.

AI presents risks and challenges that can impact our business including by posing security risks to our confidential information, proprietary information and personal data.

We incorporate AI solutions into our platform, and these applications are increasingly important to our operations over time. AI presents risks such as inaccuracy, bias, toxicity, intellectual property infringement or misappropriation, data privacy and cybersecurity and data provenance. In addition, AI utilizes machine learning and predictive analytics, which in some cases present flawed, biased, and inaccurate results, and may have errors or inadequacies that are not easily detectable and may also be subject to data herding and interconnectedness (*i.e.*, multiple market participants utilizing the same data), in each case adversely impacting our business. The regulatory framework for AI is rapidly evolving as many federal, state, and foreign government bodies and agencies have introduced or are currently considering additional laws and regulations. Additionally, existing laws and regulations could affect the operation of our AI. As a result, implementation standards and enforcement practices are likely to remain uncertain for the foreseeable future, and we cannot yet determine the impact of future laws, regulations, standards, or market perception of their requirements on our business and our business and our response. It is possible that new laws and regulations will be adopted in the United States and in other jurisdictions outside the United States, or that existing laws and regulations, including competition and antitrust laws, may be interpreted in ways that would limit our ability to use AI for our business, or require us to change the way we use AI in a manner that negatively affects the performance of our technologies, services, and business and the way in which we use AI. We may need to expend resources to adjust our technologies or services in certain jurisdictions if the laws, regulations, or decisions are not consistent across jurisdictions. Further, the cost to comply with such laws, regulations, or decisions and/or guidance interpreting existing laws, could be significant and would increase our operating expenses (such as by imposing additional reporting obligations regarding our use of AI). Such an increase in operating expenses, as well as any actual or perceived failure to comply with such laws and regulations, could adversely affect our business, financial condition, and results of operations.

These issues, combined with an uncertain regulatory environment, may further result in reputational harm, liability, or other adverse consequences to our business operations. Incorporation of generative AI tools by vendors into their offerings without disclosing this use to us would expose us to potential liability, and the providers of these generative AI tools may not meet existing or rapidly evolving regulatory or industry standards with respect to privacy and data protection and may inhibit our or our vendors' ability to maintain an adequate level of service and experience. If our vendors, or our third-party partners experience an actual or perceived breach or privacy or security incident because of the use of generative AI, we may lose valuable intellectual property and confidential information and our reputation and the public perception of the effectiveness of our security measures could be harmed. Further, bad actors around the world use increasingly sophisticated methods, including the use of AI, to engage in illegal activities involving the theft and misuse of personal information, confidential information, and intellectual property. Any of these outcomes could damage our reputation, result in the loss of valuable property and information, and adversely impact our business.

We may not be able to establish or strengthen our brand.

We believe that establishing and strengthening our brand, including our ecosystem, is important to achieving widespread acceptance of our implant systems, particularly because of the highly competitive nature of the market for similar products. We believe the quality, breadth and reliability of our systems is critical to building physician support for our systems, and any negative publicity regarding the quality or reliability of our systems or shoulder arthroplasty procedures generally could significantly damage our reputation. In the course of conducting our business, we must adequately address quality issues that may arise with our systems and technologies. Although we have established internal procedures designed to minimize risks that may arise from quality issues, we may not be able to eliminate or mitigate occurrences of these issues and associated liabilities. In addition, even in the absence of quality issues, we may be subject to claims and liability if the performance of our systems do not meet the expectations of physicians or patients or any improper use of our systems or technologies that results in injury or unsatisfactory results.

In addition, promoting and positioning our brand will depend largely on the success of our medical education efforts and our ability to educate shoulder specialists and patients. These brand promotion activities may not yield increased sales and, even if they do, any sales increases may not offset the expenses we incur to promote our brand.

If we fail to successfully promote and maintain our brand, or if we incur substantial expenses in an unsuccessful attempt to promote and maintain our brand, our shoulder solutions may not be accepted by physicians or patients, which would adversely affect our business, results of operations and financial condition.

Our inability to maintain contractual relationships with health care professionals could have a negative impact on our research and development and medical education programs.

We maintain contractual relationships with respected physicians and medical personnel in hospitals, private practice and universities who assist in clinical studies, product research, data collection for our registries and development and in the training of shoulder specialists on the safe and effective use of our products. We continue to place emphasis on the validation of the benefits of our implant systems through clinical studies, the development of proprietary products and product improvements as well as providing high quality training and support. If we are unable to maintain these relationships, our ability to develop and market new and improved products and train on the use of those products could decrease, and future operating results could be unfavorably affected. At the same time, the medical device industry's relationship with physicians is under increasing scrutiny by the U.S. Department of Health and Human Services Office of Inspector General ("OIG"), the U.S. Department of Justice ("DOJ"), the state attorneys general and other foreign and domestic government agencies. Our failure to comply with requirements governing the industry's relationships with physicians or an investigation into our compliance by the OIG, the DOJ, state attorneys general and other government agencies, could negatively affect our business, financial condition and results of operations. See the section titled "—Risks Related to Regulatory Matters—Our relationships with customers, physicians and third-party payors are subject to federal and state health care fraud and abuse laws, false claims laws, physician payment transparency laws and other health care laws and regulations. If we or our employees, independent contractors, consultants, commercial partners, or vendors violate these laws we could face substantial penalties."

We may be unable to continue to successfully demonstrate to shoulder specialists or key opinion leaders the merits of our implant systems and technologies compared to those of our competitors, which may make it difficult to achieve market acceptance.

Shoulder specialists play the primary role in determining the course of treatment and, ultimately, the type of products that will be used to treat a patient. As a result, our success depends, in large part, on our ability to effectively market and demonstrate to shoulder specialists the merits of our systems and methodologies compared to those of our competitors. Acceptance of our systems and methodologies depends on educating shoulder specialists as to the distinctive characteristics, clinical benefits, safety, cost-effectiveness and efficiencies of our systems and technologies as compared to those of our competitors, and on training surgeons in the proper use of our systems and offerings. If we are not successful in convincing surgeons of the merits of our systems and methodologies or educating them on the use of our systems, they may not use our systems or may not use them effectively and we may be unable to increase our sales, sustain our growth or achieve and sustain profitability.

Also, since certain of our offerings and products are relatively new, including our Inset 95 Humeral Stem, some surgeons may be reluctant to change their surgical treatment practices for the following reasons, among others:

- lack of experience with our systems and technologies;
- existing relationships with competitors and distributors that sell competitive products;
- lack or perceived lack of evidence supporting additional patient benefits;
- perceived liability risks generally associated with the use of new systems and technologies;
- less attractive availability of coverage and reimbursement by third-party payors compared to procedures using competitive products and other techniques;
- costs; and
- the time commitment that may be required for training.

These reasons may affect the pace of adoption of our systems and future products that we may offer.

In addition, we believe recommendations and support of our systems and technologies by surgeons, shoulder specialists and other key opinion leaders in our industry are essential for market acceptance. If we do not receive support from such surgeons, shoulder specialists and other key opinion leaders, if long-term data does not show the benefits of using our systems or if the benefits offered by our systems are not sufficient to justify their cost, shoulder specialists, hospitals, ASCs and other health care facilities may not use our systems and we might be unable to achieve market acceptance.

Our systems are designed to satisfy long term demands for patients and, given the length of time we have been commercially active, we do not have significant long term use cases available. In the orthopedic industry, participants have been able to demonstrate favorable short-term outcomes that later went on to fail and not provide adequate long term performance. Any such failure to demonstrate long-term satisfaction for our systems in the future could have adverse affect on our brand, demand for our systems and our results of operations and financial condition.

If surgeons fail to safely and appropriately use our implant systems, or if we are unable to train surgeons on the safe and appropriate use of our systems, we may be unable to achieve our expected sales, growth or profitability.

An important part of our sales process includes our ability to screen for and identify surgeons who have the requisite training and experience to safely and appropriately use our implant systems and to train a sufficient number of these surgeons and provide them with adequate instruction in the use of our systems. There is a training process involved for surgeons to become proficient in the use of our systems. This training process may take longer or be more expensive than expected and may therefore affect our ability to increase sales. Convincing surgeons to dedicate the time and energy necessary for adequate training is challenging, and we may not be successful in these efforts. Recent changes to federal guidance regarding medical education programs under the federal Anti-Kickback Statute also could limit our ability to train surgeons, and such programs could be subject to challenge under the federal Anti-Kickback Statute. See the section titled "—Risks Related to Regulatory Matters—Our relationships with customers, physicians and third-party payors are subject to federal and state health care fraud and abuse laws, false claims laws, physician payment transparency laws and other health care laws and regulations. If we or our employees, independent contractors, consultants, commercial partners, or vendors violate these laws we could face substantial penalties."

Furthermore, if clinicians are not properly trained, they may misuse or ineffectively use our systems. Any such improper use may result in unsatisfactory outcomes, patient injury, negative publicity or lawsuits against us, any of which could harm our reputation and affect future sales. Accordingly, if surgeons fail to safely and appropriately use our systems, use them for off-label purposes or if we are unable to train them on the safe and appropriate use of our systems, we may be unable to achieve our desired sales, growth or profitability. Furthermore, despite the requisite training, surgeons may continue to deliberately use our systems in ways not in conformance to our instructions or methods for which our systems are designed, which could lead to unsuccessful patient outcomes which may harm our brand, demand and results of operations.

The loss of any member on our executive management team or our inability to attract and retain highly skilled members of our dedicated commercial leadership and marketing teams as well as certain third-party engineers could have a material adverse effect on our business, financial condition and results of operations.

Our success depends on the skills, experience and performance of the members of our executive management team including Robert Ball, our Chief Executive Officer & Executive Chairman, and Matthew Ahearn, our Chief Operating Officer, as well as David Blue, Jeffrey Points and Jon Osborne. The individual and collective efforts of these executives will be important as we continue to commercialize our existing systems, develop new products and technologies and expand our commercial activities. The loss or incapacity of existing members of our executive management team could have a material adverse effect on our business, financial condition and results of operations if we experience difficulties in hiring qualified successors. We do not maintain "key person" insurance for any of our executives or key employees.

Our commercial, quality and research and development programs and operations depend on our ability to attract and retain highly skilled team members. We may be unable to attract or retain qualified team members. All of our employees are at-will, which means that either we or the employee may terminate his or her employment at any time. The loss of key employees, failure of any key employee to perform, our inability to attract and retain skilled employees, as needed, or our inability to effectively plan for and implement a succession plan for key employees could have a material adverse effect on our business, financial condition and results of operations.

Furthermore, we rely on third parties, including Genesis Innovation and Genesis Software, for engineering and design services. The loss of any such contractual arrangement with Genesis Innovation or Genesis Software, including access to the professional services of such engineers, could have a material adverse effect on our business, financial condition and results of operations.

Certain of our officers and directors also serve as directors of, and are affiliates of, Genesis Innovation, Genesis Software and cultivate(MD), which may give rise to a perceived or actual conflict of interest which may adversely impact your interests.

We have and will continue to operate under agreements with third parties that are owned or otherwise affiliated with certain of our officers and directors. For example, Robert Ball, our Chief Executive Officer and Executive Chairman, is a co-founder and director of Genesis Innovation, Genesis Software and Genesis Investment Holdings, LLC ("Genesis Investment Holdings"). Genesis Investment Holdings is affiliated with cultivate(MD) Holdings, LLC ("cultivate(MD)"), a beneficial holder of more than 5% of our capital stock. In addition, Mr. Ball, Matthew Ahearn, our Chief Operating Officer and a member of our board of directors, and David Blue, our Chief Customer Experience Officer, are directors of cultivate(MD), which is an investor in Genesis Software. Mr. Ahearn is also a director of Genesis Investment Holdings. Pursuant to our consulting agreement with Genesis Innovation, Genesis Innovation has agreed to provide consulting services for concept development, intellectual property creation, surgeon relationship management and large-scale project management. As described under "Business—Intellectual Property—Software License Agreement with Genesis Software," we have entered into a software licensing agreement with Genesis Software. As a result of these roles and affiliations with Genesis Innovation and Genesis Software, a conflict of interest could rise in which our interests are different than those of Genesis Innovation and/or Genesis Software or in which the interests of certain of our officers and directors in relation to Genesis Innovation and Genesis Software are different than the interests in us and our stockholders. The interests of these officers and directors may not be fully aligned with yours, which could lead to actions that are not in your best interests. Furthermore, such officers and directors could be seen as benefiting directly or indirectly from transactions in which we will pay Genesis Innovation and/or Genesis Software for their services and offerings under our respective agreements with each party regardless of whether such agreements are believed to be in the best interests of our stockholders. In such cases, it may be difficult to manage the conflict of interest through the oversight of our board of directors. See also the sections titled "Certain Relationships and Related Party Transactions—Agreements with Stockholders—Consulting Arrangement with Genesis Innovation" and "Certain Relationships and Related Party Transactions—Agreements with Stockholders—Software License Agreement with Genesis Software Innovations" for more information.

This could give rise to a conflict of interest in which our interests are different than those of Genesis Innovation or Genesis Software or in which the interests of our officers and directors in relation to Genesis Innovation and Genesis Software are different than the interests of the company and its stockholders. In such cases, we may be unable to effectively manage the conflict of interest through the oversight of our board of directors.

Any future international expansion will subject us to additional costs and risks that may have a material adverse effect on our business, financial condition and results of operations.

Historically, all of our sales have been to customers in the United States. Our long-term strategy may include increasing our international presence, including securing regulatory approvals in targeted countries outside the United States. This strategy may include establishing and maintaining physician outreach and education capabilities outside of the United States and expanding our relationships with international payors. To the extent we enter into

international markets in the future, there are significant costs and risks inherent in conducting business in international markets, including:

- Difficulties in staffing and managing our international operations;
- Multiple, conflicting and changing laws and regulations such as tax laws, privacy laws, export and import restrictions, employment laws, regulatory requirements and other governmental approvals, permits and licenses;
- Reduced or varied protection for intellectual property rights in some countries;
- Obtaining regulatory clearance where required for our products in various countries;
- Requirements to maintain data and the processing of that data on servers located within such countries;
- Complexities associated with managing multiple payor reimbursement regimes, government payors or patient self-pay systems;
- Limits on our ability to penetrate international markets if we are required to manufacture locally;
- Financial risks, such as longer payment cycles, difficulty collecting accounts receivable, foreign tax laws and complexities of foreign value-added tax systems, the effect of local and regional financial pressures on demand and payment for our products and exposure to foreign currency exchange rate fluctuations;
- Restrictions on the site-of-service for use of our products and the economics related thereto for physicians, providers and payors;
- Natural disasters, political and economic instability, including wars, terrorism, political unrest, outbreak of disease, pandemics and epidemics, boycotts, curtailment of trade and other market restrictions; and
- Regulatory and compliance risks that relate to maintaining accurate information and control over activities subject to regulation under the United States Foreign Corrupt Practices Act of 1977 ("FCPA"), U.K. Bribery Act of 2010 and comparable laws and regulations in other countries.

We have limited experience with regulatory environments and market practices internationally, and we may not be able to penetrate or successfully operate in new markets. We may also encounter difficulty expanding into international markets because of limited brand recognition in certain parts of the world, leading to delayed acceptance of our products by surgeons and their patients, hospitals, ASCs and payors in these international markets. For example, any perceived advantages of our products in the United States may not be perceived the same internationally. If we are unable to expand internationally and manage the complexity of international operations successfully, it could have a material adverse effect on our business, financial condition and results of operations. If our efforts to introduce our products into foreign markets are not successful, we may have expended significant resources without realizing the expected benefit. Ultimately, the investment required for expansion into foreign markets could exceed the results of operations generated from this expansion.

Risks Related to Our Intellectual Property

If we are unable to obtain and maintain significant patent or other intellectual property protection for our products, or if the scope of our patents and other intellectual property rights do not adequately protect our products, our competitors could develop and commercialize products similar or identical to ours and we may be unable to gain significant market share and be unable to operate our business profitably.

We have and will rely on patents, trade secrets, copyrights, know-how, trademarks, license agreements and contractual provisions to establish our intellectual property rights and protect our products. These legal means, however, afford only limited protection and may not completely protect our rights. Any failure to obtain or maintain patent and other intellectual property protection with respect to our products could harm our business, financial condition and results of operations. We generally seek to protect our proprietary position by filing patent

applications that are important to our business. We also seek to protect our proprietary position by acquiring or in-licensing relevant issued patents or pending patent applications or other intellectual property or proprietary rights from third parties.

We cannot assure you that our intellectual property position will not be challenged or that all patents for which we have applied will be granted. The validity and breadth of claims in patents involve complex legal and factual questions and, therefore, may be highly uncertain. Uncertainties and risks that we face include the following:

- our pending or future patent applications may not result in the issuance of patents;
- the scope of any existing or future patent protection may not exclude competitors or provide competitive advantages to us;
- our patents may not be held valid or enforceable if subsequently challenged;
- other parties may claim that our products and designs infringe the proprietary rights of others—even if we are successful in defending our patents and proprietary rights, the cost of such litigation may adversely affect our business; and
- other parties may develop similar products, duplicate our products, or design around our patents.

The patent prosecution process is expensive and time-consuming, and we may not be able to file, prosecute, maintain, enforce or license all necessary or desirable patents or patent applications at a reasonable cost or in a timely manner, or in all jurisdictions. Although we enter into non-disclosure and confidentiality agreements with parties who have access to confidential or patentable aspects of our research and development output, any of these parties may breach such agreements and disclose such output before a patent application is filed, thereby jeopardizing our ability to seek and obtain patent protection. We may choose not to seek patent protection for certain innovations and may choose not to pursue patent protection in certain jurisdictions, and under the laws of certain jurisdictions, patents or other intellectual property rights may be unavailable or limited in scope. It is also possible that we will fail to identify patentable aspects of our developments before it is too late to obtain patent protection. In addition, our ability to obtain and maintain valid and enforceable patents depends in part on whether the differences between our inventions and the prior art allow our inventions to be patented over the prior art. Furthermore, the publication of discoveries in scientific literature often lags behind the actual discoveries, and patent applications in the United States and other jurisdictions are typically not published until eighteen (18) months after filing, or in some cases not at all. Therefore, we cannot be certain that we were the first to file for patent protection of such inventions.

In addition, the laws of foreign jurisdictions may not protect our rights to the same extent as the laws of the United States. For example, most countries outside of the United States do not allow patents for methods of treating the human body. This may preclude us from obtaining method patents outside of the United States having similar scope to those we have obtained or may obtain in the future in the United States. This includes certain key method patents covering the InSet system. Changes in either the patent laws or interpretation of the patent laws in the United States and other countries may diminish our ability to protect our inventions, obtain, maintain and enforce our patent rights and, more generally, could affect the value of our patents. Additionally, we cannot predict whether the patent applications we are currently pursuing will issue as patents in any particular jurisdiction or whether the claims or any issued patents will provide sufficient protection from competitors or third parties.

Moreover, we may be subject to a third-party pre-issuance submission of prior art to the U.S. Patent and Trademark Office ("USPTO") or patent offices in foreign jurisdictions, or become involved in opposition, derivation, reexamination, *inter partes* review, post-grant review or interference proceedings challenging our patent rights or the patent rights of others. An adverse determination in any such submission, proceeding or litigation could reduce the scope of, or invalidate, our patent rights, allow third parties to commercialize our technology and compete directly with us, without payment to us.

The patent positions of companies may involve complex legal and factual questions that have been the subject of much litigation in recent years, and, therefore, the scope of any patent claims that we have or may obtain cannot

be predicted with certainty. Accordingly, we cannot provide any assurances about which of our patent applications will issue, the breadth of any resulting patent, whether any of the issued patents will be found to be infringed, invalid, or unenforceable or will be threatened or challenged by third parties, that any of our issued patents have, or that any of our currently pending or future patent applications that mature into issued patents will include, claims with a scope sufficient to protect our products, services, or technology. Our pending and future patent applications may not result in the issuance of patents or, if issued, may not issue in a form that will be advantageous to us. The coverage claimed in a patent application can be significantly reduced before the patent is issued, and its scope can be reinterpreted after issuance. We cannot offer any assurances that the breadth of our issued patents will be sufficient to stop a competitor from developing, manufacturing, and commercializing one or more products, services, or technologies in a non-infringing manner that would be competitive with one or more of our products, services, or technologies, or otherwise provide us with any competitive advantage. Furthermore, any successful challenge to these patents or any other patents owned by or licensed to us after patent issuance could deprive us of rights necessary for our commercial success. Further, there can be no assurance that we will have adequate resources to enforce our patents.

The issuance of a patent is not conclusive as to its inventorship, scope, validity or enforceability, and our patents may be challenged in the courts or patent offices in the United States and abroad, and as a result, may not provide us with adequate proprietary protection or competitive advantage against competitors with similar products. Such challenges may result in loss of exclusivity or freedom to operate or in patent claims being narrowed, invalidated or held unenforceable, in whole or in part, which could limit our ability to stop others from using or commercializing similar or identical products and techniques without payment to us, or limit the duration of the patent protection of our technology.

We cannot ensure that we do not infringe any patents or other proprietary rights held by third parties. If our products were found to infringe any proprietary right of another party, we could be required to pay significant damages or license fees to such party and/or cease production, marketing and distribution of those products. Litigation may also be necessary to defend infringement claims of third parties or to enforce patent rights we hold or to protect trade secrets or techniques we own. In addition, because patent applications can take many years to issue and because publication schedules for pending applications vary by jurisdiction, there may be applications now pending of which we are unaware and which may result in issued patents which our current or future products infringe. Also, because the claims of published patent applications can change between publication and patent grant, there may be published patent applications that may ultimately issue with claims that we infringe.

We also rely on trade secrets and other unpatented proprietary technology. There can be no assurances that we can meaningfully protect our rights in our trade secrets and other unpatented proprietary technology or that others will not independently develop substantially equivalent proprietary products or processes or otherwise gain access to our proprietary technology. We seek to protect our trade secrets and proprietary know-how and technology, in part, with confidentiality and invention assignment agreements with employees and consultants that include customary intellectual property assignment obligations. There can be no assurances, however, that the agreements will not be breached, adequate remedies for any breach would be available or competitors will not discover our trade secrets or independently develop comparable intellectual property. Further, litigation may be necessary to obtain ownership or to defend against claims challenging ownership. Even if we are successful in defending against such claims, litigation could result in substantial costs and be a distraction to our management and other employees, and such claims could harm our business, financial condition and results of operations.

Some of our patents and patent applications may in the future be jointly owned with third parties. If we are unable to obtain an exclusive license to any such third party joint owners' interest in such patents or patent applications, such co-owners may be able to license their rights to other third parties, including our competitors, and our competitors could market competing products and technology. In addition, we may need the cooperation of any such joint owners of our patents in order to enforce such patents against third parties, and such cooperation may not be provided to us. Any of the foregoing could harm our business, financial condition and results of operations.

Obtaining and maintaining intellectual property protection depends on compliance with various procedural, document submission, fee payment and other requirements imposed by governmental agencies, and our intellectual property protection could be reduced or eliminated for non-compliance with these requirements.

The USPTO, the United States Copyright Office ("USCO") and various foreign governmental agencies require compliance with a number of procedural, documentary, fee payment and other similar provisions during the intellectual property application process. In addition, periodic maintenance fees, renewal fees, annuity fees and various other government fees on intellectual property registrations often must be paid to the USPTO, USCO and foreign agencies over the lifetime of the intellectual property registration and/or application and any intellectual property rights we may obtain in the future. While an unintentional lapse of an intellectual property registration or application can in many cases be cured by payment of a late fee or by other means in accordance with the applicable rules, there are situations in which noncompliance can result in abandonment or lapse of the registration or application, resulting in partial or complete loss of intellectual property rights in the relevant jurisdiction. Non-compliance events that could result in abandonment or lapse of a registration or application include, but are not limited to, failure to respond to official actions within prescribed time limits, non-payment of fees and failure to properly legalize and submit formal documents. If we fail to maintain the intellectual property registrations and applications covering our products, we may not be able to stop a competitor from developing and marketing products that are the same as or similar to our products, which would have a material adverse effect on our business.

We rely on intellectual property licensed from others and may become dependent on patents or other intellectual property licensed from others in the future. If we lose our licenses for intellectual property that is important to our business, we may not be able to continue developing or selling our products.

We rely on certain licenses that give us rights to third party intellectual property that is necessary or useful to our business, including pursuant to our License Agreement with Genesis Software. Our rights to use such intellectual property rights in our business are or may in the future be subject to the continuation of and our compliance with the terms of the license agreements between us and each of our licensors. In addition, the agreements under which we in-license intellectual property or technology from third parties are or may in the future be complex, and certain provisions in such agreements may be susceptible to multiple interpretations. The resolution of any contract interpretation disagreement that may arise could narrow what we believe to be the scope of our rights to the relevant intellectual property or technology or increase what we believe to be our financial or other obligations under the relevant agreement, either of which could have a material adverse effect on our business, financial condition, results of operations and prospects. Moreover, if disputes over intellectual property that we have in-licensed, or in-license in the future, prevent, or impair our ability to maintain our licensing arrangements on commercially acceptable terms, we may be unable to successfully develop and commercialize the affected product candidates, which could have a material adverse effect on our business, financial condition, results of operations, and prospects. License agreements may impose various obligations on us, including the obligation to pay certain royalties. For example, we are required to pay certain royalties under our License Agreement with Genesis Software. One or more of our licensors may allege that we have breached our license agreement with them, and accordingly seek to terminate our license. As a result, we may not be able to market products that were covered by the license, which would result in the loss of significant rights, restrict our ability to commercialize certain of our products and could adversely affect our competitive business position and harm our business prospects. In addition, any claims brought against us by our licensors could be costly and time-consuming and would divert the attention of our management and key personnel from our business operations.

We may not identify relevant third-party patents or may incorrectly interpret the relevance, scope or expiration of a third-party patent, which might adversely affect our ability to develop and market our products.

We cannot guarantee that any of our patent searches or analyses, including but not limited to the identification of relevant patents, analysis of the scope of relevant patent claims or determination of the expiration of relevant patents, are complete or thorough, nor can we be certain that we have identified each and every third-party patent and pending application in the United States, Europe and elsewhere that is relevant to or necessary for the commercialization of our products in any jurisdiction. Patent applications in the United States, the European Union (the "EU") and elsewhere are published approximately 18 months after the earliest filing for which priority is claimed, with such earliest filing date being commonly referred to as the priority date. Additionally, pending patent

applications that have been published can, subject to certain limitations, be later amended in a manner that could cover our products or the use of our products. After issuance, the scope of patent claims remains subject to construction as determined by an interpretation of the law, the written disclosure in a patent and the patent's prosecution history. Our interpretation of the relevance or the scope of a patent or a pending application may be incorrect, which may negatively impact our ability to market our products. We may incorrectly determine that our products are not covered by a third-party patent or may incorrectly predict whether a third party's pending application will issue with claims of relevant scope. Our determination of the expiration date of any patent in the United States, the EU or elsewhere that we consider relevant may be incorrect, which may negatively impact our ability to develop and market our products. Our failure to identify and correctly interpret relevant patents may negatively impact our ability to develop and market our products, if approved.

We are presently party to lawsuits involving patents and other intellectual property and the possibility exists that we may in the future be party to other lawsuits or administrative proceedings involving patents or other intellectual property. If we were to lose any intellectual property lawsuits, a court could require us to pay significant damages and/or prevent us from selling our products.

The medical device industry is highly competitive and litigious with respect to patents, trademarks, trade secrets, and other intellectual property rights. Companies in the medical device industry have used intellectual property litigation to gain a competitive advantage. Our commercial success depends in part upon our ability and that of our contract manufacturers and suppliers to manufacture, market, and sell our products, and to use our proprietary technologies without infringing, misappropriating or otherwise violating the proprietary rights or intellectual property of third parties. We may in the future become party to, or be threatened with, adversarial proceedings or litigation regarding intellectual property rights with respect to our products and any future products and technology, whether or not we are actually infringing, misappropriating or otherwise violating the rights of third parties. While we take reasonable steps to ensure that we and our suppliers do not infringe upon, misappropriate, or otherwise violate the intellectual property rights of others, there may be pertinent intellectual property rights of others which we or our suppliers are presently unaware.

We are presently a party to a lawsuit involving Catalyst OrthoScience, which we describe in greater detail in "Business—Legal Proceedings." In addition, we may in the future be party to other lawsuits or other administrative proceedings, including interference proceedings, post grant review and inter partes review before the USPTO, involving our patents or other intellectual property, regardless of merit. Third parties may assert infringement claims against us based on existing patents or patents that may be granted in the future, regardless of their merit. There is a risk that third parties may choose to engage in litigation with us to enforce or to otherwise assert their patent rights against us. Even if we believe such claims are without merit, a court of competent jurisdiction could hold that these third-party patents are valid, enforceable and infringed, which could have a negative impact on our ability to commercialize our current and any future product candidates. In order to successfully challenge the validity of any such U.S. patent in federal court, we would need to overcome a presumption of validity. As this is a high burden and requires us to present clear and convincing evidence as to the invalidity of any such U.S. patent claim, there is no assurance that a court of competent jurisdiction would invalidate the claims of any such U.S. patent. Conversely, a patent owner need only prove infringement by a preponderance of the evidence, which is a lower burden of proof. Moreover, given the vast number of patents in our field of technology, we cannot be certain that we do not infringe existing patents or that we will not infringe patents that may be granted in the future. While we may decide to initiate proceedings to challenge the validity of these or other patents in the future, we may be unsuccessful, and courts or patent offices in the United States and abroad could uphold the validity of any such patent. Furthermore, because patent applications can take many years to issue and may be confidential for 18 months or more after filing, and because pending patent claims can be revised before issuance, there may be applications now pending which may later result in issued patents that may be infringed by the manufacture, use or sale of our product candidates. Regardless of when filed, we may fail to identify relevant third-party patents or patent applications, or we may incorrectly conclude that a third-party patent is invalid or not infringed by our product candidates or activities. If a patent holder believes that our product candidate or technology platform infringes its patent, the patent holder may sue us even if we have received patent protection for our technology. Moreover, we may face patent infringement claims from nonpracticing entities that have no relevant product revenue and against whom our own patent portfolio may thus have no deterrent effect. If a patent infringement suit were threatened or brought against us, we could be

forced to stop or delay research, development, manufacturing or sales of the product or product candidate that is the subject of the actual or threatened suit.

If we are found to infringe a third party's intellectual property rights, we could be required to obtain a license from such third party to continue selling, developing and marketing our products and techniques. However, we may not be able to obtain any required license on commercially reasonable terms or at all. The acquisition or licensing of third-party intellectual property rights is a competitive area, and our competitors may pursue strategies to acquire or license third party intellectual property rights that we may consider attractive or necessary. If we are unable to successfully obtain rights to required third party intellectual property rights or maintain the existing intellectual property rights we have, we may have to abandon development of the relevant products or redesign those products that contain the allegedly infringing intellectual property, which could harm our business, financial condition and results of operations. Even if we were able to obtain a license, it could be non-exclusive, thereby giving our competitors access to the same technologies licensed to us. We could be forced, including by court order, to cease commercializing the infringing technology or product. In addition, we could be found liable for monetary damages, including treble damages and attorneys' fees if we are found to have willfully infringed a patent. A finding of infringement could force us to cease some of our business operations, which could materially harm our business. Claims that we have misappropriated the confidential information or trade secrets of third parties could have a similar negative impact on our business. Intellectual property litigation may lead to unfavorable publicity that harms our reputation and causes the market price of our common shares to decline. A legal proceeding, regardless of the outcome, could drain our financial resources and divert the time and effort of our management. Protracted litigation to defend or prosecute our intellectual property rights could result in our customers or potential customers deferring or limiting their purchase or use of the affected products until resolution of the litigation.

Because competition in our industry is intense, competitors may infringe or otherwise violate our issued patents, patents of our licensors or other intellectual property. To counter infringement or unauthorized use, we may be required to file infringement claims, which can be expensive and time consuming, and could distract our technical and management personnel from their normal responsibilities. Any claims we assert against perceived infringers could provoke these parties to assert counterclaims or file administrative actions against us alleging that we infringe their patents. In addition, in a patent infringement proceeding, a court may decide that a patent of ours is invalid or unenforceable, in whole or in part, construe the patent's claims narrowly or refuse to stop the other party from using the technology at issue on the grounds that our patents do not cover the technology in question. An adverse result in any litigation proceeding or administrative action could put one or more of our patents at risk of being invalidated or interpreted narrowly and could put our related pending patent applications at risk of not issuing. Our competitors may assert invalidity on various grounds, including lack of novelty, obviousness or that we were not the first applicant to file a patent application related to our product. We may elect to enter into license agreements in order to settle patent infringement claims or to resolve disputes before litigation, and any such license agreements may require us to pay royalties and other fees that could be significant. Furthermore, because of the substantial amount of discovery required in connection with intellectual property litigation, there is a risk that some of our confidential information could be compromised by disclosure.

Our competitors, many of which have made substantial investments in patent portfolios, trade secrets, trademarks and competing technologies, may have applied for or obtained, or may in the future apply for or obtain, patents or trademarks that may prevent, limit or otherwise interfere with our ability to make, use, sell and/or export our products or to use our technologies or product names. Moreover, individuals and groups that are non-practicing entities, commonly referred to as "patent trolls," purchase patents and other intellectual property assets for the purpose of making claims of infringement in order to extract settlements. From time to time, we may receive threatening letters, notices or "invitations to license," or may be the subject of claims that our products and business operations infringe or violate the intellectual property rights of others. The defense of these matters can be time consuming, costly to defend in litigation, divert management's attention and resources, damage our reputation and brand and cause us to incur significant expenses or make substantial payments. Further, the outcome of litigation is uncertain and such litigation could result in us being forced to stop making, selling or using products or technologies that allegedly infringe the asserted intellectual property; pay substantial damages or royalties to the party whose intellectual property rights we may be found to be infringing; redesign those products that contain the allegedly infringing intellectual property, which could be costly, disruptive, or infeasible; and or attempt to obtain a license to

the relevant intellectual property from third parties, which may not be available on reasonable terms, or at all, or, from third parties. In addition, third parties may assert infringement claims against our customers. These claims may require us to initiate or defend protracted and costly litigation on behalf of our customers or indemnify our customers for any costs associated with their own initiation or defense of infringement claims, regardless of the merits of these claims. If any of these claims succeed or settle, we may be forced to pay damages or settlement payments on behalf of our customers or may be required to obtain licenses for the products they use. If we cannot obtain all necessary licenses on commercially reasonable terms, our customers may be forced to stop using our products.

Because of the substantial amount of discovery required in connection with intellectual property litigation, there is a risk that some of our confidential information could be compromised by disclosure during litigation. There could also be public announcements of the results of hearing, motions, or other interim developments. If securities analysts or investors perceive these results to be negative, it could have a material adverse effect on the price of shares of our common stock. Even if we ultimately prevail, a court may decide not to grant an injunction against further infringing activity and instead award only monetary damages, which may not be an adequate remedy. Furthermore, even if resolved in our favor, the monetary cost of such litigation and the diversion of the attention of our management could outweigh any benefit we receive as a result of the proceedings. Uncertainties resulting from the initiation and continuation of patent litigation or other proceedings could have a material adverse effect on our business. Any of the foregoing may cause us to incur substantial costs and could place a significant strain on our financial resources, divert the attention of management from our core business, and harm our reputation.

If we fail to execute invention assignment agreements with our employees and contractors involved in the development of intellectual property or are unable to protect the confidentiality of our trade secrets, the value of our products and our business and competitive position could be harmed.

In addition to patent protection, we also rely on the protection of trademarks, copyrights, trade secrets, know-how and confidential and proprietary information. We generally enter into confidentiality and invention assignment agreements with our employees, consultants and third parties upon their commencement of a relationship with us. However, we may not enter into such agreements with all employees, consultants and third parties who have been involved in the development of our intellectual property. We may also be subject to claims that our current or former employees, contractors, or other third parties have an ownership interest in our current or future patents, patent applications, or other intellectual property rights, including as an inventor or co-inventor. We may be subject to ownership or inventorship disputes in the future arising, for example, from conflicting obligations of employees, consultants, or others who were or are involved in developing our products, services, or technologies. In addition, these agreements may not provide meaningful protection against the unauthorized use or disclosure of our trade secrets or other confidential information, and adequate remedies may not exist if unauthorized use or disclosure were to occur. The exposure of our trade secrets and other proprietary information would impair our competitive advantages and could have a material adverse effect on our business, financial condition and results of operations. In particular, a failure to protect our proprietary rights may allow competitors to copy our products and procedures, which could adversely affect our pricing and market share. We may need to share our proprietary information, including trade secrets, with our current and future business partners, collaborators, contractors and others located in countries at heightened risk of theft of trade secrets, including through direct intrusion by private parties or foreign actors, and those affiliated with or controlled by state actors. As a result, we may encounter significant problems in protecting and defending our intellectual property both in the United States and abroad. The failure to obtain or maintain trade secret protection could adversely affect our competitive business position. Further, other parties may independently develop substantially equivalent know-how and technology. In addition to contractual measures, we try to protect the confidential nature of our proprietary information using commonly accepted physical and technological security measures. Such measures may not, for example, in the case of misappropriation of a trade secret by an employee or third party with authorized access, provide adequate protection for our proprietary information. Our security measures may not prevent an employee or consultant from misappropriating our trade secrets and providing them to a competitor, and recourse we take against such misconduct may not provide an adequate remedy to protect our interests fully. Unauthorized parties may also attempt to copy or reverse engineer certain aspects of our products that we consider proprietary. Enforcing a claim that a party illegally disclosed or misappropriated a trade secret can be difficult, expensive and time-consuming, and the outcome is unpredictable.

Even though we use commonly accepted security measures, trade secret violations are often a matter of state law, and the criteria for protection of trade secrets can vary among different jurisdictions. In addition, trade secrets may be independently developed by others in a manner that could prevent legal recourse by us. While we have agreements with our employees, consultants and third parties that obligate them to assign their inventions to us, these agreements may not be self-executing, not all employees or consultants may enter into such agreements, or employees or consultants may breach or violate the terms of these agreements, and we may not have adequate remedies for any such breach or violation. If any of our intellectual property or confidential or proprietary information, such as our trade secrets, were to be disclosed or misappropriated, or if any such information was independently developed by a competitor, it could have a material adverse effect on our competitive position, business, financial condition, results of operations and prospects.

If our trademarks and trade names are not adequately protected, we may not be able to build name recognition in our markets of interest and our competitive position may be harmed.

We rely on our trademarks, trade names and brand names to distinguish our systems from the products of our competitors and have registered or applied to register many of these trademarks. We cannot guarantee that our trademark applications will be approved. Third parties may also oppose our trademark applications or otherwise challenge our use of the trademarks. Our registered and unregistered trademarks, trade names, and brand names may be challenged, infringed, circumvented, declared generic or determined to be violating or infringing on other marks. We may not be able to protect our rights to these trademarks, trade names, and brand names which we rely upon to build name recognition among potential partners and customers in our markets of interest. In the event that our trademarks are successfully challenged, we could be forced to rebrand our systems, which could result in loss of brand recognition, and could require us to devote resources to advertising and marketing new brands. Further, there can be no assurance that competitors will not infringe our trademarks or that we will have adequate resources to enforce our trademarks.

Patent terms may not be sufficient to effectively protect our systems and business for an adequate period of time.

Patents have a limited lifespan. In the United States, the natural expiration of a utility patent is generally 20 years after its first effective non-provisional filing date and the natural expiration of a design patent is generally 14 years after its issue date, unless the filing date occurred on or after May 13, 2015, in which case the natural expiration of a design patent is generally 15 years after its issue date. However, the actual protection afforded by a patent varies from country to country, and depends on many factors, including the type of patent, the scope of its coverage, the availability of regulatory-related extensions, the availability of legal remedies in a particular country and the validity and enforceability of the patent. Although various extensions may be available, the term of a patent, and the protection it affords, is limited. We may also be required to disclaim a portion of a patent term in order to overcome double patenting rejections from the patent office, thus potentially shortening our exclusivity period. Even if patents covering our proprietary technologies and their uses are obtained, once the patent has expired, we may be open to competition, which may harm our business prospects. In addition, although upon issuance in the United States a patent's term can be extended based on certain delays caused by the USPTO, this extension can be reduced or eliminated based on certain delays caused by the patent applicant during patent prosecution. A patent term extension based on regulatory delay may be available in the United States. However, only a single patent can be extended for each marketing approval, and any patent can be extended only once, for a single product. Moreover, the scope of protection during the period of the patent term extension does not extend to the full scope of the claim, but instead only to the scope of the product as approved. Laws governing analogous patent term extensions in foreign jurisdictions vary widely, as do laws governing the ability to obtain multiple patents from a single patent family. Additionally, we may not receive an extension if we fail to exercise due diligence during the testing phase or regulatory review process, apply within applicable deadlines, fail to apply prior to expiration of relevant patents or otherwise fail to satisfy applicable requirements. Given the amount of time required for the development, testing and regulatory review of new products, patents protecting such products might expire before or shortly after such products are commercialized. If we do not have sufficient patent terms to protect our products, proprietary technologies and their uses, our business would be seriously harmed. As our patents expire, the scope of our patent protection will be reduced, which may reduce or eliminate any competitive advantage afforded by our patent portfolio. As a result, our reduced patent portfolio may not provide us with sufficient rights to exclude others from commercializing products similar or identical to ours.

Changes in U.S. or foreign patent laws or their interpretation may limit our ability to obtain, maintain, defend and/or enforce our patents.

Patent reform legislation could increase the uncertainties and costs surrounding the prosecution of our patent applications and the enforcement or defense of our issued patents. The Leahy-Smith America Invents Act ("Leahy-Smith Act") includes a number of significant changes to U.S. patent law. These include provisions that affect the way patent applications are prosecuted and also affect patent litigation. The USPTO has developed regulations and procedures to govern administration of the Leahy-Smith Act, and many of the substantive changes to patent law associated with the Leahy-Smith Act, and in particular, the first to file provisions, which became effective on March 16, 2013. The first to file provisions limit the rights of an inventor to patent an invention if not the first to file an application for patenting that invention, even if such invention was the first invention. Accordingly, it is not clear what, if any, impact the Leahy-Smith Act will have on the operation of our business.

However, the Leahy-Smith Act and its implementation could increase the uncertainties and costs surrounding the enforcement and defense of our issued patents. For example, the Leahy-Smith Act provides that an administrative tribunal known as the Patent Trial and Appeals Board ("PTAB"), provides a venue for challenging the validity of patents at a cost that is much lower than district court litigation and on timelines that are much faster. Although it is not clear what, if any, long-term impact the PTAB proceedings will have on the operation of our business, the initial results of patent challenge proceedings before the PTAB since its inception in 2013 have resulted in the invalidation of many U.S. patent claims. The availability of the PTAB as a lower-cost, faster and potentially more potent tribunal for challenging patents could increase the likelihood that our own patents will be challenged, thereby increasing the uncertainties and costs of maintaining and enforcing them.

The America Invents Act also includes a number of significant changes that affect the way U.S. patent applications will be prosecuted and also may affect patent litigation. These include allowing third-party submission of prior art to the USPTO during patent prosecution and additional procedures to attack the validity of a patent by the USPTO administered post-grant proceedings, including post-grant review, inter partes review and derivation proceedings.

Because of a lower evidentiary standard in USPTO proceedings compared to the evidentiary standard in U.S. federal courts necessary to invalidate a patent claim, a third party could potentially provide evidence in a USPTO proceeding sufficient for the USPTO to hold a claim invalid even though the same evidence would be insufficient to invalidate the claim if first presented in a district court action. Therefore, the America Invents Act and its implementation could increase the uncertainties and costs surrounding the prosecution of our patent applications and the enforcement or defense of our issued patents. In addition, future actions by the U.S. Congress, the federal courts and the USPTO could cause the laws and regulations governing patents to change in unpredictable ways. Any of the foregoing could harm our business, financial condition and results of operations.

In addition, on June 1, 2023, the European Union Patent Package, or the EU Patent Package, regulations were implemented with the goal of providing a single pan-European Unitary Patent and a new European Unified Patent Court (the "UPC"), for litigation involving European patents. As a result, all European patents, including those issued prior to ratification of the EU Patent Package, now by default automatically fall under the jurisdiction of the UPC, unless otherwise opted out. It is uncertain how the UPC will impact granted European patents in the biotechnology and pharmaceutical industries. Our European patent applications, if issued, could be challenged in the UPC. During the first seven years of the UPC's existence, the UPC legislation allows a patent owner to opt its European patents out of the jurisdiction of the UPC. We have opted out certain of our European patents and may decide to opt out our future European patents from the UPC, but doing so may preclude us from realizing the benefits of the UPC. Moreover, if we do not meet all of the formalities and requirements for opt-out under the UPC, our current or future European patents could remain under the jurisdiction of the UPC. The UPC will provide our competitors with a new forum to centrally revoke our European patents, and allow for the possibility of a competitor to obtain pan-European injunction. Such a loss of patent protection could have a material adverse impact on our business and our ability to commercialize our technology and our systems due to increased competition and, resultantly, on our business, financial condition, results of operations and prospects. The UPC and Unitary Patent are significant changes in European patent practice. As the UPC is a new court system, there is no precedent for the court, increasing the uncertainty of any litigation in the UPC.

We have foreign intellectual property rights and may be unable to enforce our intellectual property rights throughout the world.

We have intellectual property rights outside the United States. Filing, prosecuting and defending patents or trademarks on our systems and any future products in all countries throughout the world would be prohibitively expensive, and the laws of foreign countries may not protect our rights to the same extent as the laws of the United States. Consequently, we may not be able to prevent third parties from practicing our inventions or utilizing our trademarks in all countries outside the United States, or from selling or importing products made using our inventions in and into the United States or other jurisdictions. Competitors may use our technologies in jurisdictions where we have not obtained patent protection to develop their own products and, further, may export otherwise infringing products to territories where we have patent protection but enforcement is not as strong as that in the United States. These products may compete with our products and any future products, and our patents or other intellectual property rights may not be effective or sufficient to prevent them from competing in these jurisdictions.

The laws of some foreign countries do not protect intellectual property rights to the same extent as the laws of the United States. Many companies have encountered significant problems in protecting and defending intellectual property rights in certain foreign jurisdictions. This could make it difficult for us to stop infringement of our foreign patents, if obtained, or the misappropriation of our other intellectual property rights. For example, some foreign countries have compulsory licensing laws under which a patent owner must grant licenses to third parties. In addition, some countries limit the enforceability of patents against third parties, including government agencies or government contractors. In these countries, patents may provide limited or no benefit. Patent protection must ultimately be sought on a country-by-country basis, which is an expensive and time-consuming process with uncertain outcomes. Accordingly, we may choose not to seek patent protection in certain countries, and we will not have the benefit of patent protection in such countries.

Proceedings to enforce our patent rights in foreign jurisdictions could result in substantial costs and divert our efforts and attention from other aspects of our business. Accordingly, our efforts to protect our intellectual property rights in such countries may be inadequate. In addition, changes in the law and legal decisions by courts in the United States and foreign countries may affect our ability to obtain adequate protection for our technology and the enforcement of our intellectual property.

The legal systems in certain countries may also favor state-sponsored or companies headquartered in particular jurisdictions over our first-in-time patents and other intellectual property protection. We are aware of incidents where such entities have stolen the intellectual property of domestic companies in order to create competing products and we believe we may face such circumstances ourselves in the future. For example, through its “Annual Special 301 Report on Intellectual Property,” the Office of the United States Trade Representative has been reporting on the adequacy and effectiveness of intellectual property protection in a number of foreign countries that are U.S. trading partners and their protection and enforcement of intellectual property rights. Placement of a country on the Priority Watch List indicates that particular problems exist in that country with respect to intellectual property protection, enforcement, or market access for persons relying on intellectual property rights. Countries placed on the Priority Watch List are the focus of increased bilateral attention concerning the specific problem areas. It is possible that we will not be able to enforce our intellectual property rights against third parties that misappropriate our proprietary technology in those countries.

Moreover, geopolitical actions in the U.S. and in foreign countries could increase the uncertainties and costs surrounding the prosecution or maintenance of our patent applications or those of any current or future licensors and the maintenance, enforcement or defense of our issued patents or those of any current or future licensors. Government actions may also prevent maintenance of issued patents in Russia. These actions could result in abandonment or lapse of our future Russian patents or patent applications, resulting in partial or complete loss of patent rights in Russia. Competitors may use our technologies in jurisdictions where we have not obtained patent protection to develop their own products and may export otherwise infringing products to territories where we have patent protection, but enforcement rights are not as strong as those in the United States. These products may compete with our products and our patents or other IP rights may not be effective or sufficient to prevent them from competing.

We are and in the future may be subject to claims that we or our employees have misappropriated the intellectual property of a third party, including trade secrets or know-how, or are in breach of non-competition or non-solicitation agreements with our competitors and third parties may claim an ownership interest in intellectual property we regard as our own.

We could in the future be subject to claims that we or our employees have inadvertently or otherwise used or disclosed alleged trade secrets or other confidential information of former employers or competitors. Many of our employees and consultants were previously employed at or engaged by other medical device companies, including our competitors or potential competitors. Some of these employees, consultants and contractors may have executed proprietary rights, non-disclosure and non-competition agreements in connection with such previous employment. Although we try to ensure that our employees and consultants do not use the intellectual property, proprietary information, know-how or trade secrets of others in their work for us, we may be subject to claims that we or these individuals have, inadvertently or otherwise, misappropriated the intellectual property or disclosed the alleged trade secrets or other proprietary information, of these former employers, competitors or other third parties. Additionally, we may be subject to claims from third parties challenging our ownership interest in or inventorship of intellectual property we regard as our own, for example, based on claims that our agreements with employees or consultants obligating them to assign intellectual property to us are ineffective or in conflict with prior or competing contractual obligations to assign inventions to another employer, to a former employer, or to another person or entity. Litigation may be necessary to defend against claims, and it may be necessary or we may desire to enter into a license to settle any such claim; however, there can be no assurance that we would be able to obtain a license on commercially reasonable terms, if at all. If our defense to those claims fails, in addition to paying monetary damages or a settlement payment, a court could prohibit us from using technologies, features or other intellectual property that are essential to our products, if such technologies or features are found to incorporate or be derived from the trade secrets or other proprietary information of the former employers. An inability to incorporate technologies, features or other intellectual property that are important or essential to our products could have a material adverse effect on our business and competitive position and may prevent us from selling our products. In addition, we may lose valuable intellectual property rights or personnel. Even if we are successful in defending against these claims, litigation could result in substantial costs and could be a distraction to management. Any litigation or the threat thereof may adversely affect our ability to hire employees or contract with independent sales representatives. A loss of key personnel or their work product could hamper or prevent our ability to commercialize our products, which could materially and adversely affect our business, financial condition, operating results, cash flows and prospects.

If our third-party manufacturing partners do not respect our intellectual property and trade secrets and produce competitive products using our designs or intellectual property, our business, financial condition and results of operations would be harmed.

Although our agreements with third-party manufacturing partners generally seek to prevent them from misusing our intellectual property and trade secrets, or using our designs to manufacture products for our competitors, we may be unsuccessful in monitoring and enforcing our intellectual property rights against these manufacturing partners and may find counterfeit goods in the market being sold as our systems or future products similar to ours produced for our competitors using our intellectual property. Although we take steps to stop counterfeits, we may not be successful and network operators who purchase these counterfeit goods may experience product defects or failures, harming our reputation and brand and causing us to lose future sales. Any of the foregoing could harm our business, financial condition and results of operations.

Our use of open source software could impose limitations on our ability to commercialize our product.

We may use open source software in the future that contains modules licensed for use from third-party authors under open source licenses. Some of the software may be provided under license arrangements that allow use of the software for research or other non-commercial purposes. As a result, in the future, as we seek to use our platform in connection with commercially available products, we may be required to license that software under different license terms, which may not be possible on commercially reasonable terms, if at all. If we are unable to license software components on terms that permit its use for commercial purposes, we may be required to replace those software components, which could result in delays, additional cost and/or additional regulatory approvals.

Use and distribution of open source software may entail greater risks than use of third-party commercial software, as open source licensors generally do not provide warranties or other contractual protections regarding infringement claims or the quality of the software code. Some open source licenses contain requirements that we make available source code for modifications or derivative works we create based upon the type of open source software we use. If we combine our proprietary software with open source software in a certain manner, we could, under certain of the open source licenses, be required to release the source code of our proprietary software to the public. This could allow our competitors to create similar products with lower development effort and time, and ultimately could result in a loss of product sales for us. Although we monitor our use of open source software, the terms of many open source licenses have not been interpreted by U.S. courts, and there is a risk that those licenses could be construed in a manner that could impose unanticipated conditions or restrictions on our ability to commercialize our product candidates. We could be required to seek licenses from third parties in order to continue offering our product candidates, to re-engineer our product candidates or to discontinue the sale of our product candidates in the event re-engineering cannot be accomplished on a timely basis, any of which could materially and adversely affect our business, financial condition, results of operations and prospects.

Intellectual property rights do not necessarily address all potential threats to our competitive advantage.

The degree of future protection afforded by our intellectual property rights is uncertain because intellectual property rights have limitations and may not adequately protect our business or permit us to maintain our competitive advantage. For example:

- others may be able to make products that are similar to ours but that are not covered by the claims of the patents that we own or have exclusively licensed;
- we or our licensors or future collaborators might not have been the first to make the inventions covered by the issued patent or pending patent application that we own or have exclusively licensed;
- we or our licensors or future collaborators might not have been the first to file patent applications covering certain of our inventions;
- others may independently develop similar or alternative technologies or duplicate any of our technologies without infringing our intellectual property rights;
- it is possible that our pending patent applications will not lead to issued patents;
- issued patents that we own or have exclusively licensed may be held invalid or unenforceable, as a result of legal challenges by our competitors;
- our competitors might conduct research and development activities in countries where we do not have patent rights and then use the information learned from such activities to develop competitive products for sale in our major commercial markets;
- we may not develop additional proprietary technologies that are patentable;
- we cannot predict the scope of protection of any patent issuing based on our patent applications, including whether the patent applications that we own or in-license will result in issued patents with claims that cover our products or uses thereof in the United States or in other foreign countries;
- the claims of any patent issuing based on our patent applications may not provide protection against competitors or any competitive advantages, or may be challenged by third parties;
- if enforced, a court may not hold that our patents are valid, enforceable and infringed;
- we may need to initiate litigation or administrative proceedings to enforce and/or defend our patent rights which will be costly whether we win or lose;

- we may choose not to file a patent in order to maintain certain trade secrets or know-how, and a third party may subsequently file a patent covering such intellectual property;
- the laws of foreign countries may not protect our or our licensors', as the case may be, proprietary rights to the same extent as the laws of the United States;
- the inventors of our owned or in-licensed patents or patent applications may become involved with competitors, develop products or processes which design around our patents or become hostile to us or the patents or patent applications on which they are named as inventors;
- it is possible that our owned or in-licensed patents or patent applications omit individual(s) that should be listed as inventor(s) or include individual(s) that should not be listed as inventor(s), which may cause these patents or patents issuing from these patent applications to be held invalid or unenforceable;
- we have engaged in scientific collaborations in the past, and will continue to do so in the future. Such collaborators may develop adjacent or competing products to ours that are outside the scope of our patents;
- we may fail to adequately protect and police our trademarks and trade secrets; and
- the patents of others may have an adverse effect on our business, including if others obtain patents claiming subject matter similar to or improving that covered by our patents and patent applications.

Should any of these events occur, they could significantly harm our business, results of operations and prospects.

Risks Related to Regulatory Matters

Our devices and operations are subject to extensive government regulation and oversight in the United States, and our failure to comply with applicable requirements could harm our business.

Our systems are regulated as medical devices in the United States. Medical devices and their manufacturers and product developers are subject to extensive regulation in the United States, including by the FDA. The FDA regulates, among other things, with respect to medical devices: design, development, and manufacturing; testing, labeling, content, and language of instructions for use and storage; clinical trials; product safety; establishment registration and device listing; marketing, sales, and distribution; premarket clearance, classification, and approval or certification; recordkeeping procedures; advertising and promotion; recalls and field safety corrective actions; post-market surveillance, including reporting of deaths or serious injuries and malfunctions that, if they were to recur, could lead to death or serious injury; post-market studies; and product import and export.

The regulations to which we are subject are complex, burdensome to understand and apply, and have tended to become more stringent over time. Regulatory changes could result in restrictions on our ability to carry on or expand our operations, higher than anticipated costs or lower than anticipated sales. The FDA enforces its regulatory requirements through, among other means, periodic unannounced inspections. We do not know whether we or any of our contract manufacturers or commercial partners will be found compliant in connection with any future FDA or foreign inspections. Failure to comply with applicable regulations could jeopardize our ability to sell our systems and result in enforcement actions such as: warning letters; fines; injunctions; civil penalties; termination of distribution; import alerts; recalls or seizures of products; delays in the introduction of products into the market; total or partial suspension of production; refusal to grant future clearances or approvals; withdrawals or suspensions of clearances or approvals, resulting in prohibitions on sales of our systems; and in the most serious cases, criminal penalties.

Our relationships with customers, physicians and third-party payors are subject to federal and state health care fraud and abuse laws, false claims laws, physician payment transparency laws and other health care laws and regulations. If we or our employees, independent contractors, consultants, commercial partners, or vendors violate these laws we could face substantial penalties.

Our relationships with customers, physicians and third-party payors are subject to federal and state health care fraud and abuse laws, false claims laws, physician payment transparency laws and other health care laws and regulations. In particular, the promotion, sales and marketing of health care items and services is subject to extensive laws and regulations designed to prevent fraud, kickbacks, self-dealing and other abusive practices. These laws and regulations may restrict or prohibit a wide range of pricing, discounting, marketing and promotion, sales commission, customer incentive and other business arrangements. The U.S. health care laws and regulations that may affect our ability to operate include, but are not limited to:

- the federal Anti-Kickback Statute, which prohibits, among other things, any person or entity from knowingly and willfully, offering, paying, soliciting or receiving any remuneration, directly or indirectly, overtly or covertly, in cash or in kind, to induce or reward, or in return for, the purchasing, leasing, ordering, arranging for or recommending the purchase, lease, or order of any item or service reimbursable under Medicare, Medicaid or other federal health care programs. The term "remuneration" has been broadly interpreted to include anything of value. Although there are a number of statutory exceptions and regulatory safe harbors protecting some common activities from prosecution, the exceptions and safe harbors are drawn narrowly. Practices that may be alleged to be intended to induce the purchases or recommendations, include any payments of more than fair market value, may be subject to scrutiny if they do not qualify for an exception or safe harbor. In addition, a person or entity does not need to have actual knowledge of this statute or specific intent to violate it in order to have committed a violation;

- federal civil and criminal false claims laws, including the federal civil False Claims Act, and civil monetary penalty laws, which prohibit, among other things, individuals or entities from knowingly presenting, or causing to be presented, claims for payment or approval from Medicare, Medicaid or other federal government programs that are false or fraudulent; knowingly making, using, or causing to be made or used, a false record or statement material to a false or fraudulent claim or obligation to pay or transmit money or property to the federal government; or knowingly making a false statement or representation of a material fact to improperly avoid, decrease or conceal an obligation to pay money to the federal government, including federal health care programs. In addition, the government may assert that a claim including items or services resulting from a violation of the federal Anti-Kickback Statute constitutes a false or fraudulent claim for purposes of the federal civil False Claims Act;

- the federal Health Insurance Portability and Accountability Act of 1996 ("HIPAA") which created new federal civil and criminal statutes that prohibit knowingly and willfully executing, or attempting to execute, a scheme to defraud any health care benefit program or obtain, by means of false or fraudulent pretenses, representations, or promises, any of the money or property owned by, or under the custody or control of, any health care benefit program, including private third-party payors, and knowingly and willfully falsifying, concealing or covering up by any trick, scheme or device, a material fact or making any materially false, fictitious or fraudulent statements in connection with the delivery of, or payment for, health care benefits, items or services. Similar to the federal Anti-Kickback Statute, a person or entity does not need to have actual knowledge of the statute or specific intent to violate it in order to have committed a violation;

- federal consumer protection and unfair competition laws, which broadly regulate marketplace activities and activities that potentially harm consumers;

- the federal Physician Payments Sunshine Act, which requires certain manufacturers of drugs, devices, biologicals and medical supplies for which payment is available under Medicare, Medicaid or the Children's Health Insurance Program (with certain exceptions) to report annually to the government information related to payments or other transfers of value made to physicians (defined to include doctors, dentists, optometrists, podiatrists and chiropractors), certain other health care professionals (including

physician assistants, nurse practitioners, clinical nurse specialists, certified nurse anesthetists, anesthesiologist assistants and certified nurse midwives) and teaching hospitals, as well as ownership and investment interests held by physicians and their immediate family members; and

- state and foreign equivalents of each of the health care laws described above, among others, some of which may be broader in scope including, without limitation, state anti-kickback and false claims laws that may apply to sales or marketing arrangements and claims involving health care items or services reimbursed by non-governmental third party payors, including private insurers, or that apply regardless of payor (including cash payments); state laws that require device companies to comply with the industry's voluntary compliance guidelines and the relevant compliance guidance promulgated by the federal government; state laws that require device manufacturers to report information related to payments and other transfers of value to physicians and other health care providers, and marketing expenditures; state laws that prohibit fee-splitting arrangements between companies and physicians and other health care professionals; and state and local laws requiring the registration of device sales and medical representatives. Greater scrutiny of marketing practices in the medical device industry has resulted in numerous government investigations by various government authorities, and this industry-wide enforcement activity is expected to continue. The shifting regulatory environment, along with the requirement to comply with multiple jurisdictions with different and difficult compliance and reporting requirements, increases the possibility that we may run afoul of one or more laws. The costs to comply with these regulatory requirements are becoming more expensive and will also impact our profitability.

Because of the breadth of these laws and the narrowness of the statutory exceptions and regulatory safe harbors available, it is possible that some of our business activities, including our arrangements with physicians, independent distributors and customers could be subject to challenge under one or more of such laws. We have entered into advisory board and consulting agreements with physicians, including some who have ownership interests in us and/or influence the ordering of or use our systems in procedures they perform. Compensation under some of these arrangements includes the provision of stock or stock options. Efforts to ensure that our business arrangements will comply with applicable health care laws may involve substantial costs. It is possible that governmental and enforcement authorities will conclude that our business practices may not comply with current or future statutes, regulations or case law interpreting applicable fraud and abuse or other health care laws and regulations.

If we or our employees, agents, independent contractors, consultants, commercial partners and vendors violate these laws, we may be subject to investigations, enforcement actions and/or significant penalties, including the imposition of significant civil, criminal and administrative penalties, damages, disgorgement, monetary fines, imprisonment, possible exclusion from participation in Medicare, Medicaid and other federal health care programs, contractual damages, reputational harm, diminished profits and future earnings, additional reporting requirements and/or oversight if we become subject to a corporate integrity agreement or similar agreement to resolve allegations of non-compliance with these laws, and curtailment of our operations, any of which could adversely affect our ability to operate our business and our results of operations.

Failure to maintain marketing authorizations for our systems, or to timely obtain necessary marketing authorizations for our future products, may have a material adverse effect on our business, financial condition, results of operations, and prospects.

In the United States, before we can market a new medical device, or a new use of, or other significant modification to an existing, marketed medical device, we must first receive either clearance under Section 510(k) of the Federal Food, Drug and Cosmetic Act (the "FDCA"), approval of a premarket approval application ("PMA"), or grant of a *de novo* classification request from the FDA, unless an exemption applies. In the 510(k) clearance process, before a device may be marketed, the FDA must determine that a proposed device is "substantially equivalent" to a legally-marketed "predicate" device, which includes a device that has been previously cleared through the 510(k) process, a device that was legally marketed prior to May 28, 1976 (pre-amendments device), a device that was originally on the U.S. market pursuant to an approved PMA and later down-classified, or a 510(k)-exempt device. To be "substantially equivalent," the proposed device must have the same intended use as the predicate device, and either have the same technological characteristics as the predicate device or have different technological characteristics and not raise different questions of safety or effectiveness than the predicate device. Clinical data are

sometimes required to support substantial equivalence. In the process of obtaining PMA approval, the FDA must determine that a proposed device is safe and effective for its intended use based, in part, on extensive data, including, but not limited to, technical, pre-clinical, clinical trial, manufacturing, and labeling data. The PMA process is typically required for devices that are deemed to pose the greatest risk, such as life-sustaining, life-supporting, or implantable devices.

In the *de novo* classification process, a manufacturer whose novel device under the FDCA would otherwise be automatically classified as Class III and require the submission and approval of a PMA prior to marketing is able to request down-classification of the device to Class I or Class II on the basis that the device presents a low or moderate risk. If the FDA grants the *de novo* classification request, the applicant will receive authorization to market the device. This device type may be used subsequently as a predicate device for future 510(k) submissions.

The PMA approval, 510(k) clearance and *de novo* classification processes can be expensive, lengthy, and uncertain. The FDA's 510(k) clearance process usually takes from three to 12 months, but can take longer. The process of obtaining a PMA is much more costly and uncertain than the 510(k) clearance process and generally takes from one to three years, or even longer, from the time the application is submitted to the FDA. In addition, a PMA generally requires the performance of one or more clinical trials. Clinical data may also be required in connection with an application for 510(k) clearance or a *de novo* classification request. Despite the time, effort and cost, a device may not obtain marketing authorization by the FDA. We have obtained 510(k) clearances for our commercialized medical devices and we must obtain marketing authorization for any future devices we develop, unless they are exempt. Marketing authorizations for any of our future products, if granted, may include significant limitations on the indicated uses for the device, which may limit the potential commercial market for the device.

In the United States, any modification to a medical device for which we have obtained marketing authorization may require us to submit a new 510(k) premarket notification and obtain clearance, to submit a PMA and obtain FDA approval, or to submit a *de novo* request prior to implementing the change. For example, any modification to a 510(k)-cleared device that could significantly affect its safety or effectiveness, or that would constitute a major change in its intended use, design or manufacture, generally requires a new 510(k) clearance or other marketing authorization. The FDA requires every manufacturer to make such determinations in the first instance, but the FDA may review any manufacturer's decision. The FDA may not agree with a manufacturer's decisions regarding whether new clearances or approvals are necessary. We have made modifications to 510(k)-cleared products in the past and have determined based on our review of the applicable FDA regulations and guidance that in certain instances new 510(k) clearances or PMA approvals were not required. We may make modifications or add additional features in the future to our medical devices that we believe do not require a new 510(k) clearance, *de novo* request, or approval of a PMA. If the FDA disagrees with our determination and requires us to seek new marketing authorizations for the modifications for which we have concluded that new marketing authorizations are unnecessary, we may be required to cease marketing or to recall the modified product until we obtain such marketing authorization, and we may be subject to significant regulatory fines or penalties. If the FDA requires us to go through a lengthier, more rigorous examination for future products or modifications to existing products than we had expected, product introductions or modifications could be delayed or canceled, which could adversely affect our business.

The FDA can delay, limit or deny marketing authorization of a device for many reasons, including:

- our inability to demonstrate to the satisfaction of the FDA that our products are substantially equivalent to a predicate device or are safe and effective for their intended uses;
- the disagreement of the FDA with the design or implementation of clinical trials or the interpretation of data from preclinical studies or clinical trials;
- serious and unexpected adverse device effects experienced by participants in clinical trials;
- the data from preclinical studies and clinical trials may be insufficient to support clearance, *de novo* classification, or approval, where required;
- our inability to demonstrate that the clinical and other benefits of the device outweigh the risks;

- the manufacturing process or facilities we use may not meet applicable requirements; and
- the potential for marketing authorization regulations of the FDA to change significantly in a manner rendering our clinical data or regulatory filings insufficient for marketing authorization.

We are subject to ongoing regulatory review and scrutiny. Failure to comply with post-marketing regulatory requirements could subject us to enforcement actions, including substantial penalties, and might require us to recall or withdraw a product from the market.

We are subject to ongoing and extensive regulatory requirements governing, among other things, the manufacture, marketing, advertising, medical device reporting, sale, promotion, import, export, registration, and listing of devices. For example, medical device manufacturers must submit certain reports to the FDA and keep required records as a condition of obtaining and maintaining marketing authorization. These reports include information about failures and certain adverse events potentially associated with the device after its marketing authorization. Failure to submit such reports, or failure to submit the reports in a timely manner, could result in enforcement action by the FDA. Following its review of the periodic reports, the FDA might ask for additional information or initiate further investigation.

Regulatory changes could result in restrictions on our ability to continue or expand our operations, higher than anticipated costs, or lower than anticipated sales. We have ongoing responsibilities under FDA regulations, and the FDA and state regulatory authorities have broad enforcement powers. Our failure to comply with applicable regulatory requirements could result in enforcement action by the FDA or state regulatory authorities, which may include any of the following or other sanctions:

- untitled letters or warning letters;
- fines, injunctions, consent decrees, and civil penalties;
- recalls, termination of distribution, administrative detention, or seizure of our products;
- customer notifications or repair, replacement, or refunds;
- operating restrictions or partial suspension or total shutdown of production;
- delays in or refusal to grant our requests for future clearances, *de novo* classifications or approvals, or comparable foreign marketing authorizations of new products, new intended uses, or modifications to existing products;
- withdrawals or suspensions of any granted marketing authorizations, resulting in prohibitions on sales of our products;
- FDA refusal to issue certificates to foreign governments needed to export products for sale in other countries; and
- criminal prosecution.

Any of these sanctions could result in negative publicity, higher than anticipated costs or lower than anticipated sales and have a material adverse effect on our reputation, business, financial condition, results of operations, and prospects.

In addition, the FDA may change its marketing authorization policies which may affect future products. The FDA may adopt additional regulations or revise existing regulations, or take other actions, which may prevent or delay marketing authorization of any products under development or impact our ability to modify any products authorized for market on a timely basis. Such changes may also occur in foreign jurisdictions where we may market our products in the future. Such changes could impose additional requirements upon us that could delay our ability to obtain future marketing authorizations, increase the costs of compliance, or restrict our ability to maintain any marketing authorizations we have obtained.

Our products must be manufactured in accordance with applicable laws and regulations, and we could be forced to recall our devices or terminate production if we fail to comply with these regulations.

In the United States, the methods used in, and the facilities used for, the manufacture of medical devices must comply with the FDA's cGMPs for medical devices, known as the Quality System Regulation ("QSR"), which is a complex regulatory scheme that covers the procedures and documentation of the design, testing, production, process controls, quality assurance, labeling, packaging, handling, storage, distribution, installation, servicing, and shipping of medical devices. On February 23, 2024, the FDA issued a final rule to amend the QSR to align more closely with the International Organization for Standardization ("ISO") standards. Specifically, this final rule, which the FDA expects to go into effect on February 2, 2026, replaces the QSR with the Quality Management System Regulation ("QMSR"), and among other things, incorporates by reference the quality management system requirements of ISO 13485:2016. Although the FDA has stated that the standards contained in ISO 13485:2016 are substantially similar to those set forth in the QSR, it is unclear the extent to which this final rule, once effective, could impose additional or different regulatory requirements on us that could increase the costs of compliance or otherwise create market pressure that may negatively affect our business. Furthermore, we are required to verify that our suppliers maintain facilities, procedures, and operations that comply with our quality standards and applicable regulatory requirements. The FDA enforces the QSR, and is expected to enforce the QMSR, through periodic announced or unannounced inspections of medical device manufacturing facilities, which may include the facilities of subcontractors. Our products are also subject to similar state regulations governing manufacturing.

Despite our efforts to ensure compliance, our third-party manufacturers may not take the necessary steps to comply with applicable regulations, which could cause delays in the delivery of our medical devices. In addition, failure to comply with applicable FDA requirements or later discovery of previously unknown problems with our products or manufacturing processes could result in, among other things: warning letters or untitled letters; fines, injunctions, or civil penalties; suspension or withdrawal of marketing authorizations; seizures or recalls of our products; total or partial suspension of production or distribution; administrative or judicially imposed sanctions; the FDA's refusal to grant pending or future clearances or approvals for our products or similar decisions by foreign regulatory authorities or notified bodies; clinical holds; refusal to permit the import or export of our products; and criminal prosecution of us, our suppliers, or our employees. Any of these actions could significantly and negatively affect supply of our products. If any of these events occurs, our reputation could be harmed, we could be exposed to product liability claims and we could lose customers and experience reduced sales and increased costs.

Legislative or regulatory reforms may have a material adverse effect on us.

From time to time, legislation is drafted and introduced in Congress that could significantly change the statutory provisions governing the regulation of medical devices. In addition, the FDA may change its clearance and approval policies, adopt additional regulations, or revise existing regulations, or take other actions, which may prevent or delay marketing authorization of any future products under development or impact our ability to modify any products for which we have already obtained marketing authorizations on a timely basis. For example, on January 31, 2024, the FDA issued a final rule to amend the QSR, which establishes cGMP requirements for medical device manufacturers, to align more closely with the ISO standards. Specifically, this final rule, which the FDA expects to go into effect on February 2, 2026, replaces the QSR with the QMSR, and among other things, incorporates by reference the quality management system requirements of ISO 13485:2016. This new final rule, referred to as the QMSR, will take effect on February 2, 2026. Although the FDA has stated that the standards contained in ISO 13485:2016 are substantially similar to those set forth in the existing QSR, it is unclear the extent to which this final rule, once effective, could impose additional or different regulatory requirements on us that could increase the costs of compliance or otherwise create market pressure that may negatively affect our business. It is unclear the extent to which any other legislative or regulatory proposal, if adopted, could impose additional or different regulatory requirements on us that could increase the costs of compliance or otherwise create competition that may negatively affect our business.

In addition, FDA regulations and guidance are often revised or reinterpreted by the FDA in ways that may significantly affect our business and our products. Any new statutes, regulations or revisions or reinterpretations of existing regulations may make it more difficult and costly to manufacture, market, or distribute our commercialized products, or may impose additional costs, lengthen marketing authorization review times, or make it more difficult

to obtain marketing authorizations for any future products we develop. If we are slow or unable to adapt to changes in existing requirements or the adoption of new requirements or policies, or if we are not able to maintain regulatory compliance, we may be subject to enforcement action and we may not achieve or sustain profitability.

In response to perceived increases in health care costs in recent years there have been and continue to be proposals by the federal government, state governments, regulators and third-party payors to control these costs and, more generally, to reform the U.S. health care system. Certain of these proposals could limit the prices we will be able to charge for our products or the amount of reimbursement available for our products and could limit the acceptance and availability of our products.

In March 2010, the federal government enacted the ACA. Among other provisions, the ACA established new value-based payment programs, increased funding of comparative effectiveness research, reduced hospital payments for avoidable readmissions and hospital acquired conditions, and pilot programs to evaluate alternative payment methodologies that promote care coordination (such as bundled physician and hospital payments). Since its enactment, there have been judicial, executive and Congressional challenges to certain aspects of the ACA. On June 17, 2021, the U.S. Supreme Court dismissed the most recent judicial challenge to the ACA brought by several states without specifically ruling on the constitutionality of the ACA.

Other legislative changes have been proposed and adopted since the ACA was enacted. The Budget Control Act of 2011, among other things, reduced Medicare payments to providers, effective on April 1, 2013 and, due to subsequent legislative amendments to the statute, will remain in effect through 2032, with the exception of a temporary suspension from May 1, 2020 through March 31, 2022, unless additional Congressional action is taken.

There are additional state and federal health care reform measures under consideration that may be adopted in the future which could have a material adverse effect on our industry generally and on our customers. We cannot predict what health care programs and regulations will be ultimately implemented at the federal or state level, particularly given the recent change in administration, or the effect of any future legislation or regulation. However, any regulatory and legal changes that lower reimbursement for our products, increase taxes on our medical devices, increase cost containment pressures on us or others in the health care sector or reduce medical procedure volumes could adversely affect our business, financial condition, results of operations or cash flows.

The misuse or off-label use of our systems may result in injuries that harm patients and lead to product liability suits, harm our reputation in the marketplace, or result in costly investigations, fines, or sanctions by regulatory bodies if we are deemed to have engaged in the promotion of these uses, any of which could be costly to our business.

Our systems that we currently commercialize, and any marketing authorization we may receive for future products, are, and will be, limited to specified indications for use. Our dedicated commercial leadership team and marketing personnel are trained to not promote our devices for uses outside of the FDA-authorized indications for use, known as “off-label uses.” We cannot, however, prevent a healthcare professional from using our devices off-label, when in the healthcare professional’s independent professional judgment he or she deems it appropriate. There may be increased risk of injury to patients if healthcare professionals attempt to use our devices off-label, which could harm our reputation in the marketplace among healthcare professionals and patients.

If the FDA determines that our promotional materials or training constitute promotion of an off-label use, it could request that we modify our training or promotional materials or subject us to regulatory or enforcement actions, including the issuance or imposition of an untitled letter, which is used for violators that do not necessitate a warning letter, injunction, seizure, civil fine, or criminal penalties. It is also possible that other federal or state enforcement authorities might take action under other regulatory authority, such as false advertising and consumer protection laws, or false claims laws, if they consider our business activities to constitute promotion of an off-label use, which could result in significant penalties, including, but not limited to, criminal, civil, and administrative penalties, damages, fines, disgorgement, exclusion from participation in government healthcare programs, and the curtailment of our operations.

In addition, healthcare professionals may misuse our products or use improper techniques if they are not adequately trained, potentially leading to injury and an increased risk of product liability. If our devices are misused

or used with improper technique, we may become subject to costly litigation by our customers or their patients. As described above, product liability claims could divert management's attention from our core business, be expensive to defend, and result in sizeable damage awards against us that may not be covered by insurance, all of which would have a material adverse effect on our business, financial condition, results of operations, and prospects.

Our implant systems and other future products may cause or contribute to adverse medical events or be subject to failures or malfunctions which we may be required to report to the FDA, and if we fail to do so, we would be subject to sanctions that could harm our reputation, business, financial condition, results of operations, and prospects. In addition, the discovery of serious safety issues, or a voluntarily or required recall, could have a negative impact on us.

It is possible that there may be side effects and adverse events associated with the use of our implant systems or any future products we develop. We have experienced a modest increase in the number of direct patient complaints relative to the number of procedures performed using our systems. The FDA's medical device reporting regulations require us to assess reportability of adverse events that come to our attention and report to the FDA when we receive or become aware of information that reasonably suggests that one or more of our products may have caused or contributed to a death or serious injury or malfunctioned in a way that, if the malfunction were to recur, it could cause or contribute to a death or serious injury. The timing of our obligation to report is triggered by the date we become aware of the adverse event as well as the nature of the event. We may fail to report adverse events of which we become aware within the prescribed timeframe. We may also fail to recognize that we have become aware of a reportable adverse event, especially if it is not reported to us as an adverse event or if it is an adverse event that is unexpected or removed in time from the use or implant of the system. The FDA may also disagree with our determinations that an event was not reportable. If we fail to comply with our reporting obligations, the FDA could take action, including warning letters, untitled letters, administrative actions, criminal prosecution, imposition of civil monetary penalties, revocation of our marketing authorizations, seizure of our implant systems, or delay in obtaining marketing authorizations for our future products.

The FDA has the authority to require the recall of commercialized products based on a finding that there is reasonable probability that the device could cause serious, adverse health consequences or death. We may also choose to voluntarily recall a product if any material deficiency is found. A government-mandated or voluntary recall by us could occur as a result of an unacceptable risk to health, component failures, malfunctions, manufacturing defects, labeling or design deficiencies, packaging defects, or other deficiencies or failures to comply with applicable regulations. Product defects or other errors may occur in the future.

Depending on the corrective action we take to redress a product's deficiencies or defects, the FDA may require, or we may decide, that we will need to obtain new marketing authorizations for the device before we may market or distribute the corrected device. Seeking such clearances or approvals may delay our ability to replace the recalled devices in a timely manner. Moreover, if we do not adequately address problems associated with our devices, we may face additional regulatory enforcement action, including FDA warning letters, product seizure, injunctions, administrative penalties, or civil or criminal fines.

Companies are required to maintain certain records of recalls and corrections, even if they are not reportable to the FDA. We have and may in the future initiate voluntary withdrawals or corrections for our products that we determine do not require notification of the FDA. If the FDA disagrees with our determinations, it could require us to report those actions as recalls and we may be subject to enforcement action. A future recall announcement could harm our reputation with customers, potentially lead to product liability claims against us, and negatively affect our sales. Any corrective action, whether voluntary or involuntary, as well as defending ourselves in a lawsuit, will require the dedication of our time and capital, distract management from operating our business, and may harm our reputation, business, financial condition, results of operations, and prospects.

Disruptions at the FDA and other government agencies caused by, funding shortages, staffing limitations, or global health concerns could hinder their ability to hire, retain or deploy key leadership and other personnel, prevent new or modified products from being developed, review, approved or commercialized in a timely manner or at all, which could negatively impact our business.

The ability of the FDA and foreign regulatory authorities to review and approve new products can be affected by a variety of factors, including government budget and funding levels, statutory, regulatory, and policy changes, the FDA's or foreign regulatory authorities' ability to hire and retain key personnel and accept the payment of user fees, and other events that may otherwise affect the FDA's or foreign regulatory authorities' ability to perform routine functions. Average review times at the FDA and foreign regulatory authorities have fluctuated in recent years as a result. In addition, government funding of other government agencies that fund research and development activities is subject to the political process, which is inherently fluid and unpredictable. Disruptions at the FDA and other agencies may also slow the time necessary for new drugs or modifications to approved drugs and biologics to be reviewed and/or approved by necessary government agencies, which would adversely affect our business. For example, in recent years, the U.S. government has shut down several times and certain regulatory agencies, such as the FDA, have had to furlough critical employees and stop critical activities. In addition, the current U.S. Presidential administration has issued certain policies and Executive Orders directed towards reducing the employee headcount and costs associated with U.S. administrative agencies, including the FDA, and it remains unclear the degree to which these efforts may limit or otherwise adversely affect the FDA's ability to conduct routine activities.

Separately, in response to the COVID-19 pandemic, the FDA postponed most inspections at domestic and foreign manufacturing facilities at various points. If a prolonged government shutdown occurs, or if renewed global health concerns, funding shortages or staffing limitations hinder or prevent the FDA or other regulatory authorities from conducting their regular inspections, reviews, or other regulatory activities, it could significantly impact the ability of the FDA or other such regulatory authorities to timely review and process our regulatory submissions, which could have a material adverse effect on our business.

We and the third parties with whom we work are subject to stringent and evolving U.S. and foreign laws, regulations, and rules, contractual obligations, industry standards, policies and other obligations related to data privacy and security. Our (or the third parties with whom we work) actual or perceived failures to comply with applicable data protection, privacy and security laws, regulations, standards and other requirements could adversely affect our business, results of operations and financial condition and lead to regulatory investigations or actions; litigation (including class action claims) and mass arbitration demands; fines and penalties; disruptions of our business operations; reputational harm; loss of revenue or profits; loss of customers or sales; and other adverse business consequences.

In the course of our operations, we may collect, use, store, disclose, transfer and otherwise process an increasing volume of personal information, including from our employees and third parties with whom we conduct business. We and our partners are or may become subject to federal, state and foreign data protection laws and regulations (*i.e*., laws and regulations that address data privacy and security). In the United States, numerous federal and state laws and regulations, including data breach notification laws, health information privacy laws and consumer protection laws and regulations (e.g., Section 5 of the FTC Act), that govern the collection, use, disclosure and protection of health-related and other personal information could apply to our operations or the operations of our partners. We may also be subject to U.S. federal rules, regulations and guidance concerning data security for medical devices, including guidance from the FDA.

For example, HIPAA, as amended by the Health Information Technology for Economic and Clinical Health Act, imposes specific requirements relating to the privacy, security, and transmission of individually identifiable protected health information. Furthermore, the Federal Trade Commission ("FTC") and many state Attorneys General continue to enforce federal and state consumer protection laws against companies for online collection, use, dissemination and security practices that appear to be unfair or deceptive. The FTC expects a company's data security measures to be reasonable and appropriate in light of the sensitivity and volume of consumer information it holds, the size and complexity of its business, and the cost of available tools to improve security and reduce vulnerabilities.

The regulatory framework for data privacy and security worldwide is continuously evolving and developing and, as a result, interpretation and implementation standards and enforcement practices are likely to remain uncertain for the foreseeable future. Certain states have also adopted comparable privacy and security laws and regulations that govern the privacy, processing and protection of health-related and other personal information. Such laws and regulations will be subject to interpretation by various courts and other governmental authorities, thus creating potentially complex compliance issues for us and our future customers and strategic partners. For example, California enacted the California Consumer Privacy Act, as amended by the California Privacy Rights Act (collectively, the "CCPA"), which requires covered businesses that process the personal information of California residents to, among other things, (i) provide certain disclosures to California residents regarding the business's collection, use, and disclosure of their personal information; (ii) receive and respond to requests from California residents to access, delete, and correct their personal information, or to opt out of certain disclosures of their personal information; and (iii) enter into specific contractual provisions with service providers that process California resident personal information on the business's behalf. Washington State enacted the "My Health My Data Act," which broadly defines consumer health data, creates a private right of action to allow individuals to sue for violations of the law, imposes stringent consent requirements, and grants consumers certain rights with respect to their health data, including to request deletion of their information. Additional compliance investment and potential business process changes may be required. Similar laws have been passed in other states, and are continuing to be proposed at the state and federal level, reflecting a trend toward more stringent privacy legislation in the United States. The enactment of such laws could have potentially conflicting requirements that would make compliance challenging. In the event that we are subject to or affected by HIPAA, the CCPA or other domestic privacy and data protection laws, any liability from failure to comply with the requirements of these laws could adversely affect our financial condition.

As our operations and business grow, we may become subject to or affected by new or additional data protection laws and regulations and face increased scrutiny or attention from regulatory authorities. For example, in the EU, the European Union General Data Protection Regulation ("GDPR") imposes strict requirements for processing the personal data of individuals within the European Economic Area ("EEA"). Companies that must comply with the GDPR face increased compliance obligations and risk, including more robust regulatory enforcement of data protection requirements and potential fines for noncompliance of up to €20 million or 4% of the annual global revenues of the noncompliant company, whichever is greater.

Further, Europe and other jurisdictions have enacted laws requiring data to be localized or limiting the transfer of personal data to other countries. In particular, the EEA and the United Kingdom ("UK") have significantly restricted the transfer of personal data to the United States and other countries whose privacy laws it generally believes are inadequate. Other jurisdictions may adopt or have already adopted similarly stringent data localization and cross-border data transfer laws.

Although there are currently various mechanisms that may be used to transfer personal data from the EEA and UK to the United States in compliance with law, such as the EEA standard contractual clauses, the UK's International Data Transfer Agreement / Addendum, and the EU-U.S. Data Privacy Framework and the UK extension thereto (which allows for transfers to relevant U.S.-based organizations who self-certify compliance and participate in the EU-U.S. Data Privacy Framework), these mechanisms are subject to legal challenges, and there is no assurance that we can satisfy or rely on these measures to lawfully transfer personal data to the United States.

To the extent our business and operations expand to the EEA, the UK or other jurisdictions and there is no lawful manner for us to transfer personal data from such jurisdictions to the United States, or if the requirements for a legally-compliant transfer are too onerous, we could face significant adverse consequences, including the interruption or degradation of our operations, the need to relocate part of or all of our business or data processing activities to other jurisdictions (such as Europe) at significant expense, increased exposure to regulatory actions, substantial fines and penalties, the inability to transfer data and work with partners, vendors and other third parties, and injunctions against our processing or transferring of personal data necessary to operate our business. Additionally, companies that transfer personal data out of the EEA and the UK to other jurisdictions, particularly to the United States, are subject to increased scrutiny from regulators, individual litigants, and activist groups.

In addition to government regulation, privacy advocates and industry groups have and may in the future propose self-regulatory standards from time to time. These and other industry standards may legally or contractually apply to us, or we may elect to comply with such standards. We expect that there will continue to be new proposed laws and regulations concerning data privacy and security, and we cannot yet determine the impact such future laws, regulations and standards may have on our business. We make public statements about our use and disclosure of personal information through our privacy policies, information provided on our website and press statements. Although we endeavor to comply with our public statements and documentation, we may at times fail to do so or be alleged to have failed to do so. The publication of our privacy policies and other statements that provide promises and assurances about data privacy and security can subject us to potential government or legal action if they are found to be deceptive, unfair or misrepresentative of our actual practices. Any concerns about our data privacy and security practices, even if unfounded, could damage the reputation of our business and harm our business, financial condition and results of operations.

Although we work to comply with applicable laws, regulations and standards, our contractual obligations and other legal obligations, these requirements are evolving and may be modified, interpreted and applied in an inconsistent manner from one jurisdiction to another, and may conflict with one another or other legal obligations with which we must comply. This evolution creates uncertainty in our business, affects our ability to operate in certain jurisdictions or to collect, store, transfer use and share personal information, necessitates the acceptance of more onerous obligations in our contracts, results in liability or imposes additional costs on us. The cost of compliance with these laws, regulations and standards is high and is likely to increase in the future. Any failure or perceived failure by us or our employees, representatives, contractors, consultants, collaborators or other third parties to comply with such requirements or adequately address privacy and security concerns, even if unfounded, could result in additional cost and liability to us, damage our reputation and adversely affect our business and results of operations. In addition, if our practices are not consistent, or viewed as not consistent, with legal and regulatory requirements, including changes in laws, regulations and standards or new interpretations or applications of existing laws, regulations and standards, we may also become subject to audits, inquiries, whistleblower complaints, adverse media coverage, investigations, criminal or civil sanctions, all of which may harm our business, financial condition and results of operations.

Risks Related to This Offering and Ownership of Our Common Stock

There may not be an active trading market for our common stock, which may cause shares of our common stock to trade at a discount from the initial public offering price and make it difficult to sell the shares of common stock you purchase.

Prior to this offering, there has been no public market for our common stock. It is possible that after this offering, an active trading market will not develop or, if developed, that any market will not be sustained, which would make it difficult for you to sell your shares of common stock at an attractive price or at all. The lack of an active market may impair the value of your shares or your ability to sell your shares at the time you wish to sell them or at a price that you consider reasonable. The initial public offering price per share of common stock will be determined by agreement among us and the representatives of the underwriters and may not be indicative of the price at which shares of our common stock will trade in the public market, if any, after this offering. The market value of our common stock may decrease from the initial public offering price. Furthermore, an inactive market may also impair our ability to raise capital in the future by selling shares of our common stock.

We are an "emerging growth company" and a "smaller reporting company," and we cannot be certain if the reduced reporting requirements applicable to emerging growth companies and smaller reporting companies will make our common stock less attractive to investors.

We are an "emerging growth company," as defined in the JOBS Act. We will remain an "emerging growth company" until the earliest to occur of:

- the last day of the fiscal year during which our total annual revenue equals or exceeds $1.235 billion (subject to adjustment for inflation);
- the last day of the fiscal year following the fifth anniversary of this offering;

- the date on which we have, during the previous three-year period, issued more than $1 billion in non-convertible debt; or
- the date on which we are deemed to be a "large accelerated filer" under the Exchange Act.

As a result of our "emerging growth company" status, we may take advantage of exemptions from various reporting requirements that would otherwise be applicable to public companies including, but not limited to, reduced disclosure obligations regarding executive compensation in our periodic reports and proxy statements and exemptions from the requirements of holding a nonbinding advisory vote on executive compensation and stockholder approval of any golden parachute payments not previously approved.

We also are a "smaller reporting company," meaning that the market value of our stock held by non-affiliates plus the proposed aggregate amount of gross proceeds to us as a result of this offering is less than $700.0 million and our annual revenue is less than $100.0 million during the most recently completed fiscal year. We may continue to be a smaller reporting company after this offering if either (i) the market value of our stock held by non-affiliates is less than $250.0 million or (ii) our annual revenue is less than $100.0 million during the most recently completed fiscal year and the market value of our stock held by non-affiliates is less than $700.0 million. If we are a smaller reporting company at the time we cease to be an emerging growth company, we may continue to rely on exemptions from certain disclosure requirements that are available to smaller reporting companies. Specifically, as a smaller reporting company we may choose to present only the two most recent fiscal years of audited financial statements in our annual report on Form 10-K and, similar to emerging growth companies, smaller reporting companies have reduced disclosure obligations regarding executive compensation.

Investors may find our common stock less attractive because we may rely on these exemptions. If some investors find our common stock less attractive as a result, there may be a less active trading market for our common stock and the market price of our common stock may be adversely affected and more volatile.

We will incur increased costs and become subject to additional regulations and requirements as a result of becoming a public company, which could lower our profits or make it more difficult to run our business.

As a public company, we will incur significant legal, accounting, and other expenses that we have not incurred as a private company, including costs associated with public company reporting requirements. We have also incurred and will continue to incur costs associated with the Sarbanes-Oxley Act and related rules implemented by the SEC and the exchange on which our securities are listed. The expenses generally incurred by public companies for reporting and corporate governance purposes have been increasing. We expect these rules and regulations to increase our legal and financial compliance costs and to make some activities more time-consuming and costly, although we are currently unable to estimate these costs with any degree of certainty. These laws and regulations also could make it more difficult or costly for us to obtain certain types of insurance, including director and officer liability insurance, and we may be forced to accept reduced policy limits and coverage or incur substantially higher costs to obtain the same or similar coverage. These laws and regulations could also make it more difficult for us to attract and retain qualified persons to serve on our board of directors, on our board committees or as our executive officers. Furthermore, if we are unable to satisfy our obligations as a public company, we could be subject to delisting of our common stock, fines, sanctions, other regulatory action, and potentially civil litigation.

If we are unable to design, implement, and maintain effective internal control over financial reporting in the future, investors may lose confidence in the accuracy and completeness of our financial reports and the market price of our common stock may decline.

As a public company, we will be required to maintain internal control over financial reporting and to report any material weaknesses in such internal controls. In addition, beginning with our second annual report on Form 10-K, we will be required to furnish a report by management on the effectiveness of our internal control over financial reporting, pursuant to the rules and regulations of the SEC regarding compliance with Section 404 of the Sarbanes-Oxley Act. The process of designing, implementing and testing the internal control over financial reporting required to comply with this obligation is time consuming, costly and complicated. While we have not identified material weaknesses in the past, we may identify material weaknesses in our internal control over financial reporting in the future. Any failure to maintain internal control over financial reporting could severely inhibit our ability to

accurately report our financial condition, results of operations, or cash flows. Further, if we identify one or more material weaknesses in our internal control over financial reporting, if we are unable to comply with the requirements of Section 404 of the Sarbanes-Oxley Act in a timely manner, or if we or, if required, our auditors, are unable to assert that our internal control over financial reporting is effective, investors may lose confidence in the accuracy and completeness of our financial reports and the market price of our common stock could decline, and we could also become subject to investigations by the stock exchange on which our common stock is listed, the SEC or other regulatory authorities, which could require additional financial and management resources. Failure to remedy any material weakness in our internal control over financial reporting, or to implement or maintain other effective control systems required of public companies, could also restrict our future access to the capital markets.

We do not intend to pay dividends in the foreseeable future. As a result, your ability to achieve a return on your investment will depend on appreciation in the market price of our common stock.

We have never declared or paid cash dividends on our capital stock, and we do not currently intend to pay any cash dividends on our capital stock in the foreseeable future. We currently intend to retain all available funds and any future earnings to fund the development and expansion of our business. Any future determination related to dividend policy will be made at the discretion of our board of directors, subject to applicable laws, and will depend upon, among other factors, our results of operations, prospects, financial condition, contractual restrictions and capital requirements. In addition, our ability to pay cash dividends on our capital stock is limited by the terms of the Trinity Loan Agreement with Trinity Capital, and may be limited by the terms of any future debt or preferred securities we issue or any future credit facilities we enter into. Accordingly, investors must for the foreseeable future rely on sales of their common stock after price appreciation, which may never occur, as the only way to realize any future gains on their investments.

The market price of our common stock may be volatile, which could result in substantial losses for investors purchasing shares in this offering.

Even if an active trading market develops, the market price of our common stock may be highly volatile and could be subject to wide fluctuations. The initial public offering price for our common stock will be determined through negotiations with the underwriters. This initial public offering price may differ from the market price of our common stock after the offering. As a result, you may not be able to sell your common stock at or above the initial public offering price. Some of the factors that may cause the market price of our common stock to fluctuate include:

- delays or setbacks in the ongoing commercialization of our products and procedures;
- the success of existing or new competitive products or technologies;
- regulatory or legal developments in the United States and other countries that we pursue expansion in;
- developments or disputes concerning patent applications, issued patents or other proprietary rights;
- the recruitment or departure of key personnel;
- the commencement of litigation;
- actual or anticipated changes in estimates as to financial condition and results of operations;
- announcement or expectation of additional financing efforts;
- announcements by us or our competitors of significant business developments, acquisitions, new offerings, licenses, strategic partnerships, joint ventures or capital commitments;
- the impact of pandemics epidemics, endemics and other public health emergencies on the performance of procedures;
- the impact of political instability, natural disasters, events of terrorism and or war, such as the ongoing conflict between Ukraine and Russia and in Israel, Gaza and surrounding areas, and the corresponding

tensions created from such conflict between Russia, the United States and countries in Europe as well as other countries such as China, and broader conflicts in the Middle East;

- sales of our common stock by us, our insiders or other stockholders, or the anticipation of such sales, including if existing stockholders sell shares into the market when applicable "lock-up" periods end;
- expiration of market standoff or lock-up agreements;
- variations between our actual operating results, or those of companies that are perceived to be similar to us, and the expectations of securities analysts, investors and the financial community;
- any forward-looking financial or operating information we may provide to the public or securities analysts, any changes in this information or our failure to meet expectations based on this information;
- changes in estimates or recommendations by securities analysts, if any, that cover our stock;
- various macroeconomic events, including changes in inflation, interest rates and overall economic conditions and uncertainties;
- changes in the structure of health care payment systems;
- market conditions in the medical device sector;
- changes in the anticipated future size and growth rate of our market;
- the seasonality of our business;
- an increase in the rate of returns of our products;
- general economic, industry and market conditions, including economic recessions or slowdowns; and
- the other factors described in this "Risk Factors" section.

In recent years, the stock market in general, and the market for medical device companies in particular, has experienced significant price and volume fluctuations that have often been unrelated or disproportionate to changes in the operating performance of the companies whose stock is experiencing those price and volume fluctuations. Further, the stock market in general has been highly volatile due to various macroeconomic events. Broad market and industry factors may seriously affect the market price of our common stock, regardless of our actual operating performance. These fluctuations may be even more pronounced in the trading market for our stock shortly following this offering. Following periods of such volatility in the market price of a company's securities, securities class action litigation has often been brought against that company. Because of the potential volatility of our stock price, we may become the target of securities litigation in the future. Securities litigation could result in substantial costs and divert management's attention and resources from our business.

If our operating and financial performance in any given period does not meet any guidance that we provide to the public, the market price of our common stock may decline.

We may, but are not obligated to, provide public guidance on our expected operating and financial results for future periods. Any such guidance will be comprised of forward-looking statements subject to the risks and uncertainties described in this prospectus and in our other public filings and public statements. Our actual results may not always be in line with or exceed any guidance we have provided, especially in times of economic uncertainty. If actual circumstances differ from those in our assumptions, our operating and financial results could fall below our publicly announced guidance or the expectations of investors. If, in the future, our operating or financial results for a particular period do not meet any guidance we provide or the expectations of investment analysts or investors generally, or if we reduce our guidance for future periods, the market price of our common stock may decline. Even if we do issue public guidance, there can be no assurance that we will continue to do so in the future.

We will have broad discretion in the use of net proceeds to us from this offering and may not use them effectively.

We will have broad discretion in the application of the net proceeds to us from this offering, including for any of the purposes described in "Use of Proceeds," and you will not have the opportunity as part of your investment decision to assess whether the net proceeds are being used appropriately. Because of the number and variability of factors that will determine our use of the net proceeds from this offering, their ultimate use may vary substantially from their currently intended use. If we do not use the net proceeds that we receive in this offering effectively, our business, financial condition, results of operations, and prospects could be harmed, and the market price of our common stock could decline. Pending their use, we may invest the net proceeds from this offering in short-term, investment-grade, interest-bearing securities such as money market accounts, certificates of deposit, commercial paper, and guaranteed obligations of the U.S. government that may not generate a high yield for our stockholders. These investments may not yield a favorable return to our investors.

Investors in this offering will experience immediate and substantial dilution.

The initial public offering price of our common stock is expected to be substantially higher than the pro forma as adjusted net tangible book value per share of our common stock. Therefore, if you purchase shares of our common stock in this offering, you will pay a price per share that substantially exceeds our pro forma as adjusted net tangible book value per share after this offering. Based on the initial public offering price of $15.00 per share you will experience immediate dilution of $7.12 per share, representing the difference between our pro forma as adjusted net tangible book value per share after giving effect to this offering, including the issuance of the 2025 Convertible Notes and the related Notes Conversion. In addition, purchasers of common stock in this offering will have contributed 43.7% of the aggregate price paid by all purchasers of our common stock but will own only approximately 24.7% of our total equity outstanding after this offering. Furthermore, if the underwriters exercise their over-allotment option, or outstanding options and warrants are exercised, you could experience further dilution. For a further description of the dilution that you will experience immediately after this offering, see the section titled "Dilution."

Participation in this offering by our existing stockholders and/or their affiliated entities may reduce the public float for our common stock.

To the extent certain of our existing stockholders and their affiliated entities participate in this offering, such purchases would reduce the non-affiliate public float of our shares, meaning the number of shares of our common stock that are not held by officers, directors, and controlling stockholders. A reduction in the public float could reduce the number of shares that are available to be traded at any given time, thereby adversely impacting the liquidity of our common stock and depressing the price at which you may be able to sell shares of common stock purchased in this offering. While certain of our existing stockholders and their affiliated entities have expressed interest in potentially participating in this offering, there are no assurances that they will participate in the offering to a material extent, or at all.

Our principal stockholders and management own a significant percentage of our stock and will be able to exert significant control over matters subject to stockholder approval.

After this offering, our directors, officers, holders of more than 5% of our outstanding stock and their respective affiliates will beneficially own shares representing approximately 56.0% of our outstanding shares (assuming no exercise of the underwriters' over-allotment option and no purchases of shares in this offering by anyone of this group). As a result, these stockholders, if they act together, will be able to influence our management and affairs and all matters requiring stockholder approval, including the election of directors and approval of significant corporate transactions. This concentration of ownership may have the effect of delaying or preventing a change in control of our company and might affect the market price of our common stock.

Sales of a substantial number of shares of our common stock in the public market could cause our stock price to fall.

Sales of a substantial number of shares of our common stock in the public market could occur at any time. These sales, upon the expiration of the market standoff and lock-up agreements, the early release of these

agreements or the perception in the market that the holders of a large number of shares of our common stock intend to sell shares, could reduce the market price of our common stock. After this offering, we expect that we will have 20,228,510 shares of our common stock outstanding (assuming no exercise of the underwriters' over-allotment option). Of these shares, the 5,000,000 shares we are selling in this offering may be resold in the public market immediately, unless purchased by our affiliates. The remaining shares, or approximately 75.3 % of our outstanding shares after this offering, are currently prohibited or otherwise restricted under securities laws, or lock-up agreements entered into by our directors, officers and substantially all of our stockholders with the underwriters. However, subject to applicable securities law restrictions, prohibitions and restrictions on the sale of these shares in the public market will be lifted beginning 180 days after the date of this prospectus. Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co. may release all or some portion of the shares subject to lock-up agreements at any time and for any reason.

All of the shares of common stock sold in this offering will be freely tradable without restriction or further registration under the Securities Act of 1933, as amended (the "Securities Act"), except that any shares held by our affiliates, as that term is defined under Rule 144 of the Securities Act ("Rule 144"), including those purchased by our directors or officers pursuant to our directed share program, as described in "Underwriting—Directed Share Program," may be sold only in compliance with the limitations described in "Shares Eligible for Future Sale—Affiliate Resales of Restricted Securities."

In addition, shares issued upon the exercise of stock options outstanding under our equity incentive plans, or pursuant to future awards granted under those plans, will become available for sale in the public market to the extent permitted by the provisions of applicable vesting schedules, any applicable market standoff and lock-up agreements, and Rule 144 and Rule 701 under the Securities Act ("Rule 701"). See the section titled "Shares Eligible for Future Sale" for additional information.

In connection with this offering, we intend to file a registration statement on Form S-8 providing for the registration of all shares of our common stock that we may issue under our equity compensation plans. Once we register these shares, they can be freely sold in the public market upon issuance and once vested, subject to volume limitations applicable to affiliates and the lock-up agreements described in the section titled "Underwriting." If any of these additional shares are sold, or if it is perceived that they will be sold, in the public market, the market price of our common stock could decline.

Provisions in our charter documents and under Delaware law could discourage a takeover that stockholders may consider favorable and may lead to entrenchment of management.

Our amended and restated certificate of incorporation and amended and restated bylaws that will be in effect immediately prior to the consummation of this offering will contain provisions that could delay or prevent changes in control or changes in our management without the consent of our board of directors. These provisions will include the following:

- a classified board of directors with three-year staggered terms, which may delay the ability of stockholders to change the membership of a majority of our board of directors;
- no cumulative voting in the election of directors, which limits the ability of minority stockholders to elect director candidates;
- the exclusive right of our board of directors, unless the board of directors grants such right to the stockholders, to elect a director to fill a vacancy created by the expansion of the board of directors or the resignation, death or removal of a director, which prevents stockholders from being able to fill vacancies on our board of directors;
- the required approval of at least 66-2/3% of the shares entitled to vote to remove a director for cause, and the prohibition on removal of directors without cause;

- the ability of our board of directors to authorize the issuance of shares of preferred stock and to determine the price and other terms of those shares, including preferences and voting rights, without stockholder approval, which could be used to significantly dilute the ownership of a hostile acquiror;
- the ability of our board of directors to alter our amended and restated bylaws without obtaining stockholder approval;
- the required approval of at least 66-2/3% of the shares entitled to vote to adopt, amend or repeal our amended and restated bylaws or repeal the provisions of our amended and restated certificate of incorporation regarding the election and removal of directors;
- a prohibition on stockholder action by written consent, which forces stockholder action to be taken at an annual or special meeting of our stockholders;
- an exclusive forum provision providing that the Court of Chancery of the State of Delaware will be the exclusive forum for certain actions and proceedings;
- the requirement that a special meeting of stockholders may be called only by the board of directors, which may delay the ability of our stockholders to force consideration of a proposal or to take action, including the removal of directors; and
- advance notice procedures that stockholders must comply with in order to nominate candidates to our board of directors or to propose matters to be acted upon at a stockholders' meeting, which may discourage or deter a potential acquiror from conducting a solicitation of proxies to elect the acquiror's own slate of directors or otherwise attempting to obtain control of us.

We are also subject to the anti-takeover provisions contained in Section 203 of the General Corporation Law of the State of Delaware (the "Delaware General Corporation Law"). Under Section 203, a corporation may not, in general, engage in a business combination with any holder of 15% or more of its capital stock unless the holder has held the stock for three years or, among other exceptions, the board of directors has approved the transaction. For a description of our capital stock, see the section titled "Description of Capital Stock."

Claims for indemnification by our directors, officers, and other employees or agents may reduce our available funds to satisfy successful third-party claims against us and may reduce the amount of money available to us.

Our amended and restated certificate of incorporation and amended and restated bylaws will provide that we will indemnify our directors and officers, in each case to the fullest extent permitted by Delaware law. In addition, as permitted by Section 145 of the Delaware General Corporation Law, our amended and restated bylaws to be effective immediately prior to the completion of this offering and our indemnification agreements that we have entered into with our directors, officers and certain other employees will provide that:

- We will indemnify our directors and officers for serving us in those capacities or for serving other business enterprises at our request, to the fullest extent permitted by Delaware law. Delaware law provides that a corporation may indemnify such person if such person acted in good faith and in a manner such person reasonably believed to be in or not opposed to the best interests of the registrant and, with respect to any criminal proceeding, had no reasonable cause to believe such person's conduct was unlawful.
- We may, in our discretion, indemnify employees and agents in those circumstances where indemnification is permitted by applicable law.
- We are required to advance expenses, as incurred, to our directors and officers in connection with defending a proceeding, except that such directors or officers shall undertake to repay such advances if it is ultimately determined that such person is not entitled to indemnification.
- We will not be obligated pursuant to our amended and restated bylaws to indemnify a person with respect to proceedings initiated by that person against us or our other indemnitees, except with respect to proceedings authorized by our board of directors or brought to enforce a right to indemnification.

- The rights conferred in our amended and restated bylaws are not exclusive, and we are authorized to enter into indemnification agreements with our directors, officers, employees, and agents and to obtain insurance to indemnify such persons.
- We may not retroactively amend our amended and restated bylaws provisions to reduce our indemnification obligations to directors, officers, employees, and agents.

Our current amended and restated certificate of incorporation provides, and our amended and restated certificate of incorporation will provide, that the Court of Chancery of the State of Delaware will be the exclusive forum for substantially all disputes between us and our stockholders and that the federal district courts shall be the exclusive forum for the resolution of any complaint asserting a cause of action arising under the Securities Act, which could limit our stockholders' ability to obtain a favorable judicial forum for disputes with us or our directors, officers or employees or the underwriters or any offering giving rise to such claim.

Our current amended and restated certificate of incorporation provides, and our amended and restated certificate of incorporation that will be in effect immediately after the closing of this offering will provide, that the Court of Chancery of the State of Delaware is the exclusive forum for any derivative action or proceeding brought on our behalf, any action asserting a breach of fiduciary duty, any action asserting a claim against us arising pursuant to the Delaware General Corporation Law, our amended and restated certificate of incorporation or our amended and restated bylaws, or any action asserting a claim against us that is governed by the internal affairs doctrine; provided, that, this provision would not apply to suits brought to enforce a duty or liability created by the Exchange Act or any other claim for which the federal courts have exclusive jurisdiction. Furthermore, our amended and restated certificate of incorporation will also provide that unless we consent in writing to the selection of an alternative forum, the federal district courts of the United States shall be the exclusive forum for the resolution of any complaint asserting a cause of action arising under the Securities Act. These choice of forum provisions may result in increased costs to stockholders to bring a claim, limit a stockholder's ability to bring a claim in a judicial forum that it finds favorable for disputes with us or our directors, officers or other employees, and may generally have the effect of discouraging lawsuits against us and our directors, officers and other employees. By agreeing to this provision, however, stockholders will not be deemed to have waived our compliance with the federal securities laws and the rules and regulations thereunder. Furthermore, the enforceability of similar choice of forum provisions in other companies' certificates of incorporation has been challenged in legal proceedings, and it is possible that a court could find these types of provisions to be inapplicable or unenforceable. If a court were to find the choice of forum provisions in our amended and restated certificate of incorporation to be inapplicable or unenforceable in an action, we may incur additional costs associated with resolving such action in other jurisdictions, which could adversely affect our business and financial condition.

General Risk Factors

If we engage in acquisitions or strategic partnerships, it may increase our capital requirements, dilute our stockholders, cause us to incur debt or assume contingent liabilities, and subject us to other risks.

From time to time, we may evaluate various acquisitions and strategic partnerships, including licensing or acquiring complementary offerings, intellectual property rights, technologies, or businesses. Any acquisition or strategic partnership may entail numerous risks, including:

- increased operating expenses and cash requirements;
- the assumption of additional indebtedness or contingent liabilities;
- assimilation of operations, intellectual property, and products of an acquired company, including difficulties associated with integrating new personnel;
- the diversion of our management's attention from our existing operations in pursuing such a strategic merger or acquisition;
- loss of key personnel and uncertainties in our ability to maintain key business relationships;

- uncertainties associated with the other party to such a transaction, including the prospects of that party and their existing products or future products and regulatory approvals; and
- our inability to generate revenue from acquired technology and/or products sufficient to meet our objectives in undertaking the acquisition or even to offset the associated acquisition and maintenance costs.

In addition, if we undertake acquisitions or strategic partnerships, we may issue dilutive securities, assume or incur debt obligations, incur large one-time expenses, and acquire intangible assets that could result in significant future amortization expense. Moreover, we may not be able to locate suitable acquisition or partnership opportunities, and even if we do locate such opportunities, we may not be able to successfully bid for or obtain them due to competitive factors or lack of sufficient resources. This inability could impair our ability to grow or obtain access to technology or products that may be important to the development of our business.

Unfavorable global and regional economic, political and health conditions could adversely affect our business, financial condition or results of operations.

Our results of operations could be adversely affected by global or regional economic, political and health conditions. A global financial crisis or global or regional political and economic instability (including changes in inflation, interest rates and overall economic conditions and uncertainties), tariffs, wars, terrorism, civil unrest, pandemics, epidemics, endemics and other public health emergencies, and other unexpected events, such as supply chain constraints or disruptions, could cause extreme volatility, increase our costs and disrupt our business. Business disruptions could include, among others, disruptions to our commercial activities, including due to supply chain or distribution constraints or challenges, clinical enrollment, clinical site availability, patient accessibility and conduct of our clinical trials, as well as temporary closures of our facilities and the facilities of suppliers or contract manufacturers in our supply chain. For example, these macroeconomic factors could affect the ability of our current or potential future manufacturers, sole source or single source suppliers, licensors or licensees to remain in business, or otherwise manufacture or supply components, materials or services relevant to our systems. Any failure by any of them to remain in business could affect the manufacture of our systems or our ability to meet demand for our systems. In addition, if inflation or other factors were to significantly increase our business costs, we may be unable to pass through price increases to our customers. Interest rates and the ability to access credit markets could also adversely affect the ability of our customers to purchase our systems.

The imposition of tariffs and other orders or restrictions impacting trade could adversely impact our business, including by increasing or otherwise impacting the costs and expenses we incur in connection with our operations and supply chain, and by potentially increasing the price of our systems to purchasers. For example, certain of our suppliers import raw materials, including certain metals, from Canada. The actual impacts of any tariffs and other orders or restrictions are subject to a number of factors including the effective date and duration of such tariffs, orders and restrictions, the amount, scope and nature of such inputs, any countermeasures that the target countries may take and any mitigating actions that may become available.

In addition, during certain crises and events, patients may prioritize other items over certain or all of their treatments or delay their requisite shoulder care, which could have a negative impact on our commercial sales.

A severe or prolonged economic downturn, political disruption or adverse health conditions could result in a variety of risks to our business, including our ability to raise capital when needed on acceptable terms, if at all. Any of the foregoing could harm our business and we cannot anticipate all of the ways in which the political or economic climate and financial market conditions could adversely impact our business.

We or the third parties we depend on may be adversely affected by natural disasters and other catastrophic events, and our business continuity and disaster recovery plans may not adequately protect us from a serious natural disaster or other catastrophic event. Any interruption in our operations or the operations of third parties who supply components or other materials for our products may have a material adverse effect on our business, financial condition, results of operations, and prospects.

Severe weather, natural disasters and other catastrophic events, including pandemics or other public health crises, earthquakes, tsunamis, hurricanes, floods, fires, explosions, accidents, power outages, cyberattacks,

telecommunications failures, mechanical failures, unscheduled downtimes, civil unrest, strikes, transportation interruptions, unpermitted discharges or releases of toxic or hazardous substances, other environmental risks, wars or other conflicts (including wars in Russia and Ukraine as well as the Middle East), sabotage, terrorist attacks, or other intentional acts of vandalism or misconduct could severely disrupt our operations, or the operations of third parties who manufacture or supply components or other materials for our products, and have a material adverse effect on our business, financial condition, results of operations, and prospects.

If a natural disaster or other catastrophic event occurs that prevents us or third-party suppliers or manufacturers from using all or a significant portion of our or their headquarters or other facilities, that damages critical infrastructure or that otherwise disrupts operations, it may be difficult or, in certain cases, impossible, for us to continue our business for a substantial period of time. For example, we are highly dependent on our corporate headquarters in Grand Rapids, Michigan which has approximately 7,000 square feet of warehouse, distribution and office space as well as a 3,000 square foot distribution facility in Modesto California that we leverage to distribute our implant systems in the Western United States. Any such natural disaster or catastrophic event that impacts either of these facilities could have a material adverse impact on our business, financial condition and results of operations. Moreover, the disaster recovery and business continuity plans we have in place currently are limited and are unlikely to prove adequate in the event of a serious disaster or similar catastrophic event. The potential impact of any disruption would depend on the nature and extent of the damage caused by a disaster. We may incur substantial expenses as a result of the limited nature of our disaster recovery and business continuity plans, which, particularly when taken together with our lack of earthquake insurance, could have a material adverse effect on our business, financial condition, results of operations, and prospects.

In addition, integral parties in our supply chain are similarly vulnerable to natural disasters or other sudden, unforeseen, and severe adverse events. If such an event were to affect our supply chain, it could have a material adverse effect on our business, financial condition, results of operations, and prospects.

We are subject to risks from legal and arbitration proceedings that may prevent us from pursuing our business activities or require us to incur additional costs in defending against claims or paying damages.

We may become subject to legal disputes and regulatory proceedings in connection with our business activities involving, among other things, product liability, product defects, intellectual property infringement, employment matters, and/or alleged violations of other applicable laws in various jurisdictions. We may not be insured against all potential damages that may arise out of any claims to which we may be party in the ordinary course of our business. A negative outcome of these proceedings may prevent us from pursuing certain activities and/or require us to incur additional costs in order to do so and pay damages. In addition, securities class action litigation has often been instituted against companies following periods of volatility in the market price of a company's securities. This type of litigation, if instituted, could result in substantial costs and a diversion of management's attention and resources, which would harm our business, financial condition, results of operations and prospects. Additionally, the significant increase in the cost of directors' and officers' liability insurance may cause us to opt for lower overall policy limits or to forgo insurance that we may otherwise rely on to cover significant defense costs, settlements, and damages awarded to plaintiffs.

The outcome of pending or potential future legal and arbitration proceedings is difficult to predict with certainty. In the event of a negative outcome of any material legal or arbitration proceeding, whether based on a judgment or a settlement agreement, we could be obligated to make substantial payments, which could have a material adverse effect on our business, financial condition, results of operations, and prospects. In addition, the costs related to litigation and arbitration proceedings may be significant, and any legal or arbitration proceedings could have a material adverse effect on our business, financial condition, results of operations, and prospects.

The requirements of being a public company may divert our management's attention from our growth strategies and other business concerns.

As a public company, we will be subject to the reporting requirements of the Exchange Act and will be required to comply with the applicable requirements of the Sarbanes-Oxley Act and the Dodd- Frank Wall Street Reform and Consumer Protection Act of 2010, the listing requirements of the New York Stock Exchange and other applicable

securities rules and regulations. Compliance with these rules and regulations will increase our legal and financial compliance costs, make some activities more difficult, time consuming or costly and increase demand on our systems and resources. Among other things, the Exchange Act requires that we file annual, quarterly and current reports with respect to our business and results of operations and maintain effective disclosure controls and procedures and internal controls over financial reporting. In order to maintain and, if required, improve our disclosure controls and procedures and internal controls over financial reporting to meet this standard, significant resources and management oversight may be required. As a result, management's attention may be diverted from executing our growth strategies and managing other business concerns and, which could have a material adverse effect on our business, financial condition and results of operations. Although we intend to hire additional employees to comply with these requirements, we may need to hire even more employees in the future, which will increase our costs and expenses. Additionally, as a public company, it will be more expensive for us to obtain director and officer liability insurance, and we may be required to accept reduced coverage or incur substantially higher costs to obtain coverage. These factors could make it more difficult for us to attract and retain qualified members of our board of directors, particularly to serve on our audit committee and compensation committee, and qualified executive officers.

We will incur significant costs as a result of operating as a public company and our executive management team expects to devote substantial time to public company compliance programs.

As a public company, we will incur significant legal, accounting and other expenses due to our compliance with regulations and disclosure obligations applicable to us, including compliance with the Sarbanes-Oxley Act, as well as rules implemented by the SEC and the New York Stock Exchange. Stockholder activism, the current political environment and the current high level of government intervention and regulatory reform may lead to substantial new regulations and disclosure obligations, which may lead to additional compliance costs and impact, in ways we cannot currently anticipate, the manner in which we operate our business. Our executive management team and other personnel will devote a substantial amount of time to these compliance programs and monitoring of public company reporting obligations and as a result of the new corporate governance and executive compensation related rules, regulations and guidelines prompted by the Dodd-Frank Wall Street Reform and Consumer Protection Act, and further regulations and disclosure obligations expected in the future, we will likely need to devote additional time and costs to comply with such compliance programs and rules. These rules and regulations will cause us to incur significant legal and financial compliance costs and will make some activities more time-consuming and costly.

If securities or industry analysts do not publish research or reports about our business, or if they issue an adverse or misleading opinion regarding our stock, our stock price and trading volume could decline.

The trading market for our common stock will be influenced by the research and reports that industry or securities analysts publish about us or our business. We do not currently have and may never obtain research coverage by securities and industry analysts. If no or few securities or industry analysts commence coverage of us, the market price for our stock would be negatively impacted. In the event we obtain securities or industry analyst coverage, if any of the analysts who cover us issue an adverse or misleading opinion regarding us, our business model, our intellectual property or our stock performance, or if our results of operations fail to meet the expectations of analysts, our stock price would likely decline. If one or more of these analysts cease coverage of us or fail to publish reports on us regularly, we could lose visibility in the financial markets, which in turn could cause our stock price or trading volume to decline.

Our disclosure controls and procedures may not prevent or detect all errors or acts of fraud.

Upon the completion of this offering, we will become subject to the periodic reporting requirements of the Exchange Act. We must design our disclosure controls and procedures to reasonably assure that information we must disclose in reports we file or submit under the Exchange Act is accumulated and communicated to management, and recorded, processed, summarized, and reported within the time periods specified in the rules and forms of the SEC. We believe that any disclosure controls and procedures or internal controls and procedures, no matter how well-conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met.

These inherent limitations include the realities that judgments in decision-making can be faulty, and that breakdowns can occur because of simple error or mistake. For example, our directors or executive officers could inadvertently fail to disclose a new relationship or arrangement, causing us to fail to make a required related party transaction disclosure. Additionally, controls can be circumvented by the individual acts of some persons, by collusion of two or more people, or by an unauthorized override of the controls. Accordingly, because of the inherent limitations in our control system, misstatements due to error or fraud may occur and not be detected.

Our insurance may not cover all potential losses or liabilities that may arise.

We are not insured against all potential losses or liabilities that may arise, as insurance coverage may be unavailable, not cost-effective, or subject to significant limitations. For example, we are not insured against business interruptions suffered by third parties that we depend on, environmental liabilities or patent infringement, among other types of risks. Furthermore, no assurance can be given that an insurance carrier will not seek to cancel or deny coverage after a claim has occurred. If a loss or liability occurs that is not or not fully covered by insurance, we may be required to pay substantial amounts, which could adversely affect its cash position and results of operations.

We are subject to U.S. anti-bribery, anti-corruption, and anti-money laundering laws, including the FCPA, as well as export control, customs laws, economic sanctions and other trade laws and regulations (collectively, the "Trade Laws"). We can face serious consequences for violations.

As we grow our international presence and global operations, we will be increasingly exposed to trade and economic sanctions and other restrictions imposed by the governments of the United States and other applicable jurisdictions. Compliance with applicable regulatory requirements regarding the export of our products may create delays in the introduction of our products in international markets or, in some cases, prevent the export of our products to some countries altogether. Furthermore, U.S. export control laws and economic sanctions prohibit the provision of certain products and services to countries, governments, and persons targeted by U.S. sanctions.

We are also subject to anti-corruption laws, including the FCPA, the U.S. domestic bribery statute contained in 18 U.S.C. 201, the U.S. Travel Act, and other state and national anti-bribery laws in the countries in which we may conduct activities in the future. Anti-corruption laws are interpreted broadly and generally prohibit companies and their employees, agents and intermediaries from authorizing, promising, offering or providing, directly or indirectly, corrupt or improper payments or anything else of value to recipients in the public or private sector. The FCPA also requires public companies to make and keep books and records that accurately and fairly reflect the transactions of the corporation and to devise and maintain an adequate system of internal accounting controls. We can be held liable for the corrupt or illegal activities of our agents and intermediaries, even if we do not explicitly authorize or have actual knowledge of such activities.

Violations of Trade Laws can result in substantial criminal fines and civil penalties, imprisonment, the loss of trade privileges, debarment, tax reassessments, breach of contract and fraud litigation, reputational harm and other consequences. Likewise, any investigation of potential violations of Trade Laws could also have an adverse impact on our reputation, our business, results of operations and financial condition.

We cannot assure you that our policies and procedures are or will be sufficient or that directors, officers, employees, representatives, consultants and agents have not engaged and will not engage in conduct for which we may be held responsible, nor can we assure you that our business partners have not engaged and will not engage in conduct that could materially affect their ability to perform their contractual obligations to us or even result in our being held liable for such conduct. Violations of the FCPA, Trade Laws, or other export control, anti-corruption, anti-money laundering and anti-terrorism laws or regulations may result in severe criminal or civil sanctions, and we may be subject to other liabilities, which could negatively affect our business, financial condition and results of operations.

We, along with our suppliers, are dependent on various information technology systems. If our information technology systems or those of third parties with whom we work or our data, are or were compromised, we could experience adverse consequences resulting from such compromise, including but not limited to regulatory investigations or actions; litigation; fines and penalties; disruptions of our business operations; reputational harm; loss of revenue or profits; loss of customers or sales; and other adverse consequences.

We and our suppliers collect and maintain information in digital form that is necessary to conduct our business, and rely extensively on information technology systems, networks and services, including internet sites, data hosting and processing facilities and tools, physical security system and other hardware, software and technical applications and platforms, some of which are managed, hosted, provided or used by third-parties or their vendors, and some of which may be stored outside of the United States, to the extent we expand our operations internationally. These systems include, but are not limited to, ordering and managing materials from suppliers, converting materials to finished products (suppliers), shipping products to customers, processing transactions, summarizing and reporting results of operations, complying with regulatory, legal or tax requirements, providing data security and other processes necessary to manage our business.

A significant breakdown, invasion, corruption, destruction or interruption of critical information technology systems or infrastructure, by our workforce, others with authorized access to our systems or unauthorized persons could negatively impact operations. The use of cloud-based computing creates opportunities for the unintentional dissemination or intentional destruction of confidential information stored in our or our third-party providers' systems, portable media or storage devices. Our internal computer systems and those of our contractors, consultants and collaborators have been and are vulnerable to damage from cyberattacks, "phishing" attacks, intentional or accidental actions or omissions to act that cause vulnerabilities, computer viruses, unauthorized access, natural disasters, terrorism, war and telecommunication and electrical failures.

Attacks upon information technology systems are increasing in their frequency, levels of persistence, sophistication and intensity, and are being conducted by sophisticated and organized groups and individuals with a wide range of motives and expertise. Cyber-attacks, malicious internet-based activity, online and offline fraud, and other similar activities threaten the confidentiality, integrity, and availability of our sensitive information and information technology systems, and those of the third parties with whom we work. Our information technology systems and those of our third-party service providers, strategic partners and other contractors or consultants are vulnerable to attack, damage and interruption from computer viruses and malware (e.g. ransomware), misconfigurations, "bugs" or other vulnerabilities, malicious code, natural disasters, terrorism, war, telecommunication and electrical failures, hacking, cyberattacks, phishing attacks and other social engineering schemes, employee theft or misuse, human error, fraud, denial or degradation of service attacks and sophisticated nation-state and nation-state-supported actors.

We have also outsourced elements of our information technology infrastructure, and as a result a number of third-party vendors may or could have access to our confidential information. If we or our third-party vendors were to experience a significant cybersecurity breach of our or their information systems or data, the costs associated with the investigation, remediation and potential notification of the breach to counter-parties and data subjects could be material. In addition, our remediation efforts may not be successful. If we do not allocate and effectively manage the resources necessary to build and sustain the proper technology and cybersecurity infrastructure, we could suffer significant business disruption, including transaction errors, supply chain or manufacturing interruptions, processing inefficiencies, data loss or the loss of or damage to intellectual property or other proprietary information. There can also be no assurance that our and our third-party service providers', strategic partners', contractors', consultants', Contract Research Organizations' and collaborators' cybersecurity risk management program and processes, including policies, controls or procedures, will be fully implemented, complied with or effective in protecting our systems, networks and confidential information. For example, a third-party software provider which we have engaged as part of our advanced implant systems was recently inspected by the FDA, which identified a single observation regarding inadequate validation of the software to address active vulnerability threats. While such a finding may not be material, if not properly remediated, it could affect our brand, business, financial condition, results of operations and prospects.

We and certain of our service providers are from time to time subject to cyberattacks and security incidents. While we do not believe that we have experienced any significant system failure, accident or security breach to date, if such an event were to occur and cause interruptions in our operations, it could result in a material disruption of our development programs and our business operations, whether due to a loss, corruption or unauthorized disclosure of our trade secrets, personal information or other proprietary or sensitive information or other similar disruptions. We may also experience security breaches that may remain undetected for an extended period. If our systems are damaged or cease to function properly due to any number of causes, ranging from catastrophic events to power outages to security breaches, and our business continuity plans do not effectively compensate timely, we may suffer interruptions in our ability to manage operations, and would also be exposed to a risk of loss, including financial assets or litigation and potential liability, which could materially adversely affect our business, financial condition, results of operations and prospects.

We cannot assure you that any limitations of liability provisions in our contracts would be enforceable or adequate or would otherwise protect us from any liabilities or damages with respect to any particular claim relating to a security lapse or breach. While we maintain certain insurance coverage, including cyber insurance, our insurance may be insufficient or may not cover all liabilities incurred by such attacks. We also cannot be certain that our insurance coverage will be adequate for data handling or data security liabilities actually incurred, that insurance will continue to be available to us on economically reasonable terms, or at all, or that any insurer will not deny coverage as to any future claim. The successful assertion of one or more large claims against us that exceeds available insurance coverage, or the occurrence of changes in our insurance policies, including premium increases or the imposition of large deductible or co-insurance requirements, could have a material adverse effect on our business, including our financial condition, operating results and reputation.

Changes in tax laws or regulations or in their implementation or interpretation that are applied adversely to us or our customers may seriously harm our business.

We are or may become subject to income and non-income taxes in the United States under federal, state and local jurisdictions and in certain foreign jurisdictions in which we may operate in the future. New income, sales, use or other tax laws, statutes, rules, regulations or ordinances could be enacted at any time, which could affect the tax treatment of any of our future domestic and foreign earnings. Any new taxes could adversely affect our domestic and international business operations, and our business and financial performance. Further, existing tax laws, statutes, rules, regulations or ordinances could be interpreted, changed, modified or applied adversely to us, possibly on a retroactive basis.

Our ability to use our net operating loss carryforwards and certain other tax attributes may be subject to limitation.

As of December 31, 2024, we had U.S. federal net operating loss carryforwards of approximately $45.3 million, which may be available to offset future taxable income for U.S. income tax purposes. Under the Internal Revenue Code of 1986, as amended (the "Code"), our federal net operating losses generated in taxable years beginning after December 31, 2017 will not expire and may be carried forward indefinitely, but the deductibility of such federal net operating loss carryforwards in a taxable year is limited to 80% of current year taxable income (with certain adjustments). Our ability to utilize our federal net operating loss carryforwards and certain other tax attributes may be limited under Section 382 and 383 of the Code. The limitation applies if we experience an "ownership change," which is generally defined as a greater than 50 percentage point change (by value) in the ownership of our equity by certain stockholders over a rolling three-year period. The amount of the annual limitation is generally equal to the product of the applicable long-term tax exempt-rate (as published by the U.S. Internal Revenue Service (the "IRS") for the month in which the "ownership change" occurred) and the value of our outstanding stock immediately prior to the "ownership change." If we have a net unrealized built-in gain in our assets immediately prior to the "ownership change," the annual limitation may be increased as certain gains are, or are treated as, recognized during the five-year period beginning on the date of the "ownership change."

Similar provisions of state tax law may also apply to limit the use of any state net operating loss carryforwards. For example, California imposed limits on the usability of California state net operating losses to offset taxable income in tax years beginning on or after January 1, 2024 and before January 1, 2027. We do not know if this

offering will give rise to an “ownership change,” but even if we were to experience an “ownership change” due to this offering, our ability to use our net operating loss carryforwards and other tax attributes to offset our taxable income may not be significantly limited. However, we have undergone and may in the future undergo an “ownership change” due to transactions in our stock, which may be outside of our control. As a result, if we earn net taxable income, our ability to use our net operating loss carryforwards and other tax attributes to offset taxable income may be subject to limitations, which could potentially result in increased future tax liability to us. We cannot predict whether any future “ownership change” would result in a significant limitation on our ability to use our net operating loss carryforwards and other tax attributes to offset our taxable income and adversely affect our future cash flows. If not utilized, a portion of our net operating loss carryforwards may expire.

The increasing use of social media platforms presents new risks and challenges.

Social media is increasingly being used to communicate about our systems and technologies, and we intend to utilize appropriate social media in connection with the commercialization of our advanced implant systems. Social media practices in the medical device industry continue to evolve and regulations and regulatory guidance relating to such use are evolving and not always clear. This evolution creates uncertainty and risk of noncompliance with regulations applicable to our business, resulting in potential regulatory actions against us, along with the potential for litigation. In addition, there is a risk of inappropriate disclosure of sensitive information or negative or inaccurate posts or comments about us or our systems on any social networking website. If any of these events were to occur or we otherwise fail to comply with any applicable regulations, we could incur liability, face restrictive regulatory actions, or incur other harm to our business such as reputational damage.

We could be subject to securities class action litigation.

In the past, securities class action litigation has often been brought against a company following a decline in the market price of its securities. This risk is especially relevant for us because biotechnology and biopharmaceutical companies have experienced significant stock price volatility in recent years. If we face such litigation, even if ultimately decided in our favor, it could result in substantial costs and a diversion of our management’s attention and resources, which could harm our business.

SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS

This prospectus contains forward-looking statements about us and our industry that involve substantial risks and uncertainties. All statements other than statements of historical facts contained in this prospectus, including statements regarding our strategy, future financial condition, future operations, projected costs, prospects, plans, objectives of management, and expected market growth, are forward-looking statements. In some cases, you can identify forward-looking statements by words such as "may," "will," "shall," "should," "expects," "plans," "anticipates," "could," "intends," "target," "projects," "contemplates," "believes," "estimates," "predicts," "potential," "goal," "objective," "seeks," or "continue" or the negative of these words or other similar terms or expressions that concern our expectations, strategy, plans, or intentions. Forward-looking statements contained in this prospectus include, but are not limited to, statements about:

- the expected use of our implant systems, preoperative planning technology and instrument systems;
- the expected growth of our business and our organization;
- our expectations regarding the potential market size and size of the potential patient populations for our systems and any future products, if approved for commercial use;
- our expected uses of the net proceeds from this offering and our existing cash, cash equivalents and marketable securities;
- our expectations regarding government and third-party payor coverage and reimbursement;
- our ability to retain and recruit key personnel, including the continued development of a sales and marketing infrastructure and surgeon-to-surgeon collaboration;
- our ability to obtain an adequate supply of finished goods and components for our implant systems from our third-party suppliers, most of whom are single-source suppliers;
- our plans and expected timeline related to our implant systems, the identification and development of planned, new or improved products, and our expansion into additional indications or adjacent areas in shoulder surgical care and/or new geographic markets;
- our ability to obtain, maintain and expand regulatory clearances for our implant systems and technologies and any new products we create;
- our ability to maintain existing, and establish new, collaborations, licensing or other arrangements and the financial terms of any such agreements;
- our ability to obtain and maintain intellectual property protection;
- potential claims relating to our intellectual property and third-party intellectual property;
- our compliance with extensive requirements and government laws, rules and regulations in the United States;
- our estimates of our expenses, ongoing losses, future revenue, capital requirements and our need for, or ability to obtain, additional financing;
- our future financial performance;
- our expectations regarding the time during which we will be an emerging growth company under the JOBS Act; and
- developments and projections relating to our competitors or our industry.

We caution you that the foregoing list does not contain all of the forward-looking statements made in this prospectus.

You should not rely upon forward-looking statements as predictions of future events. We have based the forward-looking statements contained in this prospectus primarily on our current expectations, estimates, forecasts, and projections about future events and trends that we believe may affect our business, financial condition, results of operations, and prospects. Although we believe that we have a reasonable basis for each forward-looking statement contained in this prospectus, we cannot guarantee that the future results, levels of activity, performance, or events and circumstances reflected in the forward-looking statements will be achieved or occur at all. The outcome of the events described in these forward-looking statements is subject to risks, uncertainties, and other factors described in the section titled "Risk Factors" and elsewhere in this prospectus. Moreover, we operate in a very competitive and rapidly changing environment. New risks and uncertainties emerge from time to time, and it is not possible for us to predict all risks and uncertainties that could have an impact on the forward-looking statements contained in this prospectus. The results, events, and circumstances reflected in the forward-looking statements may not be achieved or occur, and actual results, events, or circumstances could differ materially from those described in the forward-looking statements.

The forward-looking statements made in this prospectus relate only to events as of the date on which the statements are made. We undertake no obligation to update any forward-looking statements made in this prospectus to reflect events or circumstances after the date of this prospectus or to reflect new information or the occurrence of unanticipated events, except as required by law. We may not actually achieve the plans, intentions, or expectations disclosed in our forward-looking statements, and you should not place undue reliance on our forward-looking statements. Our forward-looking statements do not reflect the potential impact of any future acquisitions, mergers, dispositions, joint ventures, or investments we may make.

In addition, statements that "we believe" and similar statements reflect our beliefs and opinions on the relevant subject. These statements are based upon information available to us as of the date of this prospectus, and while we believe such information forms a reasonable basis for such statements, such information may be limited or incomplete, and our statements should not be read to indicate that we have conducted an exhaustive inquiry into, or review of, all potentially available relevant information. These statements are inherently uncertain, and you are cautioned not to unduly rely upon these statements.

You should read this prospectus and the documents that we reference in this prospectus and have filed as exhibits to the registration statement, of which this prospectus is a part, completely and with the understanding that our actual future results may be materially different from what we expect. We qualify all of the forward-looking statements in this prospectus by these cautionary statements.

USE OF PROCEEDS

We estimate that the net proceeds from this offering will be approximately $64.8 million (or approximately $75.2 million if the underwriters exercise their over-allotment option in full), based on the initial public offering price of $15.00 per share after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us.

The principal purposes of this offering are to obtain capital to support our operations, to create a public market for our common stock and to facilitate our future access to the public equity markets. We currently intend to use the net proceeds from this offering as follows:

- approximately $55.0 million to scale up our commercial organization through the hiring of additional sales representatives and expansion of our commercial leadership team, as well as to invest in additional instrument sets to support volume growth and geographic expansion;
- approximately $7.5 million to fund the research and development of continued general innovation in our implant systems; and
- the remainder for working capital and other general corporate purposes.

We may also use a portion of the proceeds to acquire complementary businesses, products, services, or technologies. We periodically evaluate strategic opportunities; however, we have no current understandings or commitments to enter into any such acquisitions or make any such investments.

Based on our current operating plan, we believe that the estimated net proceeds from this offering, together with the expected cash generated from the sale of our systems, our existing cash, cash equivalents and marketable securities and amounts under our Trinity Loan Agreement and proceeds from the 2025 Convertible Notes, will be sufficient to fund our planned operating expenses and capital expenditure requirements for at least the next 12 months.

This expected use of the net proceeds from this offering represents our intentions based on our current plans and business conditions, which could change in the future as our plans and business conditions evolve. We cannot predict with certainty all of the particular uses for the proceeds of this offering or the amounts that we will actually spend on the uses set forth above. Accordingly, our management will have broad discretion in applying the net proceeds from this offering. The timing and amount of our actual expenditures will be based on many factors, including cash flows from operations and the anticipated growth of our business. Pending their use, we intend to invest the net proceeds from this offering in a variety of capital-preservation investments, including government securities and money market funds.

DIVIDEND POLICY

We have never declared or paid cash dividends on our capital stock. We currently intend to retain all available funds and any future earnings, if any, to fund the development and expansion of our business, and we do not anticipate paying any cash dividends in the foreseeable future. The terms of our Trinity Loan Agreement also limit our ability to pay dividends, and we may enter into additional credit agreements or other borrowing arrangements in the future that may restrict our ability to declare or pay cash dividends on our capital stock. Any future determinations regarding the declaration and payment of dividends, if any, will be at the discretion of our board of directors, subject to applicable law, and will depend upon then-existing conditions, including our financial condition, results of operations, contractual restrictions, general business conditions, capital requirements, and other factors our board of directors may deem relevant.

CAPITALIZATION

The following table sets forth our cash, cash equivalents and marketable securities, and our capitalization as of March 31, 2025:

- on an actual basis;
- on a pro forma basis, to reflect: (i) the receipt of approximately $20.0 million in aggregate gross proceeds from the issuance and sale of our Series E convertible preferred stock in June 2025, (ii) the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) into an aggregate of 12,320,676 shares of our common stock, (iii) the Warrant Conversion and the related reclassification of warrant liability to stockholders' equity, (iv) the issuance of the 2025 Convertible Notes and the related Notes Conversion, (v) the filing and effectiveness of our amended and restated certificate of incorporation, which will become effective immediately prior to the completion of this offering; and
- on a pro forma as adjusted basis, giving effect to the pro forma adjustments discussed above, and our receipt of estimated net proceeds from the issuance and sale of shares of common stock in this offering at the initial offering price of $15.00 per share after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us.

You should read this table together with the section titled "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our financial statements and the related notes included elsewhere in this prospectus. The pro forma as adjusted information below reflects our capitalization following the completion of this offering.

	As of March 31, 2025		
	Actual	Pro Forma	Pro Forma as Adjusted
	(unaudited)		
	(in thousands, except share and per share data)		
Cash, cash equivalents and marketable securities	$ 27,564	$ 87,564	$ 152,314
Long-term debt[1]	14,721	14,721	14,721
Warrant liabilities	970	—	—
Convertible preferred stock, par value $0.001 per share; 238,447,976 shares authorized, 205,759,356 shares issued and outstanding, actual; no shares authorized, issued or outstanding, pro forma and pro forma as adjusted	94,147	—	—
Stockholders' deficit:			
Preferred stock, par value $0.001 per share; no shares authorized, issued and outstanding, actual; 20,000,000 shares authorized, no shares issued or outstanding, pro forma and pro forma as adjusted	—	—	—
Common stock, par value $0.001 per share; 280,986,575 shares authorized, 102,785 shares issued and outstanding, actual; 730,000,000 shares authorized and 15,228,510 shares issued and outstanding, pro forma; 730,000,000 shares authorized and 20,228,510 shares issued and outstanding, pro forma as adjusted	1	15	20
Additional paid-in capital	2,307	157,409	222,154
Accumulated deficit	(61,703)	(61,703)	(61,703)
Accumulated other comprehensive income	81	81	81
Total stockholders' (deficit) equity	(59,314)	95,803	160,553
Total capitalization	$ 49,554	$ 110,524	$ 175,274

(1) Long-term debt as of March 31, 2025 excludes the 2025 Convertible Notes issued on July 18, 2025, which will automatically convert into shares of our common stock upon completion of this offering. See "Management's Discussion and Analysis of Financial Condition and Results of Operations—Liquidity and Capital Resources—Indebtedness."

If the underwriters exercise their over-allotment option in full, our pro forma as adjusted cash, cash equivalents, and marketable securities, additional paid-in capital, total stockholders' deficit, and total capitalization, and shares of common stock outstanding as of March 31, 2025 would be $162.8 million, $232.6 million, $171.0 million, $185.7 million, and 20,978,510 shares, respectively.

The information in the table above excludes:

- 1,310,108 shares of our common stock issuable upon the exercise of outstanding stock options under our Stock Option Plan as of March 31, 2025, with a weighted-average exercise price of $2.08 per share;
- 461,236 shares of our common stock issuable upon the exercise of outstanding stock options granted subsequent to March 31, 2025 under our Stock Option Plan, with a weighted-average exercise price of $2.86 per share;
- 17,827 shares of our common stock issuable upon the exercise of a warrant to purchase shares of common stock outstanding as of March 31, 2025, with an exercise price of $2.10 per share;
- 45,859 shares of our common stock issuable upon the exercise of warrants to purchase 875,000 shares of our Series B convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.372 per share of Series B convertible preferred stock;
- 130,736 shares of our common stock issuable upon the exercise of a warrant to purchase 2,494,457 shares of our Series D convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.54120 per share of Series D convertible preferred stock, which will convert into a warrant to purchase common stock immediately prior to the completion of this offering; and
- 2,743,112 shares of our common stock reserved for future issuance under our equity compensation plans, consisting of:
    - 2,493,739 shares of our common stock reserved for future issuance under the 2025 Plan, which became effective upon the effectiveness of the registration statement of which this prospectus forms a part (which number includes 46,000 shares of our common stock subject to restricted stock unit awards granted to certain of our directors pursuant to our 2025 Plan in connection with this offering; and
    - 249,373 shares of our common stock reserved for future issuance under the ESPP, which became effective upon the effectiveness of the registration statement of which this prospectus forms a part.

DILUTION

If you purchase shares of our common stock in this offering, your ownership interest will be immediately and substantially diluted to the extent of the difference between the initial public offering price per share of our common stock and the pro forma as adjusted net tangible book value per share of our common stock immediately after this offering.

As of March 31, 2025, our historical net tangible book deficit was $(60.5 million), or $(588.46) per share of our common stock, based on 102,785 shares of our common stock issued and outstanding as of such date. Our historical net tangible book value (deficit) per share represents our total tangible assets less total liabilities and convertible preferred stock, divided by the aggregate number of shares of our common stock outstanding as of March 31, 2025.

Our pro forma net tangible book value as of March 31, 2025 was $94.6 million, or $6.21 per share. Pro forma net tangible book value per share represents total tangible assets less total liabilities, after giving effect to: (i) the receipt of approximately $20.0 million in aggregate gross proceeds from the issuance and sale of our Series E convertible preferred stock in June 2025, (ii) the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) into an aggregate of 12,320,676 shares of our common stock, (iii) the Warrant Conversion and the related reclassification of warrant liability to stockholders' equity, and (iv) the issuance of the 2025 Convertible Notes and the related Notes Conversion. Pro forma net tangible book value per share represents our pro forma net tangible book value divided by the total number of shares outstanding as of March 31, 2025, after giving effect to the pro forma adjustments described above.

After giving further effect to the sale by us of 5,000,000 shares of our common stock in this offering at the initial public offering price of $15.00 per share and after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us, our pro forma as adjusted net tangible book value as of March 31, 2025 would have been $159.4 million, or $7.88 per share. This represents an immediate increase in pro forma net tangible book value to existing stockholders of $1.66 per share and an immediate dilution in pro forma net tangible book value to new investors of $7.12 per share.

We determine dilution by subtracting the pro forma as adjusted net tangible book value per share after this offering from the amount of cash that a new investor paid for a share of common stock. The following table illustrates this dilution on a per share basis:

Initial public offering price per share		$ 15.00
Historical net tangible book value (deficit) per share as of March 31, 2025	$ (588.46)	
Pro forma increase in historical net tangible book value per share as of March 31, 2025 attributable to the pro forma adjustments described above	594.67	
Pro forma net tangible book value per share as of March 31, 2025	6.21	
Increase in pro forma net tangible book value per share attributable to new investors participating in this offering	1.66	
Pro forma as adjusted net tangible book value per share after this offering		7.88
Dilution per share to new investors participating in this offering		$ 7.12

If the underwriters exercise in full their over-allotment option, the pro forma as adjusted net tangible book value per share of our common stock after this offering would be $8.10 per share, and the dilution in pro forma as adjusted net tangible book value per share to investors participating in this offering would be $6.90 per share of our common stock, assuming the initial public offering price of $15.00 per share.

The following table summarizes, as of March 31, 2025, on a pro forma as adjusted basis as described above, the number of shares of our common stock, the total consideration and the average price per share (i) paid to us by existing stockholders and (ii) to be paid by new investors acquiring our common stock in this offering at the initial public offering price of $15.00 per share before deducting estimated underwriting discounts and commissions and

estimated offering expenses payable by us. As the table below shows, investors participating in this offering will pay an average price per share substantially higher than our existing stockholders paid.

	Shares Purchased		Total Consideration		Average Price Per Share
	Number	Percent	Amount (in thousands)	Percent	
Existing stockholders before this offering[1]	15,228,510	75.3 %	$ 96,455	56.3 %	$6.46
New investors participating in this offering	5,000,000	24.7 %	$ 75,000	43.7 %	$15.00
Total	20,228,510	100.0 %	$ 171,455	100.0 %	

(1) The presentation in this table regarding ownership by existing stockholders does not give effect to any purchases that existing stockholders may make through our directed share program or otherwise purchase in this offering.

Except as otherwise indicated, the above discussion and tables assume no exercise of the underwriters' over-allotment option and excludes:

- 1,310,108 shares of our common stock issuable upon the exercise of outstanding stock options under our Stock Option Plan as of March 31, 2025, with a weighted-average exercise price of $2.08 per share;
- 461,236 shares of our common stock issuable upon the exercise of outstanding stock options granted subsequent to March 31, 2025 under our Stock Option Plan, with a weighted-average exercise price of $2.86 per share;
- 17,827 shares of our common stock issuable upon the exercise of a warrant to purchase shares of common stock outstanding as of March 31, 2025, with an exercise price of $2.10 per share;
- 45,859 shares of our common stock issuable upon the exercise of warrants to purchase 875,000 shares of our Series B convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.372 per share of Series B convertible preferred stock.
- 130,736 shares of our common stock issuable upon the exercise of a warrant to purchase 2,494,457 shares of our Series D convertible preferred stock outstanding as of March 31, 2025, with an exercise price of $0.54120 per share of Series D convertible preferred stock, which will convert into a warrant to purchase common stock immediately prior to the completion of this offering; and
- 2,743,112 shares of our common stock reserved for future issuance under our equity compensation plans, consisting of:
  - 2,493,739 shares of our common stock to be reserved for future issuance under the 2025 Plan, which became effective upon the effectiveness of the registration statement of which this prospectus forms a part (which number includes 46,000 shares of our common stock subject to restricted stock unit awards granted to certain of our directors pursuant to our 2025 Plan in connection with this offering; and
  - 249,373 shares of our common stock reserved for future issuance under the ESPP, which became effective upon the effectiveness of the registration statement of which this prospectus forms a part.

We may choose to raise additional capital through the sale of equity or convertible debt securities due to market conditions or strategic considerations even if we believe we have sufficient funds for our current or future operating plans. To the extent that any outstanding warrants or options are exercised, new options or other equity awards are issued under our equity incentive plans, or we issue additional shares or convertible securities in the future, there will be further dilution to new investors participating in this offering.

MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

The following discussion should be read in conjunction with the section titled "Summary Financial Data" and our financial statements and the related notes included elsewhere in this prospectus. In addition to historical information, this discussion contains forward-looking statements that involve risks, uncertainties and assumptions that could cause actual results to differ materially from management's expectations. See the section titled "Special Note Regarding Forward-Looking Statements." Factors that could cause or contribute to such differences include, but are not limited to, those discussed in the section titled "Risk Factors." Our historical results are not necessarily indicative of the results that may be expected for any period in the future. We are not undertaking any obligation to update any forward-looking statements or other statements we may make in the following discussion or elsewhere in this prospectus even though these statements may be affected by events or circumstances occurring after the forward-looking statements or other statements were made.

Overview

We are a commercial-stage medical technology company exclusively focused on transforming the shoulder surgical care market. We currently offer advanced implant systems for shoulder arthroplasty. These systems are a core element of our ecosystem, which we designed to improve core components of shoulder surgical care – preoperative planning, implant design and procedural efficiency – to benefit each stakeholder in the care chain. Our ecosystem is also comprised of enabling technologies, efficient instrument systems, specialized support and surgeon-to-surgeon collaboration. Together, these elements seek to address the long-standing clinical and operational challenges in the shoulder surgical care market by delivering predictable outcomes, procedural simplicity, and efficiency across all sites of care. We believe our exclusive focus on shoulder surgical care, combined with a highly specialized commercial organization and strong clinical data, positions us well to capture significant share in this large, growing market.

We believe the shoulder surgical care market today presents a significant market opportunity. Our initial focus within this broader market is on shoulder arthroplasty, and we developed our ecosystem with an approach to innovation that prioritizes ease of use, flexibility, predictability of outcomes and site of care efficiency, attributes we believe are critical to win in this market. While our primary commercial focus to date has been within the United States, we believe the international shoulder surgical care market represents a compelling opportunity for long-term growth. We plan to strategically pursue entry into certain international markets over time.

We view ourselves as specialists serving specialists, having purposefully built our commercial organization around the unique needs of shoulder surgeons. Our commercial organization is comprised of three key components: (i) a dedicated commercial leadership team, (ii) a Customer Experience and Medical Education team and (iii) a network of independent distributors. These key components work in tandem to form a commercial flywheel that is designed to build and reinforce relationships with surgeons and other stakeholders in the shoulder surgical care market, accelerate adoption, and enhance long-term retention. Our commercial organization is strategically focused on surgeons in hospital and ASC settings, with a particular focus on the high-volume surgeons who perform the vast majority of shoulder arthroplasty procedures each year.

We utilize third-party manufacturing and supply providers to manufacture our implants. We believe this outsourcing strategy provides the expertise and capacity required to effectively and efficiently scale production based on demand, and helps to ensure low-cost production and a capital efficient business model.

We have experienced significant growth in recent years, primarily driven by growth in our net revenue from the sale of our advanced implant systems sold. We generated net revenue of $31.6 million, with a gross margin of 77.0% and net loss of $15.6 million for the year ended December 31, 2024, compared to net revenue of $19.3 million, with a gross margin of 79.2% and net loss of $12.7 million for the year ended December 31, 2023. The Adjusted EBITDA loss increased from $8.7 million for the year ended December 31, 2023 to $11.4 million for the year ended December 31, 2024. We generated net revenue of $10.1 million, with a gross margin of 76.9% and net loss of $4.7 million for the three months ended March 31, 2025, compared to net revenue of $7.2 million, with a gross margin of 76.8% and net loss of $3.6 million for the three months ended March 31, 2024. The Adjusted

EBITDA loss increased to $3.5 million for the three months ended March 31, 2025 from $2.6 million for the three months ended March 31, 2024. Adjusted EBITDA is not a financial measure under GAAP. See the subsection titled "—Non-GAAP Financial Measures" below for an explanation of how we compute this non-GAAP financial measure and for the reconciliation to the most directly comparable GAAP financial measure.

As of March 31, 2025, we had cash, cash equivalents and marketable securities of $27.6 million. As of March 31, 2025, we had $15.0 million of principal outstanding under our Trinity Loan Agreement and an accumulated deficit of $61.7 million. To date, our primary sources of capital have been from net revenue received from the sale of our implant systems, proceeds from private placements of our convertible preferred stock and debt financing arrangements. Since inception, we have raised a total of $114.6 million in net proceeds from private placements of our convertible preferred stock.

Key Business Metrics

We regularly review a number of operating and financial metrics to evaluate our business, measure our performance, identify trends affecting our business, formulate our business plan and make strategic decisions. We believe that the number of implant systems sold is a useful indicator of our ability to drive demand for our implant systems, generate net revenue and expand our business. The following table sets forth the number of implant systems sold in each of the three-month periods indicated:

	Three Months Ended									
	Mar. 31, 2023	June 30, 2023	Sept. 30, 2023	Dec. 31, 2023	Mar. 31, 2024	June 30, 2024	Sept. 30, 2024	Dec. 31, 2024	Mar. 31, 2025	June 30, 2025
Implant systems sold	559	564	618	943	971	1,121	1,037	1,220	1,443	1,503

While we believe that the number of implant systems sold is a useful indicator and is helpful in tracking the progress of our current business, we anticipate this metric may be substituted for additional or different metrics as our business continues to grows and scale.

Key Factors Affecting Our Results of Operations

We believe the following important factors have impacted and will continue to impact our results of operations for the foreseeable future. While these factors may present significant opportunities for us, they also pose risks and challenges that we must address, as well as those described in the section titled "Risk Factors."

- ***Market awareness and adoption.*** The growth of our business depends on our ability to generate broader awareness of our ecosystem in an effort to drive adoption by new surgeons and to increase utilization by existing surgeons. To drive adoption, our commercial organization is strategically focused on surgeons in hospital and ASC settings, and leverages our internal business intelligence platform to appropriately target surgeons in the shoulder surgical care market. The organization uses key touchpoints, surgeon support and surgeon education initiatives to deliver high quality services and information to surgeons. We are also focused on supporting surgeons that already use our implant systems in order to further increase utilization. We intend to continue scaling our commercial organization to further drive awareness, adoption and demand. Over time, we expect to further expand and utilize our external network of independent distributors. In the future, we may increase our international presence and any such expansion may adversely affect our gross margin and results of operations. Our financial performance will be significantly impacted by the extent to which we can increase awareness of our ecosystem, as well as the timing and rate of adoption of our implant systems by key stakeholders in the shoulder surgical care market.

- ***Increasing importance of outpatient and ASC settings.*** While our ecosystem provides advantages across all shoulder surgical care settings, we believe we are particularly well positioned to address the increasing importance of outpatient and ASC settings. The number of procedures performed in outpatient and ASC settings has increased over time due to both the transition of such procedures from the hospital setting, as well as from a general increase in the number of total procedures performed, due in part to the access and availability of these settings. We generally derive a similar amount of net revenue from procedures whether

they are performed in a hospital or an ASC. As a result, we believe outpatient and ASC settings represent an important and growing opportunity to drive demand and net revenue. Approximately 30% of procedures performed with our implant systems during the month ended December 31, 2024 were performed in ASCs, as compared to approximately 10% during the month ended December 31, 2023. While we currently generate the majority of our net revenue from procedures performed in hospitals, we expect to continue to focus our efforts on serving the outpatient and ASC settings in order to capture further market growth and penetration.

- ***Continued investments in product development, innovation and growth.*** We expect to continue to focus on long-term revenue growth through investments in our ecosystem and expansion of our operations. In research and development, we continually invest in improving our technologies, developing new products and further expanding our cleared indications. For example, we began using ProVoyance in 2021 and are actively developing a fracture-specific system, a revision solution and implants tailored for patients with metal-hypersensitivity. We also believe our ecosystem can be further expanded to address adjacent markets, such as sports medicine and trauma. We anticipate we will continue to invest significantly in product development, including with respect to our supporting technologies, in order to further bolster our ecosystem. While research and development are time consuming and costly and therefore negatively impact our results of operations in the near term, we believe expanding into new areas, implementing product improvements and continuing to demonstrate the efficacy, safety and cost effectiveness of our products through clinical data and surgeon education are all critical to increasing the adoption of our implant systems and to the success of our business over the long term. As we expand our operations in line with our anticipated growth, we will be required to maintain sufficient levels of inventory and instrumentation to meet our estimated demand, which we expect will increase expenses.

- ***Reimbursement and coverage.*** Healthcare providers generally rely on third-party payors, including federal Medicare, state Medicaid and private health insurance plans, to cover and reimburse all or part of the cost of our implant systems. As a result, demand for our implant systems depends in large part on the availability of reimbursement from such payors and the rates that such payors reimburse for procedures using our implant systems, which can vary due to geographic location, nature of facility in which the procedure is performed and other factors. While we benefit from established reimbursement practice and codes applicable to partial and total shoulder arthroplasty, we also work with payors to ensure positive coverage decisions and payments rates in outpatient settings. Effective as of January 1, 2024, CMS added total shoulder arthroplasty to the ASC covered procedures list, which allows procedures that use our implant systems to be performed at ASCs and be reimbursed by Medicare. We believe this decision helped to improve demand from ASCs and supported improved payment rates in outpatient settings for the year ended December 31, 2024, which had a positive impact on net revenue during the period. We expect this trend to continue and further support our growth in outpatient settings, such as ASCs.

- ***Seasonality.*** We have experienced and expect to continue to experience seasonality in our business. While we have experienced significant growth across quarters, we expect that future demand for our advanced implant systems will typically be lower in the months in and surrounding the third calendar quarter, as is common across our industry, as a result of summer seasonality associated with warmer weather and its corresponding impact on individual lifestyles.

Non-GAAP Financial Measures

In addition to our results and measures of performance determined in accordance with U.S. GAAP, we believe that non-GAAP financial measures can be useful in evaluating and comparing our financial and operational performance over multiple periods, identifying trends affecting our business, formulating business plans and making strategic decisions. We use utilize and present Adjusted EBITDA and Adjusted EBITDA margin for these purposes. We define Adjusted EBITDA as net loss before interest expense, net, income tax expense, depreciation and amortization, stock-based compensation expense and loss on extinguishment of convertible promissory notes. Loss on extinguishment of convertible promissory notes relates to the extinguishment of derivative liability in the year ended December 31, 2023 as a result of the conversion of our then-outstanding convertible promissory notes. Adjusted EBITDA margin is calculated as Adjusted EBITDA as a percentage of net revenue.

We believe that Adjusted EBITDA, together with a reconciliation to net loss, provides meaningful supplemental information regarding our performance by excluding certain items that may not be indicative of our business, results of operations, or outlook. However, Adjusted EBITDA has limitations as an analytical tool, and you should not consider this measure in isolation or as a substitute for analysis of our financial results as reported under U.S. GAAP. Some of these potential limitations include:

- other companies, including companies in our industry which have similar business arrangements, may report Adjusted EBITDA, or similarly titled measures but calculate them differently, which reduces their usefulness as comparative measures;
- although depreciation and amortization expenses are non-cash charges, the assets being depreciated and amortized may have to be replaced in the future, and Adjusted EBITDA does not reflect cash capital expenditures for such replacements or for new capital expenditure requirements;
- Adjusted EBITDA also does not reflect changes in, or cash requirements for, our working capital needs or the potentially dilutive impact of stock-based compensation; and
- Adjusted EBITDA does not reflect the interest expense, or the cash requirements necessary to service interest or principal payments, on existing or future debt that we may incur.

Because of these and other limitations, you should consider Adjusted EBITDA only as supplemental to other GAAP-based financial measures.

The following table presents a reconciliation of Adjusted EBITDA to the most comparable GAAP financial measure, net loss, for each of the periods indicated:

	Three Months Ended March 31,		Year Ended December 31,	
	2025	2024	2024	2023
	(unaudited)			
	(in thousands)			
Net loss	$ (4,662)	$ (3,600)	$ (15,619)	$ (12,655)
Interest expense, net	367	317	1,316	761
Income tax expense	—	—	—	—
Depreciation and amortization expense	668	498	2,196	1,641
Stock-based compensation expense	127	168	754	451
Loss on extinguishment of convertible promissory notes	—	—	—	1,127
Adjusted EBITDA	$ (3,500)	$ (2,617)	$ (11,353)	$ (8,675)
Net revenue	$ 10,132	$ 7,184	$ 31,623	$ 19,274
Net loss margin	(46.0)%	(50.1)%	(49.4)%	(65.7)%
Adjusted EBITDA margin	(34.5)%	(36.4)%	(35.9)%	(45.0)%

Components of Our Results of Operations

Net Revenue

We currently derive our net revenue from the sale of our anatomic and reverse total shoulder arthroplasty implant systems, which generally consist of our InSet Glenoid and humeral stem products. We sell our implants to hospitals, outpatient centers and ASCs in the United States through dedicated commercial leadership team and a network of external independent distributors. Net revenue is recognized when the performance obligation to deliver these implant systems to our customers is satisfied and transfer control of the implants to our customers, which is generally when we have received a purchase order and appropriate notification that the procedure has been used or implanted. Revenue is recognized in the amount of the consideration received net of any sales taxes that we expect to collect from customers. We also record shipping and handling costs as revenue. Our average sales price for our

implant systems was $7,271 and $7,181 for the years ended December 31, 2024 and 2023, respectively. Our average sales price for our implant systems was $7,330 and $7,240 for the three months ended June 30, 2025 and 2024, respectively. No single customer accounted for more than 10% of our net revenue during the years ended December 31, 2024 and 2023 and the three months ended March 31, 2025 and 2024. We expect our net revenue to increase for the foreseeable future as we expand our commercial organization, add new customers, expand our sales territories, introduce new products, as existing customers perform more procedures using our systems and as we generally expand awareness of our systems with new and existing customers. While industry trends have resulted in increased downward pricing pressure on medical services and products, we have not experienced a material impact on our net revenue to date; however, we cannot assure you that our net revenue will not be impacted in the future by these industry trends. Our net revenue may fluctuate from quarter to quarter due to a variety of factors, such as the size and success of our dedicated commercial leadership team, the number of hospitals and physicians who are aware of and use our systems and seasonality.

Cost of Goods Sold, Gross Profit and Gross Margin

Cost of Goods Sold

Cost of goods sold consists primarily of the cost of components, packaging and sterilization, and obsolete inventory adjustments. Our systems are manufactured to our specifications primarily by third-party suppliers in the United States and are generally ordered on a purchase order basis. Cost of goods sold is recognized at the time the related revenue is recognized. Prior to use in surgery, the cost of our products is recorded as inventories, net of obsolescence reserve in our balance sheets. Cost of good sold does not include depreciation expense for instruments, which is included in selling, general and administrative expenses. Depreciation expense for instruments was $1.8 million and $1.3 million for the years ended December 31, 2024 and 2023, respectively, and was $0.6 million and $0.4 million for the three months ended March 31, 2025 and 2024, respectively. See Notes 1 and 4 to our audited financial statements and Note 7 to our unaudited financial statements included elsewhere in this prospectus for additional information. We expect cost of goods sold to increase as our net revenue increases and more of our implant systems are sold.

Gross Profit and Gross Margin

Gross profit is calculated as net revenue less cost of goods sold. We calculate gross margin as gross profit divided by net revenue. Our gross margin has been and will continue to be affected by a variety of factors, including average selling prices, sales mix for our implant systems, costs associated with third party manufacturing, seasonality of our business and costs of other services. We expect our gross margin to remain consistent for the foreseeable future as our net revenue grows and our related costs of goods sold increases.

Operating Expenses

Our operating expenses consist of (i) selling, general and administrative expenses and (ii) research and development expenses.

Selling, General and Administrative Expenses

Selling, general and administrative expenses consist primarily of personnel costs, including commissions, salaries, bonuses, benefits and stock-based compensation related to personnel performing selling, marketing and general and administrative functions, including the costs associated with marketing initiatives and medical education programs. All of our stock-based compensation charges are included in selling, general and administrative expenses. In addition, selling, general and administrative expenses include depreciation expense for instruments, royalty payments made to product design surgeons, royalty payments made pursuant to our License Agreement (as defined below), travel expenses, professional services fees (including consulting, legal, finance, audit and tax fees), insurance costs, allocated facility expenses and other general corporate expenses.

We expect our selling, general and administrative expenses to continue to increase for the foreseeable future as we continue to grow our business and increase our utilization of internal and external resources within our commercial organization. As we continue to invest in growth, we will be required to maintain significant levels of

instrumentation, which we expect to increase our selling, general and administrative expenses. Furthermore, the royalty payments made pursuant to our License Agreement will increase as our net revenue increases. We also expect our administrative expenses to increase as we increase our headcount and expand our organization to support our operations as a public company. Additionally, we anticipate increased expenses related to audit, legal, regulatory and tax-related services associated with being a public company, compliance with exchange listing and SEC requirements, director and officer insurance premiums and investor relations costs. We also expect to see an increase in our stock-based compensation expense with the establishment of a new publicly-traded company equity plan and to the extent of grants in the form of restricted stock units or options.

Research and Development Expenses

Research and development expense consist of costs incurred in performing or for the outsourcing of various research and development activities, including consulting fees and other expenses paid related to such activities, costs associated with our registry, any future clinical trial costs and costs related to prototypes and related supplies related to our research and development efforts. We maintain a procedurally focused approach to product development and have projects underway to add new systems and implants across multiple shoulder indications and to add additional functionality or versatility to our existing systems. We expect our research and development expenses to increase as we pursue development of new products and product enhancements.

Other (Income) Expense

Our other (income) expense consists of (i) interest expense, net, (ii) loss on extinguishment of convertible promissory notes and (iii) other income (expense), net.

Interest Expense, Net

Interest expense, net consists of interest expense related to our Trinity Loan Agreement and non-cash interest related to the amortization of debt discount, issuance costs and deferred interest associated with our indebtedness, as well as interest income earned on our cash, cash equivalents and marketable securities.

Loss on Extinguishment of Convertible Promissory Notes

We recorded a loss on extinguishment of convertible promissory notes of $1.1 million for the year ended December 31, 2023 in connection with the conversion of our convertible promissory notes into shares of our Series D convertible preferred stock in February 2023.

Other (Income) Expense, Net

Other (income) expense, net consists primarily of adjustment in the fair market value of marketable securities and change in fair value of warrant liabilities.

Change in fair value of warrant liabilities consists of gains and losses resulting from the remeasurement of the fair value of our warrant liabilities at each balance sheet date. We will continue to record adjustments to the estimated fair value of these warrants until they are exercised or at such time the warrants receive equity treatment.

Results of Operations

Comparison of the Three Months Ended March 31, 2025 and 2024

The following table sets forth the components of our statements of operations for the periods presented below:

	Three Months Ended March 31,		Change	
	2025	2024	$	%
	(unaudited)			
	($ in thousands)			
Net revenue	$ 10,132	$ 7,184	$ 2,948	41.0 %
Cost of goods sold	2,341	1,666	675	40.5 %
Gross profit	7,791	5,518	2,273	41.2 %
Operating expenses:				
Selling, general and administrative expenses[(1)]	10,502	7,704	2,798	36.3 %
Research and development expenses	1,583	1,070	513	48.0 %
Total operating expenses	12,085	8,774	3,311	37.7 %
Operating loss	(4,294)	(3,256)	(1,038)	(31.9)%
Other (income) expense:				
Interest expense, net	367	317	50	15.8 %
Other (income) expense, net	1	27	(26)	(96.3)%
Total other expense	368	344	24	7.0 %
Loss before income tax expense	(4,662)	(3,600)	(1,062)	(29.5)%
Income tax expense	—	—	—	—
Net loss	$ (4,662)	$ (3,600)	$ (1,062)	(29.5)%

(1) Includes stock-based compensation expense of $127 thousand and $168 thousand for the three months ended March 31, 2025 and 2024, respectively.
* Not meaningful

Net Revenue. Net revenue increased $2.9 million, or 41.0%, to $10.1 million for the three months ended March 31, 2025, compared to $7.2 million for the three months ended March 31, 2024. This increase in net revenue was due to an increase in demand and in the number of our systems sold, as well as an increase in the number of our customers.

Cost of Goods Sold and Gross Margin. Cost of goods sold increased $0.7 million, or 40.5%, to $2.3 million for the three months ended March 31, 2025, compared to $1.7 million for the three months ended March 31, 2024. This increase in cost of goods sold was primarily due to the increase in the number of our systems sold. Gross margin for the three months ended March 31, 2025 increased to 76.9%, compared to 76.8% for the three months ended March 31, 2024.

Selling, General and Administrative Expenses. Selling, general and administrative expenses increased $2.8 million, or 36.3%, to $10.5 million for the three months ended March 31, 2025, compared to $7.7 million for the three months ended March 31, 2024. This increase in selling, general and administrative expenses was primarily due to a $1.0 million increase in legal costs related to litigation, a $0.9 million increase in commissions due to higher sales of our systems, a $0.7 million increase in personnel-related expenses as a result of increased headcount of our commercial organization and $0.2 million increase in depreciation of surgical instruments.

Research and Development Expenses. Research and development expenses increased $0.5 million, or 48.0%, to $1.6 million for the three months ended March 31, 2025, compared to $1.1 million for the three months ended March 31, 2024. The increase in research and development expenses was due to our investment in new product development efforts, including an increase in external consulting fees of $0.3 million related to such efforts.

Interest Expense, Net. Interest expense, net increased $0.1 million, or 15.8%, to $0.4 million for the three months ended March 31, 2025, compared to $0.3 million for the three months ended March 31, 2024. This increase in interest expense, net was due to lower interest earned on marketable securities.

Other (Income) Expense, Net. Other (income) expense, net did not change materially for the three months ended March 31, 2025, compared to the three months ended March 31, 2024.

Comparison of Years Ended December 31, 2024 and 2023

The following table sets forth the components of our statements of operations for the periods presented below:

	Year Ended December 31,		Change	
	2024	2023	$	%
	($ in thousands)			
Net revenue	$ 31,623	$ 19,274	$ 12,349	64 %
Cost of goods sold	7,282	4,004	3,278	82 %
Gross profit	24,341	15,270	9,071	59 %
Operating expenses:				
Selling, general and administrative expenses[1]	34,505	23,113	11,392	49 %
Research and development expenses	4,489	3,021	1,468	49 %
Total operating expenses	38,994	26,134	12,860	49 %
Operating loss	(14,653)	(10,864)	(3,789)	35 %
Other (income) expense:				
Interest expense, net	1,316	761	555	73 %
Loss on extinguishment of convertible promissory notes	—	1,127	(1,127)	*
Other (income) expense, net	(350)	(97)	(253)	*
Total other expense	966	1,791	(825)	(46)%
Loss before income tax expense	(15,619)	(12,655)	(2,964)	23 %
Income tax expense	—	—	—	—
Net loss	$ (15,619)	$ (12,655)	$ (2,964)	23 %

(1) Includes stock-based compensation expense of $0.8 million and $0.5 million for the years ended December 31, 2024 and 2023, respectively.
* Not meaningful

Net Revenue. Net revenue increased $12.3 million, or 64%, to $31.6 million for the year ended December 31, 2024, compared to $19.3 million for the year ended December 31, 2023. This increase in net revenue was due to an increase in demand and in the number of our systems sold, as well as an increase in the number of our customers.

Cost of Goods Sold and Gross Margin. Cost of goods sold increased $3.3 million, or 82%, to $7.3 million for the year ended December 31, 2024, compared to $4.0 million for the year ended December 31, 2023. This increase in cost of goods sold was primarily due to the increase in the number of our systems sold. Gross margin for the year ended December 31, 2024 decreased to 77.0%, compared to 79.2% for the year ended December 31, 2023, primarily due to inventory adjustments.

Selling, General and Administrative Expenses. Selling, general and administrative expenses increased $11.4 million, or 49%, to $34.5 million for the year ended December 31, 2024, compared to $23.1 million for the year ended December 31, 2023. This increase in selling, general and administrative expenses was primarily due to a $3.9 million increase in personnel-related expenses as a result of increased headcount of our commercial organization, a $3.5 million increase in commissions due to higher sales of our systems, a $1.4 million increase in legal costs related to litigation and a $0.3 million increase in depreciation of surgical instruments.

Research and Development Expenses. Research and development expenses increased $1.5 million, or 49%, to $4.5 million for the year ended December 31, 2024, compared to $3.0 million for the year ended December 31, 2023. The increase in research and development expenses was due to our investment in new product development efforts, including an increase in external consulting fees of $0.5 million related to such efforts.

Interest Expense, Net. Interest expense, net increased $0.6 million, or 73%, to $1.3 million for the year ended December 31, 2024, compared to $0.8 million for the year ended December 31, 2023. This increase in interest expense was due to the increased interest incurred under our Trinity Loan Agreement.

Loss on Extinguishment of Convertible Promissory Notes. In connection with issuance and sale of shares of our Series D convertible preferred stock in February 2023, our outstanding convertible promissory notes were converted into shares of our Series D convertible preferred stock. This conversion resulted in a loss of $1.1 million for the year ended December 31, 2023.

Other (Income) Expense, Net. Other (income) expense, net decreased $(0.3) million, to $(0.4) million for the year ended December 31, 2024, compared to $(0.1) million for the year ended December 31, 2023, due to adjustment in the fair market value of our marketable securities and an increase in the fair value of our warrant liabilities.

Quarterly Results of Operations Data

The following tables set forth selected quarterly statements of operations data for each of the four fiscal quarters in each of the years ended December 31, 2024 and 2023 and the fiscal quarter ended March 31, 2025. The information for each of these quarters has been prepared in accordance with GAAP, on the same basis as our audited annual financial statements included elsewhere in this prospectus and includes, in the opinion of management, all adjustments, consisting only of normal recurring adjustments, necessary for the fair presentation of the results of operations for these periods. This data should be read in conjunction with our financial statements and the related

notes included elsewhere in this prospectus. These historical quarterly operating results are not necessarily indicative of our operating results for the full year or any future period.

	Three Months Ended								
	Mar. 31, 2023	June 30, 2023	Sept. 30, 2023	Dec. 31, 2023	Mar. 31, 2024	June 30, 2024	Sept. 30, 2024	Dec. 31, 2024	Mar. 31 2025
	(unaudited) ($ in thousands)								
Net revenue	$ 3,865	$ 4,150	$ 4,418	$ 6,841	$ 7,184	$ 8,260	$ 7,454	$ 8,725	$ 10,132
Cost of goods sold	777	902	920	1,405	1,666	1,906	1,753	1,957	2,341
Gross profit	3,088	3,248	3,498	5,436	5,518	6,354	5,701	6,768	7,791
Operating expenses:									
Selling, general and administrative expenses[1]	4,650	5,183	6,229	7,051	7,704	9,171	8,493	9,137	10,502
Research and development expenses	614	757	856	794	1,070	1,162	1,090	1,167	1,583
Total operating expenses	5,264	5,940	7,085	7,845	8,774	10,333	9,583	10,304	12,085
Operating loss	(2,176)	(2,692)	(3,587)	(2,409)	(3,256)	(3,979)	(3,882)	(3,536)	(4,294)
Other (income) expense									
Interest expense, net	213	(94)	77	565	318	292	315	391	367
Loss on extinguishment of convertible promissory notes	1,127	—	—	—	—	—	—	—	—
Other (income) expense, net	198	61	88	(444)	27	(100)	(142)	(135)	1
Total other expense	1,538	(33)	165	121	345	192	173	256	368
Loss before income tax expense	(3,714)	(2,659)	(3,752)	(2,530)	(3,601)	(4,171)	(4,055)	(3,792)	(4,662)
Income tax expense	—	—	—	—	—	—	—	—	—
Net loss	$ (3,714)	$ (2,659)	$ (3,752)	$ (2,530)	$ (3,601)	$ (4,171)	$ (4,055)	$ (3,792)	$ (4,662)

(1) Includes stock-based compensation expense of $61 thousand, $99 thousand, $143 thousand, $148 thousand, $168 thousand, $170 thousand, $237 thousand, $179 thousand and $127 thousand for the quarters ended March 31, 2023, June 30, 2023, September 30, 2023, December 31, 2023, March 31, 2024, June 30, 2024, September 30, 2024, December 31, 2024 and March 31, 2025, respectively.

Selected Quarterly Trends

Net Revenue

Our quarterly net revenue increased throughout the periods presented, primarily due to an increase in demand and in the number of our systems sold.

Cost of Goods Sold and Gross Margin

Cost of goods sold increased throughout the periods presented, primarily due to the increase in the number of our systems sold. Our gross margin fluctuated during the periods presented due primarily to inventory adjustments and sales mix.

Operating Expenses Trends

Our selling, general and administrative expenses and research and development expenses each increased throughout the periods presented, primarily due to personnel-related expenses as a result of higher headcount, increased commission payments due to higher sales of our systems, increased legal costs related to litigation, higher depreciation of surgical instruments and increased research and development costs due to our investment in new product development efforts.

Liquidity and Capital Resources

To date, our primary sources of capital have been from net revenue received from the sale of our implant systems, proceeds from private placements of our convertible preferred stock and debt financing arrangements. Since inception, we have raised a total of $114.6 million in net proceeds from private placements of our convertible preferred stock. As of March 31, 2025, we had $15.0 million of principal outstanding under our Trinity Loan Agreement.

We have generated losses from our operations since our inception as reflected in our accumulated deficit of $61.7 million and $57.0 million as of March 31, 2025 and December 31, 2024, respectively, including our net loss of $4.7 million and $15.6 million for the three months ended March 31, 2025 and year ended December 2024, respectively. Our losses primarily resulted from the costs incurred in the development, sales, and marketing of our systems and providing support for our operations. We expect to continue to incur losses for the foreseeable future and to expend significant amounts of cash for the foreseeable future as we continue to scale our business, increase selling, general and administrative expenses to support the expansion of our commercial organization and efforts, increase general and administrative expenses to support being a publicly-traded company and invest in research and development activities.

Indebtedness

As of March 31, 2025, we had cash, cash equivalents and marketable securities of $27.6 million and $15.0 million of principal outstanding under our Trinity Loan Agreement.

On August 7, 2023, we entered into the Trinity Loan Agreement, as amended on July 21, 2025, with Trinity Capital, as administrative agent and collateral agent (in such capacities, the “Agent”) and as a lender, and the other lenders from time to time party thereto, providing for term loans of up to an aggregate principal amount of $45.0 million, available in three tranches: (i) a $15.0 million tranche that was fully funded on the August 7, 2023, (ii) a $15.0 million tranche that expires on December 31, 2025 and (iii) a $15.0 million tranche available through December 31, 2026. The availability of the second tranche is subject to, among other things, our achievement of at least $30.0 million of annualized trailing 6-month revenue by December 31, 2025. The availability of the third tranche is subject to, among other things, our achievement of at least $45.0 million of annualized tailing six-month revenue by December 31, 2026. In connection with the Trinity Loan Agreement, we issued a warrant to purchase 2,494,457 shares of our Series D convertible preferred stock to Trinity Capital. The warrant has an exercise price of $0.5412 per share and expires ten years from the date of its issuance. As of March 31, 2025, the aggregate outstanding principal balance under the Trinity Loan Agreement was $15.0 million.

The term loans under the Trinity Loan Agreement bear interest at an annual rate equal to the greater of the prime rate plus 3.50% and 11.00%. Under the terms of the Trinity Loan Agreement, the prime rate is equal to the greater of 8.0% per year and the prime rate as reported in The Wall Street Journal. We are required to make monthly payments of interest only through maturity of the term loans on September 1, 2028 (“Maturity Date”). The unpaid balance of principal and accrued interest is due on the Maturity Date. The Trinity Loan Agreement provides that we can at any time prepay the term loans, in whole or in part, subject to a prepayment premium equal to: (i) 2.50% of the then-outstanding principal amount of the term loans, if such prepayment occurs on or prior to the first anniversary of the Trinity Loan Agreement; (ii) 1.50% of the then-outstanding principal amount of the advance, if such prepayment occurs after the first anniversary of the Trinity Loan Agreement and on or prior to the second anniversary of the Trinity Loan Agreement; and (iii) 1.00% of the then-outstanding principal amount of the advance, if such prepayment occurs after the second anniversary of the Trinity Loan Agreement and prior to the Maturity Date. We are required to make an end of term payment equal to 3.00% of the aggregate principal amount of the term

loans funded on the earlier of (i) the Maturity Date, (ii) the date that we prepay all of the outstanding principal in full or (iii) the date of acceleration of the balance of the outstanding term loans by the Agent. The term loans are secured by substantially all our assets, including intellectual property.

The Trinity Loan Agreement also includes customary affirmative and negative covenants and events of default. Upon the occurrence and continuance of an event of default the Agent may demand immediate repayment of all principal and unpaid interest under the Trinity Loan Agreement, and exercise remedies against us and the collateral securing the Trinity Loan Agreement. Events of default under the Trinity Loan Agreement include, among other things: (i) insolvency, bankruptcy or similar proceedings subject to a certain grace period in respect of any involuntary insolvency, bankruptcy or similar proceedings; (ii) failure to pay any debts due under the Trinity Loan Agreement or other loan documents on a timely basis; (iii) failure to observe any covenant or secured obligation under the Trinity Loan Agreement, subject to a certain cure period; (iv) occurrence of a material adverse change; (v) material misrepresentations; (vi) occurrence of any default under any material agreement (or termination thereof) or any other agreement resulting in a right by the applicable third party to accelerate debt in excess of $500,000; (vii) entry of certain final, non-appealable judgments against us in excess of $500,000 not paid or bonded within 10 days of such entry; (viii) a change of control unless as a condition to the closing of such change of control all outstanding term loans will be paid in full; and (ix) certain changes in the composition of board of directors.

As of March 31, 2025, we were in compliance with all covenants contained in the Trinity Loan Agreement.

On July 18, 2025, we issued the 2025 Convertible Notes to certain investors in an aggregate principal amount of $40.0 million. The 2025 Convertible Notes mature on September 1, 2028 and accrue interest at a rate of 5.0% per annum through June 30, 2026 and 10.0% per annum after June 30, 2026. Upon the completion of this offering, the 2025 Convertible Notes and any accrued interest will automatically convert into shares of our common stock at the applicable conversion price. The conversion price is the lower of (a) 80% of the initial public offering price per share in this offering and (b) $280.0 million divided by the number of fully diluted shares of capital stock (on an as-converted basis) outstanding immediately prior to this offering but excluding the 2025 Convertible Notes (the “Fully Diluted Capitalization”), provided, that in no event shall such conversion price be less than the quotient obtained by dividing $210.0 million by the Fully Diluted Capitalization.

Future Funding Requirements

Based on our current operating plan, we believe that the estimated net proceeds from this offering, together with the expected cash generated from the sale of our systems, our existing cash, cash equivalents and marketable securities and amounts under our Trinity Loan Agreement and the 2025 Convertible Notes, will be sufficient to fund our planned operating expenses and capital expenditure requirements for at least the next 12 months. We have based this estimate on assumptions that may prove to be wrong, and we could deplete our capital resources sooner than we expect. We may experience lower than expected cash generated from operating activities or greater than expected capital expenditures, cost of goods sold, or operating expenses, and may need to raise additional capital to fund operations, increase our commercial organization and efforts, further research and development activities, or acquire, invest in, or in-license other businesses, assets, or technologies.

Our future capital needs will depend upon many factors, including:

- the market awareness and adoption of our systems, including our InSet Glenoid and InSet humeral stem products;
- the scope, timing and costs of supporting the growth and expansion of our commercial organization and efforts;
- the cost and pace of our research and development activities;
- the costs associated with any product recall that may occur;
- the costs associated with the manufacture and supply of our products at increased production levels;
- the costs associated with securing additional suppliers and service providers;

- the scope, rate of progress and costs of our current or future clinical and registries as well as costs associated with complying with regulatory requirements;
- the cost and timing of additional regulatory clearances or approvals;
- the costs of attaining, defending, and enforcing our intellectual property rights;
- whether we acquire third-party products or technologies;
- the terms and timing of any other distribution, collaborative, licensing, and other arrangements that we may establish;
- the emergence of competing technologies or other adverse market developments;
- our ability to raise additional funds to finance our operations;
- debt service requirements;
- the rate at which we expand internationally; and
- the cost associated with being a public company.

We have based this estimate on assumptions that may prove to be wrong, and we could utilize our available capital resources sooner than we expect. We may seek to raise any necessary additional capital through public or private equity offerings or debt financings, credit or loan facilities or a combination of one or more of these or other funding sources. Additional funds may not be available to us on acceptable terms or at all. If we fail to obtain necessary capital when needed on acceptable terms, or at all, we could be forced to delay, limit, reduce or terminate our product development programs, commercialization efforts or other operations. If we raise additional funds by issuing equity securities or convertible debt, our stockholders will suffer dilution and the terms of any financing may adversely affect the rights of our stockholders. In addition, as a condition to providing additional funds to us, future investors may demand, and may be granted, rights superior to those of existing stockholders. If we raise additional capital through collaborations agreements, licensing arrangements or marketing and distribution arrangements, we may have to relinquish valuable rights, future revenue streams, research programs or product or grant licenses that may not be favorable to us. Debt financing, if available, is likely to involve restrictive covenants limiting our flexibility in conducting future business activities, and, in the event of insolvency, debt holders would be repaid before holders of our equity securities received any distribution of our corporate assets.

Cash Flows

The following table shows a summary of our cash flows for each of the periods presented:

	Three Months Ended March 31,		Change		Year Ended December 31,		Change	
	2025	2024	$	%	2024	2023	$	%
	(unaudited)							
	($ in thousands)							
Net cash (used in) provided by:								
Operating activities	$ (6,477)	$ (3,475)	$ (3,002)	(86.4)%	$ (14,143)	$ (13,144)	$ (999)	(7.6)%
Investing activities	(13,070)	(1,015)	$ (12,055)	*	13,964	(28,973)	42,937	148.20 %
Financing activities	19,794	—	$ 19,794	*	66	44,728	(44,662)	(99.9)%
Net (decrease) increase in cash and cash equivalents	$ 247	$ (4,490)	$ 4,737	105.5 %	$ (113)	$ 2,611	$ (2,724)	*

* Not meaningful

Operating Activities

For the three months ended March 31, 2025, net cash used in operating activities was $(6.5) million, consisting primarily of a net loss of $4.7 million and net cash used by changes in our operating assets and liabilities of $(2.5) million, partially offset by non-cash charges of $0.7 million. The non-cash charges primarily consisted of depreciation and amortization expense of $0.7 million and stock-based compensation expense of $0.1 million, partially offset by realized gain on marketable securities of $(0.2) million. Net cash used by changes in our operating assets and liabilities primarily consisted of a decrease of $2.1 million to accounts payable and an increase of $1.5 million in trade accounts receivable, partially offset by a decrease of $0.9 million in inventory due to lower inventory purchases than inventory sold.

For the three months ended March 31, 2024, net cash used in operating activities was $(3.5) million, consisting primarily of a net loss of $3.6 million and net cash used by changes in our operating assets and liabilities of $(0.8) million, partially offset by non-cash charges of $0.9 million. The non-cash charges primarily consisted of depreciation and amortization expense of $0.5 million, stock-based compensation expense of $0.2 million and realized loss on marketable securities of $0.1 million. Net cash used by changes in our operating assets and liabilities primarily consisted of a decrease of $0.6 million in other current liabilities and an increase of $0.5 million in inventory to support demand for our implant systems, partially offset by a decrease of $0.4 million in trade accounts receivable.

For the year ended December 31, 2024, net cash used in operating activities was $(14.1) million, consisting primarily of a net loss of $15.6 million and net cash used by changes in our operating assets and liabilities of $(1.1) million, partially offset by non-cash charges of $2.5 million. The non-cash charges primarily consisted of depreciation and amortization expense of $2.2 million, stock-based compensation expense of $0.8 million, and realized gain on marketable securities of $(0.5) million. Net cash used by changes in our operating assets and liabilities primarily consisted of an increase of $4.0 million in inventory to support demand for our implant systems and $0.6 million in trade accounts receivable, partially offset by a decrease of $3.1 million in accounts payable.

For the year ended December 31, 2023, net cash used in operating activities was $(13.1) million, consisting primarily of a net loss of $12.7 million and net cash used by changes in our operating assets and liabilities of $(4.8) million, partially offset by non-cash charges of $4.4 million. The non-cash charges primarily consisted of depreciation and amortization expense of $1.6 million, stock-based compensation expense of $0.5 million, and loss on extinguishment of convertible promissory notes of $1.1 million. Net cash used by changes in our operating assets and liabilities primarily consisted of an increase of $2.7 million in trade accounts receivable and $2.4 million in inventory, partially offset by a decrease of $1.4 million in other current liabilities.

Investing Activities

For the three months ended March 31, 2025, net cash used in investing activities was $(13.1) million, consisting primarily of purchases of $(20.1) million in marketable securities and $(0.8) million in fixed assets proceeds, partially offset by proceeds of $7.8 million from sales of our marketable securities.

For the three months ended March 31, 2024, net cash used in investing activities was $(1.0) million, consisting primarily of the purchases of $(2.8) million in marketable securities and $(1.3) million in fixed assets proceeds, partially offset by cash proceeds of $3.1 million from sales of our marketable securities.

For the year ended December 31, 2024, net cash provided in investing activities was $14.0 million, consisting primarily of proceeds of $23.2 million from sales of our marketable securities, partially offset by purchases of $(5.3) million in marketable securities and $(4.0) million in fixed assets.

For the year ended December 31, 2023, net cash used in investing activities was $(29.0) million, consisting primarily of the purchase of $(29.3) million in marketable securities and $(2.7) million in fixed assets, partially offset by cash of $3.0 million from sales of our marketable securities.

Financing Activities

For the three months ended March 31, 2025, net cash provided by financing activities was $19.8 million, consisting of proceeds of $19.6 million from the issuance and sale of shares of our Series E convertible preferred stock and proceeds from the exercise of warrants for our Series Seed preferred stock.

For the three months ended March 31, 2024, there was no net cash provided by financing activities.

For the year ended December 31, 2024, net cash provided by financing activities was $0.1 million, consisting of proceeds from the exercise of our outstanding common stock options.

For the year ended December 31, 2023, net cash provided by financing activities was $44.7 million, consisting primarily of proceeds of $34.9 million from the issuance and sale of shares of our Series D convertible preferred stock and $15.0 million in proceeds related to our Trinity Loan Agreement, partially offset by payments of $5.0 million related to the repayment of our prior loan and security agreement.

Contractual Obligations and Commitments

Our contractual commitments will have an impact on our future liquidity. These commitments include future payments on our Trinity Loan Agreement, future payments on facility leases and certain royalty obligations. Where applicable, we calculate our obligation based on termination fees that can be paid to exit the contract.

Debt

The principal outstanding under our Trinity Loan Agreement was $15.0 million as of March 31, 2025, however, we are required to make monthly payments of interest only through September 2027. Following the interest-only period, we are required to make payments of interest and principal in monthly installments through maturity of the term loans on September 1, 2028.

Leases

We have entered into an operating lease for office space in Michigan. The lease has a five-year term, which commenced in July 2021 and is renewable for one additional five-year term upon expiration. As of March 31, 2025, the operating lease obligations under this operating lease were $60 thousand.

Royalties

We have entered into a License Agreement with Genesis Software, pursuant to which we are required to pay Genesis Software certain payments, including royalty payments, until such time we have paid Genesis Software an aggregate of $7.0 million under the License Agreement. As of March 31, 2025, we have paid an aggregate of $4.1 million of the total $7.0 million, including royalties of $0.4 million and $0.3 million in the three months ended March 31, 2025 and 2024, respectively.

Off-Balance Sheet Arrangements

We did not have during the periods presented, and we do not currently have, any off-balance sheet arrangements or any relationships with unconsolidated entities or financial partnerships, such as structured finance, special purpose entities, or variable interest entities, that were established for the purpose of facilitating off-balance sheet arrangements or other contractually narrow or limited purposes.

Critical Accounting Policies and Estimates

Management’s discussion and analysis of our financial condition and results of operations is based on our financial statements, which have been prepared in accordance with U.S. generally accepted accounting principles. The preparation of these financial statements requires us to make estimates and assumptions for the reported amounts of assets, liabilities, revenue, expenses and related disclosures. Our estimates are based on our historical experience and on various other factors that we believe are reasonable under the circumstances, the results of which form the basis for making judgments about the carrying value of assets and liabilities that are not readily apparent

from other sources. Actual results may differ from these estimates under different assumptions or conditions and any such differences may be material.

While our significant accounting policies are more fully described in Note 1 of our financial statements included elsewhere in this prospectus, we believe the following discussion addresses our most critical accounting policies and estimates, which are those that are most important to our financial condition and results of operations and require our most difficult, subjective and complex judgments.

Revenue Recognition

Revenue is recognized as the performance obligations to deliver products are satisfied and are recorded based on the amount of consideration we expect to receive in exchange for satisfying the performance obligations. Our sales are recognized primarily when we transfer control to the customer, which is generally when we have received a purchase order and appropriate notification the product has been used or implanted. Products are primarily transferred to customers at a point in time.

Revenue represents the amount of consideration we expect to receive from customers in exchange for transferring products. Net revenue excludes sales taxes we collect from customers. Other costs to obtain and fulfill contracts are generally expensed as incurred due to the short-term nature of most of our sales. We extend terms of payment to our customers based on commercially reasonable terms for the markets of our customers, while also considering their credit quality. Shipping and handling costs charged to customers are included in net sales.

Our payment terms with customers are customary and vary by customer but typically range from 30 to 60 days. We have evaluated the terms of our arrangements and determined that they do not contain significant financing components.

Inventories

Inventories consist of finished goods purchased from third-party suppliers. Inventories are stated at the lower of average cost or net realizable value. We have applied these inventory cost valuation methods consistently from year to year. Two suppliers provide substantially all of our finished goods.

We record inventory reserves for estimated obsolescence or unmarketable inventory equal to the difference between the cost of inventory and the estimated net realizable value based upon the age of specific inventory on hand and assumptions about future demand and market conditions. Our inventory has an initial five year usable life and we maintain the ability to extend the usable life of inventory for an additional five years. As a result, our analysis focuses on to what extent inventory on hand is in excess of market demand, which requires assessment of various factors that require the use of judgement including forecasted sales, continued growth, impact of competitors' products and ensuring we maintain adequate inventory on hand. If actual conditions differ from our assumptions, adjustments to the reserve may be required.

Inventories presented on the balance sheets are net of reserves for estimated obsolescence or unmarketable inventory which were $253,000 and $287,000 as of March 31, 2025 and December 31, 2024, respectively. Our inventories as of March 31, 2025 included $430,000 of inventories that, as of March 31, 2025, did not have material sales in the preceding twelve months. These inventories were primarily comprised of implants of less commonly used sizes that are within various implant families made available to surgeons at the time of surgery. These inventories are included in our analysis of our reserve for estimated obsolescence or unmarketable inventory. A portion of these inventories have been reserved for, and the remaining portion of these inventories are expected to be sold through our existing distribution network or in future markets.

Stock-Based Compensation

We measure all stock options based on their fair value on the date of the grant. Those awards typically have a graded vesting schedule and compensation expense for awards with only service conditions are recognized on a straight-line basis over the requisite service period, which is generally the vesting period of the respective award. We have not issued any stock-based awards with performance-based or market-based vesting conditions.

We use the Black-Scholes option pricing model, which incorporates assumptions and estimates, to measure the fair value of its option awards on the date of grant of each stock option award. We determined the assumptions for the Black-Scholes option pricing model as discussed below. Each of these inputs described below is subjective and generally requires significant judgment to determine. Forfeitures were accounted for as they occurred.

- *Fair Value of Our Common Stock*. Prior to this offering, our common stock was not publicly traded and therefore we estimated the fair value of our common stock, as discussed in the subsection titled "—Common Stock Valuation" below.
- *Expected Term*. The expected term represents the period that the stock-based awards are expected to be outstanding. As we do not have sufficient historical exercise data to provide a reasonable basis upon which to estimate the expected term, the expected term of stock options granted has been determined using the simplified method, which is the average of the midpoints between the vesting date and the contractual term for all vesting tranches.
- *Risk-Free Interest Rate*. The risk-free interest rate is based on the rate of the U.S. Treasury yield curve in effect at the time of grant of the award for time periods approximately equal to the expected term of the award.
- *Expected Volatility*. Because we do not have a trading history of our common stock, the expected volatility was derived from the average historical stock volatilities of several public companies within our industry that we consider to be comparable to our business over a period equivalent to the expected term of the stock-based award.
- *Expected Dividend Yield*. The expected dividend yield is zero as we have not paid and do not anticipate paying any dividends for the foreseeable future.

If any of the assumptions used in the Black-Scholes option pricing model change significantly, stock-based compensation for future awards may differ materially compared with the awards granted previously. See Note 10 to our financial statements included elsewhere in this prospectus for further details.

Common Stock Valuation

Given the absence of a public trading market to date, the fair value of our common stock has been determined by our board of directors at the time of each option grant, with input from management, considering contemporaneous independent third-party valuations of common stock, and our board of directors' assessment of additional objective and subjective factors that it believed were relevant and which may have changed from the date of the most recent valuation through the date of the grant, including: the timing and prices at which we sold shares of our convertible preferred stock to outside investors in arms-length transactions, and the superior rights, preferences, and privileges of our convertible preferred stock relative to our common stock at the time of each grant; our stage of development and commercialization; our business strategy; operating and financial performance; the lack of liquidity of our common stock; trends in the broader economy and industry; the likelihood of achieving a liquidity event for our securityholders, such as an initial public offering or a sale of the company, taking into consideration prevailing market conditions; and the hiring of key personnel and the experience of management.

These independent third-party valuations were performed in accordance with the guidance outlined in the American Institute of Certified Public Accountants' Auditing and Valuation Guide, *Valuation of Privately-Held-Company Equity Securities Issued as Compensation* (the "Guide"). The methodology to determine the fair value of our common stock included estimating the fair value of the enterprise using a market approach, which estimates the fair value of a company by including an estimation of the value of the business based on guideline public companies under a number of different scenarios. The Guide identifies various available methods for allocating enterprise value across classes and series of capital stock to determine the estimated fair value of common stock at each valuation date.

In accordance with the Guide, we considered the following methods:

- *Current Value Method*. Under the current value method, once the fair value of the enterprise is established, the value is allocated to the various series of preferred and common stock based on their respective seniority, liquidation preferences or conversion values, whichever is greatest.
- *Option Pricing Method ("OPM")*. Under the OPM, shares are valued by creating a series of call options with exercise prices based on the liquidation preferences and conversion terms of each equity class. The estimated fair values of the convertible preferred stock and common stock are inferred by analyzing these options. This method is appropriate to use when the range of possible future outcomes is so difficult to predict that estimates would be highly speculative, and dissolution or liquidation is not imminent.
- *Probability-Weighted Expected Return Method ("PWERM")*. The PWERM is a scenario-based analysis that estimates value per share based on the probability-weighted present value of expected future investment returns, considering each of the possible outcomes available to us, as well as the economic and control rights of each share class.

Based on our early stage of development and commercialization, frequency of financing transactions, the difficulty in predicting the range of specific outcomes (and their likelihood), and other relevant factors, we determined the OPM scenario was most appropriate for valuations of stock options awarded through October 2024.

Once a public trading market for our common stock has been established in connection with the consummation of this offering, it will no longer be necessary for our board of directors to estimate the fair value of our common stock in connection with our accounting for equity-based awards, as the fair value of our common stock will be determined based on the trading price of our common stock on the New York Stock Exchange.

The intrinsic value of all outstanding options as of March 31, 2025, was approximately $16.9 million, based on the initial public offering price of $15.00 per share of which approximately $10.3 million was related to vested options and approximately $6.6 million was related to unvested options.

Determination of Fair Value of Warrant Liabilities

Given the absence of a public trading market to date, the fair value of our common stock has been determined by our board of directors at the time of issuance of warrants, with input from management, considering contemporaneous independent third-party valuations of common and preferred stock, and our board of directors' assessment of additional objective and subjective factors that it believed were appropriate.

Our common stock and preferred stock warrants require liability classification and accounting. The warrants are recorded at fair value upon issuance and are subject to remeasurement to fair value at each balance sheet date, with any changes in fair value recognized in other (income) expense, net. The fair value was estimated using a Black-Scholes option pricing model. The valuation model used incorporates significant assumptions and estimates, which include, but are not limited to, the fair value per share of the underlying shares, the remaining contractual term of the warrants, risk-free interest rate and expected volatility of the price of the underlying shares.

Recently Issued Accounting Pronouncements

See Note 1 to our financial statements included elsewhere in this prospectus for a description of recent accounting pronouncements applicable to our financial statements.

Qualitative and Quantitative Disclosures About Market Risk

Interest Rate Risk

The primary risk associated with fluctuating interest rates is related to our debt. The term loans under the Trinity Loan Agreement bear interest at an annual rate equal to the greater of the prime rate plus 3.50% and 11.00%. Under the terms of the Trinity Loan Agreement, the prime rate is equal to the greater of 8.0% per year and the prime rate as reported in The Wall Street Journal. In addition, we hold cash and cash equivalents as well as marketable securities,

all of which may generate interest income. The primary objectives of our investment activities are to preserve principal and provide liquidity. Since our results of operations are not dependent on investments, we believe the risk associated with fluctuating interest rates is limited. We do not believe that a hypothetical 10% increase or decrease in interest rates during any of the periods presented would have had a material effect on our financial statements included elsewhere in this prospectus. We do not currently use or plan to use financial derivatives in our investment portfolio and we do not currently engage in hedging transactions to manage our exposure to interest rate risk.

Financial Institution Risk

Substantially all of our cash, cash equivalents and marketable securities are held with two financial institutions. Cash amounts held at financial institutions are insured by the Federal Deposit Insurance Corporation up to $250,000.

Inflation Risk

Inflation generally affects us by increasing our cost of labor. Inflationary and supply chain pressures may adversely impact our future financial results. Our operating costs have increased and may continue to increase because of these pressures, and we may not be able to fully offset these cost increases by raising prices for products or other mitigation efforts, which could result in downward pressure on margins.

Emerging Growth Company and Smaller Reporting Company Status

We are an emerging growth company, as defined in the JOBS Act. The JOBS Act permits an "emerging growth company" such as us to take advantage of an extended transition period for complying with new or revised accounting standards. This provision allows an emerging growth company to delay the adoption of some accounting standards until those standards would otherwise apply to private companies. We have elected to use the extended transition period for any other new or revised accounting standards during the period in which we remain an emerging growth company; however, we may adopt certain new or revised accounting standards early. As a result, we will not be subject to the same new or revised accounting standards as other public companies that are not emerging growth companies and our financial statements may not be comparable to other public companies that comply with new or revised accounting pronouncements as of public company effective dates. The JOBS Act also exempts us from having to provide an attestation and report from our independent registered public accounting firm on the assessment of our internal control over financial reporting pursuant to the Sarbanes-Oxley Act.

We will remain an emerging growth company until the earliest of: (i) the last day of the fiscal year following the fifth anniversary of the consummation of this offering; (ii) the last day of the fiscal year in which we have total annual gross revenue of at least $1.235 billion; (iii) the last day of the fiscal year in which we are deemed to be a "large accelerated filer" as defined in Rule 12b-2 under the Exchange Act, which would occur if the market value of our common stock held by non-affiliates exceeded $700.0 million as of the last business day of the second fiscal quarter of such year; or (iv) the date on which we have issued more than $1.0 billion in non-convertible debt securities during the prior three-year period.

We are also a "smaller reporting company" as defined in the Exchange Act. We may continue to be a smaller reporting company even after we are no longer an emerging growth company. We may take advantage of certain of the scaled disclosures available to smaller reporting companies and will be able to take advantage of these scaled disclosures for so long as our voting and non-voting common stock held by non-affiliates is less than $250.0 million measured on the last business day of our second fiscal quarter, or our annual revenue is less than $100.0 million during the most recently completed fiscal year and our voting and non-voting common stock held by non-affiliates is less than $700.0 million measured on the last business day of our second fiscal quarter. We cannot predict if investors will find our shares of common stock less attractive because we may rely on these exemptions. If some investors find our shares of common stock less attractive as a result, there may be a less active trading market for shares of our common stock and our share price may be more volatile.

BUSINESS

Overview

We are a commercial-stage medical technology company exclusively focused on transforming the shoulder surgical care market. We currently offer advanced implant systems for shoulder arthroplasty. These systems are a core element of our ecosystem, which we designed to improve core components of shoulder surgical care – preoperative planning, implant design and procedural efficiency – to benefit each stakeholder in the care chain. Our ecosystem is also comprised of enabling technologies, efficient instrument systems, specialized support and surgeon-to-surgeon collaboration. Together, these elements seek to address the long-standing clinical and operational challenges in the shoulder surgical care market by delivering predictable outcomes, procedural simplicity, and efficiency across all sites of care. We believe our exclusive focus on shoulder surgical care, combined with a highly specialized commercial organization and strong clinical data, positions us well to capture significant share in this large, growing market.

Shoulder pain is highly prevalent, often chronic, and can significantly reduce quality of life. The primary conditions that can result in shoulder pain and reduced functionality include osteoarthritis, rheumatoid arthritis, rotator cuff tears and shoulder fractures. Shoulder conditions are widespread, often debilitating, are commonly experienced concurrently as interrelated musculoskeletal disorders, and are estimated to result in more than eight million physician visits annually in the United States, based on data from the National Ambulatory Medical Care Survey 2015-2016 conducted by the National Center for Health Statistics of the Centers for Disease Control and Prevention ("CDC"). Despite this prevalence, we believe there has been a historical underutilization of surgical treatments for shoulder care due to several factors including patient hesitation to pursue surgical intervention, insufficient technology to appropriately treat shoulder conditions, complex shoulder anatomy, perceived difficulty of surgical intervention and barriers to patient access of care.

We believe the shoulder surgical care market today presents a significant market opportunity. Our initial focus within this broader market is on shoulder arthroplasty. Shoulder arthroplasty is an established surgical procedure involving the reconstruction of the shoulder joint with prosthetic implants through one of two main approaches: aTSA and rTSA. Both approaches can be performed in inpatient hospital settings and in outpatient settings, including ASCs. A key competitive advantage of ours has been the emergence of ASCs as a cost-efficient site of care with positive outcomes relative to hospital-based care. We expect that future growth in the shoulder surgical care market will be significantly driven by ASCs as hospitals face capacity constraints and are more limited in their ability to meet increasing demand.

We estimate that approximately 250,000 shoulder arthroplasty procedures will be performed in the United States in 2025, which we believe represents an approximately $1.7 billion market opportunity. Based on data from a healthcare market research provider regarding the estimated number of shoulder arthroplasty procedures performed, our internal estimates and analysis of such data, as well as our knowledge of our industry, we expect the number of annual procedures performed to grow by approximately 11% annually through 2029. We also believe a significant opportunity exists outside of the United States and, based on a market research report issued by Transparency Market Research in 2025, we estimate that the total annual international shoulder arthroplasty market is approximately $1.0 billion in 2025. Together, this represents a global annual shoulder arthroplasty market of approximately $2.8 billion. We believe we have an opportunity to expand to address adjacent areas in the shoulder surgical care market over time.

We believe traditional implants used in shoulder arthroplasty procedures are hindered by several limitations, including poor biomechanical fit, suboptimal kinematics, difficult replacement and conversion procedures (aTSA to rTSA or stemless to stem), imprecise implant positioning due to limited surgical planning, inefficient and burdensome workflow designs and non-specialized case support. These limitations can result in continued pain, lack of mobility, postoperative complications, low rates of implant survivorship, necessity of revision surgeries and costly and inefficient procedures for healthcare providers.

We developed our ecosystem with an approach to innovation that prioritizes ease of use, flexibility, predictability of outcomes and site of care efficiency, attributes we believe are critical to win in this market. Our ecosystem is comprised of the following elements:

- **Advanced Implant Systems:** Our advanced implants include a diverse range of interchangeable InSet anatomic and reverse total shoulder arthroplasty systems that leverage our novel, InSet Glenoid and InSet humeral stem technologies.
- **ProVoyance Preoperative Planning Technology:** This preoperative planning technology integrates AI and ML to transform planar CT imaging into 3D renderings, allowing surgeons to create bespoke surgical plans considering patient-specific anatomy ahead of surgery.
- **Efficient Instrument System:** Our efficient instrument system supports both aTSA and rTSA procedures from start to finish with just two convenient trays, considerably less than the six to nine trays typically required by other offerings.
- **Specialized Support:** Our team of dedicated shoulder specialists enables us to deliver a highly tailored experience to surgeons operating in a complex and technically demanding procedure category.
- **Surgeon-to-Surgeon Collaboration:** Our CEME team fosters a collaborative network of expert surgeon educators and promotes surgeon-to-surgeon training and peer education.

Our ecosystem offers notable benefits that differentiate it within the shoulder surgical care market. These benefits include:

- **Improved Fixation and Stability**: Our foundational InSet Glenoid technology is biomechanically designed to specifically address what we believe is the primary problem in legacy aTSA implants: glenoid loosening.
- **Restores Full Functionality**: We engineered our three humeral stem options to provide consistent, optimized biomechanics in both aTSA and rTSA.
- **Exceptional Longevity**: Our advanced implant systems are designed for longevity. In one published study, our InSet Glenoid demonstrated no surgical complications, cases of glenoid loosening or revision surgeries at a mean follow-up time of 8.7 years in a particularly challenging patient population.
- **Improved Implant Selection and Positioning**: ProVoyance preoperative planning technology empowers surgeons to create bespoke surgical plans that facilitate consistent, effective positioning of our implants.
- **Optimized Procedural Workflow and Efficiency**: Our efficient instrument system supports our implant portfolio across both aTSA and rTSA. This can provide significant workflow advantages for our customers by minimizing the operating room footprint, reducing procedural setup time, lowering sterilization requirements and costs and reducing the risk of errors - key advantages for the ASC setting.
- **Access to Expert Advice**: Our commercial organization creates deep relationships with surgeons and allows us to support their practice with specialized customer service and case support before, during and after surgery. In addition to direct support, our commercial team connects surgeons to the broader shoulder surgical community in various settings with a goal to improve connectivity across the shoulder surgical community and contribute to better patient outcomes.

We developed our comprehensive implant portfolio to address the unique needs of patients and surgeons. Our advanced implants are comprised of our aTSA and rTSA systems, which include various, specifically designed components capable of a wide array of system configurations to facilitate different modes of operation (anatomic or reverse) that are optimized for patient-specific needs. Our InSet Glenoid technology serves as the foundation for our advanced implant systems and includes a novel “InSet” design that aims to reduce mechanical stress at the bone implant interface, improve fixation mechanics, enhance stability and reduce micromotion. Our implant systems leverage consistent surgical techniques and the same efficient, two tray instrumentation system.

In addition to our advanced implant systems, we offer a leading preoperative surgical planning technology: ProVoyance. We believe that surgeon-level engagement in preoperative planning provides for better care for patients, and that bespoke surgical plans can help facilitate consistent positioning of implants. ProVoyance integrates AI and ML to transform planar CT imaging into 3D renderings of patient-specific anatomy ahead of surgery, and is cleared by the FDA for preoperative shoulder planning. ProVoyance received 510(k) clearance in 2021 and is classified by the FDA as a Class II device. ProVoyance is listed on the FDA's AI/ML-enabled medical devices list, which is a resource maintained, published, and periodically updated by the FDA to identify AI/ML-enabled devices that have been authorized for marketing in the United States through any of the standard paths to market for medical devices, although it is not intended to be a comprehensive list of all such devices that incorporate AI/ML. We believe the differentiation and value proposition of ProVoyance is validated by high utilization rates across procedures using our advanced implant systems. For example, for the three months ended June 30, 2025, we estimate an implied utilization rate for our ProVoyance technology of approximately 98%, based on 1,478 surgical plans created using ProVoyance technology and 1,503 implant systems sold during such period. This implied utilization rate is based on real-world data from our customers during such period, as ProVoyance technology tracks and reports each surgical plan that our customers create, and the actual number of implant systems sold during the respective period.

A key component of shoulder arthroplasty procedures are instrument trays, equipped with the specific instruments, supplies, and equipment needed for the surgery. We have developed a proprietary two-tray instrument system designed to enable interoperability between our aTSA and rTSA systems and a range of humeral stem options. We believe our efficient, two tray instrument system can enable surgeons and staff to reduce operating room footprint, procedural setup time, sterilization time and expense, and procedural complexity.

To best support our surgeon customers, we have built our commercial organization around their unique needs. Our commercial organization is comprised of a dedicated commercial leadership team that drives our internal commercial efforts with an exclusive focus on shoulder care, a CEME team that enhances surgeon engagement and training and a network of independent distributors. These three key components of our commercial organization work in tandem to form a commercial flywheel that is designed to build and reinforce relationships with surgeons and other stakeholders in the shoulder surgical care market, accelerate adoption, and enhance long-term retention.

We leverage our team's decades of experience developing and launching novel shoulder surgical care technologies to identify the unmet needs of patients and surgeons and develop solutions to address those unmet needs. With respect to our advanced implant systems, we commenced development efforts with our InSet Glenoid in 2009 and received 510(k) clearance in 2011. We commercially launched an initial aTSA system with our InSet Glenoid in 2016. Since this initial launch in 2016, we have successfully launched a wide range of new technologies to enhance our ecosystem and provide surgeons with the tools and support needed to deliver quality outcomes for patients requiring shoulder surgical care. For example, we commenced development efforts for our InSet PLUS Augmented Glenoid in 2019 and received 510(k) clearance in 2020. We commenced development efforts for our rTSA system in 2019 and received 510(k) clearance in 2021. We commenced development efforts for our short stem, stemless and I-Series humeral stem system options for our aTSA and rTSA systems in 2017, 2019 and 2021, respectively, and received our primary 510(k) clearances in 2018 and 2022, with an additional 510(k) clearance in 2024 for use of our primary I-Series humeral stem for use with anatomic fractures. Each of these devices is classified by the FDA as a Class II device. We have a robust pipeline of new technologies in various stages of development and evaluation, including the anticipated expansion of our humeral stem line, indication expansions into fracture and revision, and implants tailored for metal-sensitive patients. For example, we commenced development of InSet 70, InSet 135 and InSet 185 stems to expand our I-Series humeral stem line in 2024, and are currently pursuing FDA clearance of these stems. We are also evaluating expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets.

We have contributed to numerous publications that we believe evidence and strengthen our position as a leader in shoulder surgical care. There is a significant body of clinical evidence that supports the safety, efficacy, and durability of our implants in shoulder arthroplasty, including our InSet Glenoid technology. For example, a retrospective long-term follow-up analysis of patients who received our InSet Glenoid was published in the *Journal of Shoulder and Elbow Surgery* in 2019, which demonstrated a 72-point improvement in the mean ASES outcome score, statistically significant improvements in pain scores and range of motion, with no surgical complications, no

cases of glenoid loosening and no revision surgeries performed at a mean follow-up time of 8.7 years. We are committed to continued investment in obtaining further clinical evidence with the support of surgeons who are recognized as thought leaders in shoulder surgical care. We believe these efforts will continue to generate a substantial body of clinical evidence that will drive increased awareness and adoption of our products.

We have experienced significant growth in recent years, primarily driven by growth in our net revenue from the sale of our advanced implant systems. We generated net revenue of $31.6 million for the year ended December 31, 2024, compared to net revenue of $19.3 million for the year ended December 31, 2023, representing 64.0% year-over-year growth. We recognized a gross margin of 77.0% and net loss of $15.6 million for the year ended December 31, 2024, compared to a gross margin of 79.2% and net loss of $12.7 million for the year ended December 31, 2023. We generated net revenue of $10.1 million for the three months ended March 31, 2025, compared to net revenue of $7.2 million for the three months ended March 31, 2024, representing an increase of 41.0%. We recognized a gross margin of 76.9% and net loss of $4.7 million for the three months ended March 31, 2025, compared to a gross margin of 76.8% and net loss of $3.6 million for the three months ended March 31, 2024.

Our Success Factors

We attribute our success to a combination of the following factors. We believe these attributes are central to our business outcomes and will be significant factors in our continued success and growth.

Disruptive Ecosystem to Address Existing Limitations Within Shoulder Surgical Care

Our purpose-built ecosystem is designed to directly address several of the limitations associated with shoulder surgical care, starting with shoulder arthroplasty. This ecosystem is comprised of our advanced implant systems for aTSA and rTSA, ProVoyance, efficient instrument system, specialized support and surgeon-to-surgeon collaboration. Our advanced implant systems are designed to address existing challenges, such as glenoid loosening, subscapularis tendon failure, limited range of motion, stability and eventual implant failure. Glenoid loosening is a leading cause of revision shoulder arthroplasty procedures, with studies showing that approximately 30% of implants exhibit moderate to severe loosening within 6.6 years and up to 40% require revision within ten years. Unlike traditional glenoid designs, our InSet Glenoid sits within a rim of the native bone, which we believe enhances stability and reduces micromotion that contributes to loosening and potential failure. Complementing our InSet Glenoid technology, our humeral stem technology is fully compatible across both aTSA and rTSA procedures using consistent surgical techniques and instrumentation. This design facilitates interoperability and is intended to enable simplified revision and conversion procedures, which we believe provides surgeons flexibility, preserves bone mass and reduces the procedural burden associated with switching implant systems during a procedure. Surgeons leverage ProVoyance to create bespoke surgical plans that can help ensure consistent, effective positioning of our implants. Additionally, our design has allowed us to consolidate all of the instruments necessary to complete either an aTSA or a rTSA into two convenient trays, which can reduce operational complexity. We complement our offerings with specialist support and by facilitating surgeon-to-surgeon collaboration. We believe the elements of our ecosystem work together to help surgeons reduce operational complexity and surgical time, while supporting improved patient outcomes, lower revision risk and improved long-term satisfaction. We believe our differentiated design and approach positions us as a leader in addressing the critical clinical issues facing shoulder arthroplasty, as evidenced by our commercial traction to date. Since the initial launch of our InSet Glenoid in 2016, our implants have been utilized in more than 12,000 procedures.

Strong Clinical Results and Positive Outcomes for Patients and Surgeons

Our InSet Glenoid technology is supported by a compelling body of clinical evidence, with approximately nine years of published data and four clinical studies and articles published in the *Journal of Shoulder and Elbow Surgery* demonstrating its significant impact on patient outcomes. These studies demonstrate that, following surgery using our InSet Glenoid, patients have experienced greater functionality, lower pain and higher overall satisfaction than before surgery. Clinical data has also demonstrated strong performance in key indicators, such as forward flexion, rotational movement, range of motion, pain reduction and implant durability. Notably, a retrospective long-term follow-up analysis of patients who received our InSet Glenoid was published in the *Journal of Shoulder and Elbow Surgery* in 2019, which demonstrated a 72-point improvement in the mean ASES outcome score as compared to

before surgery, reflecting a meaningful improvement in function (range and strength) and patient well-being following surgery. In addition, no surgical complications, cases of glenoid loosening or revision surgeries were reported at a mean follow-up time of 8.7 years. In a finite element analysis published in the *Journal of Shoulder and Elbow Surgery* in 2012, our InSet Glenoid demonstrated an 87% reduction in "rocking horse" motion, which is a widely known key contributor to implant loosening and failure. We have also focused on optimizing rTSA biomechanics by aiming for a more anatomic feel and aesthetic, preserving native bone and replicating the anatomic humeral positioning, which can contribute to improved postoperative range of motion and function. We believe the biomechanics of our reverse offerings are an important differentiator of our technology. Our improved biomechanics are driven by our proprietary InSet design, which was engineered to create a more anatomic relationship between the glenoid and humeral stem to enhance range of motion. We believe the existing published data and patient outcomes, coupled with growing clinical validation and positive physician feedback, support the long-term adoption and utilization of our products within the shoulder surgical care market.

Well Positioned as Shoulder Surgical Care Market Grows in Outpatient Settings

Shoulder procedures can be performed in multiple care settings, including at hospitals, outpatient care centers and ASCs. Over the past three decades, there has been significant growth in outpatient care, driven by lower overall costs and increased procedural efficiency. We believe we are well positioned to succeed in these outpatient settings, particularly ASCs, because our purpose-built ecosystem can facilitate reproducible procedures with low complication rates, intraoperative flexibility, simplified workflows and efficient use of operating room space. We believe these attributes are particularly relevant in the ASC setting, where surgeons commonly face more limited resources (cost constraints and limited space for instrumentation) and financial risk if a surgery requires escalation. In addition to procedural and clinical reliability, ASCs generally prioritize or require cost-efficient, streamlined solutions. Our efficient instrument system supports both aTSA and rTSA procedures from start to finish with just two convenient trays, considerably less than the six to nine trays typically required by other offerings. This two tray instrument system can reduce operational complexity and lower costs by, for example, increasing room turnover and surgical throughput and minimizing storage and handling, ultimately helping ASCs to treat more patients with fewer resources. As a result of our positioning in the outpatient market, we have achieved significant growth in the ASC setting. The number of ASC-based procedures using our systems increased approximately 350% from 2023 to 2024, significantly outpacing the approximately 26% growth in ASC-based shoulder arthroplasty procedures in the United States over the same period. Approximately 30% of procedures performed with our implant systems during the month ended December 31, 2024 were performed in ASCs, as compared to approximately 10% during the month ended December 31, 2023. Additionally, as of June 30, 2025, approximately one third of our surgeons performed their first surgery with our implants in an ASC setting, underscoring our capabilities within this high-growth care setting and our alignment with the evolving site-of-care dynamics in the shoulder surgical care market.

Proven and Experienced Management Team

Our proven and experienced management team collectively has decades of experience in orthopedic product development and commercialization. Members of our leadership team have held senior and executive roles at some of the most recognized companies in the medical technology industry, including publicly traded companies, and have track records of delivering strong growth and results. We are an innovation-centric team, having played instrumental roles in the development and commercialization of foundational technologies in the shoulder space, including rTSA, stemless implants and advanced preoperative planning technologies. Our commitment to innovation is further evidenced by our combined 250+ orthopedic patents developed over the course of our management team's careers, reflecting deep clinical insight, technical expertise and a nuanced understanding of the orthopedic market. As a result, we are able to leverage deep clinical insight, technical expertise and an informed understanding as to the limitations of existing technologies in the development of next-generation solutions. In addition, through decades of thought leadership in the space, our team has established strong relationships with key opinion leaders and surgeons. We believe the history and experience of our management team position us to effectively execute on our strategic objectives.

Unique Commercial Organization Dedicated to the Shoulder Surgical Care Market

Shoulder surgeries are some of the most complex procedures in the orthopedic space due to the joint's complex range of motion, surrounding soft tissue structures and difficult surgical exposure. This complexity requires a high degree of surgical precision, clinical support, depth of expertise and specialization. We view ourselves as specialists serving specialists and are exclusively focused on shoulder care, which we believe represents a level of expertise that is essential for addressing the unique challenges of these procedures. Our commercial organization is comprised of a dedicated commercial leadership team that drives our internal commercial efforts, our CEME team, and our network of independent distributors. As of June 30, 2025, we had 34 shoulder specialists across our dedicated commercial leadership team and CEME team, representing more than 500 years of combined direct selling, physician engagement and clinical education experience. These specialists collaborate closely with our distributors to direct our sales activities and specialized support services, including hands-on support for surgeons—from procedure planning to execution—and help to facilitate surgeon training and performance management. In addition, our network of independent distributors allows for prospecting, relationship management and case coverage. As of June 30, 2025, our distributor network comprised of over 41 independent groups and 150 trained agents across the United States. These three key components of our commercial organization work in tandem to form a commercial flywheel that is designed to build and reinforce relationships with surgeons and other stakeholders in the shoulder surgical care market. We estimate that there are approximately 15,000 surgeons in the United States who perform at least one shoulder arthroplasty procedure per year, of which we target the approximately 1,800 high-volume surgeons who perform the vast majority of procedures. We believe these high-volume surgeons require dedicated shoulder expertise and support. Our entrenched relationships with many high-volume surgeons, together with engagement through highly tailored support, ongoing medical education and third-party symposiums, lab events, and surgeon-to-surgeon activities, all contribute to fuel our commercial strategy and reinforce our position as a leader in the shoulder surgical care market.

AI-Enabled Business and Clinical Intelligence Technologies

We leverage AI-enabled technology solutions to support our business and surgeons. At the center of our commercial strategy is our internal business intelligence platform, which draws insights from our proprietary database to help us target the right stakeholders in the shoulder surgical care market. We also offer similar tools to physician practices to help them identify and acquire new patients through proprietary search engine and funnel optimization strategies. By engaging with surgeons early in the care process and supporting patient acquisition, we become deeply integrated across the surgical workflow. In addition, our proprietary AI powered preoperative planning technology, ProVoyance, is designed to enable consistent, effective positioning of our advanced implant systems. ProVoyance is cleared by the FDA for preoperative shoulder planning, and is listed on the FDA's AI/ML-enabled medical devices list. We believe the benefits of our preoperative planning technology are evidenced by the fact that, for the three months ended June 30, 2025, we estimate an implied utilization rate for our ProVoyance technology of approximately 98% based on the number of surgical plans reported to have been created using ProVoyance technology and the number of our implant systems sold during such period. We also maintain a patient registry that allows practices to monitor outcomes and engage with patients postoperatively. Together, these tools are designed to provide information to enable surgeons to improve care delivery, optimize practice performance and achieve better outcomes. We believe these technologies play an important role in solidifying us as a differentiated, data-enabled leader in shoulder surgical care.

Our Growth Strategies

Our goal is to leverage our purpose-built ecosystem to become the leader for shoulder surgical care. The key elements of our growth strategy include:

Increase Awareness of our Purpose-Built, Innovative Shoulder Surgical Care Ecosystem to Continue Taking Share Across Care Settings

We plan to center our marketing efforts around educating surgeons about our ecosystem's advantages across all care settings, including hospitals, outpatient care centers and ASCs. Our ecosystem has been thoughtfully developed from the ground up to address the limitations associated with current offerings and help surgeons and patients access

consistent quality care regardless of where it is delivered. We believe that by focusing on increasing awareness of our holistic, innovation-driven approach, we can drive broader adoption of our ecosystem. We believe our ecosystem provides key advantages that surgeons seek in shoulder arthroplasty solutions, including improved fixation and stability, restoration of full functionality, exceptional longevity, improved implant selection and positioning, optimized procedural workflow and efficiency and access to expert advice. The output of our innovation-first and patient-centric approach has resonated with surgeons, as demonstrated by our 62% increase in procedure volume from 2023 to 2024. As the shoulder surgical care market continues to grow, we believe increased awareness of our ecosystem's advantages will position us to continue capturing market share from less specialized offerings.

Expand our Commercial Flywheel of Shoulder Specialists, Surgeon-to-Surgeon Collaboration, and Network of Independent Distributors

Our commercial flywheel includes our dedicated commercial leadership team, CEME team and network of independent distributors. We believe our dedication to shoulder surgical care is a strategic differentiator that enables us to deliver a highly tailored experience to surgeons operating in what is a complex and technically demanding procedure category. We intend to continue scaling our commercial organization with a continued specialized focus on shoulder care. By increasing the size of our dedicated commercial leadership team, we believe we can enhance targeting and prospecting efficiency, in an effort to ensure we engage the highest-value surgeon customers. Likewise, we believe growing our CEME team will allow us to increase the frequency and reach of our educational initiatives, such as symposiums, lab events, and surgeon-to-surgeon training programs. This increased presence will aim to continue to foster and grow our collaborative network of expert surgeon educators who can serve as powerful promoters of broader adoption through surgeon-to-surgeon training and peer education. Finally, by expanding our network of independent distributors, we will strive to ensure the quality of our customer service and execution remains up to our current standard as we scale. We believe our commercial flywheel and exclusive focus on shoulder care allow us to convert high-potential accounts and nurture new surgeons into loyal users of our ecosystem. Through expansion of this integrated and specialized commercial approach, we aim to accelerate growth, drive surgeon engagement, and expand adoption of our differentiated shoulder surgical care solutions in a cost-efficient manner.

Capitalize on Our Unique Advantages to Capture Outsized ASC Growth

We intend to expand our presence in the ASC setting through a multi-pronged approach that aims to capitalize on the outsized growth of ASCs as key sites for shoulder surgical care. We will drive targeted sales initiatives to increase awareness of our unique advantages in the ASC setting, including the potential efficiency and economic benefits of our ecosystem. We will also leverage surgeon-to-surgeon collaboration to reinforce these advantages with real-world experiences and will continue to generate clinical and economic validation of our ecosystem's benefits in the ASC. In addition, we intend to continue innovating our systems with the ASC setting in mind. As ASC procedure volume grows, we believe we will be able to drive meaningful growth and establish ourselves as the leader for shoulder arthroplasty in the ASC setting.

Increase our Addressable Market Through our Commitment to Continuous Innovation and Advancing Shoulder Surgical Care

We have a strong track record of innovation and product commercialization, having successfully launched a wide range of new technologies to address unmet needs in the shoulder surgical care market. We believe this track record demonstrates our ability to identify clinical needs and bring differentiated solutions to market. We plan to continue investing in our robust product pipeline to expand our capabilities and service additional segments of the shoulder surgical care market. Our near-term development efforts in shoulder arthroplasty include a fracture-specific system, a revision solution, and implants tailored for metal-sensitive patients. Additionally, we intend to build on our existing aTSA and rTSA systems by introducing product line extensions that support continuous improvement and are assessing the implementation of robotic-assisted technology for our implant systems. We are also evaluating expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets. By leveraging our innovation capabilities and deep customer relationships, we believe we can expand our

addressable market and generate incremental sources of revenue through targeted product development. We may also evaluate entering other markets through business development opportunities.

Continue Building and Driving Marketing of our Technology Solutions

We view digital enablement as an essential component of modern shoulder surgical care and believe that our technologies can drive greater clinical value and deepen physician engagement. We see four key aspects to the growth opportunity for our technologies. First, we can continue to leverage our internal business intelligence platform to help us efficiently target high-volume surgeon prospects. Second, we can strategically deploy our enabling technologies to connect patients to those physicians. Third, ProVoyance preoperative planning technology can increase surgeon loyalty by helping to improve workflows and drive positive clinical outcomes. Fourth, we can offer surgeons a platform for outcomes measurement to assess performance. We expect our underlying algorithms and predictive capabilities to improve as we drive increased adoption of our technologies and can leverage additional data inputs, which we believe will reinforce a virtuous cycle of increased accuracy, better outcomes and greater reliance on our tools. Additionally, we intend to continue investing in and expanding our technology capabilities and we are actively developing other complementary enabling technologies to support the full continuum of care in shoulder surgery. We believe these innovations will further establish our position as a comprehensive, technology-driven solution provider within the shoulder surgical care market.

Pursue Expansion in International Markets

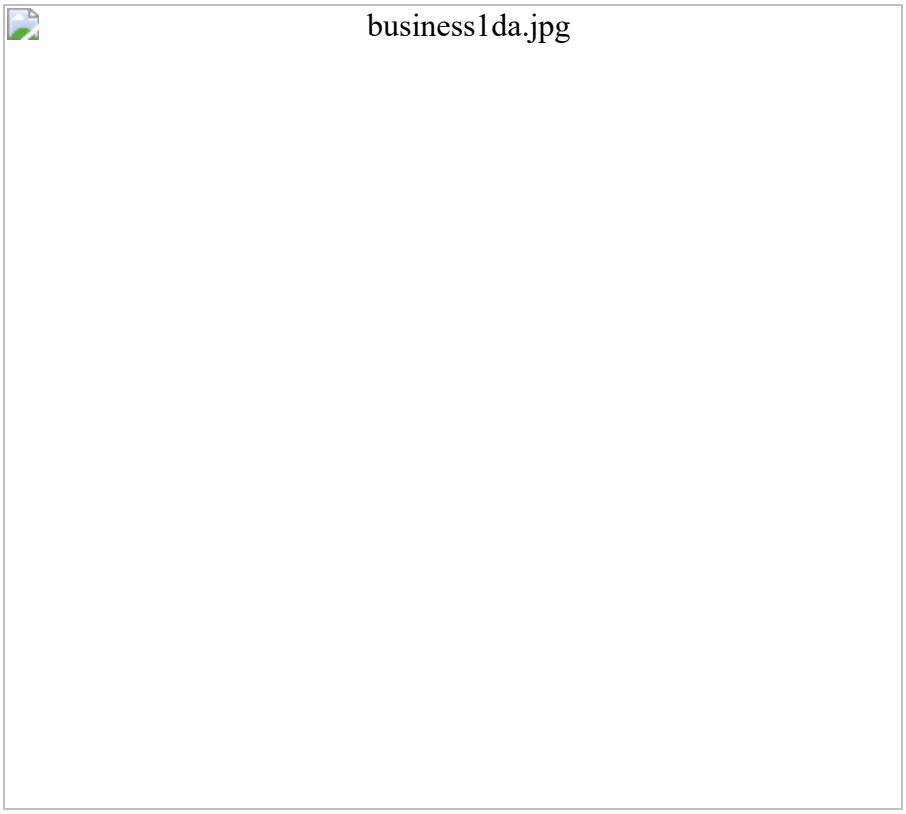
While our primary commercial focus to date has been within the United States, we believe the international shoulder surgical care market represents a compelling opportunity for long-term growth. Based on a market research report issued by Transparency Market Research, we estimate the annual international shoulder arthroplasty market opportunity outside the United States to be approximately $1.0 billion, driven by many of the same factors we see domestically. We plan to strategically pursue entry into certain international markets over time.

Market Overview

Overview of Shoulder Anatomy

The shoulder is a distinct anatomical structure comprised of bones, muscles, tendons, and ligaments that work in tandem to allow for a complex range of motion for the arm. The key bones of the shoulder include the humerus (upper arm bone), scapula (shoulder blade), and clavicle (collarbone). The humerus articulates with the glenoid fossa of the scapula to form the glenohumeral joint, while the acromion (scapular projection) and clavicle form the acromioclavicular joint. The rotator cuff includes four muscles and their tendons (infraspinatus, subscapularis, supraspinatus, and teres minor) that serve to stabilize the head of the humerus in the glenoid.

The illustration below depicts the key anatomy of the shoulder (anterior view):



Overview of Conditions Impacting the Shoulder

Shoulder pain is highly prevalent and can significantly impact and reduce quality of life as well as result in chronic pain. According to a study published in the *Journal of Shoulder and Elbow Surgery*, an estimated 38% of patients with shoulder pain reported an inability to perform activities of daily life. Furthermore, a study published in the *American Journal of Sports Medicine* estimated 89% of patients with rotator cuff tears had difficulty sleeping. Today approximately one in five adults in the United States over the age of 65 experience shoulder pain.

The primary conditions that can result in shoulder pain and reduced functionality include:

- **Osteoarthritis ("OA")**: OA is a degenerative joint disease characterized by the breakdown of cartilage and underlying bone, leading to pain, stiffness, and reduced mobility in the shoulder's glenohumeral joint. This condition results from wear-and-tear, aging, or prior injury, progressively worsening over time. OA is highly prevalent, affecting approximately one in three adults in the United States over the age of 60.

- **Rheumatoid Arthritis ("RA")**: RA is a chronic autoimmune disease characterized by synovial inflammation as well as bone and cartilage destruction and deformity. RA typically impacts the large joints more than peripheral joints in later stages of the disease leading to progressive damage in the shoulder. According to the Centers for Disease Control and Prevention, RA impacts over 1.5 million adults in the United States, with various studies estimating incidence in the shoulder as high as 90%.

- **Shoulder Fractures**: Acute and traumatic injuries may result in severe fractures of the humerus or other adjacent bones in the shoulder. Such incidents are most prevalent among elderly patients and those with chronic conditions such as OA.

- **Rotator Cuff Tears**: Rotator cuff tears refer to partial or full-thickness disruptions of the tendons stabilizing the shoulder joint. Rotator cuff tears can be caused by acute trauma, repetitive stress, or

degeneration. Rotator cuff tears are highly prevalent and impact nearly 25% of adults over the age of 40 in the United States.

Shoulder conditions are widespread, often debilitating, are commonly experienced concurrently as interrelated musculoskeletal disorders, and are estimated to result in more than eight million physician visits annually in the United States, based on data from the CDC. Despite this prevalence, we believe there has been a historical underutilization of surgical treatments for shoulder care due to several factors including patient hesitation to pursue surgical intervention, insufficient technology to appropriately treat shoulder conditions, perceived difficulty of surgical intervention, and barriers to patient access of care.

Our Addressable Market Opportunity in Shoulder Surgical Care

We believe the shoulder surgical care market today presents a significant market opportunity. Our initial focus within this broader market is on shoulder arthroplasty, and we believe we have an opportunity to expand to address adjacent areas in the shoulder surgical care market over time.

We currently serve the large, immediately addressable and rapidly growing market for shoulder arthroplasty procedures. We estimate that approximately 250,000 shoulder arthroplasty procedures will be performed in the United States in 2025, which we believe represents an approximately $1.7 billion market opportunity based on our average sales price. Based on data from a healthcare market research provider regarding the number of shoulder arthroplasty procedures performed, our internal estimates and analysis of such data, as well as our knowledge of our industry, we expect the number of annual procedures performed to grow by approximately 11% annually through 2029. While our current commercial focus is on the United States, we plan to pursue market access initiatives in other attractive, high-growth international markets. We believe a significant opportunity exists outside of the United States and, based on a market research report issued by Transparency Market Research, we estimate that the total international shoulder arthroplasty market is approximately $1.0 billion in 2025. Together, we believe these markets represent a global annual addressable market of approximately $2.8 billion. These market opportunities represent the total overall revenue opportunity that we believe is available for our systems if 100% market share is achieved by us, and are not a representation that we will achieve any such market share. The market share we achieve is subject to a number of assumptions, risks and uncertainties, including the adoption of our implant systems and our ability to execute on our commercial strategy. For more information, please see the section titled "Risk Factors—Risks Related to Our Business and Industry—Our business plan relies on certain assumptions about the market for our implant systems, however, the size and expected growth of our addressable market has not been established with precision and may be smaller than we estimate, and even if the addressable market is as large as we have estimated, we may not be able to capture additional market share."

The number of shoulder arthroplasty procedures in the United States grew at approximately 10% per year from 2019 to 2024, reflecting one of the fastest-growing segments within orthopedic reconstruction since other sectors are growing in low single digits. This growth was primarily driven by multiple, ongoing demographic factors and industry tailwinds. For example, the population in the United States is both aging and remaining more active later in life, which has contributed to a higher incidence of degenerative shoulder conditions. Additionally, greater awareness of treatment options, earlier diagnosis and broader acceptance of shoulder arthroplasty as an interventional solution have expanded the pool of potential patients. In parallel, there has been continued growth of outpatient and ASC settings, which has further increased patient access to care and care settings where surgeons can perform these procedures.

We believe the growth in shoulder arthroplasty procedures will continue to be driven by existing tailwinds. Further, we believe the number of patients who may potentially benefit from shoulder arthroplasty is significantly higher than the number of patients who currently seek treatment today. Over time, as access to care increases and clinical outcomes improve as a result of better technology, we believe the proportion of patients seeking shoulder arthroplasty will eventually approach that of the knee arthroplasty market.

Overview of Shoulder Arthroplasty

Shoulder arthroplasty is an established surgical procedure involving a reconstruction of the shoulder joint with prosthetic implants. Arthroplasty involves the resection of surfaces around the glenoid fossa and humeral head

followed by the implantation of prostheses that articulate with each other. Typically, an arthroplasty procedure includes the implantation of two devices, one “glenoid component” which attaches to the scapula and one “humeral component” which attaches to the humerus. Overall success of these procedures is measured by reduced pain, improved function, fewer post-operative complications, and increased survivorship of the implant.

In all shoulder arthroplasty procedures, appropriate biomechanical fit of the protheses is critical to ensure precise alignment with the patient’s natural shoulder anatomy to optimize stability, range of motion, and long-term functionality. Appropriate device type, size, positioning and fixation are all essential to ensuring biomechanical fit and procedure success. ASES scores, which are derived from a postoperative questionnaire used by surgeons to assess shoulder function and patient well-being following surgeries, are commonly used to measure operative success. Today, there are two approaches to shoulder arthroplasty, aTSA and rTSA:

Anatomic Total Shoulder Arthroplasty (aTSA) — aTSA imitates the natural joint anatomy by using prosthetic implant components to replace the humeral head with a prosthetic ball and stem and the glenoid with a prosthetic cup. aTSA is most often used in OA and RA patients with arthritic shoulder joints and intact rotator cuffs, using the rotator cuff to stabilize the joint and dictate motion. While aTSA procedures may reduce pain and restore function, complications such as subscapularis tendon failure and glenoid loosening have been relatively common with aTSA procedures.

Reverse Total Shoulder Arthroplasty (rTSA) — rTSA inverts the shoulder anatomy and joint configuration, reversing the ball and socket positions. rTSA transforms the humeral head into a socket and the scapula into a ball, shifting the reliance from the rotator cuff to the deltoid muscle for stability and motion. rTSA addresses a broader range of conditions, including rotator cuff arthropathy, irreparable rotator cuff tears, glenoid bone loss, acute fractures, and post-traumatic reconstruction. The development of the rTSA procedure has allowed shoulder arthroplasty to be used to address additional shoulder issues, including rotator cuff damage. While rTSA procedures have corrected some of the limitations of aTSA, historically there has been a tradeoff with patients often sacrificing shoulder motion and movement patterns in rTSA.

Anatomical

Replicates the normal mechanics and anatomy of the shoulder, replacing the ball and socket as they are normally found in the body with prosthetic implants.

business2ba.jpg

Reverse

“Reverses” the ball and socket placement, transforming the humeral head into a socket and the scapula into a ball.

business3ba.jpg

Market Dynamics in Shoulder Arthroplasty

Shoulder arthroplasty can be performed both as an inpatient hospital procedure and in outpatient settings, including ASCs. In recent years, as the volume of shoulder arthroplasty procedures has grown, the majority of the incremental procedures have been performed at ASCs. This trend is similar to what has occurred in the hip and knee joint replacement markets and tracks the broader long-term growth of surgical procedures shifting to outpatient settings. ASCs have emerged as a low-cost site of care and positive post-care outcomes. We expect that future growth in the market will be significantly driven by ASCs as hospitals face capacity constraints and are more limited in their ability to meet increasing demand. In 2024, CMS added total shoulder arthroplasty to the list of covered procedures that can be performed in ASCs, which facilitated reimbursement and further supported the growth of

ASCs as a key site of care in the shoulder surgical care market. This growth within ASCs also presents other benefits for key stakeholders, including streamlined workflows, scheduling flexibility and operational autonomy. Further, surgeons often have economic ownership of the ASCs where they operate, with as many as 80% of ASCs owned at least in part by operating surgeons according to a study published in *Global Spine Journal*, which we believe further incentivizes surgeons to utilize ASCs for shoulder arthroplasty procedures.

Shoulder arthroplasty volumes in the United States experienced a significant increase driven by clear and meaningful growth of the rTSA procedure. First approved in the United States in 2004, rTSA has expanded the market and now accounts for a significant majority of shoulder arthroplasty procedures due to their broad treatment applications, surgeon preference relative to aTSA, and advantages in clinical outcomes, such as reduced likelihood of revision surgeries. In 2020, approximately 70% of all shoulder arthroplasty procedures were rTSA procedures.

Limitations of Existing Product Offerings in Shoulder Arthroplasty

Despite their frequent use and wide adoption, we believe traditional implants used in shoulder arthroplasty procedures are hindered by several limitations, including poor biomechanical fit, suboptimal kinematics, difficult replacement and conversion procedures (aTSA to rTSA or stemless to stem), imprecise implant positioning due to limited surgical planning, inefficient and burdensome workflow designs and non-specialized case support. These shortcomings can impact both patients and surgeons and result in continued pain or discomfort, lack of mobility, postoperative complications, low rates of implant survivorship, necessity of revision surgeries and costly and inefficient procedures for healthcare providers.

We believe traditional shoulder arthroplasty products present several limitations, including:

- **Fixation of the Glenoid Component in aTSA**: Due to poor fixation of the glenoid component in aTSA, normal movement from the humerus can also rock the glenoid loose, commonly referred to as the “rocking horse” effect, creating instability within the implant. The illustration below depicts the “rocking horse” effect resulting from an implant with poor fixation:

- **Poor Product Design Resulting in Suboptimal Kinematics in rTSA:** Traditional implants in rTSA utilize an onlay design, placed above the anatomic neck of the humerus on top of a bony surface, without directly penetrating the bone. This design often results in overstuffing whereby the implant excessively fills the joint space and alters the shoulder’s natural anatomy. This can increase pressure on surrounding tissues, restrict motion, and result in pain and discomfort. Furthermore, competitor implants that utilize an inlay design inserted directly into the bone often result in poor kinematics, restricting shoulder rotation and movement. These traditional implants may also modify the shoulder anatomy and move the humerus further away from the scapula where it would naturally sit, resulting in poor aesthetic results where the shoulder does not regain its natural form post-procedure.

- **Difficult to Replace and Convert:** Traditional implant designs place the implant stem deep into the humerus, requiring the removal of the stem from the humerus for replacement and revision surgeries and for conversions to rTSAs. This design results in a high degree of bone removal and potential damage to the surrounding soft tissue region during such procedures, and significantly limits and complicates intraoperative flexibility and the ability of surgeons to switch from aTSA to rTSA. aTSA procedures are increasingly using a stemless humeral implant as a bone-preserving alternative. These implants, however, are often not easily convertible to stemmed implants or in rTSA procedures, restricting their utility.

- **Imprecise Implant Positioning Due to Limited Surgical Planning:** Shoulder arthroplasty procedures require precision and accuracy and clinical evidence demonstrates that positioning as little as five degrees off angle can lead to inferior patient outcomes. Surgeons have historically relied on basic imagery to inform their surgical approach, using tools that often lack 3D bone rendering or biomechanical simulation that do not fully capture patient-specific anatomy and properly simulate postoperative range of motion and joint stability. These existing market offerings have resulted in limited surgeon engagement as well as the outsourcing of imaging interpretation components and planning that surgeons might otherwise seek to perform themselves.

- **Inefficient and Burdensome Surgical Workflow Design:** Traditional implant systems have typically been developed and expanded product by product, indication by indication, over long periods of time. The complexity of these systems and the related equipment, including multiple trays of instruments, generally requires a wide range of surgical techniques and can considerably complicate surgical workflows across care settings. This legacy approach has led to an outsized amount of equipment in the operating room, leading to logistical challenges including an increased capital footprint, and extended procedure and operating room turnover time, thereby creating unnecessary costs to healthcare providers.

- **Non-Specialized Case Support:** Shoulder arthroplasty is a highly complex and technical procedure. Operating room staff and surgeons routinely rely on the product knowledge and know-how of orthopedic salespeople. Legacy shoulder implants are predominantly sold by non-specialized salespeople who also sell devices in other segments of orthopedics, such as hip and knee implants. The lack of specific product and technical knowledge in shoulder arthroplasty can result in suboptimal case support and patient outcomes.

These limitations of legacy shoulder arthroplasty products can result in the following shortcomings:

- **Failure to Reduce Pain and Improve Shoulder Function:** The key objectives of undertaking shoulder arthroplasty surgery – reduction in pain and increased shoulder functionality – are often not achieved post-surgery. For example, studies have shown that approximately 20% of shoulder arthroplasty patients remain in pain or discomfort beyond one to two years following surgery.

- **Frequent Post-Operative Complications:** As a result of poor fixation and suboptimal kinematics, traditional implants can lead to several complications including overstuffing, glenoid loosening, shoulder subluxation (dislocation), humeral fractures, and subscapularis tendon failure (a condition where the tendon connecting the subscapularis muscle to the humeral bone is damaged, which in one study published in the *Journal of Shoulder and Elbow Surgery* was demonstrated to occur in approximately 15% of aTSA procedures). These complications may result in further pain, reduced function for patients, and require additional intervention.

- **Low Rates of Implant Survivorship:** Glenoid implant loosening is frequent and common. According to a study published in the *Journal of Shoulder and Elbow Surgery*, approximately 30% of aTSA procedures demonstrated moderate to severe loosening at mean follow-up time of 6.6 years. The resulting pain and inflammation from excessive and continued loosening over time may require additional intervention. In a separate study of aTSA revision surgeries published in the same journal, approximately 70% of glenoid failures that resulted in revision surgery demonstrated glenoid loosening. In this study, approximately 20% demonstrated glenoid loosening within 5 years post procedure and approximately 25% demonstrated precursors to loosening within 5 years post procedure.

- **Necessity of Revision Surgeries:** Poor clinical outcomes and complications significantly increase the likelihood for a second surgery, often requiring a revision surgery to correct or replace a failed implant. Revision surgeries are common with studies showing up to 40% of glenoid implants were subject to revision surgery at 10-year follow-up. These repeat surgeries compromise patients' health and quality of life and are an unnecessary burden and avoidable cost to the healthcare system, with clinical evidence demonstrating that complication rates in revision surgeries are significantly higher. According to a study published in the *Journal of Shoulder and Elbow Surgery*, overall complication rates were 69% in revision rTSA, significantly higher than the 25% complication rate for initial rTSA procedures.

- **Costly and Inefficient for Healthcare Providers:** Current instrumentation processes require up to nine trays for a single shoulder arthroplasty procedure. Each tray requires significant storage space and must be sterilized before and after every use resulting in higher costs and burdening staff. This inefficient workflow has an outsized impact on outpatient care centers and ASCs given the inherent resource and space constraints in these settings. We believe that requiring more time and space in the operating room for maintaining surgery trays and using large quantities of unfamiliar equipment increase the risk of infection and impact patient safety.

Additional Opportunities

We plan to continue investing in our robust product pipeline to expand our capabilities in shoulder arthroplasty and serve additional, adjacent segments of the shoulder surgical care market that we do not currently address. Our near-term development efforts include a revision solution to address the $85 million annual shoulder revision market in the United States (which, based on our estimates, represents approximately 5% of the estimated $1.7 billion annual shoulder arthroplasty market in the United States). We also are developing a fracture-specific system to address the $28.9 million annual shoulder fracture market in the United States. Fracture fixation of the shoulder is a surgical intervention designed to stabilize and repair broken bones within the shoulder employing internal fixation techniques to secure fractured segments and ensure proper healing and functional recovery. We are also developing a line of humeral head and glenoid technologies for the approximately 10% to 15% of the general population who test positively for a metal hypersensitivity and may experience persistent pain, or other symptoms associated with allergic reactions from metal implants. In addition to these near-term development efforts, we are also evaluating expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets, and are assessing the implementation of robotic-assisted technology for our implant systems.

Our Solutions

Our Approach to Innovation

We are building Shoulder Innovations with the goal of addressing some of the most pressing challenges to patient outcomes in shoulder surgical care: poor biomechanical fit, suboptimal kinematics, difficult replacement and conversion procedures (aTSA to rTSA or stemless to stem), imprecise implant positioning, inefficient and burdensome workflow designs, and limitations of a generalist approach. Our innovation-first mentality centers on the patient, realizing that each stakeholder in the care chain is motivated by and benefits from focused and improved patient care. We seek to improve core components of shoulder surgical care — preoperative planning, implant design and procedural efficiency — in an effort to benefit each stakeholder in the care chain. We believe our approach has positioned us to drive long-term success in the shoulder surgical care market.

We believe that the following attributes are critical to win in this market:

- **Ease of Use:** We are simplifying the shoulder arthroplasty procedure to help surgeons achieve reproducible results with confidence and ease.

- **Flexibility:** We are developing implant systems to enable surgeons to perform both aTSA and rTSA procedures with consistent surgical techniques and instrumentation, with interchangeable systems.

- **Predictable Outcomes:** We are leveraging a leading AI-enabled technology to analyze patient anatomy and preoperatively plan surgeries that, together with thoughtful implant design, can lead to improved implant positioning and outcomes.

- **Site of Care Efficiency:** We are engineering our offerings to optimize workflow efficiency across care settings.

The image below depicts our approach to innovation and key attributes for success in the shoulder surgical care market:

business6ca.jpg

Key Elements of Our Ecosystem

Our ecosystem is comprised of the following key elements:

- **Advanced Implant Systems:** Our advanced implants include a diverse range of interchangeable InSet aTSA and rTSA systems that leverage our novel, InSet Glenoid and InSet humeral stem technologies.
- **ProVoyance Preoperative Planning Technology:** This preoperative planning technology integrates AI and ML to transform planar CT imaging into 3D renderings, allowing surgeons to create bespoke surgical plans considering patient-specific anatomy ahead of surgery.
- **Efficient Instrument System:** Our efficient instrument system supports both aTSA and rTSA procedures from start to finish with just two convenient trays.
- **Specialized Support:** Our team of dedicated shoulder specialists enable us to deliver a highly tailored experience to surgeons operating in a complex and technically demanding procedure category.
- **Surgeon-to-Surgeon Collaboration:** Our CEME team fosters a collaborative network of expert surgeon educators and promotes surgeon-to-surgeon training and peer education.

The image below depicts the key elements of our ecosystem:

Key Benefits of Our Ecosystem

Our ecosystem offers notable benefits that differentiate it within the shoulder surgical care market. These key benefits include**:**

- **Improved Fixation and Stability:** Our foundational InSet Glenoid technology is biomechanically designed to specifically address what we believe is the primary problem in legacy aTSA implants: glenoid loosening. Our design aims to reduce mechanical stress at the bone-implant interface, improve fixation mechanics, enhance stability and reduce micromotion. These biomechanical advantages are evidenced by a published finite element analysis in which our InSet Glenoid technology demonstrated an 87% reduction in "rocking horse" motion, which is a widely known key contributor to implant loosening and failure. Notably, a separate published retrospective long-term follow-up analysis of patients who received our InSet Glenoid demonstrated a 72-point improvement in the mean ASES outcome score as compared to before surgery, reflecting a meaningful improvement in function (range and strength) and patient well-being following surgery.

- **Restores Full Functionality:** We engineered our three humeral stem options to provide consistent, optimized biomechanics in both aTSA and rTSA. Using our InSet approach for rTSA, our system is designed with the goal of enabling patients to regain full range of motion, including the ability to raise their arm and reach behind (*i.e.*, forward elevation and internal rotation), and avoid lengthening and overstuffing. We believe our design supports improved postoperative functionality and results in a postoperative feel and aesthetic that more closely tracks the shoulder's natural form.

- **Exceptional Longevity:** Our advanced implant systems are designed for longevity, with minimal need for replacement. A published retrospective long-term follow-up analysis of patients who received our InSet Glenoid demonstrated no surgical complications, cases of glenoid loosening or revision surgeries at a mean follow-up time of 8.7 years in a particularly challenging patient population. We believe this durability provides surgeons and patients with confidence in our long-term implant performance and clinical outcomes.

- **Improved Implant Selection and Positioning:** ProVoyance preoperative planning technology empowers surgeons to create bespoke surgical plans that facilitate consistent, effective positioning of our implants.

The highly intuitive interface enables surgeon planning without the need for third party intervention and we believe that surgeon-level engagement in preoperative planning provides for better care for patients.

- **Optimized Procedural Workflow and Efficiency:** Our efficient instrument system supports our implant portfolio across both aTSA and rTSA. This can provide significant workflow advantages for our customers by minimizing the operating room footprint, reducing procedural setup time, lowering sterilization requirements and costs and reducing the risk of errors. This streamlined approach is enabled by leveraging our InSet Glenoid and humeral stem technologies across our aTSA and rTSA systems, minimizing the number of sets and instruments required. Our advanced implant systems leverage consistent surgical techniques, which we believe can improve procedural outcomes while allowing for intraoperative flexibility and seamless interchangeability between aTSA and rTSA procedures.

- **Access to Expert Advice:** Our commercial organization creates deep relationships with surgeons and allows us to support their practice with specialized customer service and case support before, during and after surgery. In addition to direct support, our commercial team connects surgeons to the broader shoulder surgical community in various settings with a goal to improve connectivity across the shoulder surgical community and contribute to better patient outcomes.

Our Implants Systems

We developed a comprehensive implant portfolio to address the unique needs of patients and surgeons. Our advanced implants are comprised of our aTSA and rTSA systems, which include various, specifically designed components capable of a wide array of system configurations to facilitate different modes of operation (anatomic or reverse) that are optimized for patient-specific needs. These systems leverage consistent surgical techniques and the same efficient, two tray instrumentation system. The design of our advanced implant systems helps to facilitate shoulder surgical care with significant operational flexibility and reduced equipment and operating room footprint requirements.

InSet aTSA System

Our aTSA configuration consists of two main components, a glenoid scapular implant and a humeral fixation device, or stem. Each of these components was developed with a view to the overall implant system in an effort to provide maximum benefits to patients and flexibility for surgeons. A retrospective long-term follow-up analysis of patients who underwent aTSA procedures with our InSet Glenoid was published in the *Journal of Shoulder and Elbow Surgery* in 2019, which demonstrated a 72-point improvement in the mean ASES outcome score, statistically significant improvements in pain scores and range of motion, with no surgical complications, no cases of glenoid loosening and no revision surgeries performed at a mean follow-up time of 8.7 years.

InSet Glenoid

Unlike traditional glenoid implants that utilize an onlay design, our InSet Glenoid technology features a design that positions the implant within a pocket of sclerotic or cortical bone and aims to reduce mechanical stress at the bone-implant interface, improve fixation mechanics, enhance stability and reduce micromotion, all of which are key contributors to glenoid loosening. This design also enables a less invasive surgical approach that reduces surgical steps and difficulty of tissue exposure. Our InSet Glenoid implants were developed with a patented complex articular surface that contributes to reduction of the rocking horse motion. The InSet Glenoid design also enables a significantly reduced central fixation peg to provide further surgical flexibility.

The InSet PLUS Augmented Glenoid was designed to treat advanced shoulder conditions in patients with severe bone loss. The design of the InSet PLUS Augmented Glenoid corrects defects on the articular surface (in contrast to the fixation side of the implant) with five or ten-degree options. This method allows surgeons to easily adjust the

implant's position for optimal patient fit and stability through this dialable and articular side augmentation. The image below depicts the implant characteristics of our InSet Glenoid:

Supported by a growing body of peer-reviewed studies and approximately nine years of published data, our InSet Glenoid technology has demonstrated effectiveness across a variety of clinical indications, including cases involving advanced arthritis and severe bone loss. In addition to preserving bone in patients, our InSet Glenoid technology has been shown to facilitate quick recovery times, decreased pain, and improved range of motion for patients after surgery.

We commenced development efforts with our InSet Glenoid in 2009 and received 510(k) clearance in 2011. We commercially launched an initial aTSA system with our InSet Glenoid in 2016. We commenced development efforts for our InSet PLUS Augmented Glenoid in 2019 and received 510(k) clearance in 2020. Each of these devices is classified by the FDA as a Class II device.

InSet Humeral Stems

We offer three unique InSet humeral stem system options for our aTSA and rTSA systems—Humeral Stemless, Humeral Short Stem, and I-Series Humeral Stem—each purpose-built to address the specific needs of patients and physicians. Our humeral stems feature a proprietary porous coating designed to promote bone ingrowth by allowing bone tissue to grow into the porous structure (osseointegration), which creates a strong natural biological fixation and improves implant stability. Our stems also feature what we believe is an industry-only 132.5 degree neck shaft angle, a feature that enables our simplified surgical technique focused on proximal geometry—the shape, size, and

position of the proximal humerus (upper arm bone) and its relationship to the glenoid (shoulder blade). The image below depicts our primary current humeral stems:

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We commenced development efforts for our short stem, stemless and I-Series humeral stem system options for our aTSA and rTSA systems in 2017, 2019 and 2021, respectively, and received our primary 510(k) clearances in 2018 and 2022, with an additional 510(k) clearance in 2024 for use of the InSet 95 stem for use with anatomic fractures. Each of these devices is classified by the FDA as a Class II device.

Our novel humeral stem solutions are differentiated and designed to be compatible across the full range of our implant systems. Surgery with each humeral stem involves a consistent surgical technique and instruments, which enables seamless interchangeability between our aTSA and rTSA systems. This interchangeability is important in cases where the surgeon makes an intraoperative decision to switch from a stemless to a stemmed implant, or in revision or conversion surgeries where the determination is made to switch to a different implant modality (anatomic or reverse) or alternative humeral stem design. The image below depicts the designed compatibility of our systems:

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InSet rTSA System

Our rTSA configuration leverages the same humeral fixation platform as our aTSA configuration but utilizes alternative humeral and glenoid articular components designed for the reverse modality. The humeral component is configured to allow attachment of a tray and bearing assembly, designed to articulate with the glenoid sphere (“glenosphere”). The glenosphere is attached to a baseplate configured to provide reliable fixation to the glenoid on the scapular bone.

Our rTSA system was designed to optimize biomechanics and provide patients and physicians with a novel reverse implant solution that functions more like an anatomic implant in terms of improved range of motion. The InSet design of our rTSA system allows for retention of the patient's native anatomy and aesthetic and helps to achieve desirable, impingement free range of motion by avoiding arm lengthening, overstuffing, and other common challenges.

Humeral Tray and Bearing Assembly

Our humeral tray and bearing assembly are comprised of two components—a titanium humeral tray, and a polyethylene bearing—designed for low-profile assembly using our proprietary "Twistlock" locking mechanism. The Twistlock mechanism improves bearing-to-tray pullout strength, reducing the potential for bearing disassociation, a serious complication in rTSA where the polyethylene liner ("socket") separates from the humeral tray ("ball").

Our humeral tray and bearing assembly feature a unique bowl-shaped design that, together with the low profile nature of the Twistlock design, allows for inlay biomechanics with a rTSA modality, in which the apex of the glenosphere is positioned on the articular surface. This tray and bearing assembly, a key component of our rTSA system, connects to our humeral stems in the same manner as our InSet humeral head solutions, a key component of our aTSA system. The identical connection mechanism between these key components of our aTSA and rTSA systems is a critical design feature that enables the interchangeability of our systems and the potential for the humeral stem to remain in place during a revision or conversion surgery.

Scapular Reverse Assembly

The scapular assembly for rTSA is comprised of two main components—a baseplate, and a glenosphere—assembled by attaching the baseplate to the scapula using peripheral and central screws, then connecting the glenosphere to the baseplate using our proprietary locking cap technology. We have developed a wide range of baseplate and glenosphere options, which allows for multiple configurations to support a broad range of patient anatomy. With the use of ProVoyance, the assembly is configured to a patient's unique needs. This assembly enables a lateralized center of rotation that, together with our humeral assembly, results in a reverse implant that we believe feels and functions more like an anatomic shoulder replacement, with optimized range of motion and joint stability.

A key advantage of our rTSA system is the ability to treat complex glenoid deformities with minimal bone removal, enabled by our augmented baseplates developed with a concave backside design. The augmented baseplate is positioned within the subchondral bone which allows for correction of complex glenoid defects, which we believe improves implant stability and can preserve bone. Unlike typical augmentation options, our InSet reverse scapular components enable complete, flush contact between the implant and the native bone, without the need for a bone graft and eliminating multiple bone preparation steps.

We commenced development efforts for our rTSA system in 2019 and received 510(k) clearance in 2021. Our rTSA system is classified by the FDA as a Class II device.

ProVoyance Preoperative Planning Technology

We believe that surgeon-level engagement in preoperative planning—that is, detail-by-detail procedural planning performed by surgeons themselves—is a key element in providing effective surgical care for the patient. Hands-on preoperative planning by the surgeon provides for intimate familiarization with the patient's unique anatomy and surgical needs and reveals key learnings which guide preoperative and intraoperative decision making. For these reasons, we offer a highly approachable, easy to use preoperative planning technology with an intuitive interface designed to engage surgeons at levels we believe are industry leading.

ProVoyance integrates AI and ML to transform planar CT imaging into 3D renderings of patient-specific anatomy ahead of surgery. Its AI and ML algorithms empower surgeons to independently create bespoke surgical plans that can help ensure consistent, effective positioning of our implants.

ProVoyance received 510(k) clearance from the FDA in 2021 and is classified by the FDA as a Class II device. ProVoyance is listed on the FDA's AI/ML-enabled medical devices list, which is a resource maintained, published, and periodically updated by the FDA to identify AI/ML-enabled devices that have been authorized for marketing in the United States through any of the standard paths to market for medical devices, although it is not intended to be a comprehensive list of all such devices that incorporate AI/ML. This is highly differentiated from other preoperative planning solutions, where the implant manufacturer or another third-party creates a surgical plan that is sent to the surgeon for their consideration.

We believe the differentiation and value proposition of ProVoyance is validated by high utilization rates of ProVoyance across procedures using our advanced implant systems. For example, for the three months ended June 30, 2025, we estimate an implied utilization rate for our ProVoyance technology of approximately 98%, based on 1,478 surgical plans created using ProVoyance technology and 1,503 implant systems sold during such period. This implied utilization rate is based on real-world data from our customers during such period, as ProVoyance technology tracks and reports each surgical plan that our customers create, and the actual number of implant systems sold during the respective period. We believe this high surgeon utilization rate of ProVoyance is due to its ease of use and technical feature set. Ease of use is facilitated by the platform's highly intuitive, Unity-based interface with full user control and high-quality graphics capabilities. Technically, ProVoyance delivers true full-depth 3D rendering, 2D/3D bone density analysis, and biplanar glenoid deformity correction. ProVoyance provides an enhanced preoperative planning experience for surgeons and allows them to approach procedures with high confidence in their surgical plan.

The image below depicts the ProVoyance interface:

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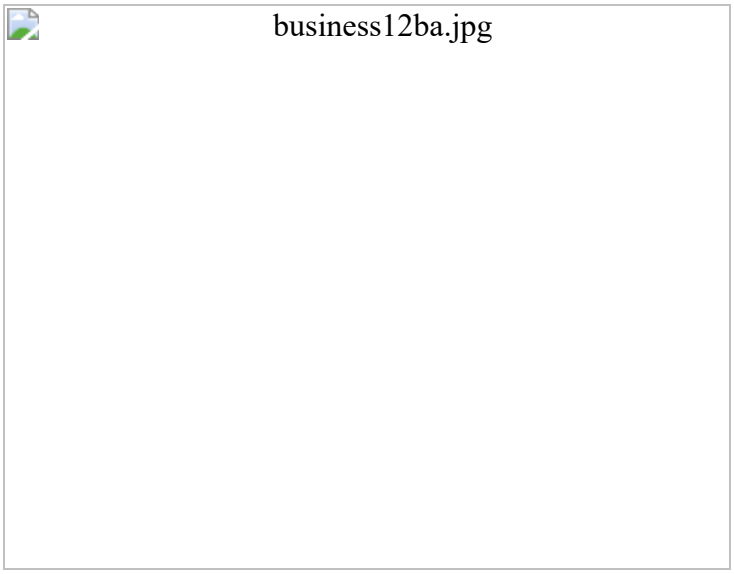
Efficient Instrument System

We re-engineered the components and technique for shoulder arthroplasty in order to create our proprietary InSet instrument system. This system was designed to enable interoperability between our novel aTSA and rTSA systems and a range of humeral stem options, while minimizing complexity and streamlining workflows.

Our instrument system is comprised of many components that fit within two trays, that we have specifically designed to facilitate the surgical technique used in connection with our implant systems. The components of each tray were designed for our implant characteristics and function. For example, the shape of the fixation fins on the

humeral components were specifically designed to facilitate a surgical technique that can be used across procedure types and systems, and each instrument was designed with a view to maximize utility and reduce the overall quantity of instruments used in procedures involving our systems.

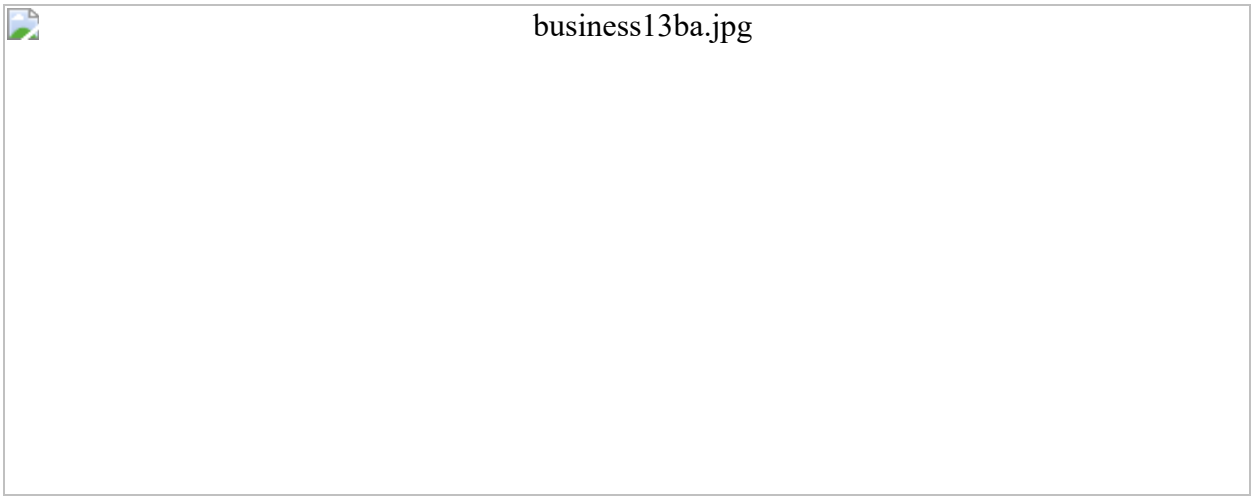
We believe our efficient, two tray instrument system can enable surgeons and staff to reduce operating room footprint, procedural setup time, sterilization time and expense and procedural complexity. We believe these benefits make it an optimal solution for aTSA and rTSA procedures performed across care settings and meaningfully reduce the per procedure capital outlay in shoulder surgical care. This advantage is particularly evident in ASCs, which generally have fewer resources and smaller footprints to handle or coordinate significant operative tools and complex instrument systems. The image below depicts our efficient instrumentation system for aTSA and rTSA:



Our Commercial Approach

We view ourselves as specialists serving specialists, having purposefully built our commercial organization around the unique needs of shoulder surgeons. Our commercial organization is comprised of three key components: (a) a dedicated commercial leadership team, (b) a CEME team and (c) a network of independent distributors. These key components work in tandem to form a commercial flywheel that is designed to build and reinforce relationships with surgeons and other stakeholders in the shoulder surgical care market, accelerate adoption, and enhance long-term retention.

The image below depicts our the three key components of our commercial organization:



We estimate that approximately 15,000 surgeons in the United States perform at least one shoulder arthroplasty procedure per year, with approximately 1,800 high-volume surgeons performing the vast majority of procedures. We believe these high-volume surgeons require dedicated expertise and support. To optimize our commercial strategy, we have developed proprietary business intelligence tools that enable us to identify and engage with the high-volume surgeons most likely to adopt our solutions. These tools analyze key data points such as surgeon location, procedure mix and care settings, enabling us to prioritize outreach and allocate commercial resources efficiently. By understanding where and how these high-volume surgeons practice, we can tailor our engagement strategies to align with their clinical and operational needs.

Once a surgeon is integrated into our ecosystem, we focus on expanding utilization by increasing the percentage of procedures performed with our solutions. To support our targeting and commercial efforts, we have classified surgeon customers into three categories: prospect surgeons, who perform between one and two shoulder arthroplasty procedures using our implant systems per quarter ("Prospect surgeons"); contender surgeons, who perform between three and eight shoulder arthroplasty procedures using our implant systems per quarter ("Contender surgeons"); and core surgeons, who perform at least nine shoulder arthroplasty procedures using our implant systems per quarter ("Core surgeons"). A key driver of increasing adoption within our existing surgeon base over time (helping Prospect surgeons become Contender surgeons, and Contender surgeons to become Core surgeons) is our surgeon-to-surgeon education program facilitated by our CEME team. We are committed to fostering a collaborative community among shoulder surgeons where expertise is shared, new ideas are exchanged and best practices are disseminated to enhance clinical outcomes. By facilitating these peer connections, we strengthen engagement within our ecosystem and reinforce its real-world success. As surgeons share their positive experiences with our ecosystem and the impact on patient outcomes, we believe this organic advocacy further accelerates adoption, positioning us to benefit from the natural network effects within the shoulder surgical care community.

Using this commercial approach, we aim to drive adoption and momentum with both existing and new surgeons and are continuing to increase our number of high-volume surgeons over time. For example, our number of Core and Contender surgeons has increased over time. For the quarter ended December 31, 2022, 2023 and 2024, we had 41, 60 and 83 Core and Contender surgeons, respectively, who performed 486, 888 and 1,145 shoulder arthroplasty procedures using our implant systems, respectively, in each of the respective periods. Over the course of 2022, 2023 and 2024, Core and Contender surgeons performed 1,408, 2,482 and 4,102 shoulder arthroplasty procedures, respectively, using our implant systems. For the quarter ended June 30, 2025, we had 107 Core and Contender surgeons who performed 1,423 shoulder arthroplasty procedures using our implant systems during this period. We have efficiently leveraged our commercial organization and approach to drive additional growth and to increase utilization of our implant systems by existing and new surgeons, and aim to continue to execute on this strategy going forward.

Dedicated Commercial Leadership Team

As of June 30, 2025, our dedicated commercial leadership team was comprised of 27 specialized representatives, organized into two commercial territories in the United States. Our team is focused on identifying high-volume surgeons specializing in shoulder surgical care. Their main responsibilities include: (a) target prospecting, by leveraging our business intelligence tools to identify and engage high-volume surgeons; (b) manage our distributor relationships, which helps to ensure alignment across our commercial organization and network of third party distributors; and (c) facilitate case coverage for each shoulder surgery performed with our systems, either directly or through our network of distributors.

Dedicated Customer Experience and Medical Education Team

As of June 30, 2025, our CEME team was comprised of seven expert surgeon educators who facilitate rewarding and meaningful experiences for surgeons focused on clinical value. Key responsibilities of our CEME team include key account conversions, expanding surgeon utilization of our solutions, and importantly, facilitating quality surgeon-to-surgeon educational opportunities. This team hosts medical education events and activities such as symposiums and lab events where like-minded surgeons can exchange ideas and nurture a culture of innovation in shoulder care. These in-person engagements are designed to build long-term relationships that extend well beyond the events themselves. For that purpose, we have developed online forums where high-volume users of our products can collaborate, share case experiences, and consult one another in real time. This community of surgeons often becomes the most passionate advocates for our solutions, often educating, training, and engaging other surgeons in the benefits of joining our ecosystem.

Independent Distributor Network

As of June 30, 2025, our network was comprised of 41 independent distributors with an aggregate of 150 trained representatives. Unlike typical stocking distributors, this network plays a critical role in expanding our market reach and driving adoption of our solutions in a cost-effective and scalable manner. In most cases, these distributors are focused on the orthopedic marketplace and sign exclusive agreements to carry our systems as their dedicated shoulder solution. They provide valuable feedback from the marketplace and are responsible for prospecting new surgeon customers, managing relationships, servicing existing accounts, and providing case coverage to support surgical execution. By maintaining strong, localized relationships with surgeons and healthcare facilities, our distributors help facilitate initial engagement, ongoing support, and long-term retention. Our dedicated commercial leadership team actively manages these relationships, ensuring distributors are meeting performance expectations and driving sales growth. We continuously evaluate distributor performance, maintaining relationships with those who generate the most sales while upgrading distribution partners where commercial execution can be improved.

Product Development and Pipeline

We leverage our team's decades of collective experience developing and launching novel shoulder surgical care technologies to identify solutions addressing the unmet needs of patients and surgeons. Since the initial launch of our InSet Glenoid in 2016, we have successfully commercialized a wide range of new technologies to enhance our ecosystem and provide surgeons with the tools and support needed to deliver quality outcomes for patients requiring shoulder surgical care. We believe our commitment to continuous innovation has been demonstrated by our track record of developing leading new technologies.

We continue to see clinical outcomes in the shoulder surgical care space that are inferior to those seen in more mature orthopedic markets and intend to continue working with our surgeon advisors to further improve clinical outcomes in shoulder surgical care. We believe this continued commitment to innovation will further expand our addressable market opportunity and improve our competitive position in shoulder surgical care.

We have a robust pipeline of new technologies in various stages of development and evaluation, including the following select projects.

I-Series Expansion and New Indications

We engineered the I-Series humeral stem line to provide a novel stem option for patients suffering from arthritis and a wide spectrum of proximal humeral bone loss or density. Our I-Series humeral stems feature a 2-fin design, specifically engineered for superior rotational control and stability. We launched our initial I-Series system in 2023, the InSet 95 Humeral Stem, which has been well received by the marketplace. We are developing additional stems, including InSet 70, InSet 135 and InSet 185 stems to expand our I-Series humeral stem line. We commenced development efforts for these additional InSet 70, InSet 135 and InSet 185 stems in 2024 and we anticipate pursuing FDA clearance of these stems, as needed, over the next twelve months. These additional clearances, if received from the FDA, will extend the range of available stem sizes and include expanded indications into fracture and revision surgeries.

Technologies for Metal Hypersensitive Patients

Market awareness of the risks and prevalence of metal hypersensitivity has risen in recent years. We are developing a line of humeral head and glenoid technologies for the approximately 10% to 15% of the general population who test positively for a metal hypersensitivity and may experience persistent pain, or other symptoms associated with allergic reactions from metal implants. These development technologies include our Humeral Head and Glenosphere, each constructed from alternative materials which do not incorporate elements that typically represent higher allergic risk for patients who test positively for a metal hypersensitivity. We anticipate pursuing FDA clearance of these solutions over the next twelve months.

Adjacent Market Expansion

We are also evaluating expansion into adjacent areas in shoulder surgical care, which may include sports medicine and shoulder trauma markets.

Clinical Overview

There is a significant body of clinical evidence that supports the safety, efficacy, and durability of our implants in shoulder arthroplasty, including our InSet Glenoid technology. We are committed to continued investment in obtaining further clinical evidence with the support of surgeons who are recognized as thought leaders in shoulder surgical care. We believe these efforts will continue to generate a substantial body of additional clinical evidence that will drive increased awareness and adoption of our products.

Since our founding, we have contributed to numerous publications that we believe evidence and strengthen our position as a leader in our industry. These studies focus on various patient outcome measures related to pain relief, range of motion, and complication rates, such as ASES outcome scores, Visual Analog Scale ("VAS") scores and Single-Assessment Numeric Evaluation ("SANE") scores. An ASES score is a metric derived from a postoperative questionnaire used by surgeons to assess shoulder function and patient well-being. ASES scores are based on a 100-point scale with higher scores indicating better function and less pain. A VAS score is a metric used to measure patient-perceived pain intensity with 10 indicating highest level of pain and 0 indicating no pain. A SANE score is a metric used to measure patient improvement where patients rate their shoulder on a scale from 0 to 100, with 100 being the patient's normal function. In addition, these studies focus on the most frequent complication associated with shoulder arthroplasty, failure of the glenoid component and published research demonstrates superior implant survivorship (*i.e.*, lack of revision).

As described in greater detail in the table below, the following is a summary of certain of these supportive, peer-reviewed publications:

- A study published in *The Journal of Shoulder and Elbow Surgery* in 2011 evaluated clinical outcomes of our InSet Glenoid implants at a mean follow-up time of 4.3 years. The results demonstrated statistically significant improvements in the mean ASES outcome score (68-point improvement), VAS scores (6.8-point improvement) and range of motion, with no surgical complications.

- A study published in *The Journal of Shoulder and Elbow Surgery* in 2012 evaluated the fixation strength and stress distribution of our InSet Glenoid fixation technique. In this study, our InSet Glenoid implants demonstrated no signs of loosening and up to an 87% reduction in displacement.
- A study published in *The Journal of Shoulder and Elbow Surgery* in 2019 evaluated clinical outcomes of aTSA procedures with our InSet Glenoid at a mean follow-up time of 8.7 years. The results indicated statistically significant improvements in the mean ASES outcome score (72-point improvement), VAS scores (7.6-point improvement) and range of motion, with no surgical complications, no cases of glenoid loosening and no revision surgeries performed at follow-up.
- A study published in *The Journal of Shoulder and Elbow Surgery* in 2023 evaluated clinical outcomes of aTSA procedures with our InSet Glenoid at a mean follow-up time of 28.7 months. There results indicated significant improvements in ASES scores (43-point improvement), VAS scores (4.2-point improvement) and SANE scores (51.6-point improvement, and statistically significant improvement in range of motion). Additionally, there were low rates of central peg radiographic lucency, asymptomatic loosening of the InSet Glenoid implant, and no patients were revised for implant-related complications.

The table below sets forth additional information regarding these peer-reviewed publications. Some of these studies were conducted with small sample sizes, did not control for clinical variables or have other design limitations (e.g., the studies may be retrospective and are not randomized controlled trials). Except as otherwise noted in the table below, we did not fund or sponsor any of these peer-reviewed publications nor were our employees involved in the studies or publications.

Reference	Source	Study Summary
Gunther et al. (2011)	*The Journal of Shoulder and Elbow Surgery*	Title: Total shoulder replacement surgery with custom glenoid implants for severe bone deficiency Authors: Stephen B. Gunther, MD; Tennyson L. Lynch, BS Description: A retrospective review of seven consecutive patients (three men, four women) treated with shoulder arthroplasty surgery with custom inset glenoid implants for severe bone deficiency at a minimum of three years. Conclusions: This study documented for the first time the use of custom-made inset glenoid implants in the treatment of severely deficient bone defects for which standard implants are contraindicated. The surgical technique has allowed glenoid implantation to be performed safely in this small group of patients, and the inset technique has been an effective short-term solution for these patients. At a mean follow-up of 4.3 years, there were statistically significant improvements ($P < 0.02$) in VAS scores (6.9 to 0.1), ASES outcome scores (26 to 94), and range of motion. No surgical complications occurred. Independent radiographic analysis determined all implants were classified as "low risk" for glenoid loosening.
Gunther et al. (2012)	*The Journal of Shoulder and Elbow Surgery*	Title: Finite element analysis and physiologic testing of a novel, inset glenoid fixation technique Authors: Stephen B. Gunther, MD; Tennyson L. Lynch, BS; Desmond O'Farrell, B.Eng (Hons) MM; Christian Calyore; Andrew Rodenhouse, BS Description: A comparative scientific analysis of glenoid loosening using an inset glenoid fixation technique vs the standard onlay technique used with a keel or pegged implant. The analysis consists of two separate methods: First, physiologic in vivo cyclic loading of glenoid implants was simulated using the dynamic model described by Anglin et al and American Society for Testing and Materials (ASTM) F2028-08 Standard Test Methods for Dynamic Evaluation of Glenoid Loosening or Disassociation. Second, finite element analysis was performed to estimate the glenohumeral joint stress and displacement for both standard onlay implants and an inset glenoid implant. Conclusions: The mechanical testing in this study of the inset glenoid design and fixation technique showed a significant reduction in post-testing distraction after physiologic loading compared with standard onlay design and fixation. The results of the finite element analysis support the concept of inset glenoid fixation based on the significant reduction in stresses on the backside surface on the inset implant compared with the standard onlay implant. Also, the nodal displacement at the edges of the implants under rim loading conditions was significantly reduced for the inset implants, and there was more uniformity of stress distribution along the inset polyethylene surface. This testing shows increased glenoid fixation strength using an inset technique, which may be beneficial in minimizing clinical glenoid loosening.

Reference	Source	Study Summary
Gunther et al. (2019)*	*The Journal of Shoulder and Elbow Surgery*	Title: Long-term follow-up of total shoulder replacement surgery with inset glenoid implants for arthritis with deficient bone Authors: Stephen B. Gunther, MD; Sterling K. Tran, BS Description: A retrospective analysis was performed on 21 of 24 consecutive patients treated with inset glenoid implants for severe glenohumeral joint arthritis with bone deficiency with prospectively collected data. Inclusion criteria were patients with shoulder arthritis and severe glenoid bone deficiency, defined by perpendicular glenoid vault depth less than 15 mm. No bone grafts were used. All patients were evaluated preoperatively and after surgery with physical examination, radiographic studies and outcome measures. There were 10 males and 11 females, 17 cases with osteoarthritis and 4 with inflammatory arthritis, and 5 patients with rotator cuff tears (3 full thickness and 2 partial tears). Mean age was 68 years. Conclusions: This inset glenoid fixation technique offers an innovative approach to a difficult clinical conundrum of shoulder arthritis with deficient glenoid bone. In this series, there were no complications, no cases of glenoid implant loosening and no revision surgeries performed at a mean 8.7-year follow-up. This technique is also safe, because there is only minimal penetration of the glenoid surface bone. Finally, the technique is simple and easily reproducible. It is a reasonable alternative to other current techniques available for patients with shoulder joint arthritis, severe bone deficiency, and an intact rotator cuff. At a mean follow-up of 8.7 years, there were statistically significant improvements ($P < 0.001$) in VAS scores (7.7 to 0.1), ASES outcome scores (23 to 95) and range of motion.
Johnston et al. (2023)	*The Journal of Shoulder and Elbow Surgery*	Title: Clinical and radiographic outcomes following anatomic total shoulder arthroplasty utilizing an inset glenoid component at 2-year minimum follow-up: a dual center study Authors: Peter S. Johnston, MD; John T. Strony, MD; Jessica L. Churchill, MD; Roma Kankaria, BS; Benjamin W. Sears, MD; Grant E. Garrigues, MD; Robert J. Gillespie, MD Description: A retrospective review of 75 patients undergoing aTSA using an inset glenoid component by two fellowship-trained shoulder surgeons at two separate institutions from August 2016 to August 2019 with minimum follow-up of two years. Range of motion and ASES, VAS and SANE scores were obtained. Radiographic outcomes, including central peg lucency and glenoid loosening, were assessed by three independent reviewers on the postoperative Grashey and axillary radiographs obtained at the final follow-up. Conclusions: At 2-year minimum follow-up, there were significant improvements in range of motion, VAS, SANE and ASES scores with low rates of central peg radiographic lucency and glenoid loosening. Additional long-term follow-up is needed to further clarify the advantages of InSet glenoid components when used in anatomic shoulder arthroplasty.

*As part of this publication, Mr. Gunther received 125,000 Warrants to purchase our Series B convertible preferred stock in January 2020.

In addition, we are investing in a registry to collect observational non-randomized standard-of-care data to document survival of our implant and evaluate the short- and long-term clinical and radiographic outcomes associated with real world use of our implant systems. The primary purpose of the registry will be to assess survival of our implant with or without revision at 2 years, 5 years and 10 years post-operatively. We believe this registry will provide further clinical support and additional clinical evidence that will drive increased awareness and adoption of our products. We expect the first publication of this registry to be available as early as 2026.

Coverage and Reimbursement

We sell our systems directly to hospitals, outpatient care centers and ASCs. These customers in turn bill various third-party payors, such as Medicare, Medicaid and private health insurance plans, for the total healthcare services required to treat the patient. Government agencies, private insurers and other payors determine whether to provide coverage for a particular procedure and to reimburse hospitals, outpatient care centers and ASCs at rates based on a prospective payment system. For procedures performed during a hospital inpatient stay, Medicare generally reimburse hospitals a single bundled payment that is based on the patient's principal diagnosis, up to 24 additional diagnoses and up to 25 procedures performed during the stay. Cases are classified into Medicare Severity-Diagnosis Related Groups ("MS-DRGs"), for payment under the Medicare Inpatient Prospective Payment System, for all items and services provided to the patient during a single hospitalization, regardless of whether procedures utilizing our products are performed during such hospitalization. Medicare rates for the same or similar procedures vary due to geographic location, nature of facility in which the procedure is performed and other factors. With respect to procedures performed in a hospital outpatient setting, all items and services paid under the Medicare outpatient prospective payment system, are assigned to payment groups called Ambulatory Payment Classifications, which group together items and services that are similar clinically and in terms of resource use. With respect to ASCs, covered procedures are assigned to ASC payment groups, which then determines the amount that Medicare pays for facility services furnished in connection with a covered procedure. Effective January 1, 2024, CMS added total shoulder arthroplasty to the list of covered procedures that can be performed in ASCs, which facilitated reimbursement and further supported the growth of ASCs as a key site of care in the shoulder surgical care market. While private payors vary in their coverage and payment policies, most use coverage and payment by Medicare as a benchmark by which to make their own decisions.

Reimbursement for professional services performed by physicians are also reported using CPT codes and based on a different payment methodology. Procedures using our systems are reported under CPT code 23470 for partial shoulder arthroplasty and CPT Code 23472 for total shoulder arthroplasty:

- *CPT 23470 (Arthroplasty, glenohumeral joint; hemiarthroplasty)* – Customers may categorize a procedure using our system as a partial shoulder arthroplasty when the reconstruction procedure is limited to the humeral articulation, and no replacement of the scapular articular surface (glenoid) is made.
- *CPT 23472 (Arthroplasty, glenohumeral joint; total shoulder (glenoid and proximal humeral replacement (e.g., total shoulder)))* – Customers may categorize a procedure using our system as a total shoulder arthroplasty when either aTSA or rTSA reconstruction is performed and the articular surfaces of both the humerus and the scapula are replaced by artificial components.

Intellectual Property

Intellectual property, including patents, trade secrets, trademarks and copyrights, is important to our business. Our commercial success depends in part on our ability to obtain and maintain proprietary intellectual property protection for our current products as well as for future product candidates and novel discoveries, product development technologies, and know-how. Our commercial success also depends in part on our ability to operate without infringing the proprietary rights of others and to prevent others from infringing our proprietary rights. We seek to maintain our proprietary position by, among other means, filing United States and foreign patent applications to obtain issued patents with claims directed to our products, product candidates, technology, inventions, and improvements that are important to the development and implementation of our business. We may also license from

third parties certain patent rights and proprietary know-how that we believe to be necessary or useful to our business. Additionally, we protect our proprietary know-how that may not be patentable, and other confidential information, by maintaining and implementing appropriate policies and procedures with respect to secrecy and confidentiality.

We are currently seeking and maintaining patent protection in the United States and key foreign jurisdictions where we intend to market our products, and plan to do so with respect to any of our future product candidates. Our patent portfolio includes a combination of patents and pending patent applications solely held by us.

As of May 16, 2025, our owned patent estate contains eleven (11) patent families comprising eleven (11) issued U.S. patents, seven (7) issued foreign patents, eleven (11) pending U.S. non-provisional patent applications and twenty (20) pending foreign patent applications. The eleven (11) issued U.S. patents are expected to expire between April 2025 and October 2041, after accounting for potentially available patent term adjustments, and assuming payment of appropriate maintenance, renewal, annuity and other governmental fees. Any patents that may issue from the eleven (11) pending U.S. patent applications are expected to expire between March 2030 and January 2045, without accounting for potentially available patent term adjustments, term-limiting effects of terminal disclaimers and assuming payment of appropriate maintenance, renewal, annuity and other governmental fees. The seven (7) issued foreign patents include one or more issued patents in jurisdictions such as Australia, Canada and Europe (validated in one or more of Germany, France, Switzerland and the U.K. and/or as having unitary effect), and are expected to expire between February 2026 and April 2039, without accounting for potentially available patent term extensions and assuming payment of appropriate maintenance, renewal, annuity and other governmental fees. The twenty (20) pending foreign patent applications include one or more pending applications in jurisdictions such as Australia, China, Europe and Japan, and are expected to expire between April 2038 and March 2044, without accounting for potentially available patent term extensions and assuming payment of appropriate maintenance, renewal, annuity and other governmental fees. Calculation of the expiration of issued patents is complex, varies by country and is based upon many factors. Accordingly, the following expiration dates are estimates.

Our owned U.S. and foreign patents and patent applications generally relate to anatomic total shoulder arthroplasty, reverse total shoulder arthroplasty, and instrumentation systems for anatomic total shoulder arthroplasty and reverse total shoulder arthroplasty. Our owned issued U.S. and foreign patents are set forth in the table below, which are directed to our relevant technologies, including glenoid implants, humeral implants, reverse shoulder replacement systems and instrumentation for shoulder replacement.

The table below sets forth information regarding our owned non-provisional patents and pending patent applications:

Country	Status	Application No.	Patent No.	Estimated Expiration	Description	Related Technology	Types
US	Issued	11/066978	8007538	8/9/2027	SHOULDER IMPLANT FOR GLENOID REPLACEMENT	Glenoid Implants	Devices
US	Pending	18/786,205	n/a	n/a	SHOULDER IMPLANT FOR GLENOID REPLACEMENT	Glenoid Implants	Devices
EP	Issued	11753831.4	2544632	3/3/2031	METHODS AND DEVICES FOR LESS INVASIVE GLENOID REPLACEMENT	Glenoid Implants	Devices
AU	Issued	2006218936	2006218936	2/22/2026	SHOULDER IMPLANT FOR GLENOID REPLACEMENT AND METHODS OF USE THEREOF	Glenoid Implants	Methods
CA	Issued	2598659	2598659	2/22/2026	SHOULDER IMPLANT FOR GLENOID REPLACEMENT AND METHODS OF USE THEREOF	Glenoid Implants	Devices

Country	Status	Application No.	Patent No.	Estimated Expiration	Description	Related Technology	Types
EP	Issued	6735834.1	1858453	2/22/2026	SHOULDER IMPLANT FOR GLENOID REPLACEMENT	Glenoid Implants	Devices
US	Issued	13/308221	9381086	3/25/2032	STEM FOR USE IN JOINT ARTHROPLASTY	Humeral Implants	Devices
US	Issued	15/202364	10143559	11/30/2031	STEM FOR USE IN JOINT ARTHROPLASTY	Humeral Implants	Devices
US	Issued	14/845136	10492926	1/25/2037	ALIGNMENT GUIDE FOR HUMERAL OR FEMORAL STEM REPLACEMENT PROSTHESES	Instrumentation	Methods
US	Issued	16/701118	12109126	5/13/2036	ALIGNMENT GUIDE FOR HUMERAL OR FEMORAL STEM REPLACEMENT PROSTHESES	Instrumentation	Devices
US	Pending	18/824638	n/a	n/a	ALIGNMENT GUIDE FOR HUMERAL OR FEMORAL STEM REPLACEMENT PROSTHESES	Instrumentation	Devices and Methods
US	Issued	15/952063	11065125	5/1/2038	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices and Methods
AU	Issued	2018251815	2018251815	4/11/2038	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices and Methods
AU	Pending	2024203972	n/a	n/a	CONVERTIBLE TOTAL SHOULDER PROSTHESIS	Humeral Implants	Devices and Methods
AU	Pending	2025200941	n/a	n/a	CONVERTIBLE TOTAL SHOULDER PROSTHESIS	Humeral Implants	Devices
AU	Pending	2024201672	n/a	n/a	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices and Methods
US	Pending	17/379894	n/a	n/a	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices

Country	Status	Application No.	Patent No.	Estimated Expiration	Description	Related Technology	Types
EP	Issued	18783984	3609440	4/11/2038	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices
EP	Pending	19767708.1	n/a	n/a	CONVERTIBLE TOTAL SHOULDER PROSTHESIS	Humeral Implants	Devices
EP	Pending	25152107.6	n/a	n/a	TOTAL SHOULDER PROSTHESIS HAVING INSET GLENOID IMPLANT CONVERTIBLE FROM ANATOMIC TO REVERSE	Glenoid Implants	Devices
US	Pending	17/652046	n/a	n/a	TRI-FLANGE HUMERAL PROSTHESIS	Humeral Implants	Devices
AU	Pending	2025202154	n/a	n/a	INSET/ONLAY GLENOID, POROUS COATED CONVERTIBLE GLENOID, AND HUMERAL HEADS WITH TEXTURED UNDERSIDES	Glenoid Implants	Devices
US	Pending	18/762437	n/a	n/a	INSET/ONLAY GLENOID, POROUS COATED CONVERTIBLE GLENOID, AND HUMERAL HEADS WITH TEXTURED UNDERSIDES	Glenoid Implants	Devices and Methods
EP	Issued	19796484.4	3787562	4/30/2039	INSET/ONLAY GLENOID, POROUS COATED CONVERTIBLE GLENOID, AND HUMERAL HEADS WITH TEXTURED UNDERSIDES	Glenoid Implants	Devices
EP	Pending	24194270.5	n/a	n/a	INSET/ONLAY GLENOID, POROUS COATED CONVERTIBLE GLENOID, AND HUMERAL HEADS WITH TEXTURED UNDERSIDES	Glenoid Implants	Devices
US	Issued	17/052126	12138172	10/27/2041	INSET/ONLAY GLENOID, POROUS COATED CONVERTIBLE GLENOID, AND HUMERAL HEADS WITH TEXTURED UNDERSIDES	Glenoid Implants	Devices and Methods
AU	Pending	2020237088	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices

Country	Status	Application No.	Patent No.	Estimated Expiration	Description	Related Technology	Types
US	Issued	18/058058	11771561	3/11/2040	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
US	Issued	18/605361	12023254	3/11/2040	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
US	Issued	18/761202	12268611	3/11/2040	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
US	Pending	19/064439	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
CN	Pending	202080020804.1	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
EP	Pending	20770328.1	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
JP	Pending	2021-555077	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
US	Pending	17/435333	n/a	n/a	TOTAL REVERSE SHOULDER SYSTEMS AND METHODS	Reverse Shoulder Replacement Systems	Devices
US	Issued	29/683368	D977643	2/7/2038	HUMERAL STEM IMPLANT	Humeral Implants	Design
US	Pending	29/870666	n/a	n/a	HUMERAL STEM IMPLANT	Humeral Implants	Design
AU	Pending	2023240964	n/a	n/a	STEMLESS, CONVERTIBLE, HUMERAL SHOULDER PROSTHESIS	Glenoid Implants	Devices and Methods
US	Pending	18/883873	n/a	n/a	STEMLESS, CONVERTIBLE, HUMERAL SHOULDER PROSTHESIS	Glenoid Implants	Devices and Methods
CN	Pending	202380041809.6	n/a	n/a	STEMLESS, CONVERTIBLE, HUMERAL SHOULDER PROSTHESIS	Glenoid Implants	Devices
EP	Pending	23716967.7	n/a	n/a	STEMLESS, CONVERTIBLE, HUMERAL SHOULDER PROSTHESIS	Glenoid Implants	Devices
JP	Pending	2024-556274	n/a	n/a	STEMLESS, CONVERTIBLE, HUMERAL SHOULDER PROSTHESIS	Glenoid Implants	Devices
EP	Pending	23847398.7	n/a	n/a	REUSABLE, CUSTOMIZABLE SHOULDER REPLACEMENT INSTRUMENTATION	Instrumentation	Devices
JP	Pending	2025-504571	n/a	n/a	REUSABLE, CUSTOMIZABLE SHOULDER REPLACEMENT INSTRUMENTATION	Instrumentation	Devices

Country	Status	Application No.	Patent No.	Estimated Expiration	Description	Related Technology	Types
US	Pending	18/998988	n/a	n/a	REUSABLE, CUSTOMIZABLE SHOULDER REPLACEMENT INSTRUMENTATION	Instrumentation	Devices and Methods
AU	Pending	2022229902	n/a	n/a	IMPLANT STEM WITH SUTURE RETENTION STRUCTURE	Humeral Implants	Devices and Methods
EP	Pending	22764119.8	n/a	n/a	IMPLANT STEM WITH SUTURE RETENTION STRUCTURE	Humeral Implants	Devices
EP	Pending	25166640	n/a	n/a	IMPLANT STEM WITH SUTURE RETENTION STRUCTURE	Humeral Implants	Devices
US	Pending	18/548059	n/a	n/a	IMPLANT STEM WITH SUTURE RETENTION STRUCTURE	Humeral Implants	Devices and Methods

The term of individual patents in our portfolio depends upon the legal term of patents in the countries in which they are obtained. In most countries in which we file, including the United States, the patent term is 20 years from the earliest date of filing a non-provisional patent application. In the United States, the term of a patent may be reduced due to terminal disclaimer made to overcome a double patenting rejection, or may be lengthened by patent term adjustment, which permits patent term restoration as compensation for delays incurred at the USPTO during the patent prosecution process.

The patent positions of companies like ours are generally uncertain and involve complex legal and factual questions. The relevant patent laws and their interpretation outside of the United States is also uncertain. Changes in either the patent laws or their interpretation in the United States and other countries may diminish our ability to protect our technology or product candidates and could affect the value of such intellectual property. In particular, our ability to stop third parties from making, using, selling, offering to sell or importing products that infringe our intellectual property will depend in part on our success in obtaining and enforcing patent claims that cover our technology, inventions and improvements. We cannot guarantee that patents will be granted with respect to any of our pending patent applications or with respect to any patent applications we may file in the future, nor can we be sure that any patents that may be granted to us in the future will be commercially useful in protecting our products, the methods of use or manufacture of those products. Moreover, issued patents do not guarantee the right to practice our technology in relation to the commercialization of our products. Issued patents only allow us to block potential competitors from practicing the claimed inventions of the issued patents in the countries in which such patents are issued.

Further, patents and other intellectual property rights in the medical device space are evolving and involve many risks and uncertainties. For example, third parties may have blocking patents that could be used to prevent us from commercializing our product candidates and practicing our proprietary technology. Our issued patents may be challenged, invalidated or circumvented, which could limit our ability to stop competitors from marketing related products or could limit the term of patent protection that otherwise may exist for our product candidates. In addition, the scope of the rights granted under any issued patents may not provide us with protection or competitive advantages against competitors with similar technology. Furthermore, our competitors may independently develop similar technologies that are outside the scope of the rights granted under any issued patents. For these reasons, we may face competition with respect to our product candidates. Moreover, because of the extensive time required for development, testing and regulatory review of a potential product, it is possible that, before any particular product candidate can be commercialized, any patent protection for such product may expire or remain in force for only a short period following commercialization, thereby reducing the commercial advantage the patent provides.

We also rely on trade secrets relating to our discovery programs and product candidates, and seek to protect and maintain the confidentiality of proprietary information to protect aspects of our business that are not amenable to, or that we do not consider appropriate for, patent protection. It is our policy and practice to require our employees,

consultants, outside scientific collaborators, sponsored researchers and other advisors to execute confidentiality agreements upon the commencement of employment or consulting relationships with us, and for employees and consultants to enter into invention assignment agreements with us. These agreements provide that all confidential information developed or made known to the individual during the course of the individual's relationship with us is to be kept confidential and not disclosed to third parties except in specific circumstances. Where applicable, the agreements provide that all inventions to which the individual contributed as an inventor shall be assigned to us, and as such, will become our property. There can be no assurance, however, that these agreements will provide meaningful protection or adequate remedies for our trade secrets in the event of unauthorized use or disclosure of such information.

Software License Agreement with Genesis Software

On October 22, 2020, we entered into a software license agreement with Genesis Software, which was amended and restated on January 1, 2023 and subsequently amended and restated on June 10, 2025 (as amended and restated, the "License Agreement"). Pursuant to the License Agreement, Genesis Software granted us an exclusive, worldwide, transferable, sublicensable and royalty-bearing license under Genesis Software's SaaS surgery planning platform software, including all intellectual property rights therein, updates, upgrades and other modifications thereto, and object code, source code and other surgery planning software developed or controlled by Genesis Software during the term of the License Agreement (collectively, the "Licensed Software"), to store, use copy, make derivative works from, modify, extend, enhance and improve the Licensed Software in the field of shoulder surgical procedures primarily related to replacement of the shoulder joint articulation (*i.e.*, anatomic shoulder arthroplasty, reverse shoulder arthroplasty, revision shoulder arthroplasty, and hemi-shoulder arthroplasty) (the "Field of Use"). The Licensed Software is utilized in connection with ProVoyance. We granted Genesis Software a limited, non-exclusive, non-transferable license to use data generated by us through our use of the Licensed Software solely in connection with customization of an implant other software products outside the Field of Use. Genesis Software granted us a limited, non-exclusive, non-transferable license under identified data collected by Genesis Software through its use of the Licensed Software outside of the Field of Use for our use within the Field of Use. We also agreed to compensate Genesis Software for certain development, consulting, and support services related to the Licensed Software that we may request from time to time.

Pursuant to the License Agreement, we paid Genesis Software an upfront license fee of $1.0 million and a milestone payment of $0.5 million upon receipt of FDA clearance of the Licensed Software. We are also required to pay Genesis Software royalties of a mid-single-digit percentage on net sales of shoulder replacement products that incorporate or use the Licensed Software until the earlier of (i) the tenth anniversary of the effective date of the License Agreement and (ii) such time as the aggregate amount of royalties, upfront license fee and milestone due upon receipt of FDA clearance paid to Genesis Software reaches $7.0 million. At any time prior to the tenth anniversary of the effective date of the License Agreement, we have the right to buy out our royalty obligations by paying Genesis Software the difference between the aggregate amounts already paid to Genesis Software and $7.0 million, upon which our rights under the License Agreement shall become irrevocable and perpetual and shall be royalty-free and fully-paid. In 2025, we are obligated to pay Genesis Software $1.7 million for certain development, consulting and support services.

The term of the License Agreement shall continue until earlier terminated in accordance with the terms therein. The License Agreement may be terminated for cause by either party: (i) if the other party breaches a material provision of the License Agreement and fails to cure such breach within a specified notice and cure period or (ii) upon the bankruptcy, insolvency or certain dissolution or liquidation events of other party.

For each of the years ended December 31, 2024 and 2023, we paid Genesis Software $3.1 million and $3.2 million, respectively, pursuant to the License Agreement, including royalties of $1.2 million and $0.6 million in the years ended December 31, 2024 and 2023, respectively. For the three months ended March 31, 2025 and 2024, we paid Genesis Software $0.9 million and $0.7 million, respectively, including royalties of $0.4 million and $0.3 million in the three months ended March 31, 2025 and 2024, respectively.

Manufacturing and Supply

We utilize third-party manufacturing and supply providers to manufacture our implants. We believe this outsourcing strategy provides the expertise and capacity required to effectively and efficiently scale production based on demand, and helps to ensure low-cost production and a capital efficient business model. Almost all of our products are provided by single-source suppliers. For example, Avalign Technologies manufactures and supplies our humeral stems, Micropulse is our sole source for Inset Glenoid, Trifecta Medical Group manufactures and supplies our trays and RMD supplies surgical instruments used during procedures involving our systems. We also utilize a single supplier for significant majority of our sterilization needs.

We generally do not have long-term contracts with these third parties and our supply arrangements generally do not include minimum manufacturing or purchase obligations. We primarily order products through the use of purchase orders. As such, we generally do not have any obligation to purchase any given quantity of products, and our suppliers generally have no obligation to sell to us or to manufacture for us any given quantity of our products or components of our systems. We maintain strong working relationships with our suppliers and we believe our current network of third-party manufacturing and supply providers provides for sufficient capacity to meet projected market demand for our products for the foreseeable future.

While there are other suppliers that could make or provide any one of our products, we seek to manage single-source supplier risk by regularly assessing the quality and capacity of our suppliers, implementing supply and quality control protocols where appropriate and actively managing lead times and inventory levels. In addition, we are currently in the process of identifying and approving alternative suppliers to dual or multi-source certain of our products. We generally seek to maintain sufficient supply levels to help mitigate any supply interruptions and enable us to find and qualify another source of supply. For certain products, we estimate that it could take between six and twelve months to find and qualify a second source. Order quantities and lead times are based on internal forecasts, which are derived from historical demand and anticipated future demand. Lead times vary depending on the size of the order, time required to manufacture, specific supplier requirements and current market demand and dynamics for the raw materials, sub-assemblies and parts related to our products.

Our third-party manufacturing and supply providers are evaluated, qualified and approved through our supplier quality program, which includes verification and monitoring procedures to help ensure that our suppliers comply with FDA and ISO standards, as well as our own specifications and requirements. We inspect and verify products under strict processes supported by internal policies and procedures. We maintain a rigorous change control policy to assure that no product or process changes are implemented without our prior review and approval.

We operate a distribution facility from our corporate headquarters located in Grand Rapids, Michigan. This facility has approximately 7,000 square feet of warehouse, distribution and office space. The lease related to this facility expires in January 2026 with the option to extend an additional five years. We also leverage a third-party 3,000 square foot distribution facility in Modesto, California that distributes our implant systems in the western United States. We believe these facilities are sufficient to meet our current and anticipated needs in the near term and that suitable additional space is available as needed to accommodate expansion of our operations and distribution activities.

Competition

Our industry is competitive, subject to technological change and significantly affected by new product introductions and market activities of other industry participants. Our competition includes medical device manufacturers in the shoulder surgical care market and in the shoulder arthroplasty market in particular. For example, companies operating in our market include Arthrex, Enovis, Exectech, Johnson & Johnson, Smith & Nephew, Stryker, and Zimmer Biomet as well as a number of smaller companies. There are a significant number of approved devices for shoulder surgical care and shoulder arthroplasty, and we expect competition will intensify over time.

Many of our competitors have longer, more established operating histories, and significantly greater name recognition and financial, technical, marketing, sales, distribution and other resources. In addition, certain competitors have several competitive advantages, including international operations with significant scale and

established relationships with hospitals, outpatient care centers and ASCs and surgeons who use their devices and are familiar with existing devices on the market. In addition to competing for market share, we also compete against these companies for personnel, including qualified personnel that are necessary to grow our business.

We believe the principal competitive factors in our market include:

- product features and design;
- patient experience, including recovery time, level of discomfort and post-operative results;
- acceptance by surgeons and other key stakeholders in the shoulder surgical care market;
- surgeon learning curves and willingness to adopt new techniques;
- ease of use and reliability, including preoperative planning tools;
- economic benefits and efficiencies for hospitals, outpatient care centers, ASCs and surgeons;
- effective distribution and marketing to surgeons and potential patients, including physician education and information sharing programs;
- product quality and standards, including our reputation with customers and surgeons;
- intellectual property; and
- customer service and support capabilities.

We believe we have established a compelling value proposition to compete favorably in this market. However, conditions in our market could change rapidly and significantly as a result of technological advancements, partnerships, or acquisitions by competitors or continuing market consolidation and we expect the competitive environment to remain intense. For example, we have seen and continue to see consolidation amongst our competitors and have seen and continue to see our competitors innovate and improve upon traditional implants and technologies. For example, certain competitors have pursued the utilization of new materials (metal backing) for traditional and onlay glenoid implant structures and have developed stand-alone single stems that can be used in anatomic and reverse procedures. If our competitors have greater resources and access to funding than us, they may be able to finance the development of new technologies and products before we are able to do so, which may allow them to gain market share or enter new markets before us or provide lower-priced or better-quality offerings.

Government Regulation

Our products and our operations are subject to extensive regulation by the FDA, and other federal and state authorities in the United States. Our products are subject to regulation as medical devices in the United States under the FDCA, as implemented and enforced by the FDA.

United States Regulation of Medical Devices

The FDA regulates the development, design, non-clinical and clinical research, manufacturing, safety, efficacy, labeling, packaging, storage, installation, servicing, recordkeeping, premarket clearance or approval, adverse event reporting, advertising, promotion, marketing and distribution, and import and export of medical devices, including software such as AI/ML to ensure that medical devices distributed domestically are safe and effective for their intended uses and otherwise meet the requirements of the FDCA.

FDA Marketing Authorization Requirements

Unless an exemption applies, each medical device commercially distributed in the United States requires either FDA clearance of a premarket notification submitted under Section 510(k) of the FDCA, approval of a PMA, or classification under the *de novo* classification process. Under the FDCA, medical devices are classified into one of three classes—Class I, Class II or Class III—depending on the degree of risk associated with each medical device

and the extent of manufacturer and regulatory control needed to ensure its safety and effectiveness. Class I includes devices with the lowest risk to the patient and are those for which safety and effectiveness can be assured by adherence to the FDA's General Controls for medical devices, which include compliance with the applicable portions of the QSR (and once it goes into effect in 2026, the QMSR, which amended the QSR), facility registration and product listing, reporting of adverse medical events, and truthful and non-misleading labeling, advertising, and promotional materials. Class II devices are subject to the FDA's General Controls, and special controls as deemed necessary by the FDA to ensure the safety and effectiveness of the device. These special controls can include performance standards, post-market surveillance, patient registries and FDA guidance documents.

While most Class I devices are exempt from the 510(k) premarket notification requirement, manufacturers of most Class II devices are required to submit to the FDA a premarket notification under Section 510(k) of the FDCA requesting permission to commercially distribute the device. The FDA's permission to commercially distribute a device subject to a 510(k) premarket notification is generally known as 510(k) clearance. Devices deemed by the FDA to pose the greatest risks, such as life sustaining, life supporting or some implantable devices, or devices that have a new intended use, or use advanced technology that is not substantially equivalent to that of a legally marketed device, are placed in Class III, requiring approval of a PMA. Some pre-amendment devices are unclassified, but are subject to FDA's premarket notification and clearance process in order to be commercially distributed. Where software, including AI/ML, qualifies as a device, this same classification scheme and regulatory path to market applies.

510(k) Clearance Marketing Pathway

To obtain 510(k) clearance, a company must submit to the FDA a premarket notification submission demonstrating that the proposed device is "substantially equivalent" to a legally marketed predicate device. A predicate device is a legally marketed device that is not subject to premarket approval, *i.e.*, a device that was legally marketed prior to May 28, 1976 (pre-amendments device) and for which a PMA is not required, a device that has been reclassified from Class III to Class II or I, or a device that was found substantially equivalent through the 510(k) process. The FDA's 510(k) clearance process usually takes from three to twelve months, but may take longer. The FDA may require additional information, including clinical data, to make a determination regarding substantial equivalence. In addition, FDA collects user fees for certain medical device submissions and annual fees and for medical device establishments.

If the FDA agrees that the device is substantially equivalent to a predicate device currently on the market, it will grant 510(k) clearance to commercially market the device. If the FDA determines that the device is "not substantially equivalent" to a previously cleared device, the device is automatically designated as a Class III device. The device sponsor must then fulfill more rigorous PMA requirements, or can request a risk-based classification determination for the device in accordance with the *de novo* classification process, which is a route to market for novel medical devices that are low to moderate risk and are not substantially equivalent to a predicate device.

After a device receives 510(k) clearance, any modification that could significantly affect its safety or effectiveness, or that would constitute a major change or modification in its intended use, will require a new 510(k) clearance or, depending on the modification, PMA approval. The FDA requires each manufacturer to determine whether the proposed change requires submission of a 510(k) or a PMA in the first instance, but the FDA can review any such decision and disagree with a manufacturer's determination. If the FDA disagrees with a manufacturer's determination, the FDA can require the manufacturer to cease marketing and/or request the recall of the modified device until such marketing authorization has been granted. Also, in these circumstances, the manufacturer may be subject to significant regulatory fines or penalties.

PMA Approval Pathway

Class III devices require PMA approval before they can be marketed, although some pre-amendment Class III devices for which FDA has not yet required a PMA are cleared through the 510(k) process. The PMA process is more demanding than the 510(k) premarket notification process. In a PMA, the manufacturer must demonstrate that the device is safe and effective, and the PMA must be supported by extensive data, including data from preclinical studies and human clinical trials. The PMA must also contain a full description of the device and its components, a

full description of the methods, facilities, and controls used for manufacturing, and proposed labeling. Following receipt of a PMA, the FDA determines whether the application is sufficiently complete to permit a substantive review. If FDA accepts the application for review, it has 180 days under the FDCA to complete its review of a PMA, although in practice, the FDA's review often takes significantly longer, and can take up to several years. An advisory panel of experts from outside the FDA may be convened to review and evaluate the application and provide recommendations to the FDA as to the approvability of the device. The FDA may or may not accept the panel's recommendation. In addition, the FDA will generally conduct a pre-approval inspection of the applicant or its third-party manufacturers' or suppliers' manufacturing facility or facilities to ensure compliance with the QSR. PMA applications are also subject to the payment of user fees, which are higher than in the 510(k) process.

The FDA will approve the new device for commercial distribution if it determines that the data and information in the PMA constitute valid scientific evidence and that there is reasonable assurance that the device is safe and effective for its intended use(s). The FDA may approve a PMA with post-approval conditions intended to ensure the safety and effectiveness of the device, including, among other things, restrictions on labeling, promotion, sale and distribution, and collection of long-term follow-up data from patients in the clinical study that supported PMA approval or requirements to conduct additional clinical studies post-approval. The FDA may condition PMA approval on some form of post-market surveillance when deemed necessary to protect the public health or to provide additional safety and efficacy data for the device in a larger population or for a longer period of use. In such cases, the manufacturer might be required to follow certain patient groups for a number of years and to make periodic reports to the FDA on the clinical status of those patients. Failure to comply with the conditions of approval can result in material adverse enforcement action, including withdrawal of the approval.

Certain changes to an approved device, such as changes in manufacturing facilities, methods, or quality control procedures, or changes in the design performance specifications, which affect the safety or effectiveness of the device, require submission of a PMA supplement. PMA supplements often require submission of the same type of information as a PMA, except that the supplement is limited to information needed to support any changes from the device covered by the original PMA and may not require as extensive clinical data or the convening of an advisory panel. Certain other changes to an approved device require the submission of a new PMA, such as when the design change causes a different intended use, mode of operation, and technical basis of operation, or when the design change is so significant that a new generation of the device will be developed, and the data that were submitted with the original PMA are not applicable for the change in demonstrating a reasonable assurance of safety and effectiveness.

De Novo Classification

Medical device types that the FDA has not previously classified as Class I, II, or III are automatically classified into Class III regardless of the level of risk they pose. The Food and Drug Administration Modernization Act of 1997 established a route to market for low-to-moderate risk medical devices that are automatically placed into Class III due to the absence of a predicate device, called the "Request for Evaluation of Automatic Class III Designation," or the *de novo* classification procedure. This procedure allows a manufacturer whose novel device is automatically classified into Class III to request down-classification of its medical device into Class I or Class II on the basis that the device presents low or moderate risk, rather than requiring the submission and approval of a PMA application. Pursuant to the Food and Drug Administration Safety and Innovation Act ("FDASIA"), manufacturers may request *de novo* classification directly without first submitting a 510(k) pre-market notification to the FDA and receiving a not-substantially-equivalent determination.

Under FDASIA, FDA is required to classify the device within 120 days following receipt of the *de novo* request, although the process may take significantly longer. If the manufacturer seeks reclassification into Class II, the manufacturer must include a draft proposal for special controls that are necessary to provide a reasonable assurance of the safety and effectiveness of the medical device. If FDA grants the *de novo* request, the device may be legally marketed in the United States. However, the FDA may reject the request if the FDA identifies a legally marketed predicate device that would be appropriate for a 510(k) notification, determines that the device is not low-to-moderate risk, or determines that General Controls would be inadequate to control the risks and/or special controls cannot be developed. After a device receives *de novo* classification, any modification that could significantly affect its safety or efficacy, or that would constitute a major change or modification in its intended use,

will require a new 510(k) clearance or, depending on the modification, another *de novo* request or even PMA approval.

Clinical Trials

Clinical trials are almost always required to support a PMA and *de novo* classification, and are sometimes required to support a 510(k) submission. All clinical investigations of devices to determine safety and effectiveness must be conducted in accordance with the FDA's investigational device exemption ("IDE") regulations which govern investigational device labeling, prohibit promotion of the investigational device, and specify an array of recordkeeping, reporting and monitoring responsibilities of study sponsors and study investigators. If the device presents a "significant risk" to human health, as defined by the FDA, the FDA requires the device sponsor to submit an IDE application to the FDA, which must become effective prior to commencing human clinical trials. If the device under evaluation does not present a significant risk to human health, then the device sponsor is not required to submit an IDE application to the FDA before initiating human clinical trials, but must still comply with abbreviated IDE requirements when conducting such trials. A significant risk device is one that presents a potential for serious risk to the health, safety or welfare of a patient and either is implanted, used in supporting or sustaining human life, substantially important in diagnosing, curing, mitigating or treating disease or otherwise preventing impairment of human health, or otherwise presents a potential for serious risk to a subject. An IDE application must be supported by appropriate data, such as animal and laboratory test results, showing that it is safe to test the device in humans and that the testing protocol is scientifically sound. The IDE will automatically become effective 30 days after receipt by the FDA unless the FDA notifies the company that the investigation may not begin. If the FDA determines that there are deficiencies or other concerns with an IDE for which it requires modification, the FDA may permit a clinical trial to proceed under a conditional approval.

Regardless of the degree of risk presented by the medical device, clinical studies must be approved by, and conducted under the oversight of, an Institutional Review Board ("IRB") for each clinical site. The IRB is responsible for the initial and continuing review of the IDE, and may impose additional requirements for the conduct of the study. If an IDE application is approved by the FDA and one or more IRBs, human clinical trials may begin at a specific number of investigational sites with a specific number of patients, as approved by the FDA. If the device presents a non-significant risk to the patient, a sponsor may begin the clinical trial after obtaining approval for the trial by one or more IRBs without separate approval from the FDA, but must still follow abbreviated IDE requirements, such as monitoring the investigation, ensuring that the investigators obtain informed consent, and complying with labeling and record-keeping requirements. In some cases, an IDE supplement must be submitted to, and approved by, the FDA before a sponsor or investigator may make a change to the investigational plan that may affect its scientific soundness, study plan or the rights, safety or welfare of human subjects.

During a study, the sponsor is required to comply with the applicable FDA requirements, including, for example, trial monitoring, selecting clinical investigators and providing them with the investigational plan, ensuring IRB review, adverse event reporting, record keeping and prohibitions on the promotion of investigational devices or on making safety or effectiveness claims for them. The clinical investigators in the clinical study are also subject to FDA's regulations and must obtain patient informed consent, rigorously follow the investigational plan and study protocol, control the disposition of the investigational device, and comply with all reporting and recordkeeping requirements. Additionally, after a trial begins, the sponsor, the FDA or the IRB could suspend or terminate a clinical trial at any time for various reasons, including a belief that the risks to study subjects outweigh the anticipated benefits.

Post-Market Regulation of Medical Devices

After a product is placed on the market, numerous regulatory requirements continue to apply. These requirements include:

- product listing and establishment registration, which helps facilitate FDA inspections and other regulatory action;

- the QSR, which requires manufacturers, including third-party manufacturers, to follow stringent design, validation, testing, control, documentation and other quality assurance procedures during all aspects of the design and manufacturing process, and once it goes into effect, the QMSR;
- labeling regulations and FDA prohibitions against the promotion of products for uncleared or unapproved use or indication;
- authorization of marketing of certain modifications that could significantly affect safety or effectiveness or that would constitute a major change in intended use;
- medical device reporting regulations, which require that a manufacturer report to the FDA if a device it markets may have caused or contributed to a death or serious injury, or has malfunctioned and the device or a similar device that it markets would be likely to cause or contribute to a death or serious injury, if the malfunction were to recur;
- correction, removal and recall reporting regulations, which require that manufacturers report to the FDA field corrections and product recalls or removals if undertaken to reduce a risk to health posed by the device or to remedy a violation of the FDCA that may present a risk to health;
- post-market restrictions or conditions, including post-market study commitments;
- post-market surveillance regulations, which apply, when necessary, to protect the public health or to provide additional safety and effectiveness data for the medical product;
- the FDA's recall authority, whereby it can ask, or under certain conditions order, device manufacturers to recall from the market a product that is in violation of governing laws and regulations; and
- regulations pertaining to voluntary recalls.

Manufacturing processes for medical devices are required to comply with the applicable portions of the QSR (and once it goes into effect, the QMSR), which cover the methods and the facilities and controls for the design, manufacture, testing, production, processes, controls, quality assurance, labeling, packaging, distribution, installation and servicing of finished devices intended for human use. The QSR and QMSR also require, among other things, maintenance of a device master file, device history file, and complaint files. Manufacturers are subject to periodic scheduled and unscheduled inspections by the FDA. Failure to maintain compliance with the QSR and, once in effect, the QMSR requirements could result in the shut-down of, or restrictions on, manufacturing operations and the recall or seizure of marketed products. The discovery of previously unknown problems with any marketed products, including unanticipated adverse events or adverse events of increasing severity or frequency, whether resulting from the use of the device within the scope of its clearance or approval, or off-label by a physician in the practice of medicine, could result in restrictions on the device, including the removal of the product from the market or voluntary or mandatory device recalls.

The FDA has broad regulatory compliance and enforcement powers. If the FDA determines that a manufacturer has failed to comply with applicable regulatory requirements, it can take a variety of compliance or enforcement actions, which may result in any of the following sanctions:

- warning letters, untitled letters, fines, injunctions, consent decrees and civil penalties;
- recalls, withdrawals, or administrative detention or seizure of our products;
- operating restrictions or partial suspension or total shutdown of production;
- refusing or delaying requests for 510(k) clearance, *de novo* classifications, or PMA approvals of new products or modified products;
- withdrawing 510(k) clearances, *de novo* classifications, or PMA approvals that have already been granted;
- refusal to grant export approvals for our products; or

- criminal prosecution.

Foreign Government Regulation

In addition to U.S. regulations, medical devices are subject to a variety of foreign government regulations.

Foreign Regulation of Medical Devices

Regulation of Medical Devices in the European Union

The EU has adopted specific directives and regulations regulating the design, manufacture, clinical investigation, conformity assessment, labeling and adverse event reporting for medical devices.

Until May 25, 2021, medical devices were regulated by the Council Directive 93/42/EEC ("EU Medical Devices Directive"), which has been repealed and replaced by Regulation (EU) No 2017/745 ("EU Medical Devices Regulation"). Unlike directives, regulations are directly applicable in all EU member states without the need for member states to implement into national law.

In the EU, there is currently no premarket government review of medical devices. However, all medical devices placed on the EU market must meet general safety and performance requirements, including the requirement that a medical device must be designed and manufactured in such a way that, during normal conditions of use, it is suitable for its intended purpose. Medical devices must be safe and effective and must not compromise the clinical condition or safety of patients, or the safety and health of users and – where applicable – other persons, provided that any risks which may be associated with their use constitute acceptable risks when weighed against the benefits to the patient and are compatible with a high level of protection of health and safety, taking into account the generally acknowledged state of the art.

Compliance with the general safety and performance requirements is a prerequisite for European conformity marking ("CE mark"), without which medical devices cannot be marketed or sold in the EU. To demonstrate compliance with the general safety and performance requirements medical device manufacturers must undergo a conformity assessment procedure, which varies according to the type of medical device and its (risk) classification. Except for low-risk medical devices (Class I), where the manufacturer can self-assess the conformity of its products with the general safety and performance requirements (except for any parts which relate to sterility, metrology or reuse aspects), a conformity assessment procedure requires the intervention of a notified body. Notified bodies are independent organizations designated by EU member states to assess the conformity of devices before being placed on the market. A notified body would typically audit and examine a product's technical dossiers and the manufacturer's quality system. If satisfied that the relevant product conforms to the relevant general safety and performance requirements, the notified body issues a certificate of conformity, which the manufacturer uses as a basis for its own declaration of conformity. The manufacturer may then apply the CE mark to the device, which allows the device to be placed on the market throughout the EU.

Throughout the term of the certificate of conformity, the manufacturer will be subject to periodic surveillance audits to verify continued compliance with the applicable requirements. In particular, there will be a new audit by the notified body before it will renew the relevant certificate(s).

All manufacturers placing medical devices into the market in the EU must comply with the EU medical device vigilance system. Under this system, serious incidents and Field Safety Corrective Actions ("FSCAs"), must be reported to the relevant authorities of the EU member states. Manufacturers are required to take FSCAs defined as any corrective action for technical or medical reasons to prevent or reduce a risk of a serious incident associated with the use of a medical device that is made available on the market. An FSCA may include the recall, modification, exchange, destruction or retrofitting of the device.

The aforementioned EU rules are generally applicable in the EEA, which consists of the 27 EU Member States plus Norway, Liechtenstein and Iceland.

Regulation of Medical Devices in the United Kingdom

The EU Medical Devices Regulation is not applicable in Great Britain due to Brexit. Existing EU directives governing all medical devices have been given effect in domestic law through the Medical Devices Regulations 2002 (SI 2002 No 618, as amended) ("UK Medical Devices Regulations"). This means that since January 1, 2021, the Great Britain route to market is still based on the requirements derived from the pre-existing EU legislation. As a standalone regulatory body, the Medicines and Healthcare products Regulatory Agency ("MHRA") is responsible for regulating medical devices in Great Britain (England, Scotland and Wales) and Northern Ireland, though Northern Ireland is aligned with EU legislation and regulations regarding medical devices as a result of the Northern Ireland Protocol that took effect in January 2021. The UK government has passed the Medicines and Medical Devices Act which came into force on 11 February 2021 and which allows the secretary of state or an 'appropriate authority' to amend or supplement existing regulations in the area of medical devices including the UK Medical Devices Regulations. In addition, the Trade Deal between the UK and the EU generally provides for cooperation and exchange of information between the parties in the areas of product safety and compliance, including market surveillance, enforcement activities and measures, standardization-related activities, exchanges of officials, and coordinated product recalls. As such, processes for compliance and reporting should reflect requirements from regulatory authorities.

Under the UK Medical Devices Regulations, in order to be lawfully placed on the Great Britain market, certain medical devices need to be "UKCA" certified by a UK approved body. However, certain medical devices in compliance with: (1) the EU Medical Devices Directive can continue to be placed on the Great Britain market until the sooner of certificate expiration or June 30, 2028; or (2) the EU Medical Devices Regulation can continue to be placed on the Great Britain market until the sooner of certificate expiration or June 30, 2030. The EU no longer recognizes conformity assessment activities performed by UK notified bodies for medical devices placed on the market since January 1, 2021. Notified bodies must be located in an EU member state, or territory where there is a Mutual Recognition Agreement ("MRA"), that allows the marketing of medical devices that meet EU requirements, there is (currently) no such MRA between the UK and the EU.

Furthermore, on December 16, 2024, the UK government published an amendment to UK Medical Devices Regulations to clarify and strengthen the post-market surveillance requirements for medical devices in Great Britain. This amendment will come into force on June 16, 2025 and aims to facilitate greater traceability of incidents and trends enabling the MHRA to act swiftly when needed to address safety issues and support the entire health system in better protecting patients. In addition, the MHRA launched a consultation from November 14, 2024 to January 5, 2025 on proposals to update the pre-market requirements for medical devices in Great Britain, covering four topics, namely: (1) a new international reliance scheme to enable swifter market access for certain devices that have already been approved in a comparable regulator country; (2) the new UK Conformity Assessed ("UKCA") mark and, in particular, proposals to remove the requirement to place such UKCA marking on devices; (3) conformity assessment procedures for in vitro diagnostic devices; and (4) maintaining in UK law certain pieces of "assimilated" EU law which are due to sunset in 2025. This consultation builds on the MHRA's previous consultation between September and November 2021, and the UK government's response to that consultation which was published on June 26, 2022. The MHRA has stated that it will incorporate feedback to its recent consultation into new legislation on pre-market requirements for medical devices in Great Britain. The new legislation is expected to be implemented in 2026 and aims to enable greater international collaboration and practices, with more patient-centered, proportionate requirements for medical devices which are responsive to technological advances.

Coverage and Reimbursement

In the United States, our currently approved products are commonly treated as general supplies utilized in orthopedic surgery and if covered by third-party payors, are paid for as part of the surgical procedure. Our commercial success depends in part on the extent to which governmental authorities, private health insurers and other third-party payors provide coverage for and establish adequate reimbursement levels for the procedures during which our products are used. Failure by physicians, hospitals, ASCs and other users of our products to obtain sufficient coverage and reimbursement from third-party payors for procedures in which our products are used, or adverse changes in government and private third-party payors' coverage and reimbursement policies.

Based on our experience to date, third-party payors generally reimburse for the surgical procedures in which our products are used only if the patient meets the established medical necessity criteria for surgery. Some payors are moving toward a managed care system and control their healthcare costs by limiting authorizations for surgical procedures, including elective procedures using our devices. Although no uniform policy of coverage and reimbursement among payors in the United States exists and coverage and reimbursement for procedures can differ significantly from payor to payor, reimbursement decisions by particular third-party payors may depend upon a number of factors, including the payor's determination that use of a product is:

- a covered benefit under its health plan;
- appropriate and medically necessary for the specific indication;
- cost effective; and
- neither experimental nor investigational.

Third-party payors are increasingly auditing and challenging the prices charged for medical products and services with concern for upcoding, miscoding, using inappropriate modifiers, or billing for inappropriate care settings. Some third-party payors must approve coverage for new or innovative devices or procedures before they will reimburse healthcare providers who use the products or therapies. Even though a new product may have been cleared for commercial distribution by the FDA, we may find limited demand for the product unless and until reimbursement approval has been obtained from governmental and private third-party payors.

We believe the overall escalating cost of medical products and services being paid for by the government and private health insurance has led to, and will continue to lead to, increased pressures on the healthcare and medical device industry to reduce the costs of products and services. All third-party reimbursement programs are developing increasingly sophisticated methods of controlling healthcare costs through prospective reimbursement and capitation programs, group purchasing, redesign of benefits, requiring second opinions prior to major surgery, careful review of bills, encouragement of healthier lifestyles and other preventative services and exploration of more cost-effective methods of delivering healthcare.

In addition to uncertainties surrounding coverage policies, there are periodic changes to reimbursement levels. Third-party payors regularly update reimbursement amounts and also from time to time revise the methodologies used to determine reimbursement amounts. This includes routine updates to payments to physicians, hospitals and ASCs for procedures during which our products are used. These updates could directly impact the demand for our products.

Other U.S. Healthcare Laws

Device manufacturers are subject to additional healthcare regulation and enforcement by the federal government and by authorities in the states and foreign jurisdictions in which they conduct their business. Such laws include, without limitation, U.S. federal and state anti-kickback, fraud and abuse, false claims, consumer fraud, and transparency laws and regulations.

For example, the federal Anti-Kickback Statute prohibits, among other things, individuals or entities from knowingly and willfully offering, paying, soliciting or receiving remuneration, directly or indirectly, overtly or covertly, in cash or in kind to induce or in return for purchasing, leasing, ordering or arranging for or recommending the purchase, lease or order of any item or service reimbursable under Medicare, Medicaid or other federal healthcare programs. A person or entity does not need to have actual knowledge of this statute or specific intent to violate it in order to have committed a violation.

The federal civil and criminal false claims laws, including the civil False Claims Act, prohibit, among other things, any individual or entity from knowingly presenting, or causing to be presented, a false claim for payment to the federal government or knowingly making, using or causing to be made or used a false record or statement material to a false or fraudulent claim to the federal government. In addition, the government may assert that a claim

including items or services resulting from a violation of the federal Anti-Kickback Statute constitutes a false or fraudulent claim for purposes of the civil False Claims Act.

The Civil Monetary Penalty Laws impose penalties against any person or entity that, among other things, is determined to have presented or caused to be presented a claim to a federal healthcare program that the person knows or should know is for an item or service that was not provided as claimed or is false or fraudulent, or offering or transferring remuneration to a federal healthcare beneficiary that a person knows or should know is likely to influence the beneficiary's decision to order or receive items or services reimbursable by the government from a particular provider or supplier.

HIPAA created additional federal civil and criminal statutes that prohibit, among other things, knowingly and willfully executing a scheme to defraud any healthcare benefit program. Similar to the federal Anti-Kickback Statute, a person or entity does not need to have actual knowledge of the healthcare fraud statute implemented under HIPAA or specific intent to violate it in order to have committed a violation.

The federal Physician Payments Sunshine Act requires certain manufacturers of drugs, devices, biologics and medical supplies for which payment is available under Medicare, Medicaid or the Children's Health Insurance Program, with specific exceptions, to report annually to the CMS information related to payments or other transfers of value made to physicians (defined to include doctors, dentists, optometrists, podiatrists and chiropractors), certain non-physician practitioners (defined to include physician assistants, nurse practitioners, clinical nurse specialists, certified nurse anesthetists, anesthesiology assistants and certified nurse midwives) and teaching hospitals, and further requires applicable manufacturers and applicable group purchasing organizations to report annually to CMS ownership and investment interests held by physicians and their immediate family members.

Similar state and local laws and regulations may also restrict business practices in the medical device industries, such as state anti-kickback and false claims laws, which may apply to business practices, including but not limited to, research, distribution, sales and marketing arrangements and claims involving healthcare items or services reimbursed by non-government third-party payors, including private insurers, or by patients themselves; state laws that require device companies to comply with the industry's voluntary compliance guidelines and the relevant compliance guidance promulgated by the federal government, or otherwise restrict payments that may be made to healthcare providers and other potential referral sources; state laws and regulations that require manufacturers to file reports relating to pricing and marketing information or which require tracking gifts and other remuneration and items of value provided to physicians, other healthcare providers and entities; state laws that prohibit fee-splitting arrangements between companies and physicians and other healthcare professionals; and state and local laws that require the registration of sales representatives.

Violation of any of such laws or any other government regulations that apply may result in penalties, including, without limitation, civil and criminal penalties, damages, fines, additional reporting obligations, the curtailment or restructuring of operations, exclusion from participation in government healthcare programs and individual imprisonment.

Healthcare Reform

The United States is considering or has enacted a number of legislative and regulatory proposals to change the healthcare system in ways that could affect our ability to sell our products profitably. Among policy makers and payors in the United States, there is significant interest in promoting changes in healthcare systems with the stated goals of containing healthcare costs, improving quality or expanding access. Current and future legislative proposals to further reform healthcare or reduce healthcare costs may limit coverage of or lower reimbursement for the procedures associated with the use of our products. The cost containment measures that payors and providers are instituting and the effect of any healthcare reform initiative implemented in the future could impact our revenue from the sale of our products.

The implementation of the ACA in the United States, for example, has changed healthcare financing and delivery by both governmental and private insurers substantially, and affected medical device manufacturers significantly. The ACA imposed, among other things, provided incentives to programs that increase the federal government's comparative effectiveness research, and implemented payment system reforms including a national

pilot program on payment bundling to encourage hospitals, physicians and other providers to improve the coordination, quality and efficiency of certain healthcare services through bundled payment models. Additionally, the ACA has expanded eligibility criteria for Medicaid programs and created a new Patient-Centered Outcomes Research Institute to oversee, identify priorities in, and conduct comparative clinical effectiveness research, along with funding for such research.

In addition, other legislative changes have been proposed and adopted since the ACA was enacted. For example, the Budget Control Act of 2011, among other things, reduced Medicare payments to providers, effective on April 1, 2013 and, due to subsequent legislative amendments to the statute, will remain in effect through 2032 unless additional Congressional action is taken. Additionally, the American Taxpayer Relief Act of 2012, among other things, reduced Medicare payments to several providers, including hospitals, and increased the statute of limitations period for the government to recover overpayments to providers from three to five years.

The Medicare Access and CHIP Reauthorization Act of 2015 repealed the formula by which Medicare made annual payment adjustments to physicians and replaced the former formula with fixed annual updates and a new system of incentive payments which began in 2019 that are based on various performance measures and physicians' participation in alternative payment models, such as accountable care organizations. Each year, CMS updates Medicare payments for physician services through rulemaking, based on parameters established under law. In November 2024, CMS finalized a 2.83% decrease in the physician fee schedule conversion factor, a key aspect of physician payment rates under the Medicare program. This resulted in an average payment cut of 2.93% to physicians and other clinicians, which took effect on January 1, 2025 and remains in effect today, unless Congress takes additional action.

We expect additional state and federal healthcare reform measures to be adopted in the future, particularly given the recent change in administration, any of which could limit the amounts that federal and state governments will pay for healthcare products and services, which could result in reduced demand for our products or additional pricing pressure.

Data Privacy and Security Laws

Numerous state, federal and foreign laws, regulations and standards govern the collection, use, access to, confidentiality and security of health-related and other personal information, and could apply now or in the future to our operations or the operations of our partners. In the United States, numerous federal and state laws and regulations, including data breach notification laws, health information privacy and security laws and consumer protection laws and regulations govern the collection, use, disclosure, and protection of health-related and other personal information. In addition, certain foreign laws govern the privacy and security of personal data, including health-related data. Privacy and security laws, regulations, and other obligations are constantly evolving, may conflict with each other to complicate compliance efforts, and can result in investigations, proceedings, or actions that lead to significant civil and/or criminal penalties and restrictions on data processing.

Employees and Human Capital Resources

As of June 30, 2025, we had 61 full-time employees. None of our employees are represented by a labor union or party to a collective bargaining agreement. We consider our relationship with our employees to be good.

Our human capital resources objectives include, as applicable, identifying, recruiting, retaining, incentivizing, and integrating our existing and additional employees. The principal purposes of our equity incentive plans are to attract, retain, and motivate selected employees, consultants, and directors through the granting of stock-based compensation awards and cash-based performance bonus awards.

Facilities

Our corporate headquarters is located in Grand Rapids, Michigan where we lease a 7,000 square foot facility pursuant to a lease agreement that expires in January 2026. This facility includes warehouse, distribution and office space. We also leverage a third-party 3,000 square foot distribution facility in Modesto, California that distributes our implant systems in the western United States. We believe that this facility is sufficient to meet our current and

anticipated needs in the near term and that suitable additional space is available as needed to accommodate expansion of our operations and distribution activities.

Legal Proceedings

From time to time, we may be involved in various legal proceedings arising from ordinary course business activities. For example, we may face various claims brought by third parties and we may make claims or take legal actions to assert our rights, including intellectual property rights and claims relating to employment matters and our products. Any of these claims could cause us to incur substantial costs and, while we believe that we have a typical level of insurance coverage for our industry, our insurance carriers may deny coverage, may be inadequately capitalized to pay on valid claims, or our policy limits may be inadequate to fully satisfy any associated costs, damage awards or settlements. If this were to happen, the payment of any such awards could have a material adverse effect on our operations, cash flows and financial position. Additionally, any such claims, whether or not successful, could damage our reputation and business.

On February 28, 2024, we filed a complaint against Catalyst Orthoscience Inc. ("Catalyst") in the United States District Court for the District of Delaware, Case No. 1:24-cv-00266-JPM, alleging that Catalyst has infringed certain claims of our patents. In response to our suit, Catalyst filed a counterclaim claiming patent infringement of certain claims of a patent owned by Catalyst based on certain of our products. We believe we have substantial and meritorious defenses to Catalyst's counterclaim and intend to vigorously defend our position, including through the trial and appellate stages if necessary. As the counterclaim is ongoing, we are unable to determine the likelihood of an outcome or estimate a range of reasonably possible settlement, if any. Accordingly, we have not made an accrual for any possible loss. The outcome of any litigation, however, is inherently uncertain, and an adverse judgment or settlement in the counterclaim proceeding, if any, could have a material and adverse effect on our business, financial position, results of operations or cash flows. See "Risk Factors—Risks Related to our Intellectual Property" for further description of risks associated with this litigation and our intellectual property rights generally.

MANAGEMENT

The following table sets forth information regarding our executive officers and directors, including their ages as of June 30, 2025:

Name	Age	Position(s)
Executive Officers and Employee Directors		
Robert Ball	53	Chief Executive Officer and Executive Chairman
Jeffrey Points	48	Chief Financial Officer
Matthew Ahearn	56	Chief Operating Officer and Director
David L. Blue	64	Chief Customer Experience Officer
Non-Employee Directors and Director Nominee		
Paul Buckman[(1)(3)]	69	Director
Michael Carusi	60	Director
Geoff Pardo[(2)(3)]	53	Director
Kevin Sidow[(1)(2)]	67	Director
Casey Tansey[(2)(3)]	67	Director
Richard J. Buchholz[(1)(4)]	57	Director Nominee

(1) Member of our audit committee.
(2) Member of our compensation committee.
(3) Member of our nominating and corporate governance committee.
(4) Joined our board of directors immediately upon the effectiveness of the registration statement of which this prospectus is a part.

Executive Officers and Employee Directors

Robert Ball has served as our Chief Executive Officer since October 2020 and as our executive chairman of the board of directors since April 2015. Prior to joining Shoulder Innovations, Mr. Ball was a Principal at Imascap SA, a privately-held software company focusing on preoperative planning for shoulder arthroplasty, from July 2013 until its acquisition by Wright Medical Group, Inc. (now Stryker Corporation (NYSE: SYK)) in 2017. Prior to this, Mr. Ball held positions at Tornier NV, a global orthopedics company that was acquired by Wright Medical Group, Inc. (now Stryker Corporation), from August 2006 to December 2013, including Vice President, Research & Development, Regulatory and Clinical. Mr. Ball also previously held positions at Kinetikos Medical Inc., a privately-held company focusing on orthopedic implants, and DePuy, the Orthopaedics Company of Johnson & Johnson (NYSE: JNJ). Since its inception in January 2013, he has served as Chairman of Genesis Innovation, a company focused on early-stage medical device investing and development. Mr. Ball currently serves as a director for Genesis Innovation, Genesis Software, Genesis Investment Holdings, NSite LLC, Strados Labs LLC, and cultivate(MD) Holdings, LLC. Mr. Ball received a Bachelor of Science in Mechanical Engineering and a Master of Science in Mechanical Engineering from Kettering University. We believe Mr. Ball's extensive experience with medical technology companies and thorough knowledge of our industry and strategy as our Chief Executive Officer qualifies him to serve on our board of directors.

Jeffrey Points has served as our Chief Financial Officer since September 2023. Prior to joining Shoulder Innovations, Mr. Points worked at Cardiovascular Systems, Inc. (Nasdaq: CSII), a medical device company that developed solutions for treating peripheral and coronary artery disease, from 2007 to 2023, where he held various roles, including most recently as Chief Financial Officer from 2018 to 2023. Mr. Points received an M.B.A. from the University of St. Thomas and a B.A. in Business focusing on Finance and Accounting from Bethel University.

Matthew Ahearn has served as our Chief Operating Officer since October 2020 and a member of our board of directors since February 2017. Prior to this role, Mr. Ahearn was our Chief Executive Officer and President from February 2017 to October 2020 and our Chief Financial Officer and Chief Operating Officer from November 2020 to September 2023. Mr. Ahearn also currently serves as a director for HAPPE Spine, LLC, SpinTech MRI, Inc.,

CPD to SPD, LLC, cultivate(MD) Holdings, LLC, and Genesis Investment Holdings. He received a Bachelor of Arts in Economics and Management from Albion College. We believe that Mr. Ahearn's deep understanding of our business and extensive experience in financial management and operational leadership within the healthcare and technology sectors qualifies him to serve on our board of directors.

David L. Blue has served as our Chief Customer Experience Officer since October 2023 and previously served as our Chief Commercial Officer from May 2017 to October 2023. From May 2017 to October 2021, Mr. Blue served as Director, VP, Sales & Marketing at Genesis Innovation and as Chief Commercial Officer at Conventus Orthopedics from July 2013 to May 2017. Since January 2018, Mr. Blue has also been a member of the board of directors of cultivate(MD) Capital Funds and currently serves as a director of cultivate(MD) Holdings, LLC. He received a Bachelor of Arts in Business Administration and Marketing from Augustana University.

Non-Employee Directors and Director Nominee

Paul Buckman has served as a member of our board of directors since October 2020. Mr. Buckman has served as the Chief Executive Officer and a director of Rhythmlink International, LLC, a privately-held medical equipment manufacturer, since April 2024. He previously served as President of North America for LivaNova PLC (Nasdaq: LIVN), a publicly-held global medical technology company, from May 2017 to March 2023, and as the Chief Executive Officer of Coventus Orthopaedics, Inc. from 2013 to 2016. Mr. Buckman also currently serves as a director for Ablative Solutions, Inc., Helius Medical Technologies, Inc. (Nasdaq: HSDT), MiroMatrix Medical, Inc. and NeuroOne Medical Technologies Corporation (Nasdaq: NMTC). He received a Bachelor of Business Administration and a Master of Business Administration from Western Michigan University. We believe that Mr. Buckman's extensive background in the healthcare industry, directorship and management experience qualifies him to serve on our board of directors.

Michael Carusi has served on our board of directors since October 2020. Since 2013, Mr. Carusi has served as a Managing Partner at Lightstone Ventures. Concurrently, Mr. Carusi serves as a member of the board of directors for several private companies, including Cala Health, Inc., Willow, Inc., Tallac Therapeutics, Inc., Allay Therapeutics, Inc., Apreo Health, Inc., Medisix Therapeutics, Inc., and Endogastric Solutions, Inc. Mr. Carusi received a B.S. in Mechanical Engineering from Lehigh University and an M.B.A. from the Tuck School of Business. We believe that Mr. Carusi's extensive experience in venture capital and his background in healthcare and medical technology investments qualifies him to serve on our board of directors.

Geoff Pardo has served as a member of our board of directors since February 2023. Since June 2011, he has also served as the President and General Partner of Gilde Healthcare US Inc. Concurrently, he serves on the boards of Nalu Medical, Mainstay Medical Holding plc, Ablative Solutions Inc., GT Medical Technologies, Inc. and Alleviant Medical. Previously, Mr. Pardo also served on the board of directors, including various committees, for several public and formerly public companies, including Inari Medical Inc. from March 2018 to May 2021, Axonics Inc. (Nasdaq: AXNX) from July 2017 to April 2019, Eargo Inc. (Nasdaq: EAR) from July 2020 to July 2021, Vapotherm, Inc. from March 2014 to October 2020 and CVRx Inc. (Nasdaq: CVRX) from August 2016 to June 2022. He received a B.A. from Brown University and an M.B.A from The Wharton School of Business. We believe Mr. Pardo's experience leading and managing a medical technology company, as well as his healthcare industry knowledge and his experience serving on the board of directors of other companies, qualifies him to serve on our board of directors.

Kevin Sidow has served as a member of our board of directors since October 2020. From February 2008 until his retirement in July 2017, Mr. Sidow was the President and Chief Executive Officer of Moximed, Inc., a medical device company focused on knee osteoarthritis. He currently serves on the board of directors of Carlsmed Inc. (Nasdaq: CARL) and several private companies in the healthcare industry, including Hip Innovation Technology, LLC, and Mantis Health Inc. He received a B.S. in Accounting from West Virginia University. We believe Mr. Sidow's extensive experience in the medical device industry and leadership positions qualifies him to serve on our board of directors.

Casey Tansey has served on our board of directors since October 2020. Since 2014, Mr. Tansey has served as the Managing Partner of USVP Management Company. Mr. Tansey has been a member of the board of directors of

Inspire Medical Systems, Inc.(NYSE: INSP) since 2007 and of UCSF MTVP Investment Advisory since 2017. He has also served on the boards of several other medical technology and healthcare-related companies, including Cagent Vascular, HeartFlow, Inc., Highlife Medical, Inc., Luminopia, Inc., MicroTransponder, Inc., Neuros Medical, NeoChord, Inc., and ShiraTronics. He received a B.S. in Business and a M.B.A. in Business from the College of Notre Dame. We believe Mr. Tansey's significant investing experience and knowledge of our industry qualifies him to serve on our board of directors.

Richard J. Buchholz joined our board of directors upon the effectiveness of the registration statement of which this prospectus is a part. Mr. Buchholz has served as Chief Financial Officer of Inspire Medical Systems, Inc. (NYSE: INSP) since 2014. Prior to that role, Mr. Buchholz served as the Chief Financial Officer, Secretary and Treasurer of superDimension, Ltd., a medical device manufacturer (which was acquired by Covidien plc in 2012), from 2006 to 2013. Mr. Buchholz holds a B.B.A from the University of Wisconsin, Eau Claire and is a Certified Public Accountant (inactive). We believe Mr. Buchholz's extensive knowledge of our industry, financial management skills and experience as an executive of a publicly-traded medical technology company qualifies him to serve on our board of directors.

Family Relationships

There are no family relationships among any of our directors or executive officers.

Board Composition

Director Independence

Our board of directors currently consists of seven members and will consist of eight members following the completion of this offering. Our board of directors has determined that all of our directors, other than Mr. Ball and Mr. Ahearn, qualify as "independent" directors in accordance with the rules of the New York Stock Exchange. Mr. Ball and Mr. Ahearn are not considered independent by virtue of their position as our Chief Executive Officer and Chief Operating Officer, respectively. Under the New York Stock Exchange rules, the definition of independence includes a series of objective tests, such as that the director is not, and has not been for at least three years, one of our employees and that neither the director nor any of his or her family members has engaged in various types of business dealings with us. In addition, as required by the New York Stock Exchange rules, our board of directors has made a subjective determination as to each independent director that no relationship exists that, in the opinion of our board of directors, would interfere with the exercise of independent judgment in carrying out the responsibilities of a director. In making these determinations, our board of directors reviewed and discussed information provided by the directors and us with regard to each director's business and personal activities and relationships as they may relate to us and our management.

Classified Board of Directors

In accordance with our amended and restated certificate of incorporation, which will be effective immediately prior to the completion of this offering, our board of directors will be divided into three classes with staggered three-year terms. At each annual general meeting of stockholders, the successors to directors whose terms then expire will be elected to serve from the time of election and qualification until the third annual meeting following election. Our directors will be divided among the three classes as follows:

- The Class I directors will be Mr. Ball, Mr. Carusi and Mr. Pardo, and their terms will expire at the annual meeting of stockholders to be held in 2026;
- The Class II directors will be Mr. Buckman, Mr. Sidow and Mr. Tansey, and their terms will expire at the annual meeting of stockholders to be held in 2027; and
- The Class III directors will be Mr. Ahearn and Mr. Buchholz, and their terms will expire at the annual meeting of stockholders to be held in 2028.

We expect that any additional directorships resulting from an increase in the number of directors will be distributed among the three classes so that, as nearly as possible, each class will consist of one-third of the directors.

The division of our board of directors into three classes with staggered three-year terms may delay or prevent a change of our management or a change in control.

Leadership Structure of the Board

Our amended and restated bylaws and corporate governance guidelines to be in place immediately prior to the consummation of this offering will provide our board of directors with flexibility to combine or separate the positions of Chair of the board of directors and Chief Executive Officer and to implement a lead director in accordance with its determination regarding which structure would be in the best interests of our company.

Our board of directors has concluded that our current leadership structure is appropriate at this time. However, our board of directors will continue to periodically review our leadership structure and may make such changes in the future as it deems appropriate.

Role of Board in Risk Oversight Process

Risk assessment and oversight are an integral part of our governance and management processes. Our board of directors encourages management to promote a culture that incorporates risk management into our corporate strategy and day-to-day business operations. Management discusses strategic and operational risks at regular management meetings and conducts specific strategic planning and review sessions during the year that include a focused discussion and analysis of the strategic risks facing us. Throughout the year, senior management reviews these strategic risks with the board of directors at regular board meetings as part of management presentations that focus on particular business functions, operations, or strategies, and presents the steps taken by management to mitigate or eliminate such risks.

Our board of directors does not have a standing risk management committee, but rather administers this oversight function directly through our board of directors as a whole, as well as through various standing committees of our board of directors that address risks inherent in their respective areas of oversight. While our board of directors is responsible for monitoring and assessing strategic risk exposure, our audit committee is responsible for overseeing company-wide and information security risk assessment processes, our major financial risk exposures, regulatory compliance, and the steps our management has taken to monitor and control these risks and exposures. The audit committee then reviews these matters with the full board of directors as part of the audit committee's reports at regular board meetings. The audit committee also approves or disapproves any related-party transactions. Our nominating and corporate governance committee monitors the effectiveness of our corporate governance guidelines and risks related to environmental, social, and governance issues and oversees management succession planning. Our compensation committee assesses and monitors whether any of our compensation policies and programs has the potential to encourage excessive risk-taking.

Board Committees

Our board of directors has established an audit committee, a compensation committee, and a nominating and corporate governance committee. Our board of directors may establish other committees to facilitate the management of our business. The composition and functions of each committee are described below. Members serve on these committees until their resignation or until otherwise determined by our board of directors. Each committee has a written charter that satisfies the applicable rules and regulations of the Securities and Exchange Commission and New York Stock Exchange rules, which we will post on our website at www.shoulderinnovations.com upon the completion of this offering.

Audit Committee

Our audit committee oversees our corporate accounting and financial reporting process. Among other matters, the audit committee is responsible for:

- appointing, compensating, retaining, evaluating, terminating, and overseeing our independent registered public accounting firm;
- discussing with our independent registered public accounting firm their independence from management;

- reviewing with our independent registered public accounting firm the scope and results of their audit;
- approving all audit and permissible non-audit services to be performed by our independent registered public accounting firm;
- overseeing the financial reporting process and discussing with management and our independent registered public accounting firm the interim and annual financial statements that we file with the SEC;
- reviewing and monitoring our accounting principles, accounting policies, financial and accounting controls, and compliance with legal and regulatory requirements;
- reviewing our policies on risk assessment and risk strategy and management, including risk policies and risk mitigation strategies by management;
- reviewing related party transactions; and
- establishing procedures for the confidential anonymous submission of concerns regarding questionable accounting, internal controls, or auditing matters.

Our audit committee consists of Mr. Buchholz, Mr. Buckman, and Mr. Sidow. Our board of directors has determined that Mr. Buchholz, Mr. Buckman and Mr. Sidow are independent under the New York Stock Exchange rules and Rule 10A-3(b)(1) of the Exchange Act. The chair of our audit committee is Mr. Buchholz. Our board of directors has determined that Mr. Buchholz, Mr. Buckman and Mr. Sidow are each financially literate. In addition, our board of directors has determined that Mr. Buchholz, Mr. Buckman and Mr. Sidow are each an "audit committee financial expert" as such term is currently defined in Item 407(d)(5) of Regulation S-K. Our board of directors has also determined that each member of our audit committee can read and understand fundamental financial statements, in accordance with applicable requirements.

Compensation Committee

Our compensation committee oversees policies relating to compensation and benefits of our officers and employees. Among other matters, the compensation committee is responsible for:

- reviewing and approving, or recommending that our board of directors approve, the compensation of our Chief Executive Officer and executive officers;
- appointing and overseeing any compensation consultants;
- reviewing and approving or making recommendations to our board of directors regarding our incentive compensation and equity-based plans;
- reviewing and making recommendations to our board of directors with respect to director compensation;
- reviewing and discussing annually with management our "Compensation Discussion and Analysis" disclosure if and to the extent then required by SEC rules; and
- preparing the compensation committee report if and to the extent then required by SEC rules.

Our compensation committee consists of Mr. Pardo, Mr. Sidow, and Mr. Tansey. Our board of directors has determined that all members are independent under the New York Stock Exchange rules and that Mr. Sidow is a non-employee director, as defined by Rule 16b-3 promulgated under the Exchange Act. The chair of our compensation committee is Mr. Sidow.

Nominating and Corporate Governance Committee

Our nominating and corporate governance committee oversees policies relating to our corporate governance. Among other matters, the nominating and corporate governance committee is responsible for:

- identifying individuals qualified to become members of our board of directors, consistent with criteria approved by our board of directors;
- reviewing and making recommendations to our board with respect to management succession planning;
- evaluating the overall effectiveness of our board of directors and its committees; and
- reviewing developments in corporate governance compliance and developing and recommending to our board of directors a set of corporate governance guidelines and principles.

Our nominating and corporate governance committee consists of Mr. Buckman, Mr. Pardo, and Mr. Tansey. Our board of directors has determined that all members are independent under the New York Stock Exchange rules. The chair of our nominating and corporate governance committee is Mr. Pardo.

Compensation Committee Interlocks and Insider Participation

None of the members of our compensation committee is currently, or has been at any time, one of our executive officers or employees. None of our executive officers currently serves, or has served during the last year, as a member of the board of directors or compensation committee of any entity that has one or more executive officers serving as a member of our board of directors or on our compensation committee.

Code of Business Conduct and Ethics

In connection with this offering, our board of directors adopted a written code of business conduct and ethics that applies to all of our directors, officers, and employees, including our principal executive officer, principal financial officer, and principal accounting officer or controller, or persons performing similar functions, and agents and representatives, which will be effective upon the completion of this offering. The full text of our code of business conduct and ethics will be posted on our website at www.shoulderinnovations.com upon the completion of this offering. The audit committee of our board of directors will be responsible for overseeing our code of business conduct and ethics and any waivers applicable to any director, executive officer, or employee. We intend to disclose any future amendments to certain provisions of our code of business conduct and ethics, or waivers of such provisions applicable to our directors, officers, and employees, including our principal executive officer, principal financial officer, principal accounting officer or controller, or persons performing similar functions, and agents and representatives, on our website identified above or in public filings.

EXECUTIVE AND DIRECTOR COMPENSATION

Executive Compensation

This section discusses the material components of the executive compensation program for our executive officers who are named in the "2024 Summary Compensation Table" below. In 2024, our "named executive officers" and their positions were as follows:

- Robert Ball, our Chief Executive Officer;
- Jeffrey Points, our Chief Financial Officer; and
- David Blue, our Chief Commercial Officer.

This discussion may contain forward-looking statements that are based on our current plans, considerations, expectations and determinations regarding future compensation programs. Actual compensation programs that we adopt following the completion of this offering may differ materially from the currently planned programs summarized in this discussion.

2024 Summary Compensation Table

The following table sets forth information concerning the compensation of our named executive officers for the year ended December 31, 2024.

Name and Principal Position	Year	Salary ($)	Non-Equity Incentive Plan Compensation[1] ($)	All Other Compensation[2] ($)	Total ($)
Robert Ball	2024	449,613	164,701	13,800	628,114
Chief Executive Officer					
Jeffrey Points	2024	363,440	134,912	9,460	507,812
Chief Financial Officer					
David Blue	2024	349,939	133,533	13,800	497,272
Chief Commercial Officer					

(1) Amounts reflect actual cash bonuses earned for the 2024 performance year, based on the achievement of goals under our quarterly bonus program for 2024.
(2) Amounts reflect company-paid matching contributions to 401(k) plan accounts.

Narrative to Summary Compensation Table

2024 Salaries

The named executive officers each receive a base salary to compensate them for services rendered to our company. The base salary payable to each named executive officer is intended to provide a fixed component of compensation reflecting the executive's skill set, experience, role and responsibilities.

For fiscal year 2024, Messrs. Ball, Points and Blue had annual salaries of $449,613, $363,440 and $349,939, respectively. Their fiscal year 2025 salaries are $467,943, $378,144 and $364,455, respectively.

The actual base salaries paid to each named executive officer in 2024 are set forth above in the Summary Compensation Table in the column titled "Salary."

2024 Bonuses

In 2024, our named executive officers were eligible to earn quarterly cash incentive bonuses based on our achievement of pre-determined revenue and adjusted EBITDA goals, as well as the achievement of individual performance goals. Under the 2024 bonus program, the revenue, adjusted EBITDA and individual performance

goals were weighted as 60%, 20% and 20% of each executive's quarterly bonus opportunity. The executives were eligible to receive up to 200% of the executives' target bonus opportunity upon achievement of maximum financial goals.

For 2024, the target bonuses for Messrs. Ball, Points and Blue were 50% of each officer's respective quarterly base salary.

The actual annual cash bonuses awarded to each executive for 2024 performance are set forth above in the Summary Compensation Table in the column titled "—Non-Equity Incentive Plan Compensation."

Equity Compensation

We have historically used stock options as the primary incentive for long-term compensation to our employees (including our named executive officers) because they are able to profit from stock options only if our stock price increases relative to the stock option's exercise price, which generally is set at or above the fair market value of our common stock as of the applicable grant date. Prior to this offering, we maintained the Stock Option Plan. As mentioned below, in connection with this offering and the adoption of the 2025 Plan, no further awards will be granted under the Stock Option Plan.

In 2024, we did not grant stock options or any other equity awards to our named executive officers. However, in July 2024, we repriced certain of our outstanding stock options granted in 2023, including those granted to our named executive officers, that had a per share exercise price greater than or equal to $0.13, which was determined by our board of directors to be the per-share fair market value of our common stock at the time of the repricing.

In connection with this offering, we adopted a 2025 Incentive Award Plan, referred to below as the 2025 Plan, in order to facilitate the grant of cash and equity incentives to directors, employees (including our named executive officers) and consultants of our company and certain of its affiliates and to enable our company and certain of its affiliates to obtain and retain services of these individuals, which is essential to our long-term success. For additional information about the 2025 Plan, please see the section titled "—Equity Incentive Plans" below.

Other Elements of Compensation

Retirement Plan

We currently maintain a 401(k) retirement savings plan, or the 401(k) plan, for our employees, including our named executive officers, who satisfy certain eligibility requirements. Our named executive officers are eligible to participate in the 401(k) plan on the same terms as other full-time employees. The Code allows eligible employees to defer a portion of their compensation, within prescribed limits, through contributions to the 401(k) plan. Currently, we provide discretionary matching contributions equal to 100% of a participant's salary deferrals up to 3% of the participant's compensation and equal to 50% of the next 2% of the employee's contributions, subject to limits provided in the Code. These matching contributions are fully vested as of the date on which the contribution is made. We believe that providing a vehicle for retirement savings through our 401(k) plan adds to the overall desirability of our compensation package and further incentivizes our employees, including our named executive officers, in accordance with our compensation policies.

The amounts of such company matching contributions paid by us on behalf of each named executive officer are set forth above in the Summary Compensation Table in the column titled "—All Other Compensation."

Health and Welfare and Perquisites

All of our full-time employees, including our named executive officers, are eligible to participate in our health and welfare plans, including medical, dental and vision benefits; health and dependent care flexible spending accounts; short-term and long-term disability insurance; accident insurance; and life and AD&D insurance.

We did not provide any perquisites or special personal benefits to our named executive officers in fiscal year 2024, but our compensation committee may from time to time approve them in the future when our compensation committee determines that such perquisites are necessary or advisable to fairly compensate or incentivize our

employees. We believe the perquisites and benefits described above are necessary and appropriate to provide a competitive compensation package to our named executive officers.

No Tax Gross-Ups Paid

We did not make gross-up payments to cover our named executive officers' personal income taxes that may pertain to any of the compensation paid or provided by us during 2024.

Clawback Policy

In connection with this offering, our board of directors adopted a compensation recovery policy that is compliant with the New York Stock Exchange rules, as required by the Dodd-Frank Act.

Outstanding Equity Awards at Fiscal Year-End

The following table summarizes the number of shares of common stock underlying outstanding equity incentive plan awards for each named executive officer as of December 31, 2024, as adjusted for the reverse stock split as described in Note 1 to our financial statements included elsewhere in this prospectus. Each equity award listed in the following table was granted under the Stock Option Plan and covers our common stock.

			Option Awards			
Name	**Grant Date**	**Vesting Commencement Date**	**Number of Securities Underlying Unexercised Options Exercisable (#)**	**Number of Securities Underlying Unexercised Options Unexercisable(#)**	**Option Exercise Price ($)**	**Option Expiration Date**
Robert Ball	4/30/2019[(1)]	--	15,723	--	$1.07	4/30/2029
	2/11/2020[(1)]	--	1,648	--	$1.03	2/11/2030
	12/16/2020[(1)]	--	183,438	--	$2.10	12/16/2030
	7/12/2024[(2)]	5/17/2023	130,528	199,228	$2.48	5/17/2033
Jeffrey Points	7/12/2024[(2)]	9/25/2023	6,632	72,952	$2.48	11/16/2033
David L. Blue	12/16/2020[(1)]	--	78,616	--	$2.10	12/16/2030
	7/12/2024[(2)]	4/19/2023	11,687	16,362	$2.48	5/9/2033

(1) This option is fully vested and exercisable as of December 31, 2024.

(2) This option vests and becomes exercisable with respect to 1/4 of the total number of shares underlying the option on the one-year anniversary of the vesting commencement date, with the remaining number of shares vesting in equal increments on the last day of each month over the next 36 months. The grant date set forth in the table above reflects the date on which the option was repriced; the option originally was granted for Mr. Ball on May 17, 2023, for Mr. Points on November 16, 2023 and for Mr. Blue on May 9, 2023.

Executive Compensation Arrangements

Employment Agreements

The following is a description of the employment agreements that were in effect in 2024 with each of our named executive officers.

Each of the agreements is an "at will" employment agreement, pursuant to which Messrs. Ball, Points and Blue are entitled to receive a base salary of $400,000, $360,000 and $300,000, respectively, as well as an annual performance bonus of up to 50% of base salary. This annual performance bonus is paid on a quarterly basis and is based on both individual and company performance, as determined by our board of directors, and subject to the applicable executive's continued employment through the payment date. Each executive is also eligible to participate in our standard employee benefit plans.

If the employment of any executive is terminated by us without "cause" or by the executive for "good reason" (each, as defined in the applicable employment agreement), he will be entitled to receive (i) continued payments of base salary for 12 months following the date of termination and (ii) reimbursement for the cost of

COBRA premiums for up to 12 months. These severance payments and benefits are subject to the executive's timely executive and non-revocation of a release of claims in our favor.

Pursuant to the terms of his employment agreement, we granted Mr. Points an option to purchase 106,113 shares of our common stock (on a post-reverse stock split basis). The option vests and becomes exercisable with respect to one fourth of the total number of shares underlying the option on the one-year anniversary of the effective date of his employment agreement, with the remaining number of shares vesting in equal increments on the last day of each month over the next 36 months. In addition, it will become fully vested and exercisable immediately prior to a change in control, subject to Mr. Points' timely execution and non-revocation of a general release of claims.

Each employment agreement includes a non-disclosure of confidential information provision, which applies indefinitely. The employment agreements for Messrs. Ball and Blue also include a non-competition provision and an employee and customer non-solicitation provision, both of which apply during employment and for either twelve months thereafter, in the event of a termination by us for "cause" or by the executive without "good reason," or six month months thereafter, in the event of a termination by us without "cause" or by the executive for "good reason." The employment agreement for Mr. Points includes an employee and customer non-solicitation provision, which applies during employment and for twelve months thereafter.

Further, the employment agreement with Mr. Points provides that in the event any "parachute payments" (within the meaning of Section 280G of the Code) become payable to Mr. Points that are subject to excise taxes under Section 4999 of the Code, we will provide Mr. Points with an additional amount sufficient to cover any such taxes up to $500,000.

Option Acceleration Policy

We maintain a policy that provides options granted under our Stock Option Plan will become fully vested and exercisable if the optionholder's employment is terminated by us without "cause" or by the optionholder for "good reason", in either case, within the period starting three months prior to, and ending 12 months after, a "change in control" (each, as defined in the policy).

Director Compensation

During the fiscal year ended December 31, 2024, our non-employee directors were Messrs. Buckman, Carusi, Pardo, Sidow and Tansey. In 2024, Messrs. Buckman and Sidow each earned $40,000 for services rendered as members of our board of directors. In addition, Mr. Sidow received an option covering 8,909 shares (on a post-reverse stock split basis). This option vests and becomes exercisable in 12 substantially equal installments, with the first installment vesting on September 30, 2025 and the remaining installments vesting on the last day of each fiscal quarter thereafter. Messrs. Carusi, Pardo and Tansey did not receive any compensation for their services in 2024 and did not hold any outstanding equity awards as of December 31, 2024.

2024 Director Compensation Table

The table below shows compensation paid to Messrs. Buckman and Sidow for services rendered during 2024.

Name	Fees Earned or Paid in Cash ($)	Option Awards[1] ($)	Total ($)
Paul Buckman	40,000	--	40,000
Kevin Sidow	40,000	18,069	58,069

(1) Amounts reflect the full grant-date fair value of unit options granted during 2024 computed in accordance with ASC Topic 718, rather than the amounts paid to or realized by the named individual. We provide information regarding the assumptions used to calculate the value of all option awards made to executive officers in Note 11 to our audited financial statements included elsewhere in this prospectus.

The table below shows the aggregate numbers of option awards (exercisable and unexercisable) and unvested stock awards held as of December 31, 2024 by each non-employee director who was serving as of December 31,

2024, as adjusted for the reverse stock split as described in Note 1 to our financial statements included elsewhere in this prospectus.

Name	Options Outstanding at Fiscal Year End
Paul Buckman	46,748
Kevin Sidow	55,658

Director Compensation Program

In connection with this offering, we adopted, and our stockholders approved, a compensation program (the "Director Compensation Program"), which will become effective upon the closing of this offering. The Director Compensation Program consists of annual retainer fees and long-term equity awards for certain of our non-employee directors (each, an "Eligible Director"). The material terms of the Director Compensation Program are described below.

The Director Compensation Program consists of the following components:

Cash Compensation:

- Annual Retainer: $45,000
- Annual Committee Chair Retainer:
  - Audit: $20,000
  - Compensation: $15,000
  - Nominating and Governance: $10,000
- Annual Committee Member (Non-Chair) Retainer:
  - Audit: $10,000
  - Compensation: $7,500
  - Nominating and Corporate Governance: $5,000
- Lead Independent Director: $35,000
- Non-Executive Chair: $40,000

The annual cash retainers will be paid in quarterly installments in arrears. Annual cash retainers will be pro-rated for any partial calendar quarter of service.

Equity Compensation:

- *2025 Awards:* Our board of directors has approved the grant of RSU awards pursuant to the 2025 Plan to our non-employee directors, which grants will become effective in connection with the completion of this offering (the "2025 Awards"). The dollar-denominated value of each award is $115,000 and the aggregate dollar-denominated value of these awards is $690,000. The aggregate number of shares of our common stock that will be subject to the 2025 Awards will be 46,000 shares of common stock based on the $15.00 initial public offering price per share of our common stock in this offering.

  The 2025 Awards will vest in full on the date of the annual meeting of stockholders to be held in 2026, or, if earlier, the first anniversary of the closing of this offering, subject to continued service through the applicable vesting date.

- *Annual Grant*: An Eligible Director who is serving on our board of directors as of the date of an annual meeting of stockholders (beginning with calendar year 2026) automatically shall be granted, on the date of such annual meeting, an RSU award with an aggregate value of $115,000. The number of RSUs subject to the award will be determined by dividing the value of the award by the price per share of the company's common stock on the applicable grant date.

  Each Annual Grant will vest in full on the earlier to occur of the first anniversary of the grant date and the date of the next annual meeting following the grant date, subject to continued service.

In addition, each equity award granted to an Eligible Director under the Director Compensation Program will vest in full immediately prior to the occurrence of a "change in control" (as defined in the 2025 Plan). Compensation under the Director Compensation Program will be subject to the annual limits on non-employee director compensation set forth in the 2025 Plan.

Equity Incentive Plans

The following summarizes the material terms of the Stock Option Plan and the 2025 Plan, which is the long-term incentive compensation plan in which our named executive officers will be eligible to participate following the consummation of this offering. In addition, the following summarizes the material terms of the ESPP, with is a tax-qualified employee stock purchase plan in which our named executive officers will be eligible to participate following the consummation of this offering.

Stock Option Plan

Prior to this offering, we maintained the Stock Option Plan. In connection with this offering, we amended and restated the Stock Option Plan. The material terms of the Stock Option Plan, as amended and restated, are summarized below. Following the effectiveness of the 2025 Plan, the Stock Option Plan terminated, and we will not make any further awards under the Stock Option Plan.

However, any outstanding awards granted under the Stock Option Plan will remain outstanding, subject to the terms of the Stock Option Plan and applicable award agreements. Shares of our common stock subject to awards granted under the Stock Option Plan that expire unexercised or are cancelled, terminated or forfeited in any manner without issuance of shares thereunder following the effective date of the 2025 Plan will become available for issuance under the 2025 Plan in accordance with its terms.

A total of 2,185,560 shares of our common stock are reserved for issuance under the Stock Option Plan.

Eligibility and Administration. Employees and other individuals selected by our board of directors are eligible to receive awards under the Stock Option Plan. The Stock Option Plan is administered by our board of directors, which may delegate its duties and responsibilities to a committee as it deems appropriate. Following the completion of this offering, our board of directors will delegate its duties to administer the plan to our compensation committee with respect to participants who are not non-employee directors (references in this summary to "board of directors" will mean the compensation committee, to the extent authority has been delegated). The board of directors has the authority to determine who will be granted options, what type of options will be granted and in what amount, when and how options will be granted, the provisions of each option and the exercise price applicable to each option; to approve forms of award agreement for use under the Stock Option Plan; to accelerate the vesting or exercisability of any award; to submit amendments to the Stock Option Plan for stockholder approval; to adopt any procedures or subplans necessary to permit non-U.S. participation in the Stock Option Plan; to construe and interpret the Stock Option Plan; and to prescribe, amend and rescind rules and regulations relating to the Stock Option Plan.

Awards. The Stock Option Plan provides for the grant of nonqualified stock options and incentive stock options. Each option under the Stock Option Plan is evidenced by a separate agreement between our company and the participant, which evidences the terms and conditions of the options, including any applicable vesting and payment terms and post-termination exercise limitations. The following types of options have been granted under the Stock Option Plan:

- Nonqualified Stock Options. Nonqualified stock options provide for the purchase of shares of our common stock in the future at an exercise price set on the grant date. The exercise price of a stock option is fixed by the board of directors and may not be less than 100% of the fair market value of the underlying share on the date of grant. The term of a stock option is determined by our board of directors, but may not exceed ten years. Vesting conditions determined by our board of directors may apply to stock options and may include the occurrence of certain events, the passage of a specified period of time, and/or other fulfillment of certain conditions.

- Incentive Stock Options. Incentive stock options are designed to comply with the provisions of the Code and are subject to specified restrictions contained in the Code applicable to incentive stock options. Among such restrictions, incentive stock options must have an exercise price of not less than the fair market value of a share of common stock on the date of grant, may only be granted to employees, lose ISO status within a specified period of time following the participant's termination of employment, and must be exercised within ten years after the date of grant. In the case of an incentive stock option granted to an individual who owns (or is deemed to own) more than 10% of the total combined voting power of all classes of our capital stock on the date of grant, the exercise price must be at least 110% of the fair market value of a share of common stock on the date of grant and the incentive stock option must expire on the fifth anniversary of the date of its grant.

Certain Transactions. In the event of certain transactions and events affecting our common stock, such as stock dividends, stock splits, or a combination or other change in shares of our common stock, our board of directors may, in order to prevent the diminution or enlargement of benefits or potential benefits, adjust the number and class of shares that may be issued under the Stock Option Plan, and adjust the number, class and price of shares underlying outstanding options. In the event of a dissolution or liquidation, all outstanding awards will terminate. In the event of a merger or a change in control affecting the ownership of our company, all outstanding awards may be converted, replaced or substituted by the successor or acquiring corporation. If the successor or acquiring corporation refuses to assume, convert, replace or substitute options granted under the Stock Option Plan, then all outstanding awards will automatically become fully vested and, for a period of 15 days, exercisable, after which time any outstanding awards with automatically terminate.

Repricing. The administrator may, without approval of the stockholders, reduce the exercise price of any stock option, or cancel any stock option in exchange for cash, other awards or stock options with an exercise price per share that is less than the exercise price per share of the original stock options.

Plan Amendment and Termination. Our board of directors may suspend or terminate the Stock Option Plan or any portion thereof at any time and may also amend the Stock Option Plan at any time in such respects as our board of directors may deem necessary or advisable, provided that no such amendment shall be made without stockholder approval to the extent such approval is necessary and desirable to comply with applicable law. Further, no such amendment, suspension or termination shall impair the rights of participants under outstanding options without the consent of the affected participants.

As described above, the Stock Option Plan terminated as of the effective date of the 2025 Plan.

2025 Incentive Award Plan

In connection with this offering, we adopted, and our stockholders approved, the 2025 Plan, under which we may grant equity and cash incentive awards to eligible service providers in order to attract, motivate and retain the talent for which we compete.

Eligibility and Administration. Our employees, consultants and directors, and employees, consultants and directors of our subsidiaries, are eligible to receive awards under the 2025 Plan. The 2025 Plan is administered by our board of directors with respect to awards to non-employee directors and by our compensation committee with respect to other participants, each of which may delegate its duties and responsibilities to committees of our directors and/or officers (referred to collectively as the plan administrator below), subject to certain limitations that may be imposed under the 2025 Plan, Section 16 of the Exchange Act, stock exchange rules and/or other applicable laws. The plan administrator has the authority to take all actions and make all determinations under the 2025 Plan, to interpret the 2025 Plan and award agreements and to adopt, amend and repeal rules for the administration of the 2025 Plan as it deems advisable. The plan administrator also has the authority to determine which eligible service providers receive awards, grant awards and set the terms and conditions of all awards under the 2025 Plan, including any vesting and vesting acceleration provisions, subject to the conditions and limitations in the 2025 Plan.

Limitation on Awards and Shares Available. The initial aggregate number of shares of our common stock that is available for issuance under the 2025 Plan is equal to 2,493,739 shares.

In addition, the number of shares of our common stock available for issuance under the 2025 Plan is subject to an annual increase on the first day of each calendar year beginning January 1, 2026 and ending on and including January 1, 2035, equal to the lesser of (i) 5% of the aggregate number of shares of our common stock outstanding on the final day of the immediately preceding calendar year and (ii) such smaller number of shares as is determined by our board of directors. The maximum number of shares that may be issued pursuant to the exercise of incentive stock options ("ISOs"), granted under the 2025 Plan, is 100,000,000. Any shares issued pursuant to the 2025 Plan may consist, in whole or in part, of authorized and unissued common stock, treasury common stock or common stock purchased on the open market.

If an award under the 2025 Plan or Stock Option Plan expires, lapses or is terminated, exchanged for or settled in cash, any shares subject to such award (or portion thereof) may, to the extent of such expiration, lapse, termination or cash settlement, be used again for new grants under the 2025 Plan. Shares delivered to us to satisfy the applicable exercise or purchase price of an award under the 2025 Plan or the Stock Option Plan and/or to satisfy any applicable tax withholding obligations (including shares retained by us from the award under the 2025 Plan or the Stock Option Plan being exercised or purchased, and/or creating the tax obligation) will become or again be available for award grants under the 2025 Plan. Further, the payment of dividend equivalents in cash in conjunction with any awards under the 2025 Plan will not reduce the shares available for grant under the 2025 Plan. However, the following shares may not be used again for grant under the 2025 Plan: (i) shares subject to stock appreciation rights ("SARs") that are not issued in connection with the stock settlement of the SAR on exercise, and (ii) shares purchased on the open market with the cash proceeds from the exercise of options.

Awards granted under the 2025 Plan upon the assumption of, or in substitution for, awards authorized or outstanding under a qualifying equity plan maintained by an entity with which we enter into a merger or similar corporate transaction will not reduce the shares available for grant under the 2025 Plan but will count against the maximum number of shares that may be issued upon the exercise of ISOs.

The 2025 Plan provides that the sum of any cash compensation and the aggregate grant date fair value (determined as of the date of the grant under Financial Accounting Standards Board Accounting Standards Codification Topic 718, or any successor thereto) of all awards granted to a non-employee director as compensation for services as a non-employee director during any fiscal year, or director limit, may not exceed an amount equal to $500,000 (increased to $1,000,000 in the calendar year of a non-employee director's initial service as a non-employee director or any calendar year during which a non-employee director serves as chairman of our board of directors or lead independent director), which limits will not apply to the compensation for any non-employee director who serves in any capacity in addition to that of a non-employee director for which he or she receives additional compensation or any compensation paid prior to the calendar year following the calendar year in which the 2025 Plan becomes effective.

Awards. The 2025 Plan provides for the grant of stock options, including ISOs, and nonqualified stock options ("NSOs"), restricted stock, dividend equivalents, restricted stock units ("RSUs"), SARs, and other stock or cash based awards. Certain awards under the 2025 Plan may constitute or provide for a payment of "nonqualified deferred compensation" under Section 409A of the Code, which may impose additional requirements on the terms and conditions of such awards. All awards under the 2025 Plan will be evidenced by award agreements, which will detail the terms and conditions of the awards, including any applicable vesting and payment terms and post-termination exercise limitations. Awards other than cash awards generally will be settled in shares of our common stock, but the applicable award agreement may provide for cash settlement of any award. A brief description of each award type follows.

- Stock Options and SARs. Stock options provide for the purchase of shares of our common stock in the future at an exercise price set on the grant date. ISOs, in contrast to NSOs, may provide tax deferral beyond exercise and favorable capital gains tax treatment to their holders if certain holding period and other requirements of the Code are satisfied. SARs entitle their holder, upon exercise, to receive from us an amount equal to the appreciation of the shares subject to the award between the grant date and the exercise date. Unless otherwise determined by our board, the exercise price of a stock option or SAR may not be less than 100% of the fair market value of the underlying share on the grant date (or 110% in the case of ISOs granted to certain significant stockholders), except with respect to certain substitute awards granted in

connection with a corporate transaction. The term of a stock option or SAR may not be longer than ten years (or five years in the case of ISOs granted to certain significant stockholders). Conditions applicable to stock options and/or SARs may be based on continuing service, the attainment of performance goals and/or such other conditions as the plan administrator may determine.

- Restricted Stock and RSUs. Restricted stock is an award of nontransferable shares of our common stock that are subject to certain vesting conditions and other restrictions. RSUs are contractual promises to deliver shares of our common stock in the future, which may also remain forfeitable unless and until specified conditions are met and may be accompanied by the right to receive the equivalent value of dividends paid on shares of our common stock prior to the delivery of the underlying shares (*i.e.*, dividend equivalent rights). The plan administrator may provide that the delivery of the shares underlying RSUs will be deferred on a mandatory basis or at the election of the participant. The terms and conditions applicable to RSUs will be determined by the plan administrator, subject to the conditions and limitations contained in the 2025 Plan. Conditions applicable to restricted stock and RSUs may be based on continuing service, the attainment of performance goals and/or such other conditions as the plan administrator may determine.

- Other Stock or Cash Based Awards. Other stock or cash based awards are awards of cash, fully vested shares of our common stock and other awards valued wholly or partially by referring to, or otherwise based on, shares of our common stock. Other stock or cash based awards may be granted to participants and may also be available as a payment form in the settlement of other awards, as standalone payments and as payment in lieu of compensation to which a participant is otherwise entitled.

- Dividend Equivalents. Dividend equivalents represent the right to receive the equivalent value of dividends paid on shares of our common stock and may be granted alone or in tandem with awards other than stock options or SARs. Dividend equivalents are credited as of the dividend record dates during the period between the date an award is granted and the date such award vests, is exercised, is distributed or expires, as determined by the plan administrator. Dividend equivalents payable with respect to an award prior to the vesting of such award instead will be paid out to the participant only to the extent that the vesting conditions are subsequently satisfied and the award vests.

Certain Transactions. The plan administrator has broad discretion to take action under the 2025 Plan, as well as make adjustments to the terms and conditions of existing and future awards, to prevent the dilution or enlargement of intended benefits and facilitate necessary or desirable changes in the event of certain transactions and events affecting our common stock, such as stock dividends, stock splits, mergers, acquisitions, consolidations and other corporate transactions. In addition, in the event of certain non-reciprocal transactions with our stockholders known as “equity restructurings,” the plan administrator will make equitable adjustments to the 2025 Plan and outstanding awards. In the event of a change in control (as defined in the 2025 Plan), to the extent that the surviving entity declines to continue, convert, assume or replace outstanding awards, then all such awards will become fully vested and exercisable in connection with the transaction. If, however, the surviving entity assumes outstanding awards and, on or within 12 months following such change in control, a participant’s employment is terminated by the company (or the surviving entity or its affiliates) for any reason other than for “cause” or by the participant for “good reason” (each such term as defined in the 2025 Plan), then all such awards will become fully vested and exercisable as of the date of such termination. Upon or in anticipation of a change of control, the plan administrator may cause any outstanding awards to terminate at a specified time in the future and give the participant the right to exercise such awards during a period of time determined by the plan administrator in its sole discretion. Individual award agreements may provide for additional accelerated vesting and payment provisions.

Repricing. The plan administrator may, without approval of the stockholders, reduce the exercise price of any stock option or SAR, or cancel any stock option or SAR in exchange for cash, other awards or stock options or SARs with an exercise price per share that is less than the exercise price per share of the original stock options or SARs.

Foreign Participants, Claw-Back Provisions, Transferability, and Participant Payments. The plan administrator may modify award terms, establish subplans and/or adjust other terms and conditions of awards, subject to the share limits described above, in order to facilitate grants of awards subject to the laws and/or stock exchange rules of

countries outside of the United States. All awards will be subject to any company clawback policy as set forth in such clawback policy or the applicable award agreement. Awards under the 2025 Plan are generally non-transferrable, except by will or the laws of descent and distribution, or, subject to the plan administrator's consent, pursuant to a domestic relations order, and are generally exercisable only by the participant. With regard to tax withholding, exercise price and purchase price obligations arising in connection with awards under the 2025 Plan, the plan administrator may, in its discretion, accept cash or check, shares of our common stock that meet specified conditions, a "market sell order" or such other consideration as it deems suitable.

Plan Amendment and Termination. Our board of directors may amend or terminate the 2025 Plan at any time; however, no amendment, other than an amendment that increases the number of shares available under the 2025 Plan, may materially and adversely affect an award outstanding under the 2025 Plan without the consent of the affected participant, and stockholder approval will be obtained for any amendment to the extent necessary to comply with applicable laws. The 2025 Plan will remain in effect until terminated by the plan administrator in accordance with the 2025 Plan. No awards may be granted under the 2025 Plan after its termination.

2025 Employee Stock Purchase Plan

In connection with this offering, we adopted, and our stockholders approved, the ESPP.

Shares Available; Administration. The initial share reserve under the ESPP is equal to 249,373 shares.

In addition, the number of shares available for issuance under the ESPP will be annually increased on the first day of each calendar year beginning January 1, 2026 and ending on and including January 1, 2035, equal to (i) 1% of the aggregate number of shares of common stock outstanding on the final day of the immediately preceding calendar year and (ii) such smaller number of shares as is determined by our board of directors. In no event will more than 100,000,000 shares of our common stock be available for issuance under the ESPP.

Our board of directors or a committee designated by our board of directors has authority to interpret the terms of the ESPP and determine eligibility of participants. The compensation committee is the administrator of the ESPP.

Eligibility. The plan administrator may designate certain of our subsidiaries as participating "designated subsidiaries" in the ESPP and may change these designations from time to time. Employees of our company and our designated subsidiaries are eligible to participate in the ESPP if they meet the eligibility requirements under the ESPP established from time to time by the plan administrator. However, an employee may not be granted rights to purchase stock under the ESPP if such employee, immediately after the grant, would own (directly or through attribution) stock possessing 5% or more of the total combined voting power or value of all classes of our common or other class of stock.

If the grant of a purchase right under the ESPP to any eligible employee who is a citizen or resident of a foreign jurisdiction would be prohibited under the laws of such foreign jurisdiction or the grant of a purchase right to such employee in compliance with the laws of such foreign jurisdiction would cause the ESPP to violate the requirements of Section 423 of the Code, as determined by the plan administrator in its sole discretion, such employee will not be permitted to participate in the ESPP.

Eligible employees become participants in the ESPP by enrolling and authorizing payroll deductions by the deadline established by the plan administrator prior to the relevant offering date. Directors who are not employees, as well as consultants, are not eligible to participate. Employees who choose to not participate, or are not eligible to participate at the start of an offering period but who become eligible thereafter, may enroll in any subsequent offering period.

Participation in an Offering. We intend for the ESPP to qualify under Section 423 of the Code and stock will be offered under the ESPP during offering periods. The length of offering periods under the ESPP will be determined by the plan administrator and may be up to 27 months long. Employee payroll deductions will be used to purchase shares on each purchase date during an offering period. The number of purchase periods within, and purchase dates during, each offering period will be established by the plan administrator. Offering periods under the ESPP will

commence when determined by the plan administrator. The plan administrator may, in its discretion, modify the terms of future offering periods.

The ESPP will permit participants to purchase our common stock through payroll deductions of up to 20% of their eligible compensation, unless otherwise determined by the plan administrator, which will include a participant's gross base compensation for services to us, including overtime payments, and excluding sales commissions, periodic bonuses, one-time bonuses, expense reimbursements, fringe benefits and other special payments. The plan administrator will establish a maximum number of shares that may be purchased by a participant during any offering period or purchase period, which, in the absence of a contrary designation, will be 10,000 shares for an offering period and/or a purchase period. In addition, no employee will be permitted to accrue the right to purchase stock under the ESPP at a rate in excess of $25,000 worth of shares during any calendar year during which such a purchase right is outstanding (based on the fair market value per share of our common stock as of the first day of the offering period).

On the first trading day of each offering period, each participant automatically will be granted an option to purchase shares of our common stock. The option will be exercised on the applicable purchase date(s) during the offering period, to the extent of the payroll deductions accumulated during the applicable purchase period. The purchase price of the shares, in the absence of a contrary determination by the plan administrator, will be 85% of the lower of the fair market value of our common stock on the first trading day of the offering period or on the applicable purchase date, which will be the final trading day of the applicable purchase period.

Participants may voluntarily end their participation in the ESPP at any time at least two weeks prior to the end of the applicable offering period (or such longer or shorter period specified by the plan administrator), and will be paid their accrued payroll deductions that have not yet been used to purchase shares of common stock. Participation ends automatically upon a participant's termination of employment.

Transferability. A participant may not transfer rights granted under the ESPP other than by will, the laws of descent and distribution or as otherwise provided in the ESPP.

Certain Transactions. In the event of certain transactions or events affecting our common stock, such as any stock dividend or other distribution, change in control, reorganization, merger, consolidation or other corporate transaction, the plan administrator will make equitable adjustments to the ESPP and outstanding rights. In addition, in the event of the foregoing transactions or events or certain significant transactions, including a change in control, the plan administrator may provide for (i) either the replacement of outstanding rights with other rights or property or termination of outstanding rights in exchange for cash, (ii) the assumption or substitution of outstanding rights by the successor or survivor corporation or parent or subsidiary thereof, (iii) the adjustment in the number and type of shares of stock subject to outstanding rights, (iv) the use of participants' accumulated payroll deductions to purchase stock on a new purchase date prior to the next scheduled purchase date and termination of any rights under ongoing offering periods or (v) the termination of all outstanding rights. Under the ESPP, a change in control has the same definition as given to such term in the 2025 Plan.

Plan Amendment; Termination. The plan administrator may amend, suspend or terminate the ESPP at any time. However, stockholder approval of any amendment to the ESPP must be obtained for any amendment which increases the aggregate number or changes the type of shares that may be sold pursuant to rights under the ESPP, ESPP in any manner that would be considered the adoption of a new plan within the meaning of Treasury regulation Section 1.423-2(c)(4), or changes the ESPP in any manner that would cause the ESPP to no longer be an employee stock purchase plan within the meaning of Section 423(b) of the Code.

CERTAIN RELATIONSHIPS AND RELATED-PARTY TRANSACTIONS

The following includes a summary of transactions since January 1, 2022 and any currently proposed transaction to which we have been a party or are to be a participant in which the amount involved exceeded or will exceed the lesser of $120,000 or one percent of the average of our total assets at year-end for the last two completed fiscal years, and in which any of our directors, executive officers or beneficial owners of more than 5% of our capital stock or any member of the immediate family of any of the foregoing persons had or will have a direct or indirect material interest, other than equity and other compensation, termination, change in control and other arrangements, which are described in the section titled "Executive and Director Compensation." We also describe below certain other transactions with our directors, executive officers and stockholders.

Convertible Promissory Note Financing

From September 2022 to November 2022, we issued and sold to investors in private placements an aggregate principal amount of $6.9 million of convertible promissory notes. The convertible promissory notes bore interest at a rate of 8.0% per annum. In connection with the closing of our Series D convertible preferred stock financing in February 2023, the aggregate principal amount and accrued interest on the convertible promissory notes were converted into 16,464,493 shares of our Series D convertible preferred stock at an aggregate conversion price of $8.9 million. Each share of our Series D convertible preferred stock will convert into 0.052410901 shares of common stock immediately prior to the completion of this offering.

The following table sets forth the aggregate principal amount of convertible promissory notes purchased by our directors, executive officers and beneficial owners of more than 5% of our capital stock, and any member of the immediate family of any of the foregoing persons.

Participants[1]	Convertible Promissory Notes Purchased
cultivate(MD) Capital Accelerator Fund, L.P.	$ 2,213,756
Entities affiliated with U.S. Venture Partners	1,356,513
Entities affiliated with Lightstone Ventures	1,356,513
Robert Ball	300,000
Paul Buckman	50,000
Kevin Sidow	100,000

(1) Additional details regarding these stockholders and their equity holdings are provided in this prospectus under the caption "Principal Stockholders" and in "—Preferred Stock Financings."

Preferred Stock Financings

Series D Convertible Preferred Stock Financing

In February 2023 and March 2023, we issued and sold to investors in private placements an aggregate of 80,909,169 shares of our Series D convertible preferred stock at a purchase price of $0.54 per share, for aggregate consideration of approximately $43.8 million, inclusive of the issuance of 16,464,493 shares of our Series D convertible preferred stock in connection with the conversion of the convertible promissory notes as described above at an aggregate conversion price of $8.9 million.

The following table sets forth the aggregate purchase price paid and the aggregate number of shares of Series D convertible preferred stock purchased by our directors, executive officers and beneficial owners of more than 5% of our capital stock, and any member of the immediate family of any of the foregoing persons. Each share of our Series

D convertible preferred stock identified in the following table will convert into 0.052410901 shares of common stock immediately prior to the completion of this offering.

Participants[1]	Shares of Series D Convertible Preferred Stock	Aggregate Purchase Price ($)
Coöperatieve Gilde Healthcare V U.A.	33,259,424	18,000,000
Entities affiliated with U.S. Venture Partners[2]	6,732,238	3,643,487
Entities affiliated with Lightstone Ventures[3]	6,732,238	3,643,487
Gilmartin Capital Fund I L.P.	9,238,729	5,000,000

(1) Additional details regarding these stockholders and their equity holdings are provided in this prospectus under the caption "Principal Stockholders" and in "—Preferred Stock Financings."
(2) Excludes the aggregate principal amount and accrued interest of $1.4 million that was converted into 3,217,970 shares of Series D convertible preferred stock issued upon conversion of the convertible promissory notes as described in "—Convertible Promissory Note Financing" above.
(3) Excludes the aggregate principal amount and accrued interest of $1.4 million that was converted into 3,217,970 shares of Series D convertible preferred stock issued upon conversion of the convertible promissory notes as described in "—Convertible Promissory Note Financing" above.

Series E Convertible Preferred Stock Financing

In March 2025, we entered into a Series E convertible preferred stock purchase agreement, pursuant to which we agreed to issue and sell to investors in a private placement an aggregate of up to 58,774,312 shares of our Series E convertible preferred stock at a purchase price of $0.68 per share, for aggregate consideration of up to approximately $40.1 million. The Series E convertible preferred stock was issued and sold in two tranches. The first tranche of 29,455,169 shares of Series E convertible preferred stock, representing aggregate consideration of approximately $20.1 million, closed in March 2025. On June 20, 2025, we closed the second tranche of 29,319,143 shares of Series E convertible preferred stock, representing aggregate consideration of approximately $20.0 million.

The following table sets forth details regarding the aggregate purchase price paid and the aggregate number of shares of Series E convertible preferred stock purchased by our directors, executive officers and beneficial owners of more than 5% of our capital stock, and any member of the immediate family of any of the foregoing persons. Each share of our Series E convertible preferred stock identified in the following table will convert into 0.052410901 shares of common stock immediately prior to the completion of this offering.

	Initial Tranche		Second Tranche	
Participants[1]	Shares of Series E Convertible Preferred Stock	Aggregate Purchase Price ($)	Shares of Series E Convertible Preferred Stock	Aggregate Purchase Price ($)
Coöperatieve Gilde Healthcare V U.A.	5,126,094	3,499,999	5,126,094	3,499,999
Entities affiliated with U.S. Venture Partners	10,252,188	6,999,999	10,252,188	6,999,999
Gilmartin Capital Fund I L.P.	1,303,493	890,000	1,303,492	890,000
Arboretum Ventures VI, LP	9,519,889	6,499,999	9,519,889	6,499,999
Robert Ball	83,691	57,143	62,768	42,857
The David Lawrence Blue Living Trust	20,923	14,286	15,691	10,714

(1) Additional details regarding these stockholders and their equity holdings are provided in this prospectus under the caption "Principal Stockholders" and in "—Preferred Stock Financings."

Series Seed Convertible Preferred Stock Warrant

In February 2025, Genesis Investment Holdings, an entity affiliated with cultivate(MD), exercised its Series Seed preferred stock warrant for 2,233,960 shares of our Series Seed convertible preferred stock at an exercise price

of $0.08434 per share for total proceeds of approximately $0.2 million. Each share of our Series Seed convertible preferred stock will convert into 0.052410901 shares of common stock immediately prior to the completion of this offering.

Some of our directors and officers are associated with our principal stockholders as indicated in the table below:

Director / Officer	Principal Stockholder
Matthew Ahearn	Entities affiliated with cultivate(MD)
Robert Ball	Entities affiliated with cultivate(MD)
David L. Blue	Entities affiliated with cultivate(MD)
Michael Carusi	Entities affiliated with Lightstone Ventures
Geoff Pardo	Coöperatieve Gilde Healthcare V U.A.
Casey Tansey	Entities affiliated with U.S. Venture Partners

Agreements with Stockholders

Fourth Amended and Restated Investor Rights Agreement

We entered into a Fourth Amended and Restated Investors' Rights Agreement (the "Investor Rights Agreement") on March 6, 2025 with the holders of our preferred stock, including entities with which certain of our directors are related. The agreement provides, among other things, for certain rights relating to the registration of such holders' common stock, including shares issuable upon conversion of preferred stock. Certain investor rights contained in the Investor Rights Agreement will terminate upon the consummation of this offering. See the section titled "Description of Capital Stock—Registration Rights" for additional information.

Fourth Amended and Restated Voting Agreement

We entered into a Fourth Amended and Restated Voting Agreement (the "Voting Agreement") on March 6, 2025, with the holders of our preferred stock, including entities with which certain of our directors are related. Pursuant to the Voting Agreement, the signatories thereto have agreed to vote their respective shares to elect (i) one person designated by Cultivate MD Capital Fund II, LP as the Series Seed/A/B director, which is currently Matthew Ahearn, (ii) one person designated by each of U.S. Venture Partners XII, L.P. and Lightstone Ventures II, L.P. as the Series C directors, which are currently Casey Tansey and Michael Carusi, respectively, (iii) one person designated by Coöperatieve Gilde Healthcare V U.A. and its affiliates as the Series D director, which is currently Geoff Pardo and (iv) two joint directors and the CEO director, which are currently Paul Buckman, Kevin Sidow and Robert Ball, respectively.

The Fourth Amended Voting Agreement will terminate upon the completion of this offering and none of our stockholders will have any continuing voting rights, including special rights regarding the election or designation of members of our board of directors, following this offering. Members previously elected to our board of directors pursuant to this agreement will continue to serve as directors until they resign, are removed, or their successors are duly elected by the holders of our common stock. The composition of our board of directors after this offering is described in more detail in the section titled "Management—Board Composition."

Fourth Right of First Refusal and Co-Sale Agreement

We entered into a Fourth Amended and Restated Right of First Refusal and Co-Sale Agreement on March 6, 2025 with certain holders of our preferred stock, including entities with which certain of our directors are related. This agreement provides for rights of first refusal and co-sale relating to the shares of our common stock held by the parties to the agreement. This agreement will terminate upon the completion of this offering.

Consulting Arrangement with Genesis Innovation

On April 30, 2015, we entered into a consulting agreement (the "Genesis Consulting Agreement") with Genesis Innovation pursuant to which Genesis Innovation agreed to provide consulting services for concept development,

intellectual property creation, surgeon relationship management and large-scale project management. Any such services to be provided are set out and delivered pursuant to specified work plans at the agreed-upon hourly chargeable rate then in effect. The Genesis Consulting Agreement was subsequently amended to increase the hourly chargeable rate on January 1, 2022 and January 1, 2025. Either party may terminate the Genesis Consulting Agreement upon 60 days written notice, and the agreement will terminate on December 31, 2025. Robert Ball, our Chief Executive Officer and Executive Chairman, is a co-founder and director of Genesis Innovation. For each of the years ended December 31, 2024 and 2023, we paid Genesis Innovation $3.6 million and $2.8 million pursuant to the Genesis Consulting Agreement. For the three months ended March 31, 2025 we paid Genesis Innovation $1.2 million pursuant to the Genesis Consulting Agreement.

Software License Agreement with Genesis Software Innovations

On October 22, 2020, we entered into a software license agreement with Genesis Software which was amended and restated on January 1, 2023. Please see the section titled "Business—Intellectual Property—Software License Agreement with Genesis Software" for more information. Robert Ball, our Chief Executive Officer and Executive Chairman, is a co-founder and director of Genesis Software. Mr. Ball and Matthew Ahearn, our Chief Operating Officer and a member of our board of directors, are directors of Genesis Investment Holdings, which has an ownership interest in Genesis Software and is an entity affiliated with cultivate(MD). In addition, entities affiliated with cultivate(MD), a beneficial owner of more than 5% of our capital stock, are investors in Genesis Software. For each of the years ended December 31, 2024 and 2023, we paid Genesis Software $3.1 million and $3.2 million pursuant to the License Agreement, including royalties of $1.2 million and $0.6 million in the years ended December 31, 2024 and 2023, respectively. For the three months ended March 31, 2025 we paid Genesis Software $0.9 million pursuant to the License Agreement, including royalties of $0.3 million.

Supply Agreement with RMD

On June 3, 2024, we entered into a new supply agreement (the "RMD Supply Agreement") with RMD, pursuant to which RMD manufactures the surgical instruments used during procedures involving our implant systems. During such periods, Robert Ball, our Chief Executive Officer and Executive Chairman, was an investor in RMD. As of the date of this prospectus, Mr. Ball is no longer an investor in RMD. For each of the years ended December 31, 2024 and 2023, we paid RMD $0.9 million and $2.3 million pursuant to the RMD Supply Agreement. There were no payments to RMD during the three months ended March 31, 2025.

Directed Share Program

At our request, the underwriters have reserved for sale, at the initial public offering price, up to approximately 300,000 shares of our common stock being offered to certain of our directors, officers and employees and others as part of a directed share program. The directed share program will not limit the ability of our directors, officers or holders of more than 5% of our common stock, to purchase more than $120,000 in value of our common stock. We do not currently know the extent to which these related persons will participate in our directed share program, if at all, or the extent to which they will purchase more than $120,000 in value of our common stock.

Director and Officer Indemnification Agreements

We have entered into indemnification agreements with each of our directors and executive officers. These agreements, among other things, require us to indemnify each director and executive officer to the fullest extent permitted by Delaware law, including indemnification of expenses such as attorneys' fees, judgments, fines and settlement amounts incurred by the director or executive officer in any action or proceeding, including any action or proceeding by or in right of us, arising out of the person's services as a director or executive officer. For further information, see the section titled "Description of Capital Stock—Limitations on Liability and Indemnification Matters."

Policies and Procedures for Related Person Transactions

Our board of directors has adopted a written related person transaction policy, to be effective upon the completion of this offering, setting forth the policies and procedures for the review and approval or ratification of

related person transactions. This policy covers, with certain exceptions set forth in Item 404 of Regulation S-K under the Securities Act, any transaction, arrangement or relationship, or any series of similar transactions, arrangements or relationships, in which we were or are to be a participant, where the amount involved exceeds $120,000 in any fiscal year, or, for so long as we qualify as a smaller reporting company, the lesser of $120,000 or one percent of the average of our total assets at year-end for the last two completed fiscal years, and a related person had, has or will have a direct or indirect material interest, including without limitation, purchases of goods or services by or from the related person or entities in which the related person has a material interest, indebtedness, guarantees of indebtedness and employment by us of a related person. In reviewing and approving any such transactions, our audit committee will be tasked to consider all relevant facts and circumstances, including, but not limited to, whether the transaction is on terms comparable to those that could be obtained in an arm's length transaction and the extent of the related person's interest in the transaction. All of the transactions described in this section occurred prior to the adoption of this policy.

PRINCIPAL STOCKHOLDERS

The following table sets forth, as of June 30, 2025, information regarding beneficial ownership of our capital stock by:

- each person, or group of affiliated persons, known by us to beneficially own more than 5% of our common stock;
- each of our named executive officers;
- each of our directors; and
- all of our executive officers and directors as a group.

The number of shares beneficially owned by each stockholder is determined under rules issued by the SEC. Under these rules, beneficial ownership includes any shares as to which the individual or entity has sole or shared voting power or investment power. Applicable percentage ownership before the offering is based on 12,502,236 shares of common stock outstanding as of June 30, 2025, assuming the conversion of all outstanding shares of our convertible preferred stock into shares of our common stock. Applicable percentage ownership after the offering assumes the sale of 5,000,000 shares of our common stock in this offering and the issuance of 2,805,049 shares of our common stock pursuant to the Notes Conversion, which is based on the initial public offering price of $15.00 per share, without giving effect to accrued interest.

In computing the number of shares beneficially owned by an individual or entity and the percentage ownership of that person, shares of common stock subject to options, warrants or other rights held by such person that are currently exercisable or will become exercisable within 60 days of June 30, 2025 are considered outstanding, although these shares are not considered outstanding for purposes of computing the percentage ownership of any other person. The following table does not reflect any shares of our common stock that may be purchased pursuant to our directed share program described under "Underwriting—Directed Share Program." Unless otherwise indicated, the persons or entities identified in this table have sole voting and investment power with respect to all shares shown as beneficially owned by them, subject to applicable community property laws.

Unless otherwise noted below, the address for each beneficial owner listed in the table below is c/o Shoulder Innovations, Inc., 1535 Steele Avenue SW, Suite B, Grand Rapids, MI 49507.

	Beneficially Owned Before the Offering		Beneficially Owned After the Offering	
Name of Beneficial Owner	**Number of Shares**	**Percentage of Shares**	**Number of Shares**	**Percentage of Shares**
5% and Greater Stockholders				
Entities affiliated with U.S. Venture Partners[1]	2,693,121	21.5 %	2,693,121	13.3 %
Coöperatieve Gilde Healthcare V U.A.[2]	2,280,482	18.2 %	2,280,482	11.2 %
Entities affiliated with Lightstone Ventures[3]	1,618,470	12.9 %	1,618,470	8.0 %
Entities affiliated with cultivate(MD)[4]	1,363,395	10.9 %	1,363,395	6.7 %
Arboretum Ventures VI, LP[5]	997,891	8.0 %	997,891	4.9 %
FMR LLC[6]	—	— %	2,805,049	13.8 %
Named Executive Officers, Directors and Director Nominee				
Robert Ball[7]	459,082	3.6 %	459,082	2.2 %
Matthew Ahearn[8]	228,690	1.8 %	228,690	1.1 %
David L. Blue[9]	138,915	*	138,915	*
Jeffrey Points[10]	88,426	*	88,426	*
Kevin Sidow[11]	55,098	*	55,098	*
Paul Buckman[12]	48,381	*	48,381	*
Michael Carusi[3]	1,618,470	12.9 %	1,618,470	8.0 %
Geoff Pardo[2]	2,280,482	18.2 %	2,280,482	11.2 %
Casey Tansey[1]	2,693,121	21.5 %	2,693,121	13.3 %
Richard J. Buchholz	—	*	—	*
All current directors and executive officers as a group (10 persons)[13]	7,566,730	60.5 %	7,566,730	36.0 %

* Indicates beneficial ownership of less than 1% of the total outstanding common stock.

(1) Consists of (i) 1,043,988 shares of Series C Preferred Stock, 496,309 shares of Series D Preferred Stock and 146,106 shares of Series E Preferred Stock held by U.S. Venture Partners XII, L.P. ("USVP XII"); (ii) 52,983 shares of Series C Preferred Stock, 25,189 shares of Series D Preferred Stock, and 7,415 shares of Series E Preferred Stock held by U.S. Venture Partners XII-A, L.P ("USVP XII-A"); and (iii) 921,131 shares of Series E Preferred Stock held by U.S. Venture Partners Select Fund I, L.P. on its own behalf and as a nominee for U.S. Venture Partners Select Fund I-A, L.P. ("USVP Select"). Presidio Management Group XII, L.L.C. ("PMG XII") is the general partner of USVP XII and USVP XII-A and has sole voting and dispositive power with respect to the shares held by USVP XII and USVP XII-A. Steven M. Krausz, Richard W. Lewis, Jonathan D. Root, Casey M. Tansey (a member of our board of directors) and Dafina Toncheva are the managing members of PMG XII, and share voting and dispositive power with respect to the shares held by USVP XII and USVP XII-A. Presidio Management Group Select Fund I, L.L.C. ("PMG Select") is the general partner of USVP Select and has sole voting and dispositive power with respect to the shares held by USVP Select. Richard W. Lewis, Jonathan D. Root, Casey M. Tansey (a member of our board of directors) and Dafina Toncheva are the managing members of PMG Select, and share voting and dispositive power with respect to the shares held by USVP Select. Each of the managing members of PMG XII and PMG Select disclaims beneficial ownership of such holdings, except to the extent of their pecuniary interest in the shares. The address for each of these entities is 1460 El Camino Real, Suite 100, Menlo Park, California 94025.

(2) Consists of 1,743,156 shares of Series D Preferred Stock and 537,326 shares of Series E Preferred Stock held directly by Coöperatieve Gilde Healthcare V U.A. ("Gilde"). Gilde Healthcare V Management B.V. is the managing director of Gilde and has sole voting and dispositive power with respect to the shares held by Gilde. Gilde Healthcare V Management B.V. is managed by Marc Perret, Edwin de Graaf and Pieter van der Meer, who share voting and dispositive power with respect to the shares held of record by Gilde. Geoff Pardo (a member of our board of directors) is a partner at Gilde and may be deemed to share voting and dispositive power with respect to the shares held of record by Gilde. Each of Mr. Perret, Mr. de Graaf, Mr. van der Meer and Mr. Pardo disclaims beneficial ownership of such holdings, except to the extent of their pecuniary interest in the shares. The address of Gilde is Stadsplateu 36 3521 AZ Utrecht, The Netherlands, c/o Gilde Healthcare Partner.

(3) Consists of (i) 1,036,638 shares of Series C Preferred Stock, and 492,833 shares of Series D Preferred Stock held by Lightstone Ventures II, L.P. ("LSV II"); and (ii)60,333 shares of Series C Preferred Stock and 28,666 shares of Series D Preferred Stock held by Lightstone Ventures II (A), L.P. ("LSV II (A)"). LSV Associates II, LLC is the general partner of LSV II and LSV II (A) and has sole voting and

dispositive power with respect to the shares held by LSV II and LSV II (A). Jean George, Michael A. Carusi (a member of our board of directors) and Henry A. Plain, Jr. are the managing directors of LSV Associates II, LLC, and share voting and dispositive power with respect to the shares held of record by LSV II and LSV II (A). Mr. Carusi disclaims beneficial ownership of such holdings, except to the extent of his pecuniary interest in the shares. The address of each of LSV II and LSV II (A) is 420 Boylston St., Suite 602, Boston, MA 02116.

(4) Consists of (i) 185,623 shares of Series A Preferred Stock held by Cultivate MD Capital Fund I, LLC ("CMD I"); (ii) 154,970 shares of Series A Preferred Stock and 154,978 shares of Series B Preferred Stock held by Cultivate MD Capital Fund II, LP ("CMD II"); (iii) 508,566 shares of Series Seed Preferred Stock, 69,932 shares of Series A Preferred Stock and 14,088 shares of Series B Preferred Stock held by Genesis Investment Holdings; and (iv) 275,238 shares of Series D Preferred Stock held by cultivate(MD) Capital Accelerator Fund, L.P. ("CMD Accelerator"). cultivate(MD) Holdings, LLC is the general partner or manager of CMD I, CMD II, Genesis Investment Holdings and CMD Accelerator and has sole voting and dispositive power with respect to the shares held by CMD I, CMD II, Genesis Investment Holdings and CMD Accelerator. Each of David Blue (one of our executive officers), Matthew Ahearn (one of our executive officers and a member of our board of directors) and Robert Ball (one of our executive officers and a member of our board of directors) are directors, and R. Sean Churchill is a managing director, of cultivate(MD) Holdings, LLC and share voting and dispositive power with respect to the shares held of record by CMD I, CMD II, Genesis Investment Holdings and CMD Accelerator. Each of such directors and the managing director disclaims beneficial ownership of such holdings, except to the extent of his pecuniary interest in the shares. The address of cultivate(MD) Holdings, LLC is 2851 Charlevoix Drive SE, Suite 327, Grand Rapids, MI 49546.

(5) Consists of 997,891 shares of Series E Preferred Stock held by Arboretum Ventures VI, LP ("AV VI"). Arboretum Investment Manager VI, LLC ("AIM VI") serves as the general partner of AV VI. Thomas M. Shehab, Jan L. Garfinkle and Daniel M. Kidle are managing members of AIM VI and share voting and dispositive power with regards to the shares held by AV VI. The address of the principal place of business of each of the entities and individuals is 303 Detroit Street, Suite 301, Ann Arbor, Michigan 48104.

(6) Consists of (i) 873,288 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share without giving effect to accrued interest held by Fidelity Select Portfolios: Select Medical Technology and Devices Portfolio; (ii) 255,701 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Fidelity Advisor Series VII: Fidelity Advisor Health Care Fund; (iii) 447,727 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Fidelity Select Portfolios: Select Health Care Portfolio; (iv) 73,309 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Variable Insurance Products Fund IV: VIP Health Care Portfolio; (v) 660,350 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Fidelity Securities Fund: Fidelity Small Cap Growth Fund; (vi) 329,670 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Fidelity Securities Fund: Fidelity Small Cap Growth K6 Fund; and (vii) 165,000 shares of common stock issuable upon the Notes Conversion, based on the initial public offering price of $15.00 per share held by Fidelity Venture Capital Fund I LP. These funds and accounts are managed by direct or indirect subsidiaries of FMR LLC. The shares held by these funds and accounts are beneficially owned, or may be deemed to be beneficially owned, by FMR LLC, certain of its subsidiaries and affiliates, and other companies. Abigail P. Johnson is a Director, the Chairman, and the Chief Executive Officer of FMR LLC. Members of the Johnson family, including Abigail P. Johnson, are the predominant owners, directly or through trusts, of Series B voting common shares of FMR LLC, representing 49% of the voting power of FMR LLC. The Johnson family group and all other Series B shareholders have entered into a shareholders' voting agreement under which all Series B voting common shares will be voted in accordance with the majority vote of Series B voting common shares. Accordingly, through their ownership of voting common shares and the execution of the shareholders' voting agreement, members of the Johnson family may be deemed, under the Investment Company Act of 1940, to form a controlling group with respect to FMR LLC. The address of FMR LLC is 245 Summer Street, Boston, MA 02210.

(7) Consists of (i) 11,651 shares of Series Seed Preferred Stock, 9,228 shares of Series A Preferred Stock, 7,198 shares of Series B Preferred Stock, 37,032 shares of Series D Preferred Stock, and 7,676 shares of Series E Preferred Stock and (ii) 386,297 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

(8) Consists of (i) 40,793 shares of common stock, (ii) 13,139 shares of Series Seed Preferred Stock, 42,667 shares of Series A Preferred Stock and 11,436 shares of Series B Preferred Stock and (iii) 120,655 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

(9) Consists of (i) 5,241 shares of Series Seed Preferred Stock, 18,151 shares of Series A Preferred Stock, 16,190 shares of Series B Preferred Stock, 2,437 shares of Series C Preferred Stock, and 1,918 shares of Series E Preferred Stock held by David Lawrence Blue Living Trust and (ii) 94,978 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025 held by David L. Blue.

(10) Consists of (i) 37,581 shares of common stock and (ii) 50,845 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

(11) Consists of (i) 12,344 shares of Series D Preferred Stock and (ii) 42,754 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

(12) Consists of (i) 6,198 shares of Series D Preferred Stock and (ii) 42,183 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

(13) Consists of (i) 78,374 shares of common stock, (ii) 30,031 shares of Series Seed Preferred Stock, 70,046 shares of Series A Preferred Stock, 34,824 shares of Series B Preferred Stock, 2,196,379 shares of Series C Preferred Stock, 2,841,727 shares of Series D Preferred Stock, 1,621,572 shares of Series E Preferred Stock and (iii) 737,712 shares of common stock subject to options exercisable upon conversion within 60 days of June 30, 2025.

DESCRIPTION OF CAPITAL STOCK

The following description of our capital stock and provisions of our amended and restated certificate of incorporation and amended and restated bylaws are summaries. You should also refer to the amended and restated certificate of incorporation, the amended and restated bylaws, and the amended and restated investors' rights agreement, which are filed as exhibits to the registration statement of which this prospectus is a part.

General

Upon the completion of this offering and the filing of our amended and restated certificate of incorporation, our authorized capital stock will consist of 750,000,000 shares, all with a par value of $0.001 per share, of which:

- 730,000,000 shares are designated as common stock; and
- 20,000,000 shares are designated as preferred stock.

Common Stock

Outstanding Shares

As of March 31, 2025, we had 15,228,510 shares of common stock outstanding, held of record by 158 stockholders, after giving effect to the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025) and the Notes Conversion into an aggregate of 15,125,725 shares of our common stock.

Voting Rights

Each holder of common stock is entitled to one vote for each share on all matters submitted to a vote of the stockholders, including the election of directors. Our stockholders do not have cumulative voting rights in the election of directors. Accordingly, holders of a majority of the voting shares are able to elect all of the directors. In addition, the affirmative vote of holders of 66-2/3% of the voting power of all of the then outstanding voting stock will be required to take certain actions, including amending certain provisions of our amended and restated certificate of incorporation, including the provisions relating to amending our amended and restated bylaws, the classified board and director liability.

Dividends

Subject to preferences that may be applicable to any then outstanding preferred stock, holders of our common stock are entitled to receive ratably any dividends that our board of directors may declare out of funds legally available.

Liquidation

In the event of our liquidation, dissolution, or winding up, holders of our common stock will be entitled to share ratably in the net assets legally available for distribution to stockholders after the payment of all of our debts and other liabilities, subject to the satisfaction of any liquidation preference granted to the holders of any then outstanding shares of preferred stock.

Rights, Preferences and Privileges

Holders of our common stock have no preemptive, conversion, or subscription rights, and there are no redemption or sinking-fund provisions applicable to our common stock. The rights, preferences, and privileges of the holders of our common stock are subject to, and may be adversely affected by, the rights of the holders of shares of any series of our preferred stock that we may designate and issue in the future.

Preferred Stock

Immediately prior to the completion of this offering, all of our currently outstanding shares of convertible preferred stock will convert into common stock, and we will not have any preferred shares outstanding. Immediately prior to the completion of this offering, our certificate of incorporation will be amended and restated to delete all references to such shares of convertible preferred stock. Under the amended and restated certificate of incorporation, our board of directors will have the authority, without further action by the stockholders, to issue up to 20,000,000 shares of preferred stock in one or more series, to establish from time to time the number of shares to be included in each such series, to fix the rights, preferences, and privileges of the shares of each wholly unissued series and any qualifications, limitations, or restrictions thereon, and to increase or decrease the number of shares of any such series, but not below the number of shares of such series then outstanding.

Our board of directors may authorize the issuance of preferred stock with voting or conversion rights that could adversely affect the voting power or other rights of the holders of common stock. The issuance of preferred stock, while providing flexibility in connection with possible acquisitions and other corporate purposes, could, among other things, have the effect of delaying, deferring, or preventing a change in control of our company that may otherwise benefit holders of our common stock and may adversely affect the market price of the common stock and the voting and other rights of the holders of common stock. We have no current plans to issue any shares of preferred stock.

Stock Options

As of March 31, 2025, 1,310,108 shares of common stock were issuable upon the exercise of outstanding stock options, at a weighted-average exercise price of $2.08 per share. For additional information regarding terms of our equity incentive plans, see the section titled "Executive and Director Compensation—Equity Incentive Plans."

Common Stock Warrant

As of March 31, 2025, we had a warrant to purchase an aggregate of 17,827 shares of our common stock, with an exercise price of $2.10 per share. Unless exercised earlier, the warrant will expire in April 2031. The warrant contains provisions for the adjustment of the exercise price and the number of shares issuable upon the exercise of the warrant in the event of certain stock dividends, stock splits, reorganizations, reclassifications and consolidations. The warrant has a net exercise provision under which its holder may, in lieu of payment of the exercise price in cash, surrender the warrant and receive a net amount of shares based on the fair market value of the shares at the time of exercise of the warrant after deduction of the aggregate exercise price.

Preferred Stock Warrants

As of March 31, 2025, we had warrants to purchase an aggregate of up to 875,000 shares of our Series B convertible preferred stock outstanding, with an exercise price of $0.372 per share. The warrants were issued in connection with our Series B convertible preferred stock financing. Unless exercised earlier, these warrants will expire in January 2030; provided that the warrants are subject to early termination immediately prior to the completion of this offering if such warrants are not exercised prior to such time. The warrants contain provisions for the adjustment of the exercise price and the number of shares issuable upon the exercise of the warrant in the event of certain stock dividends, stock splits, reorganizations, reclassifications and consolidations. The warrants have a net exercise provision under which the respective holder may, in lieu of payment of the exercise price in cash, surrender the warrant and receive a net amount of shares based on the fair market value of the shares at the time of exercise of the warrant after deduction of the aggregate exercise price.

As of March 31, 2025, we had a warrant to purchase an aggregate of up to 2,494,457 shares of our Series D convertible preferred stock outstanding, with an exercise price of $0.5412 per share. The warrant was issued in connection with the Trinity Loan Agreement. Unless exercised earlier, the warrant will expire in August 2033. The warrant vests and becomes exercisable in three equal installments upon the drawing of each of the three tranches under the Trinity Loan Agreement. See the section titled "Management's Discussion and Analysis of Financial Condition and Results of Operations—Liquidity and Capital Resources—Indebtedness." The warrant contains provisions for the adjustment of the exercise price and the number of shares issuable upon the exercise of the

warrant in the event of certain stock dividends, stock splits, reorganizations, reclassifications and consolidations. The warrant has a net exercise provision under which its holder may, in lieu of payment of the exercise price in cash, surrender the warrant and receive a net amount of shares based on the fair market value of the shares at the time of exercise of the warrant after deduction of the aggregate exercise price. Immediately prior to the completion of this offering, the warrant will automatically convert into a warrant to purchase 130,736 shares of our common stock.

2025 Convertible Notes

On July 18, 2025, we issued the 2025 Convertible Notes to certain investors in an aggregate principal amount of $40.0 million. The 2025 Convertible Notes mature on September 1, 2028 and accrue interest at a rate of 5.0% per annum through June 30, 2026 and 10.0% per annum after June 30, 2026. Upon the completion of this offering, the 2025 Convertible Notes and any accrued interest will automatically convert into shares of our common stock at the applicable conversion price. The conversion price is the lower of (a) 80% of the initial public offering price per share and (b) $280.0 million divided by the number of fully diluted shares of capital stock (on an as-converted basis) outstanding immediately prior to this offering but excluding the 2025 Convertible Notes (the "Fully Diluted Capitalization"), provided, that in no event shall such conversion price be less than the quotient obtained by dividing $210.0 million by the Fully Diluted Capitalization.Based on the initial public offering price of $15.00 per share the 2025 Convertible Notes will convert into an aggregate of 2,805,049 shares of our common stock.

Registration Rights

The fourth amended and restated investor rights agreement grants the parties thereto certain registration rights in respect of the "registrable securities" held by them, which securities include (i) the shares of our common stock issued or issuable upon the conversion of shares of our convertible preferred stock (ii) shares of our common stock issued or issuable upon conversion or exercise of any other company securities, and (iii) any common stock issued as a dividend or other distribution to or in exchange for or in replacement of the shares referenced in clauses (i) or (ii). The registration of shares of our common stock pursuant to the exercise of these registration rights would enable the holders thereof to sell such shares without restriction under the Securities Act when the applicable registration statement is declared effective. Under the fourth amended and restated investor rights agreement, we will pay expenses relating to such registrations, including up to $35,000 of the reasonable fees of one special counsel for the participating holders, and the holders will pay all underwriting discounts and commissions relating to the sale of their shares. The fourth amended and restated investor rights agreement also includes customary indemnification and procedural terms.

Demand Registration Rights

Upon the completion of this offering, holders of up to approximately 12.3 million shares of our common stock issuable upon conversion of our outstanding convertible preferred stock will be entitled to certain demand registration rights. Beginning on the earlier of (i) March 6, 2029 and (ii) six months following the effectiveness of the registration statement of which this prospectus is a part, investors holding, collectively, not less than 30% of registrable securities may, on not more than two occasions, request that we register all or a portion of their shares, subject to certain specified exceptions. Such request for registration must cover securities the anticipated aggregate offering price of which is at least $20.0 million. Following such a request, we will as soon as practicable, but in any event no more than 90 days, use good faith commercially reasonable best efforts to effect such registration.

Piggyback Registration Rights

In connection with this offering, holders of up to approximately 12.3 million shares of our common stock issuable upon conversion of our outstanding convertible preferred stock are entitled to their rights to notice of this offering and to include their shares of registrable securities in this offering. The requisite percentage of these stockholders have waived all such stockholders' rights to notice of this offering and to include their shares of registrable securities in this offering. In the event that we propose to register any of our securities under the Securities Act in another offering, either for our own account or for the account of other security holders, the holders of registrable securities will be entitled to certain "piggyback" registration rights allowing them to include their shares in such registration, subject to specified conditions and limitations.

S-3 Registration Rights

Upon the completion of this offering, the holders of up to approximately 12.3 million shares of our common stock issuable upon conversion of our outstanding convertible preferred stock will initially be entitled to certain Form S-3 registration rights. Any holder or holders of registrable securities may, with respect to not more than two such registrations within any 12-month period, request that we register all or a portion of their shares on Form S-3 if we are qualified to file a registration statement on Form S-3, subject to specified exceptions. Such request for registration on Form S-3 must cover securities with aggregate proceeds, net of underwriting discounts and expenses related to the issuance, which equal or exceed $2.0 million. The right to have such shares registered on Form S-3 is further subject to other specified conditions and limitations.

Anti-Takeover Provisions of Delaware Law and Our Amended and Restated Certificate of Incorporation and Amended and Restated Bylaws

Some provisions of Delaware law, our amended and restated certificate of incorporation and our amended and restated bylaws could make the following transactions more difficult: an acquisition of us by means of a tender offer; an acquisition of us by means of a proxy contest or otherwise; or the removal of our incumbent officers and directors. It is possible that these provisions could make it more difficult to accomplish or could deter transactions that stockholders may otherwise consider to be in their best interest or in our best interests, including transactions which provide for payment of a premium over the market price for our shares.

These provisions, summarized below, are intended to discourage coercive takeover practices and inadequate takeover bids. These provisions are also designed to encourage persons seeking to acquire control of us to first negotiate with our board of directors. We believe that the benefits of the increased protection of our potential ability to negotiate with the proponent of an unfriendly or unsolicited proposal to acquire or restructure us outweigh the disadvantages of discouraging these proposals because negotiation of these proposals could result in an improvement of their terms.

Undesignated Preferred Stock

The ability of our board of directors, without action by the stockholders, to issue up to 20,000,000 shares of undesignated preferred stock with voting or other rights or preferences as designated by our board of directors could impede the success of any attempt to change control of us. These and other provisions may have the effect of deferring hostile takeovers or delaying changes in control or management of our company.

Stockholder Meetings

Our amended and restated bylaws provide that a special meeting of stockholders may be called only by our chairman of the board, chief executive officer or president (in the absence of a chief executive officer), or by a resolution adopted by a majority of our board of directors.

Requirements for Advance Notification of Stockholder Nominations and Proposals

Our amended and restated bylaws establish advance notice procedures with respect to stockholder proposals to be brought before a stockholder meeting and the nomination of candidates for election as directors, other than nominations made by or at the direction of the board of directors or a committee of the board of directors.

Elimination of Stockholder Action by Written Consent

Our amended and restated certificate of incorporation eliminates the right of stockholders to act by written consent without a meeting.

Staggered Board

Our board of directors is divided into three classes. The directors in each class will serve for a three-year term, one class being elected each year by our stockholders. For more information on the classified board, see the section titled "Management—Board Composition." This system of electing and removing directors may tend to discourage

a third-party from making a tender offer or otherwise attempting to obtain control of us, because it generally makes it more difficult for stockholders to replace a majority of the directors.

Removal of Directors

Our amended and restated certificate of incorporation provides that no member of our board of directors may be removed from office by our stockholders except for cause and, in addition to any other vote required by law, upon the approval of the holders of at least two-thirds in voting power of the outstanding shares of stock entitled to vote in the election of directors.

Stockholders Not Entitled to Cumulative Voting

Our amended and restated certificate of incorporation does not permit stockholders to cumulate their votes in the election of directors. Accordingly, the holders of a majority of the outstanding shares of our common stock entitled to vote in any election of directors can elect all of the directors standing for election, if they choose, other than any directors that holders of our preferred stock may be entitled to elect.

Section 203 of the Delaware General Corporation Law

We are subject to Section 203 of the Delaware General Corporation Law, which prohibits persons deemed to be "interested stockholders" from engaging in a "business combination" with a publicly held Delaware corporation for three years following the date these persons become interested stockholders unless the business combination is, or the transaction in which the person became an interested stockholder was, approved in a prescribed manner or another prescribed exception applies. Generally, an "interested stockholder" is a person who, together with affiliates and associates, owns, or within three years prior to the determination of interested stockholder status did own, 15% or more of a corporation's voting stock. Generally, a "business combination" includes a merger, asset or stock sale, or other transaction resulting in a financial benefit to the interested stockholder. The existence of this provision may have an anti-takeover effect with respect to transactions not approved in advance by the board of directors.

Choice of Forum

Our amended and restated certificate of incorporation provides that, unless we consent in writing to the selection of an alternative form, the Court of Chancery of the State of Delaware will be the sole and exclusive forum for: (i) any derivative action or proceeding brought on our behalf; (ii) any action asserting a claim of breach of a fiduciary duty or other wrongdoing by any of our directors, officers, or stockholders to us or our stockholders; (iii) any action asserting a claim against us arising pursuant to any provision of the Delaware General Corporation Law or our certificate of incorporation or bylaws; (iv) any action to interpret, apply, enforce or determine the validity of our certificate of incorporation or bylaws; or (v) any action asserting a claim governed by the internal affairs doctrine; provided that the exclusive forum provision will not apply to suits brought to enforce any liability or duty created by the Securities Act or the Exchange Act, or to any claim for which the federal courts have exclusive jurisdiction. For instance, the provision would not apply to actions arising under federal securities laws, including suits brought to enforce any liability or duty created by the Securities Act, Exchange Act, or the rules and regulations thereunder. Our amended and restated certificate of incorporation further provides that, unless we consent in writing to the selection of an alternative forum, the federal district courts of the United States of America shall, to the fullest extent permitted by law, be the sole and exclusive forum for the resolution of any complaint asserting a cause of action arising under the Securities Act. Our amended and restated certificate of incorporation also provides that any person or entity purchasing or otherwise acquiring any interest in shares of our capital stock will be deemed to have notice of and to have consented to this choice of forum provision. It is possible that a court of law could rule that the choice of forum provision contained in our amended and restated certificate of incorporation is inapplicable or unenforceable if it is challenged in a proceeding or otherwise.

Amendment of Charter Provisions

The amendment of any of the above provisions, except for the provision making it possible for our board of directors to issue preferred stock and the provision prohibiting cumulative voting, would require approval by holders of at least two-thirds in voting power of the outstanding shares of stock entitled to vote thereon.

The provisions of Delaware law, our amended and restated certificate of incorporation and our amended and restated bylaws could have the effect of discouraging others from attempting hostile takeovers and, as a consequence, they may also inhibit temporary fluctuations in the market price of our common stock that often result from actual or rumored hostile takeover attempts. These provisions may also have the effect of preventing changes in the composition of our board and management. It is possible that these provisions could make it more difficult to accomplish transactions that stockholders may otherwise deem to be in their best interests.

Limitations on Liability and Indemnification Matters

Our amended and restated certificate of incorporation provides that no director will be personally liable to us or our stockholders for monetary damages for breach of fiduciary duty as a director, except as required by applicable law, as in effect from time to time. Section 102(b)(7) of the Delaware General Corporation Law permits a corporation to provide in its certificate of incorporation that a director of the corporation shall not be personally liable to the corporation or its stockholders for monetary damages for breach of fiduciary duty as a director, except for liability for:

- any breach of the director's duty of loyalty to our company or our stockholders;
- any act or omission not in good faith or which involved intentional misconduct or a knowing violation of law;
- unlawful payments of dividends or unlawful stock repurchases or redemptions as provided in Section 174 of the Delaware General Corporation Law; and
- any transaction from which the director derived an improper personal benefit.

As a result, neither we nor our stockholders have the right, through stockholders' derivative suits on our behalf, to recover monetary damages against a director for breach of fiduciary duty as a director, including breaches resulting from grossly negligent behavior, except in the situations described above.

Our amended and restated certificate of incorporation also provides that, to the fullest extent permitted by law, we will indemnify any officer or director of our company against all damages, claims and liabilities arising out of the fact that the person is or was our director or officer, or served any other enterprise at our request as a director or officer. Amending this provision will not reduce our indemnification obligations relating to actions taken before an amendment.

Listing

We have been approved to list our common stock on the New York Stock Exchange under the trading symbol "SI."

Transfer Agent and Registrar

Upon completion of this offering, the transfer agent and registrar for our common stock will be Broadridge Corporate Issuer Solutions, LLC. The transfer agent and registrar's address is 51 Mercedes Way, Edgewood, New York 11717.

SHARES ELIGIBLE FOR FUTURE SALE

Prior to this offering, there has been no public market for our common stock, and a liquid trading market for our common stock may not develop or be sustained after this offering. Future sales of our common stock, including shares issued upon the exercise of outstanding options or warrants, in the public market after the completion of this offering, or the perception that those sales may occur, could adversely affect the prevailing market price for our common stock from time to time or impair our ability to raise equity capital in the future.

As described below, only a limited number of shares of our common stock will be available for sale in the public market for a period of several months after the completion of this offering due to contractual and legal restrictions on resale described below. Future sales of our common stock in the public market either before or after restrictions lapse, or the perception that those sales may occur, could adversely affect the prevailing market price of our common stock at such time and our ability to raise equity capital at a time and price we deem appropriate.

Sale of Restricted Shares

Upon the completion of this offering, we will have outstanding an aggregate of 20,228,510 shares of common stock, based on the number of shares outstanding as of March 31, 2025, assuming (i) the conversion of all outstanding shares of our convertible preferred stock (including 29,319,143 shares of our Series E convertible preferred stock issued in June 2025), (ii) the issuance of the 2025 Convertible Notes and the related Notes Conversion, based on the initial public offering price of $15.00 per share (iii) no exercise of the underwriters' over-allotment option, and (iv) no exercise of outstanding options or warrants.

Of these shares, all of the shares of common stock to be sold in this offering will be freely tradable in the public market without restriction or further registration under the Securities Act, unless the shares are held by any of our "affiliates" as such term is defined in Rule 144 and any shares purchased by our directors or officers pursuant to our directed share program will be subject to the lock-up agreements described below. All remaining shares of common stock held by existing stockholders will be "restricted securities," as such term is defined in Rule 144. These restricted securities are eligible for public sale only if they are registered under the Securities Act or if they qualify for an exemption from registration under Rules 144 or 701 under the Securities Act, which are summarized below. We expect that substantially all of these shares will be subject to the 180-day lock-up period under the lock-up agreements described below. Upon expiration of the lock-up period, we estimate that substantially all of such remaining shares will be available for sale in the public market, subject in some cases to applicable volume limitations under Rule 144.

These restricted securities were issued and sold by us in private transactions and are eligible for public sale only if registered under the Securities Act or if they qualify for an exemption from registration under the Securities Act, including the exemptions provided by Rule 144 or Rule 701. These rules are summarized below.

In addition, the shares of common stock reserved for future issuance under the 2025 Plan will become eligible for sale in the public market to the extent permitted by the provisions of various vesting schedules, lock-up agreements, market standoff restrictions, a registration statement under the Securities Act, or an exemption from registration, including Rule 144 and Rule 701.

Rule 144

Affiliate Resales of Restricted Securities

In general, beginning 90 days after the effective date of the registration statement of which this prospectus is a part, a person who is an affiliate of ours, or who was an affiliate at any time during the 90 days before a sale, who has beneficially owned shares of our common stock for at least six months would be entitled to sell in "broker's transactions" or certain "riskless principal transactions" or to market makers, a number of shares within any three-month period that does not exceed the greater of:

- 1% of the number of shares of our common stock then outstanding, which will equal approximately 202,285 shares immediately after this offering; or

- the average weekly trading volume in our common stock on the New York Stock Exchange during the four calendar weeks preceding the filing of a notice on Form 144 with respect to such sale.

Affiliate resales under Rule 144 are also subject to the availability of current public information about us. In addition, if the number of shares being sold under Rule 144 by an affiliate during any three-month period exceeds 5,000 shares or has an aggregate sale price in excess of $50,000, the seller must file a notice on Form 144 with the SEC and the New York Stock Exchange concurrently with either the placing of a sale order with the broker or the execution directly with a market maker.

Non-Affiliate Resales of Restricted Securities

In general, beginning 90 days after the effective date of the registration statement of which this prospectus is a part, a person who is not an affiliate of ours at the time of sale, and has not been an affiliate at any time during the three months preceding a sale, and who has beneficially owned shares of our common stock for at least six months but less than a year, is entitled to sell such shares subject only to the availability of current public information about us. If such person has held our shares for at least one year, such person can resell under Rule 144(b)(1) without regard to any Rule 144 restrictions, including the 90-day public company requirement and the current public information requirement.

Non-affiliate resales are not subject to the manner of sale, volume limitation or notice filing provisions of Rule 144.

Rule 701

In general, under Rule 701, any of an issuer's employees, directors, officers, consultants or advisors who purchases shares from the issuer in connection with a compensatory stock or option plan or other written agreement before the effective date of a registration statement under the Securities Act is entitled to sell such shares 90 days after such effective date in reliance on Rule 144. An affiliate of the issuer can resell shares in reliance on Rule 144 without having to comply with the holding period requirement, and non-affiliates of the issuer can resell shares in reliance on Rule 144 without having to comply with the current public information and holding period requirements.

The SEC has indicated that Rule 701 will apply to typical stock options granted by an issuer before it becomes subject to the reporting requirements of the Exchange Act, along with the shares acquired upon exercise of such options, including exercises after an issuer becomes subject to the reporting requirements of the Exchange Act.

Lock-Up Agreements and Market Standoff Restrictions

We and each of our directors and executive officers and holders of substantially all of our outstanding capital stock, have agreed that, without the prior written consent of Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co., and subject to certain exceptions, we and they will not, and will not publicly disclose an intention to, during the period ending 180 days after the date of this prospectus, offer, pledge, sell, contract to sell, sell any option or contract to purchase, purchase any option or contract to sell, grant any option, right or warrant to purchase, lend, or otherwise transfer or dispose of, directly or indirectly, any shares of common stock or any securities convertible into or exercisable or exchangeable for shares of common stock; file any registration statement with the SEC relating to the offering of any shares of common stock or any securities convertible into or exercisable or exchangeable for common stock; enter into any swap or other arrangement that transfers to another, in whole or in part, any of the economic consequences of ownership of the common stock; or make any demand for, or exercise any right with respect to, the registration of any shares of common stock or any security convertible into or exercisable or exchangeable for common stock.

Upon the expiration of the applicable lock-up periods, substantially all of the shares subject to such lock-up restrictions will become eligible for sale, subject to the limitations discussed above. Any shares purchased by our directors or officers pursuant to our directed share program shall also be subject to the lock-up agreements described above. For a further description of these lock-up agreements, see the section titled "Underwriting."

Registration Rights

Upon the completion of this offering, the holders of 12,320,676 shares of our common stock, which includes all of the shares of common stock issuable upon the conversion of our convertible preferred stock immediately prior to the completion of this offering, or their transferees will be entitled to various rights with respect to the registration of these shares under the Securities Act. Registration of these shares under the Securities Act would result in these shares becoming fully tradable without restriction under the Securities Act immediately upon the effectiveness of the registration, except for shares purchased by affiliates. See the section titled "Description of Capital Stock—Registration Rights" for additional information. Shares covered by a registration statement will be eligible for sale in the public market upon the expiration or release from the terms of the lock-up agreement described above.

Equity Incentive Plans

We intend to file with the SEC a registration statement on Form S-8 under the Securities Act covering the shares of common stock subject to issuance upon the exercise of outstanding stock options under the Stock Option Plan and reserved for issuance under the 2025 Plan and the ESPP. Such registration statement is expected to be filed and become effective as soon as practicable after the completion of this offering. Accordingly, shares registered under such registration statement will be available for sale in the open market following its effective date, subject to Rule 144 volume limitations, vesting restrictions, and the lock-up agreements and market standoff restrictions described above, if applicable.

Rule 10b5-1 Trading Plans

Certain of our employees, directors, and stockholders may adopt written plans, known as Rule 10b5-1 trading plans, in which they will contract with a broker to buy or sell shares of our common stock on a periodic basis to diversify their assets and investments. Under these Rule 10b5-1 trading plans, a broker may execute trades pursuant to parameters established by the employee, director, or stockholder when entering into the plan, without further direction from such employee, director, or stockholder. Such sales would not commence until the expiration of the applicable market standoff restrictions or lock-up agreements entered into by such employee, director, or stockholder in connection with this offering.

MATERIAL U.S. FEDERAL INCOME TAX CONSEQUENCES TO NON-U.S. HOLDERS

The following discussion is a summary of the material U.S. federal income tax consequences to Non-U.S. Holders (as defined below) of the ownership and disposition of our common stock issued pursuant to this offering, but does not purport to be a complete analysis of all potential tax effects. The effects of other U.S. federal tax laws, such as estate and gift tax laws, and any applicable state, local, or non-U.S. tax laws are not discussed. This discussion is based on the Code, Treasury Regulations promulgated thereunder, judicial decisions, and published rulings and administrative pronouncements of the IRS, in each case in effect as of the date hereof. These authorities may change or be subject to differing interpretations. Any such change or differing interpretation may be applied retroactively in a manner that could adversely affect a Non-U.S. Holder. We have not sought and will not seek any rulings from the IRS regarding the matters discussed below. There can be no assurance the IRS or a court will not take a contrary position to that discussed below regarding the tax consequences of the ownership and disposition of our common stock.

This discussion is limited to Non-U.S. Holders that hold our common stock as a "capital asset" within the meaning of Section 1221 of the Code (generally, property held for investment). This discussion does not address all U.S. federal income tax consequences relevant to a Non-U.S. Holder's particular circumstances, including the impact of the Medicare contribution tax on net investment income, the alternative minimum tax or the special tax accounting rules under Section 451(b) of the Code. In addition, it does not address consequences relevant to Non-U.S. Holders subject to special rules, including, without limitation:

- U.S. expatriates and former citizens or long-term residents of the United States;
- persons holding our common stock as part of a hedge, straddle, or other risk reduction strategy or as part of a conversion transaction or other integrated investment;
- banks, insurance companies, and other financial institutions;
- brokers, dealers, or traders in securities;
- "controlled foreign corporations," "passive foreign investment companies," and corporations that accumulate earnings to avoid U.S. federal income tax;
- partnerships or other entities or arrangements treated as partnerships for U.S. federal income tax purposes (and investors therein);
- tax-exempt organizations or governmental organizations;
- persons deemed to sell our common stock under the constructive sale provisions of the Code;
- persons who hold or receive our common stock pursuant to the exercise of an employee stock option or otherwise as compensation;
- tax-qualified retirement plans; and
- "qualified foreign pension funds" as defined in Section 897(l)(2) of the Code and entities all of the interests of which are held by qualified foreign pension funds.

If an entity or arrangement treated as a partnership for U.S. federal income tax purposes holds our common stock, the tax treatment of a partner in the partnership will depend on the status of the partner, the activities of the partnership, and certain determinations made at the partner level. Accordingly, partnerships holding our common stock and the partners in such partnerships should consult their tax advisors regarding the U.S. federal income tax consequences to them of the purchase, ownership and disposition of our common stock.

THIS DISCUSSION IS FOR INFORMATIONAL PURPOSES ONLY AND IS NOT TAX ADVICE. PROSPECTIVE INVESTORS SHOULD CONSULT THEIR TAX ADVISORS WITH RESPECT TO THE APPLICATION OF THE U.S. FEDERAL INCOME TAX LAWS TO THEIR PARTICULAR SITUATIONS AS WELL AS ANY TAX CONSEQUENCES OF THE PURCHASE, OWNERSHIP, AND DISPOSITION OF OUR COMMON STOCK ARISING UNDER THE U.S. FEDERAL ESTATE OR GIFT TAX LAWS OR UNDER THE LAWS OF ANY STATE, LOCAL, OR NON-U.S. TAXING JURISDICTION OR UNDER ANY APPLICABLE INCOME TAX TREATY.

Definition of a Non-U.S. Holder

For purposes of this discussion, a "Non-U.S. Holder" is any beneficial owner of our common stock that is neither a "U.S. person" nor an entity or arrangement treated as a partnership for U.S. federal income tax purposes. A U.S. person is any person that, for U.S. federal income tax purposes, is or is treated as any of the following:

- an individual who is a citizen or resident of the United States;
- a corporation created or organized under the laws of the United States, any state thereof, or the District of Columbia;
- an estate, the income of which is subject to U.S. federal income tax regardless of its source; or
- a trust that (i) is subject to the primary supervision of a U.S. court and the control of one or more "United States persons" (within the meaning of Section 7701(a)(30) of the Code), or (ii) has a valid election in effect to be treated as a United States person for U.S. federal income tax purposes.

Distributions

As described in the section titled "Dividend Policy," we do not anticipate declaring or paying dividends to holders of our common stock in the foreseeable future. However, if we do make distributions of cash or property on our common stock, such distributions will constitute dividends for U.S. federal income tax purposes to the extent paid from our current or accumulated earnings and profits, as determined under U.S. federal income tax principles. Amounts not treated as dividends for U.S. federal income tax purposes will constitute a return of capital and first be applied against and reduce a Non-U.S. Holder's adjusted tax basis in its common stock, but not below zero. Any excess will be treated as capital gain and will be treated as described below under "—Sale or Other Taxable Disposition."

Subject to the discussion below on effectively connected income, dividends paid to a Non-U.S. Holder will be subject to U.S. federal withholding tax at a rate of 30% of the gross amount of the dividends (or such lower rate specified by an applicable income tax treaty, provided the Non-U.S. Holder furnishes a valid IRS Form W-8BEN or W-8BEN-E (or other applicable documentation) certifying qualification for the lower treaty rate). A Non-U.S. Holder that does not timely furnish the required documentation, but that qualifies for a reduced treaty rate, may obtain a refund of any excess amounts withheld by timely filing an appropriate claim for refund with the IRS. Non-U.S. Holders should consult their tax advisors regarding their entitlement to benefits under any applicable income tax treaty.

If dividends paid to a Non-U.S. Holder are effectively connected with the Non-U.S. Holder's conduct of a trade or business within the United States (and, if required by an applicable income tax treaty, the Non-U.S. Holder maintains a permanent establishment in the United States to which such dividends are attributable), the Non-U.S. Holder will be exempt from the U.S. federal withholding tax described above. To claim the exemption, the Non-U.S. Holder must furnish to the applicable withholding agent a valid IRS Form W-8ECI, certifying that the dividends are effectively connected with the Non-U.S. Holder's conduct of a trade or business within the United States.

Any such effectively connected dividends generally will be subject to U.S. federal income tax on a net income basis at the regular rates applicable to U.S. persons. A Non-U.S. Holder that is a corporation also may be subject to a branch profits tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on such

effectively connected dividends, as adjusted for certain items. Non-U.S. Holders should consult their tax advisors regarding any applicable tax treaties that may provide for different rules.

Sale or Other Taxable Disposition

A Non-U.S. Holder will not be subject to U.S. federal income tax on any gain realized upon the sale or other taxable disposition of our common stock unless:

- the gain is effectively connected with the Non-U.S. Holder's conduct of a trade or business within the United States (and, if required by an applicable income tax treaty, the Non-U.S. Holder maintains a permanent establishment in the United States to which such gain is attributable);
- the Non-U.S. Holder is a nonresident alien individual present in the United States for 183 days or more during the taxable year of the disposition and certain other conditions are met; or
- our common stock constitutes a U.S. real property interest ("USRPI") by reason of our status as a U.S. real property holding corporation ("USRPHC") for U.S. federal income tax purposes.

Gain described in the first bullet point above generally will be subject to U.S. federal income tax on a net income basis at the regular rates applicable to U.S. persons. A Non-U.S. Holder that is a corporation also may be subject to a branch profits tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on such effectively connected gain, as adjusted for certain items.

A Non-U.S. Holder described in the second bullet point above will be subject to U.S. federal income tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on gain realized upon the sale or other taxable disposition of our common stock, which may be offset by U.S. source capital losses of the Non-U.S. Holder (even though the individual is not considered a resident of the United States), provided the Non-U.S. Holder has timely filed U.S. federal income tax returns with respect to such losses.

With respect to the third bullet point above, we believe we currently are not, and do not anticipate becoming, a USRPHC. Because the determination of whether we are a USRPHC depends, however, on the fair market value of our USRPIs relative to the fair market value of our non-U.S. real property interests and our other business assets, there can be no assurance we currently are not a USRPHC or will not become one in the future. Even if we are or were to become a USRPHC, gain arising from the sale or other taxable disposition of our common stock by a Non-U.S. Holder will not be subject to U.S. federal income tax if our common stock is "regularly traded," as defined by applicable Treasury Regulations, on an established securities market and such Non-U.S. Holder owned, actually and constructively, 5% or less of our common stock throughout the shorter of the five-year period ending on the date of the sale or other taxable disposition or the Non-U.S. Holder's holding period.

Non-U.S. Holders should consult their tax advisors regarding potentially applicable income tax treaties that may provide for different rules.

Information Reporting and Backup Withholding

Payments of dividends on our common stock to a Non-U.S. Holder will not be subject to backup withholding, provided the holder either certifies its non-U.S. status, such as by furnishing a valid IRS Form W-8BEN, W-8BEN-E, or W-8ECI, or otherwise establishes an exemption and the applicable withholding agent does not have actual knowledge or reason to know the holder is a United States person. However, information returns are required to be filed with the IRS in connection with any distributions on our common stock paid to the Non-U.S. Holder, regardless of whether such distributions constitute dividends or whether any tax was actually withheld. In addition, proceeds of the sale or other taxable disposition of our common stock within the United States or conducted through certain U.S.-related brokers generally will not be subject to backup withholding or information reporting if the applicable withholding agent receives the certification described above and does not have actual knowledge or reason to know that such holder is a United States person or the holder otherwise establishes an exemption. Proceeds of a disposition of our common stock conducted through a non-U.S. office of a non-U.S. broker generally will not be subject to backup withholding or information reporting.

Copies of information returns that are filed with the IRS may also be made available under the provisions of an applicable treaty or agreement to the tax authorities of the country in which the Non-U.S. Holder resides or is established.

Backup withholding is not an additional tax. Any amounts withheld under the backup withholding rules may be allowed as a credit against a Non-U.S. Holder's U.S. federal income tax liability, if any, or refunded, provided the required information is timely furnished to the IRS. Non-U.S. Holders should consult their tax advisors regarding the application of the information reporting and backup withholding rules to them.

Additional Withholding Tax on Payments Made to Foreign Accounts

Withholding taxes may be imposed under Sections 1471 to 1474 of the Code (such Sections commonly referred to as the Foreign Account Tax Compliance Act ("FATCA")) on certain types of payments made to non-U.S. financial institutions and certain other non-U.S. entities. Specifically, a 30% withholding tax may be imposed on dividends on, or (subject to the proposed Treasury Regulations discussed below) gross proceeds from the sale or other disposition of, our common stock paid to a "foreign financial institution" or a "non-financial foreign entity" (each as defined in the Code), unless (1) the foreign financial institution undertakes certain diligence and reporting obligations, (2) the non-financial foreign entity either certifies it does not have any "substantial United States owners" (as defined in the Code) or furnishes identifying information regarding each substantial United States owner, or (3) the foreign financial institution or non-financial foreign entity otherwise qualifies for an exemption from these rules. If the payee is a foreign financial institution and is subject to the diligence and reporting requirements in (1) above, it must enter into an agreement with the U.S. Department of the Treasury requiring, among other things, that it undertake to identify accounts held by certain "specified United States persons" or "United States owned foreign entities" (each as defined in the Code), annually report certain information about such accounts, and withhold 30% on certain payments to non-compliant foreign financial institutions and certain other account holders. Foreign financial institutions located in jurisdictions that have an intergovernmental agreement with the United States governing FATCA may be subject to different rules.

Under the applicable Treasury Regulations and administrative guidance, withholding under FATCA generally applies to payments of dividends on our common stock. While withholding under FATCA would have applied also to payments of gross proceeds from the sale or other disposition of stock, proposed Treasury Regulations eliminate FATCA withholding on payments of gross proceeds entirely. Taxpayers generally may rely on these proposed Treasury Regulations until final Treasury Regulations are issued.

Prospective investors should consult their tax advisors regarding the potential application of withholding under FATCA to their investment in our common stock.

PROSPECTIVE INVESTORS SHOULD CONSULT THEIR TAX ADVISORS WITH RESPECT TO THE APPLICATION OF THE U.S. FEDERAL INCOME TAX LAWS TO THEIR PARTICULAR SITUATIONS AS WELL AS ANY TAX CONSEQUENCES OF THE PURCHASE, OWNERSHIP, AND DISPOSITION OF OUR COMMON STOCK ARISING UNDER THE U.S. FEDERAL ESTATE OR GIFT TAX LAWS OR UNDER THE LAWS OF ANY STATE, LOCAL, OR NON-U.S. TAXING JURISDICTION OR UNDER ANY APPLICABLE INCOME TAX TREATY.

UNDERWRITING

Under the terms and subject to the conditions in an underwriting agreement dated the date of this prospectus, the underwriters named below, for whom Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co. are acting as representatives, have severally agreed to purchase, and we have agreed to sell to them, severally, the number of shares indicated below:

Name	Number of Shares
Morgan Stanley & Co. LLC	1,600,000
Goldman Sachs & Co. LLC	1,400,000
Piper Sandler & Co.	1,000,000
Jefferies LLC	600,000
BTIG, LLC	400,000
Total	5,000,000

The underwriters and the representatives are collectively referred to as the "underwriters" and the "representatives," respectively. The underwriters are offering the shares of common stock subject to their acceptance of the shares from us and subject to prior sale. The underwriting agreement provides that the obligations of the several underwriters to pay for and accept delivery of the shares of common stock offered by this prospectus are subject to the approval of certain legal matters by their counsel and to certain other conditions. The underwriters are obligated to take and pay for all of the shares of common stock offered by this prospectus if any such shares are taken. However, the underwriters are not required to take or pay for the shares covered by the underwriters' over-allotment option described below. The offering of the shares of common stock by the underwriters is subject to receipt and acceptance and subject to the underwriters' right to reject any order in whole or in part.

The underwriters initially propose to offer part of the shares of common stock directly to the public at the offering price listed on the cover page of this prospectus and part to certain dealers at a price that represents a concession not in excess of $0.63 per share under the public offering price. After the initial offering of the shares of common stock, the offering price and other selling terms may from time to time be varied by the representatives.

In addition, we have requested that the underwriters make issuer directed allocations in the aggregate of 944,332 shares of our common stock to certain investors.

We have granted to the underwriters an option, exercisable for 30 days from the date of this prospectus, to purchase up to 750,000 additional shares of common stock at the public offering price listed on the cover page of this prospectus, less underwriting discounts and commissions. The underwriters may exercise this option solely for the purpose of covering over-allotments, if any, made in connection with the offering of the shares of common stock offered by this prospectus. To the extent the option is exercised, each underwriter will become obligated, subject to certain conditions, to purchase about the same percentage of the additional shares of common stock as the number listed next to the underwriter's name in the preceding table bears to the total number of shares of common stock listed next to the names of all underwriters in the preceding table.

The following table shows the per share and total public offering price, underwriting discounts and commissions, and proceeds before expenses to us. These amounts are shown assuming both no exercise and full exercise of the underwriters' over-allotment option.

		Total	
	Per Share	No Exercise	Full Exercise
Public offering price	$ 15.00	$ 75,000,000	$ 86,250,000
Underwriting discounts and commissions to be paid by us	$ 1.05	$ 5,250,000	$ 6,037,500
Proceeds, before expenses, to us	$ 13.95	$ 69,750,000	$ 80,212,500

The estimated offering expenses payable by us, exclusive of the underwriting discounts and commissions, are approximately $5.0 million. We have also agreed to reimburse the underwriters for expenses relating to clearance of this offering with the Financial Industry Regulatory Authority ("FINRA") up to $40,000.

The underwriters have informed us that they do not intend sales to discretionary accounts to exceed 5% of the total number of shares of common stock offered by them.

We have been approved to list our common stock on the New York Stock Exchange under the trading symbol "SI."

We and all of our directors and executive officers and the holders of substantially all of our outstanding securities have agreed that, without the prior written consent of Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co. on behalf of the underwriters, we and they will not, and will not publicly disclose an intention to, during the period ending 180 days after the date of this prospectus (the "restricted period"):

- offer, pledge, sell, contract to sell, sell any option or contract to purchase, purchase any option or contract to sell, grant any option, right or warrant to purchase, lend or otherwise transfer or dispose of, directly or indirectly, any shares of common stock or any securities convertible into or exercisable or exchangeable for shares of common stock;
- submit or file any registration statement with the SEC relating to the offering of any shares of common stock or any securities convertible into or exercisable or exchangeable for common stock; or
- enter into any swap or other arrangement that transfers to another, in whole or in part, any of the economic consequences of ownership of the common stock;

whether any such transaction described above is to be settled by delivery of common stock or such other securities, in cash or otherwise. In addition, we and each such person have agreed that, without the prior written consent of Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co. on behalf of the underwriters, we or such other person will not, during the restricted period, make any demand for, or exercise any right with respect to, the registration of any shares of common stock or any security convertible into or exercisable or exchangeable for common stock.

With respect to us, the restrictions described in the immediately preceding paragraph do not apply to:

(i) the shares to be sold in this offering;

(ii) the issuance by us of shares of common stock upon the exercise of an option or warrant or the conversion of a security outstanding on the date of this prospectus as described in the registration statement and this prospectus;

(iii) grants of compensatory equity-based awards, and/or the issuance of shares of common stock or securities with respect thereto, made pursuant to compensatory equity-based plans as described in this prospectus; provided each recipient of such grant shall enter into a lock-up agreement;

(iv) shares of common stock or any securities convertible into, or exercisable or exchangeable for, shares of common stock, or the entrance into an agreement to issue shares of common stock or any securities convertible into, or exercisable or exchangeable for, shares of common stock, in connection with any merger, joint venture, strategic alliances, commercial or other collaborative transaction or the acquisition or licenses of the business, property, technology or other assets of another individual or entity or the assumption of an employee benefit plan in connection with a merger or acquisition; provided that the aggregate number of shares of common stock or any other securities convertible into, or exercisable or exchangeable for, shares of common stock that we may issue or agree to issue pursuant to this clause (iv) shall not exceed 5% of our total outstanding share capital immediately following the issuance of such common stock; and provided further, that the recipients of any such shares of common stock and securities issued pursuant to this clause (iv) during the restricted period shall enter into a lock-up agreement on or prior to such issuance;

(v) facilitating the establishment of a trading plan on behalf of any of our stockholders, officers or directors pursuant to Rule 10b5-1 under the Exchange Act for the transfer of shares of our common stock, provided that (A) such plan does not provide for the transfer of common stock during the restricted period and (B) to the extent a public announcement or filing under the Exchange Act, if any, is required of or voluntarily made by us regarding the establishment of such plan, such announcement or filing shall include a statement to the effect that no transfer of common stock may be made under such plan during the restricted period; or

(vi) the filing of a registration statement on Form S-8 to register shares of common stock issuable pursuant to any employee benefit plans, qualified stock option plans or other employee compensation plans, described in this prospectus.

With respect to our directors, executive officers and securityholders, the restrictions described above do not apply to:

(a) transactions relating to shares of common stock or other securities acquired in open market transactions after the completion of this offering;

(b) transfers of shares of common stock or any security convertible into or exercisable or exchangeable for common stock (i) as a bona fide gift, or for bona fide estate planning purposes, (ii) upon death or by will, testamentary document or intestate succession, (iii) to any immediate family member or to any trust for the direct or indirect benefit of such person or member of their immediate family, if such person is a trust, to a trustor or beneficiary of the trust or the estate of a beneficiary of such trust, (iv) to a corporation, partnership, limited liability company or other entity of which such person and their immediate family are the legal and beneficial owner of all of the outstanding equity securities or similar interests, or (v) to a nominee or custodian of a person or entity to whom a disposition or transfer would be permissible under clauses (b)(i) through (iv) above;

(c) if such person is a corporation, partnership, limited liability company or other business entity, any transfers (i) to another corporation, partnership, limited liability company or other business entity that is an affiliate (as defined in Rule 405 under the Securities Act of 1933, as amended) of such person, or to any investment fund or other entity which fund or entity is controlled or managed by such person or affiliates of such person, or (ii) as part of a distribution to its stockholders, partners, members or other equityholders or to the estate of any such stockholders, partners, members or other equityholders;

(d) any transfers by operation of law, such as pursuant to a qualified domestic order, divorce settlement, divorce decree or separation agreement;

(e) any transfers to us from an employee upon death, disability or termination of employment, in each case, of such employee;

(f) any transfers to us in connection with the vesting, settlement or exercise of restricted stock units, options, warrants or other rights to purchase shares of common stock (including, in each case, by way of "net" or "cashless" exercise), including any transfer to us for the payment of tax withholdings or remittance payments due as a result of the vesting, settlement or exercise of such restricted stock units, options, warrants or other rights, or in connection with the conversion of convertible securities, in all such cases pursuant to equity awards granted under a stock incentive plan or other equity award plan, or pursuant to the terms of convertible securities, each as described in this prospectus;

(g) distributions of shares of common stock or any security convertible into common stock to limited partners or stockholders of such person;

(h) the entry or amendment of a trading plan on behalf of a shareholder, officer or director pursuant to Rule 10b5-1 under the Exchange Act for the transfer of shares of common stock, provided that such plan does not provide for the transfer of common stock during the restricted period;

(i) transfers of securities to us pursuant to an agreement under which we have the option to repurchase shares or a right of first refusal with respect to transfer of such shares;

(j) conversions of outstanding preferred stock into shares of common stock, provided that any such shares received upon such conversion shall remain subject to the provisions of a lock-up agreement;

(k) conversions or reclassifications of a class or series of common stock into another class or series of common stock (including the conversion of shares of non-voting common stock into shares of voting common stock and vice versa), it being understood that any such shares received upon such conversion shall be subject to the provisions of a lock-up agreement;

(l) transfers of common stock or other security in connection with a bona fide third-party tender offer, merger, consolidation or other similar transaction that is approved by our board of directors and made to all holders of our capital stock involving a change of control in one transaction or a series of related transactions, to a person or group of affiliated persons, of shares of capital stock if, after such transfer, such person or group of affiliated persons would hold at least a majority of our outstanding voting securities (or the surviving entity); provided that in the event that such tender offer, merger, consolidation or other similar transaction is not completed, such common stock or other security shall remain subject to the provisions of a lock-up agreement;

provided that (1) in the case of clauses (b)(i), (ii), (iii), (iv) and (v) and (c) above, such transfer or distribution shall not involve a disposition for value, (2) in the case of clauses (b)(i), (ii), (iii), (iv) and (v), (c), (d) and (g) above, it shall be a condition to the transfer or distribution that the donee, devisee, transferee or distributee, as the case may be, shall sign and deliver a lock-up agreement, (3) in the case of clauses (b)(ii), (iii), (iv), and (v) and (c) and (g) above, no filing by any party (including, without limitation, any donor, donee, devisee, transferor, transferee, distributor or distributee) under the Exchange Act, or other public filing, report or announcement reporting a reduction in beneficial ownership of lock-up securities shall be required or shall be voluntarily made in connection with such transfer or distribution, and (4) in the case of clauses (a), (b)(i), (d), (e), (f), (h) and (i) above, no filing under the Exchange Act or other public filing, report or announcement shall be voluntarily made, and if any such filing, report or announcement shall be legally required during the restricted period, such filing, report or announcement shall clearly indicate in the footnotes thereto (1) the circumstances of such transfer or distribution and (2) in the case of a transfer or distribution pursuant to clauses (b)(i) or (d) above, that the donee, devisee, transferee or distributee has agreed to be bound by a lock-up agreement.

Morgan Stanley & Co. LLC, Goldman Sachs & Co. LLC and Piper Sandler & Co., in their sole discretion, may release the common stock and other securities subject to the lock-up agreements described above in whole or in part at any time.

In order to facilitate the offering of the common stock, the underwriters may engage in transactions that stabilize, maintain or otherwise affect the price of the common stock. Specifically, the underwriters may sell more shares than they are obligated to purchase under the underwriting agreement, creating a short position. A short sale is covered if the short position is no greater than the number of shares available for purchase by the underwriters under the over-allotment option. The underwriters can close out a covered short sale by exercising the over-allotment option or purchasing shares in the open market. In determining the source of shares to close out a covered short sale, the underwriters will consider, among other things, the open market price of shares compared to the price available under the over-allotment option. The underwriters may also sell shares in excess of the over-allotment option, creating a naked short position. The underwriters must close out any naked short position by purchasing shares in the open market. A naked short position is more likely to be created if the underwriters are concerned that there may be downward pressure on the price of the common stock in the open market after pricing that could adversely affect investors who purchase in this offering. As an additional means of facilitating this offering, the underwriters may bid for, and purchase, shares of common stock in the open market to stabilize the price of the common stock. These activities may raise or maintain the market price of the common stock above independent market levels or prevent or retard a decline in the market price of the common stock. The underwriters are not required to engage in these activities and may end any of these activities at any time.

We and the underwriters have agreed to indemnify each other against certain liabilities, including liabilities under the Securities Act.

A prospectus in electronic format may be made available on websites maintained by one or more underwriters, or selling group members, if any, participating in this offering. The representatives may agree to allocate a number of shares of common stock to underwriters for sale to their online brokerage account holders. Internet distributions will be allocated by the representatives to underwriters that may make Internet distributions on the same basis as other allocations.

Directed Share Program

At our request, the underwriters have reserved up to 6% of the shares of common stock to be issued by us and offered by this prospectus for sale, at the initial public offering price, to certain of our directors, officers and employees and others. The sales will be made at our direction by Morgan Stanley & Co. LLC and its affiliates through a directed share program. Any shares sold in the directed share program to our directors or officers who have entered into lock-up agreements described above will be subject to the provisions of such lock-up agreements. The number of shares of common stock available for sale to the general public will be reduced to the extent such persons purchase such reserved shares. Any reserved shares that are not so purchased will be offered by the underwriters to the general public on the same basis as the other shares offered by this prospectus.

Other Relationships

The underwriters and their respective affiliates are full service financial institutions engaged in various activities, which may include securities trading, commercial and investment banking, financial advisory, investment management, investment research, principal investment, hedging, financing and brokerage activities. Certain of the underwriters and their respective affiliates have, from time to time, performed, and may in the future perform, various financial advisory and investment banking services for us, for which they received or will receive customary fees and expenses.

In addition, in the ordinary course of their various business activities, the underwriters and their respective affiliates may make or hold a broad array of investments and actively trade debt and equity securities (or related derivative securities) and financial instruments (including bank loans) for their own account and for the accounts of their customers and may at any time hold long and short positions in such securities and instruments. Such investment and securities activities may involve our securities and instruments. The underwriters and their respective affiliates may also make investment recommendations or publish or express independent research views in respect of such securities or instruments and may at any time hold, or recommend to clients that they acquire, long or short positions in such securities and instruments.

Pricing of the Offering

Prior to this offering, there was no public market for our common stock. The initial public offering price was determined by negotiations between us and the representatives. Among the factors considered in determining the initial public offering price were our future prospects and those of our industry in general, our sales, earnings and certain other financial and operating information in recent periods, and the price-earnings ratios, price-sales ratios, market prices of securities, and certain financial and operating information of companies engaged in activities similar to ours.

Selling Restrictions

Canada

The shares may be sold only to purchasers purchasing, or deemed to be purchasing, as principal that are accredited investors, as defined in National Instrument 45-106 *Prospectus Exemptions* or subsection 73.3(1) of the *Securities Act* (Ontario), and are permitted clients, as defined in National Instrument 31-103 *Registration Requirements, Exemptions and Ongoing Registrant Obligations*. Any resale of the shares must be made in

accordance with an exemption from, or in a transaction not subject to, the prospectus requirements of applicable securities laws.

Securities legislation in certain provinces or territories of Canada may provide a purchaser with remedies for rescission or damages if this prospectus (including any amendment thereto) contains a misrepresentation, provided that the remedies for rescission or damages are exercised by the purchaser within the time limit prescribed by the securities legislation of the purchaser's province or territory. The purchaser should refer to any applicable provisions of the securities legislation of the purchaser's province or territory for particulars of these rights or consult with a legal advisor.

Pursuant to section 3A.3 of National Instrument 33-105 *Underwriting Conflicts* (NI 33-105), the underwriters are not required to comply with the disclosure requirements of NI 33-105 regarding underwriter conflicts of interest in connection with this offering.

European Economic Area

In relation to each Member State of the European Economic Area (each, a "Relevant State"), no shares have been offered or will be offered pursuant to the offering to the public in that Relevant State prior to the publication of a prospectus in relation to the shares which has been approved by the competent authority in that Relevant State or, where appropriate, approved in another Relevant State and notified to the competent authority in that Relevant State, all in accordance with the Prospectus Regulation (as defined below), except that offers of securities may be made to the public in that Relevant State at any time under the following exemptions under the Prospectus Regulation:

(i) to any legal entity which is a qualified investor as defined under the Prospectus Regulation;

(ii) to fewer than 150 natural or legal persons (other than qualified investors as defined under the Prospectus Regulation), subject to obtaining the prior consent of the representatives; or

(iii) in any other circumstances falling within Article 1(4) of the Prospectus Regulation;

provided that no such offer of the shares shall require us or any of the representatives to publish a prospectus pursuant to Article 3 of the Prospectus Regulation or supplement a prospectus pursuant to Article 23 of the Prospectus Regulation.

For the purposes of this provision, the expression an "offer to the public" in relation to the shares in any Relevant State means the communication in any form and by any means of sufficient information on the terms of the offer and any shares to be offered so as to enable an investor to decide to purchase any shares, and the expression "Prospectus Regulation" means Regulation (EU) 2017/1129 (as amended).

United Kingdom

Each underwriter has represented and agreed that:

(a) it has only communicated or caused to be communicated and will only communicate or cause to be communicated an invitation or inducement to engage in investment activity (within the meaning of Section 21 of the Financial Services and Markets Act 2000(the "FSMA")) received by it in connection with the issue or sale of the shares in circumstances in which Section 21(1) of the FSMA does not apply to us; and

(b) it has complied and will comply with all applicable provisions of the FSMA with respect to anything done by it in relation to the shares in, from or otherwise involving the United Kingdom.

No shares have been offered or will be offered pursuant to the offering to the public in the United Kingdom prior to the publication of a prospectus in relation to the shares which has been approved by the Financial Conduct Authority, except that the shares may be offered to the public in the United Kingdom at any time:

(i) to any legal entity which is a qualified investor as defined under Article 2 of the UK Prospectus Regulation (as defined below);

(ii) to fewer than 150 natural or legal persons (other than qualified investors as defined under Article 2 of the UK Prospectus Regulation), subject to obtaining the prior consent of the representatives for any such offer; or

(iii) in any other circumstances falling within Section 86 of the FSMA;

provided that no such offer of the shares shall require us or any of the representatives to publish a prospectus pursuant to Section 85 of the FSMA or supplement a prospectus pursuant to Article 23 of the UK Prospectus Regulation.

For the purposes of this provision, the expression an "offer to the public" in relation to the shares in the United Kingdom means the communication in any form and by any means of sufficient information on the terms of the offer and any shares to be offered so as to enable an investor to decide to purchase or subscribe for any shares and the expression "UK Prospectus Regulation" means Regulation (EU) 2017/1129 as it forms part of domestic law by virtue of the European Union (Withdrawal) Act 2018.

This prospectus is only for distribution to and directed at: (i) in the United Kingdom, persons having professional experience in matters relating to investments falling within Article 19(5) of the FSMA (Financial Promotion) Order 2005, as amended, or the Order, and high net worth entities falling within Article 49(2)(a) to (d) of the Order; (ii) persons who are outside the United Kingdom; and (iii) any other person to whom it can otherwise be lawfully distributed (all such persons together, "Relevant Persons"). Any investment or investment activity to which this prospectus relates is available only to and will be engaged in only with Relevant Persons, and any person who is not a Relevant Person should not rely on it.

Hong Kong

The securities have not been offered or sold and will not be offered or sold in Hong Kong, by means of any document, other than (a) to "professional investors" as defined in the Securities and Futures Ordinance (Cap. 571) of Hong Kong and any rules made under that Ordinance or (b) in other circumstances which do not result in the document being a "prospectus" as defined in the Companies Ordinance (Cap. 32) of Hong Kong or which do not constitute an offer to the public within the meaning of that Ordinance. No advertisement, invitation or document relating to the securities has been or may be issued or has been or may be in the possession of any person for the purposes of issue, whether in Hong Kong or elsewhere, which is directed at, or the contents of which are likely to be accessed or read by, the public of Hong Kong (except if permitted to do so under the securities laws of Hong Kong) other than with respect to securities which are or are intended to be disposed of only to persons outside Hong Kong or only to "professional investors" as defined in the Securities and Futures Ordinance and any rules made under that Ordinance.

Japan

The securities have not been and will not be registered under the Financial Instruments and Exchange Law of Japan (Law No. 25 of 1948, as amended) and, accordingly, will not be offered or sold, directly or indirectly, in Japan, or for the benefit of any Japanese Person or to others for re-offering or resale, directly or indirectly, in Japan or to any Japanese Person, except in compliance with all applicable laws, regulations and ministerial guidelines promulgated by relevant Japanese governmental or regulatory authorities in effect at the relevant time. For the purposes of this paragraph, "Japanese Person" means any person resident in Japan, including any corporation or other entity organized under the laws of Japan.

Singapore

This prospectus has not been registered as a prospectus with the Monetary Authority of Singapore. Accordingly, the securities were not offered or sold or caused to be made the subject of an invitation for subscription or purchase and will not be offered or sold or caused to be made the subject of an invitation for subscription or purchase, and this prospectus or any other document or material in connection with the offer or sale, or invitation for subscription or purchase, of the securities, has not been circulated or distributed, nor will it be circulated or distributed, whether directly or indirectly, to any person in Singapore other than (i) to an institutional investor (as defined in Section 4A

of the Securities and Futures Act (Chapter 289) of Singapore, as modified or amended from time to time, (the "SFA") pursuant to Section 274 of the SFA, (ii) to a relevant person (as defined in Section 275(2) of the SFA) pursuant to Section 275(1) of the SFA, or any person pursuant to Section 275(1A) of the SFA, and in accordance with the conditions specified in Section 275 of the SFA, or (iii) otherwise pursuant to, and in accordance with the conditions of, any other applicable provision of the SFA.

Where the securities are subscribed or purchased under Section 275 of the SFA by a relevant person which is:

(a) a corporation (which is not an accredited investor (as defined in Section 4A of the SFA)) the sole business of which is to hold investments and the entire share capital of which is owned by one or more individuals, each of whom is an accredited investor; or

(b) a trust (where the trustee is not an accredited investor) whose sole purpose is to hold investments and each beneficiary of the trust is an individual who is an accredited investor;

securities or securities-based derivatives contracts (each term as defined in Section 2(1) of the SFA) of that corporation or the beneficiaries' rights and interest (howsoever described) in that trust shall not be transferred within six months after that corporation or that trust has acquired the securities pursuant to an offer made under Section 275 of the SFA except:

(a) to an institutional investor or to a relevant person, or to any person arising from an offer referred to in Section 275(1A) or Section 276(4)(i)(B) of the SFA;

(b) where no consideration is or will be given for the transfer;

(c) where the transfer is by operation of law; or

(d) as specified in Section 276(7) of the SFA.

Switzerland

The shares may not be publicly offered in Switzerland and will not be listed on the SIX Swiss Exchange (the "SIX"), or on any other stock exchange or regulated trading facility in Switzerland. This prospectus has been prepared without regard to the disclosure standards for issuance prospectuses under art. 652a or art. 1156 of the Swiss Code of Obligations or the disclosure standards for listing prospectuses under art. 27 ff. of the SIX Listing Rules or the listing rules of any other stock exchange or regulated trading facility in Switzerland. Neither this prospectus nor any other offering or marketing material relating to the shares or the offering may be publicly distributed or otherwise made publicly available in Switzerland.

Neither this prospectus nor any other offering or marketing material relating to us, the offering, or the shares have been or will be filed with or approved by any Swiss regulatory authority. In particular, this prospectus will not be filed with, and the offering of shares will not be supervised by, the Swiss Financial Market Supervisory Authority ("FINMA", or the "FINMA"), and the offering of shares has not been and will not be authorized under the Swiss Federal Act on Collective Investment Schemes (the "CISA"). The investor protection afforded to acquirers of interests in collective investment schemes under the CISA does not extend to acquirers of the shares.

Dubai International Financial Centre

This prospectus relates to an Exempt Offer in accordance with the Offered Securities Rules of the Dubai Financial Services Authority (the "DFSA"). This prospectus is intended for distribution only to persons of a type specified in the Offered Securities Rules of the DFSA. It must not be delivered to, or relied on by, any other person. The DFSA has no responsibility for reviewing or verifying any documents in connection with Exempt Offers. The DFSA has not approved this prospectus nor taken steps to verify the information set forth herein and has no responsibility for the prospectus. The shares to which this prospectus relates may be illiquid and/or subject to restrictions on their resale. Prospective purchasers of the shares offered should conduct their own due diligence on the shares. If you do not understand the contents of this prospectus you should consult an authorized financial advisor.

Australia

No placement document, prospectus, product disclosure statement or other disclosure document has been lodged with the Australian Securities and Investments Commission, in relation to the offering. This prospectus does not constitute a prospectus, product disclosure statement or other disclosure document under the Corporations Act 2001, or the Corporations Act, and does not purport to include the information required for a prospectus, product disclosure statement or other disclosure document under the Corporations Act.

Any offer in Australia of the shares may only be made to persons, or Exempt Investors, who are "sophisticated investors" (within the meaning of section 708(8) of the Corporations Act), "professional investors" (within the meaning of section 708(11) of the Corporations Act) or otherwise pursuant to one or more exemptions contained in section 708 of the Corporations Act so that it is lawful to offer the shares without disclosure to investors under Chapter 6D of the Corporations Act.

The shares applied for by Exempt Investors in Australia must not be offered for sale in Australia in the period of 12 months after the date of allotment under the offering, except in circumstances where disclosure to investors under Chapter 6D of the Corporations Act would not be required pursuant to an exemption under section 708 of the Corporations Act or otherwise or where the offer is pursuant to a disclosure document which complies with Chapter 6D of the Corporations Act. Any person acquiring the shares must observe such Australian on-sale restrictions.

This prospectus contains general information only and does not take account of the investment objectives, financial situation or particular needs of any particular person. It does not contain any securities recommendations or financial product advice. Before making an investment decision, investors need to consider whether the information in this prospectus is appropriate to their needs, objectives and circumstances, and, if necessary, seek expert advice on those matters.

Israel

In the State of Israel this prospectus shall not be regarded as an offer to the public to purchase shares of common stock under the Israeli Securities Law, 5728 - 1968, which requires a prospectus to be published and authorized by the Israel Securities Authority, if it complies with certain provisions of Section 15 of the Israeli Securities Law, 5728 - 1968, including, inter alia, if: (i) the offer is made, distributed or directed to not more than 35 investors, subject to certain conditions, or the Addressed Investors; or (ii) the offer is made, distributed or directed to certain qualified investors defined in the First Addendum of the Israeli Securities Law, 5728 - 1968, subject to certain conditions (the "Qualified Investors"). The Qualified Investors shall not be taken into account in the count of the Addressed Investors and may be offered to purchase securities in addition to the 35 Addressed Investors. The company has not and will not take any action that would require it to publish a prospectus in accordance with and subject to the Israeli Securities Law, 5728 - 1968. We have not and will not distribute this prospectus or make, distribute or direct an offer to subscribe for our common stock to any person within the State of Israel, other than to Qualified Investors and up to 35 Addressed Investors.

Qualified Investors may have to submit written evidence that they meet the definitions set out in of the First Addendum to the Israeli Securities Law, 5728 - 1968. In particular, we may request, as a condition to be offered common stock, that Qualified Investors will each represent, warrant and certify to us and/or to anyone acting on our behalf: (i) that it is an investor falling within one of the categories listed in the First Addendum to the Israeli Securities Law, 5728 - 1968; (ii) which of the categories listed in the First Addendum to the Israeli Securities Law, 5728 - 1968 regarding Qualified Investors is applicable to it; (iii) that it will abide by all provisions set forth in the Israeli Securities Law, 5728 - 1968 and the regulations promulgated thereunder in connection with the offer to be issued common stock; (iv) that the shares of common stock that it will be issued are, subject to exemptions available under the Israeli Securities Law, 5728 - 1968: (a) for its own account; (b) for investment purposes only; and (c) not issued with a view to resale within the State of Israel, other than in accordance with the provisions of the Israeli Securities Law, 5728 - 1968; and (d) that it is willing to provide further evidence of its Qualified Investor status. Addressed Investors may have to submit written evidence in respect of their identity and may have to sign and submit a declaration containing, inter alia, the Addressed Investor's name, address and passport number or Israeli identification number.

LEGAL MATTERS

The validity of the issuance of our common stock offered in this prospectus will be passed upon for us by Latham & Watkins LLP. Certain legal matters in connection with this offering will be passed upon for the underwriters by Cooley LLP.

EXPERTS

The financial statements of Shoulder Innovations, Inc. as of December 31, 2024 and 2023 and for each of the two years in the period ended December 31, 2024, included in this prospectus have been audited by Deloitte & Touche LLP, an independent registered public accounting firm, as stated in their report. Such financial statements are included in reliance upon the report of such firm given upon their authority as experts in accounting and auditing.

WHERE YOU CAN FIND ADDITIONAL INFORMATION

We have filed with the SEC a registration statement on Form S-1, including exhibits and schedules, under the Securities Act, with respect to the shares of common stock being offered by this prospectus. This prospectus, which constitutes part of the registration statement, does not contain all of the information in the registration statement and its exhibits. For further information with respect to us and the common stock offered by this prospectus, we refer you to the registration statement and its exhibits. Statements contained in this prospectus as to the contents of any contract or any other document referred to are summary in nature and not necessarily complete, and in each instance, we refer you to the copy of the contract or other document filed as an exhibit to the registration statement. Each of these statements is qualified in all respects by reference to the full text of such contract or other document.

You may read our SEC filings, including this registration statement, over the internet at the SEC's website at www.sec.gov. Upon the completion of this offering, we will be subject to the information reporting requirements of the Exchange Act, and we will file reports, proxy statements, and other information with the SEC. These reports, proxy statements, and other information will be available for review at the SEC's website referred to above. We also maintain a website at www.shoulderinnovations.com, at which, following the completion of this offering, you may access these materials free of charge as soon as reasonably practicable after they are electronically filed with, or furnished to, the SEC. Information contained on or accessible through our website is not a part of this prospectus or the registration statement of which it forms a part, and the inclusion of our website address in this prospectus is an inactive textual reference only. You should not consider the contents of our website in making an investment decision with respect to our common stock.

Shoulder Innovations, Inc.

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Condensed Balance Sheets
(Unaudited)
(in thousands, except share and per share amounts)

	March 31, 2025	December 31, 2024
Assets		
Current Assets		
Cash and cash equivalents	$ 6,370	$ 6,123
Marketable securities	21,194	8,921
Trade accounts receivable, net of allowance for credit losses	6,581	5,122
Inventories, net	13,103	13,955
Prepaid expenses	981	431
Other current assets	1,381	573
Total Current Assets	49,610	35,125
Property and equipment, net	7,532	7,487
Operating lease right-of-use asset	57	68
Intangible assets, net	325	400
Total Assets	57,524	43,080
Liabilities, Convertible Preferred Stock, and Stockholders' Deficit		
Current Liabilities		
Accounts payable ($595 and $813 to related parties, respectively)	2,546	4,860
Current operating lease obligations	48	47
Accrued liabilities	4,394	2,740
Total Current Liabilities	6,988	7,647
Long-Term Liabilities		
Long-term debt	14,721	14,658
Other long-term liabilities	982	995
Total Long-Term Liabilities	15,703	15,653
Total Liabilities	22,691	23,300
Commitments and contingencies (note 8)		
Convertible Preferred Stock		
Series Seed, $0.001 par value, 16,840,400 shares authorized, and 16,840,400 and 15,851,401 issued and outstanding as of March 31, 2025 and December 31, 2024, respectively	1,420	1,337
Series A, $0.001 par value, 22,399,370 shares authorized, issued and outstanding as of March 31, 2025 and December 31, 2024, respectively	5,600	5,600
Series B, $0.001 par value, 6,913,964 shares authorized, and 6,038,964 and 5,913,964 issued and outstanding as of March 31, 2025 and December 31, 2024, respectively	2,247	2,200
Series C, $0.001 par value, 50,116,284 shares authorized, issued, and outstanding as of March 31, 2025 and December 31, 2024, respectively	21,550	21,550
Series D, $0.001 par value, 83,403,626 shares authorized, and 80,909,169 shares issued and outstanding as of March 31, 2025 and December 31, 2024, respectively	43,788	43,788
Series E, $0.001 par value, 58,774,332 shares authorized, and 29,455,169 and 0 shares issued and outstanding as of March 31, 2025 and December 31, 2024, respectively	19,542	—
Total Convertible Preferred Stock	94,147	74,475
Stockholders' Deficit		

Shoulder Innovations, Inc.
Condensed Balance Sheets
(Unaudited)
(in thousands, except share and per share amounts)

	March 31, 2025	December 31, 2024
Common stock, $0.001 par value, 280,986,575 shares authorized, and 102,785issued and outstanding as of March 31, 2025 and 212,366,763 shares authorized and 83,882 issued and outstanding as of December 31, 2024	1	1
Additional paid-in capital	2,307	2,148
Accumulated deficit	(61,703)	(57,041)
Accumulated other comprehensive income	81	197
Total Stockholders' Deficit	(59,314)	(54,695)
Total Liabilities, Convertible Preferred Stock, and Stockholders' Deficit	$ 57,524	$ 43,080

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Condensed Statements of Operations and Comprehensive Loss
(Unaudited)
(in thousands, except share and per share amounts)

Three Months Ended March 31,	2025	2024
Net Revenue	$ 10,132	$ 7,184
Cost of Goods Sold	2,341	1,666
Gross Profit	7,791	5,518
Selling, General, and Administrative Expenses	10,502	7,704
(Includes $849 and $781 to related parties, respectively)		
Research and Development	1,583	1,070
(Includes $1,235 and $942 to related parties, respectively)		
Operating Loss	(4,294)	(3,256)
Other Expense		
Interest expense, net	367	317
Other expense, net	1	27
Total Other Expense	368	344
Loss before income tax expense	(4,662)	(3,600)
Income Tax Expense	—	—
Net Loss	$ (4,662)	$ (3,600)
Other Comprehensive (loss), net		
Unrealized (loss) on marketable securities	(116)	(42)
Total Other Comprehensive (loss), net	(116)	(42)
Comprehensive loss	$ (4,778)	$ (3,642)
Net loss per share attributed to common stock – basic and diluted:		
Net loss per share	$ (52.13)	$ (62.77)
Weighted average shares outstanding:		
Weighted average common shares outstanding – basic and diluted	89,438	57,354

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Condensed Statements of Changes in Convertible Preferred Stock and Stockholders' Deficit
(Unaudited)
(in thousands, except shares and per share amounts)

	Preferred Stock	Common Stock	Additional Paid-in Capital	Accumulated Other Comprehensive Income	Accumulated Deficit	Total Stockholders' Deficit
Balance, December 31, 2023	$ 74,475	$ 1	$ 1,328	$ 356	$ (41,422)	$ (39,737)
Issuance of common stock	—	—	—	—	—	—
Stock compensation expense	—	—	168	—	—	168
Net loss	—	—	—	—	(3,600)	(3,600)
Comprehensive income	—	—	—	42	—	42
Balance, March 31, 2024	$ 74,475	$ 1	$ 1,496	$ 398	$ (45,022)	$ (43,127)
Balance, December 31, 2024	$ 74,475	$ 1	$ 2,148	$ 197	$ (57,041)	$ (54,695)
Issuance of preferred stock in private placement, net of issuance cost of $352	19,542	—	—	—	—	—
Exercise of preferred stock warrant	130	—	—	—	—	—
Issuance of common stock	—	—	32	—	—	32
Stock compensation expense	—	—	127	—	—	127
Net Loss	—	—	—	—	(4,662)	(4,662)
Comprehensive (loss)	—	—	—	(116)	—	(116)
Balance, March 31, 2025	$ 94,147	$ 1	$ 2,307	$ 81	$ (61,703)	$ (59,314)

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Condensed Statements of Cash Flows

(Unaudited)

(amounts in thousands)

Three Months Ended March 31,	2025	2024
Cash Flows from Operating Activities		
Net loss	$ (4,662)	$ (3,600)
Adjustments to reconcile net loss to net cash used in operating activities		
Depreciation and amortization	668	498
Amortization of debt discounts	63	95
Stock compensation expense	127	168
Realized gain (loss) on marketable securities	(162)	117
Change in Operating Assets and Liabilities		
Trade accounts receivable	(1,459)	431
Inventory	851	(544)
Prepaid expenses	(550)	26
Accounts payable	(2,122)	(46)
Other current liabilities	769	(620)
Net Cash Used in Operating Activities	(6,477)	(3,475)
Cash Flows from Investing Activities		
Fixed asset purchases	(843)	(1,307)
Purchases of marketable securities	(20,060)	(2,815)
Sales of marketable securities	7,833	3,107
Net Cash Provided (Used) in Investing Activities	(13,070)	(1,015)
Cash Flows from Financing Activities		
Proceeds from exercise of common stock options	32	—
Proceeds from exercise of preferred stock warrants	130	—
Proceeds from series E convertible preferred stock, net of issuance cost	19,632	—
Net Cash Provided in Financing Activities	19,794	—
Net Change in Cash for Period	247	(4,490)
Cash and Cash Equivalents, beginning of year	6,123	6,236
Cash and Cash Equivalents, end of year	6,370	$ 1,746
Supplemental Cash Flows information		
Cash paid for interest	$ 431	$ 450
Non-Cash Investing and Financing Activities		
Series E convertible preferred stock issuance cost in other current liabilities	$ 90	$ —
Capital expenditures included in accounts payable	$ 263	$ 322

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Notes to the Unaudited Condensed Financial Statements
(in thousands, except share, per share data and percentages)

1. Summary of Significant Accounting Policies

Business Activity

Shoulder Innovations, Inc. (the "Company") is principally involved in developing next generation shoulder replacement implants, utilizing contract manufacturing partners, and distributing them nationwide for surgeries through a network of employed and contracted sales representatives. The Company is headquartered in Grand Rapids, Michigan and markets and sells its products throughout the United States.

Basis of Presentation

The accompanying unaudited condensed financial statements have been prepared in accordance with generally accepted accounting principles in the United States of America ("GAAP") and the applicable rules and regulations of the Securities and Exchange Commission ("SEC") for interim reporting. Certain information and note disclosures included in the Company's annual financial statements prepared in accordance with GAAP have been condensed or omitted pursuant to such rules and regulations. These unaudited condensed financial statements should be read in conjunction with the audited annual financial statements for the year ended December 31, 2024. There have been no material changes in our significant accounting policies as described in our audited annual financial statements for the year ended December 31, 2024. In the opinion of management, these unaudited condensed financial statements reflect all adjustments, including those of a normal and recurring nature, which are necessary for a fair presentation of the results for the interim period presented.

Reverse Stock Split

On July 23, 2025, the Company amended its amended and restated certificate of incorporation to effect a reverse stock split of shares of the Company's common stock on a 1-for-19.08 basis (the "Reverse Stock Split"). The common stock warrants and options to purchase common stock were subsequently adjusted as a result of the Reverse Stock Split. All impacted share and per-share information included in these condensed financial statements and notes thereto have been retroactively adjusted to give effect to the Reverse Stock Split.

New Accounting Pronouncements

In December 2023, the FASB issued ASU 2023-09*, Income Taxes (Topic 740) ("ASC 740")*. The update requires all public business entities on an annual basis to (1) disclose specific categories in the rate reconciliation and (2) provide additional information for reconciling items that meet a quantitative threshold and an explanation, if not otherwise evident, of the individual reconciling items disclosed, such as the nature, effect, and underlying causes of the reconciling items and the judgment used in categorizing the reconciling items. In addition, the update requires certain new disclosures of the amount of income taxes paid (net of refunds received) disaggregated by federal, state, and foreign taxes and the amount of income taxes paid (net of refunds received) disaggregated by individual jurisdictions in which income taxes paid is equal to or greater than five percent of total income taxes paid (net of refunds received). Other new disclosures required include income (or loss) from continuing operations before income tax expense (or benefit) disaggregated between domestic and foreign and income tax expense (or benefit) from continuing operations disaggregated by federal, state, and foreign. The new guidance is effective for annual periods beginning after December 15, 2024. Early adoption is permitted. The amendments are to be applied on a prospective basis, with retrospective application permitted. The Company is currently evaluating the impact of the new standard on its financial statements and related disclosures.

In November 2024, the FASB issued ASU 2024-03, Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses requires additional, disaggregated disclosure around certain income statement expense line items. This ASU mandates that entities, at each interim and annual period, disclose the amounts of (a) inventory purchases, (b) employee compensation, (c) depreciation, (d) intangible asset amortization, and (e) depletion, depreciation, and amortization for oil and gas activities included within each relevant expense caption presented on the income statement within continuing operations. Entities are also required to (1) combine certain disclosures already mandated under GAAP with these new requirements, (2) provide qualitative descriptions of expenses that are not disaggregated quantitatively, and (3) disclose total selling expenses and, annually, the definition of selling expenses. The guidance

is effective for annual periods beginning after December 15, 2026, and interim periods beginning after December 15, 2027. Early adoption is permitted. The Company is currently evaluating the potential impact of adopting this new guidance on the Company's financial statements and related disclosures.

Shoulder Innovations, Inc.

Notes to the Unaudited Condensed Financial Statements
(in thousands, except share, per share data and percentages)

2. Fair Value Measurements

The fair value of marketable securities as of March 31, 2025 and December 31, 2024 are summarized below:

	March 31, 2025		
	Level 1	Level 2	Level 3
Assets:			
Cash Equivalents			
Money market funds	$ 2,173	$ —	$ —
Short-term marketable securities at fair value			
U.S. treasury and government agencies	14,222	—	—
Corporate and international bonds	—	6,972	—
Total assets	$ 16,395	$ 6,972	$ —
Liabilities:			
Preferred stock warrant liability			$ 970
Series E purchase option			218
Total liabilities			$ 1,188

	December 31, 2024		
	Level 1	Level 2	Level 3
Assets:			
Cash Equivalents			
Money market funds	$ 85	$ —	$ —
Short-term marketable securities at fair value			
U.S. treasury and government agencies	6,053	—	—
Corporate and international bonds	—	2,868	—
Total assets	$ 6,138	$ 2,868	$ —
Liabilities:			
Preferred stock warrant liability			$ 970
Series E purchase option			—
Total liabilities			$ 970

3. Accrued Liabilities and Other Current Liabilities

Accrued liabilities and other current liabilities consist of the following:

	March 31, 2025	December 31, 2024
Commissions payable	$ 988	$ 836
Accrued payroll	902	738
Accrued royalties	734	612
Other	1,680	554
Accrued liabilities and other liabilities	$ 4,304	$ 2,740

4. Related Party Transactions

On October 22, 2020, the Company entered into a software license agreement with Genesis Software which was amended and restated on January 1, 2023. Robert Ball, the Company's Chief Executive Officer and Executive Chairman, is a co-founder and director of Genesis Software. Mr. Ball and Matthew Ahearn, the Company's Chief Operating Officer and a director, are directors of Genesis Investment Holdings, LLC, which has an ownership interest in Genesis Software. In addition, cultivate(MD) Capital Accelerator Fund L.P. and Genesis Investment Holdings, LLC, each beneficial owners of more than 5% of our capital stock, are investors in Genesis Software. The software licensing agreement is a 5-year term. The agreement required an upfront payment of $1,000, an incremental $500 payment when FDA clearance was obtained, and quarterly payments of royalties equal to 4% of the net sales price of each licensed product sold. For each of the three months ended March 31, 2025 and 2024, the Company paid Genesis Software $343 and $270, respectively, pursuant to the license agreement.

For the three months ended March 31, 2025 and 2024, the Company paid $571 and $454, respectively for software development to Genesis Software. Amounts payable of $150 and $276 are included in accounts payable on the balance sheets at March 31, 2025 and December 31, 2024, respectively.

The Company has entered into a consulting agreement with an entity under common ownership. The consulting agreement is currently on a year-to-year basis. The agreement requires compensation for services performed. If services performed are on an hourly basis, the Company shall be responsible to pay for hours actually worked by the consultant's employees. The Company will reimburse the consultant for all reasonable expenses incurred in connection with performing services for the Company. For each of the three months ended March 31, 2025 and 2024, the Company paid Genesis Innovation Group $1,235 and $942, respectively. Amounts payable of $445 and $537 are included in accounts payable on the balance sheets at March 31, 2025 and December 31, 2024, respectively.

During the first three months of 2024, Robert Ball, the Company's Chief Executive Officer and Executive Chairman, was an investor in RMD. Mr. Ball is no longer an investor in RMD. For the three months ended March 31, 2025 and 2024, the Company paid RMD $0 and $859, respectively, pursuant to the RMD Supply Agreement.

5. Convertible Preferred Stock

On March 6, 2025, the Company entered into a Series E Preferred Stock Purchase Agreement pursuant to the Series E Preferred Stock Financing, whereby it received a total commitment amount of $40,130 for the issuance and sale of 58,774,332 shares of Series E Preferred Stock pursuant to two separate closing tranches, the first of which closed on March 6, 2025 whereby the Company issued 29,455,169 shares of series E preferred stock resulting in total net proceeds to the Company of $20,111. The second tranche can occur any time prior to September 15, 2025, for the issuance and sale of 29,319,143 shares of Series E Preferred Stock for an aggregate purchase price of $20,019. The Company determined that the second tranche constitutes a free standing financial instrument and allocated $218 of the Series E proceeds to the purchase option, which is included in accrued liabilities in the Condensed Balance Sheets.

On March 4, 2025, Genesis Investment Holdings, LLC exercised 988,999 series Seed warrants resulting in proceeds of $83.

On March 17, 2025, 125,000 Series B warrants were exercised resulting in proceeds of $47.

For all series of preferred stock, the initial conversion price equals their original issuance price per share, at March 31, 2025 and December 31, 2024, adjusted for the Reverse Stock Split as described in Note 1.

Issued and Outstanding Preferred Shares

Authorized, issued, and outstanding Preferred Stock were as follows:

March 31, 2025

Series	Par Value	Authorized	Issued and Outstanding	Issuance Price Per Share	Liquidation Value
Seed	$ 0.001	16,840,400	16,840,400	$ 0.0843	$ 1,420
A	0.001	22,399,370	22,399,370	0.2500	5,600
B	0.001	6,913,964	6,038,964	0.3720	2,247
C	0.001	50,116,284	50,116,284	0.4300	21,550
D	0.001	83,403,626	80,909,169	0.5412	43,788
E	0.001	58,774,332	29,455,169	0.6828	$ 20,111
Total		238,447,976	205,759,356		94,716

December 31, 2024

Series	Par Value	Authorized	Issued and Outstanding	Issuance Price Per Share	Liquidation Value
Seed	$ 0.001	16,840,400	15,851,401	$ 0.0843	$ 1,337
A	0.001	22,399,370	22,399,370	0.2500	5,600
B	0.001	6,913,964	5,913,964	0.3720	2,200
C	0.001	50,116,284	50,116,284	0.4300	21,550
D	0.001	83,403,626	80,909,169	0.5412	43,788
Total		179,673,644	175,190,188		$ 74,475

6. Loss Per Share

Basic loss per share is computed by dividing the net loss after tax attributable to common stockholders by the weighted average shares outstanding during the period. Diluted loss per share is computed by including potentially dilutive securities outstanding during the period in the calculation of weighted average shares outstanding. The Company did not have any dilutive securities during the periods presented; therefore, diluted loss per share is equal to basic loss per share.

Presented in the table below is a reconciliation of the numerator and denominator for the basic and diluted loss per share ("LPS") calculations for the three months ended March 31:

March 31,	2025	2024
Net loss	$ (4,662)	$ (3,600)
Basic and diluted weighted average shares outstanding	89,438	57,354
Net loss attributable to common shareholders, basic and diluted	$ (52.13)	$ (62.77)

The following securities were not included in the computation of diluted shares outstanding because the effect would be anti-dilutive (convertible preferred stock outstanding is presented on an as-converted basis):

	March 31, 2025	March 31, 2024
Series Seed	882,620	830,786
Series A	1,173,971	1,173,971
Series B	316,507	309,956
Series C	2,626,639	2,626,639
Series D	4,240,522	4,240,522
Series E	1,543,771	—
Common Options	1,311,346	1,163,691
Common Warrants	17,827	17,827
Series Seed Warrants	—	51,834
Series B Warrants	45,859	52,410
Series D Warrants	130,736	130,736
Total	12,289,798	10,598,372

7. Segment Information

The Company reports segment information based on how the Company's chief operating decision maker ("CODM") is the Chief Executive Officer, regularly reviews operating results, allocates resources and makes decisions regarding business operations. For segment reporting purposes the Company's business structure is comprised of one operating and reportable segment. The CODM uses segment gross margin and net loss for determining the allocation of resources (including employees, financial, or capital resources) to the segment to achieve the Company's strategic plan and to assess the performance of the segment by monitoring actual results against performance targets established in the Company's annual budget and forecasting process.

All revenue for the three months ended March 31, 2025 and 2024 was generated from customers located in the United States. No customers represent 10% or more of the Company's total revenue for the three months ended March 31, 2025 and 2024. The measure of segment assets is reported on the balance sheets as total assets.

The table below is a summary of the segment net loss, including significant segment expenses:

March 31,	2025	2024
Total Revenue	$ 10,132	$ 7,184
Cost of goods Sold	2,341	1,666
Gross margin	7,791	5,518
Operating expenses:		
General and administrative	5,213	3,325
Sales and marketing	4,733	3,501
Medical education	556	878
Research and development	1,583	1,070
Interest/other expense	368	344
Segment net loss and total net loss	$ (4,662)	(3,600)

Depreciation expense for the three months ended March 31, 2025 and 2024 totaled $593 and $423, respectively. Depreciation expense of $575 and $415 for instruments is in sales and marketing and $18 and $8 for computer equipment, furniture and fixtures, and leasehold improvements are included in general, and administrative expenses in the statements of operations and comprehensive loss for the three months ended March 31, 2025 and 2024, respectively.

Amortization expense related to the Company's software license for the three months ended March 31 2025 and 2024 totaled $75 for both periods, and is included in general and administrative expense.

8. Commitments and Contingencies

Litigation

The Company from time to time is involved in legal matters incidental to the conduct of its business. Legal fees and other costs associated with such actions are expensed as incurred. The Company assesses the need to record a liability for litigation and contingencies. Accrual estimates are recorded when and if it is determinable that such a liability for litigation and contingencies are both probable and reasonably estimable.

In February 2024, the Company filed a complaint against Catalyst Orthoscience Inc. ("Catalyst") claiming patent infringement through Catalyst's making, using, selling, offering for sale in the United States, and/or importing into the United States, reverse shoulder systems. In response to the Company's suit, Catalyst filed a counterclaim claiming patent infringement of certain of the Company's products. The Company believes that it has substantial and meritorious defenses to Catalyst's claims and intend to vigorously defend its position, including through the trial and appellate stages if necessary. As the counterclaim is ongoing, the Company is unable to determine the likelihood of an outcome or estimate a range of reasonably possible settlement, if any. Accordingly, the Company has not made an accrual for any possible loss. The outcome of any litigation, however, is inherently uncertain, and an adverse judgment or settlement in the counterclaim proceeding, if any, could have a material and adverse effect the Company's business, financial position, results of operations or cash flows.

9. Subsequent Events

The Company evaluated subsequent events from March 31, 2025, the date of these financial statements, through May 23, 2025, which represents the date the financial statements were issued, and with respect to the Reverse Stock Split, described in Note 1, through July 24, 2025, the date the financial statements were reissued, for events requiring recording or disclosure in the financial statements for the three months ended March 31, 2025.

On April 23, 2025, the Company granted 453,898 common stock options to certain employees and officers of the Company with an exercise price of $2.86. Common stock options vest over a four year period. The Company estimates it will recognize $0.8 million in stock compensation expense over the vesting period.

In June 2025 the Company called the second tranche of its Series E Preferred Stock for the issuance of 29,319,143 shares of Series E Preferred Stock resulting in total net proceeds to the Company of $20.0 million.

On July 18, 2025, the Company issued $40.0 million aggregate principal amount of convertible notes. The convertible notes mature on September 1, 2028 and accrue interest at a rate of 5.0% per annum through June 30, 2026, at which time the interest rate shall increase to 10.0% per annum. Upon the consummation of an initial public offering by the Company prior to the maturity date, the convertible notes will automatically convert into a number of shares of common stock equal to the outstanding principal amount of the notes and any accrued interest divided by the applicable conversion price. The conversion price is the lower of (i) a 20% discount to the initial public offering price and (ii) $280 million divided by the number of fully diluted shares of capital stock (on an as-converted basis) outstanding immediately prior to the initial public offering but excluding the convertible notes (the "IPO Capitalization"); provided that, in no event shall such conversion price be less than the quotient obtained by dividing $210 million by the IPO Capitalization. Upon a liquidation event, the holders of the convertible notes shall receive a cash payment equal to the sum of 1.3 times the outstanding principal amount of the convertible notes plus any accrued interest.

On July 21, 2025, the Company amended the Trinity Loan Agreement to extend the commitment date of the tranche (ii) $15,000 draw from before December 31, 2024 to before December 31, 2025 and to extend the commitment date of the tranche (iii) $15,000 draw from before December 31, 2025 to before December 31, 2026. The interest rate on the instrument was amended from a floor of 11.50% to 11.00%, along with increasing the interest only period from an initial term of 48-months to 60-months. Additionally, warrants originally vesting upon the funding of the tranche (iii) $15,000 draw were cancelled and are no longer outstanding.

On July 23, 2025, the Company amended its amended and restated certificate of incorporation to effect the Reverse Stock Split of shares of the Company's common stock on a 1-for-19.08 basis. The number of authorized shares was increased to 730,000,000 shares of common stock. The par values of the common stock and redeemable convertible preferred stock were not adjusted as a result of the Reverse Stock Split. In connection with the Reverse Stock Split, the conversion ratio for the Company's outstanding redeemable convertible preferred stock was adjusted. All references to common stock, common stock warrants, and options to purchase common stock share data, per share data, and related information contained in the financial statements have been retroactively adjusted to reflect the effect of the Reverse Stock Split for all periods presented.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and Board of Directors of Shoulder Innovations, Inc.

Opinion on the Financial Statements

We have audited the accompanying balance sheets of Shoulder Innovations, Inc. (the "Company") as of December 31, 2024 and 2023, the related statements of operations and comprehensive loss, changes in convertible preferred stock and stockholders' deficit, and cash flows, for each of the two years in the period ended December 31, 2024, and the related notes (collectively referred to as the "financial statements"). In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the two years in the period ended December 31, 2024, in conformity with accounting principles generally accepted in the United States of America.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. The Company is not required to have, nor were we engaged to perform, an audit of its internal control over financial reporting. As part of our audits, we are required to obtain an understanding of internal control over financial reporting but not for the purpose of expressing an opinion on the effectiveness of the Company's internal control over financial reporting. Accordingly, we express no such opinion.

Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

/s/ Deloitte & Touche LLP

Grand Rapids, Michigan
April 17, 2025 (July 24, 2025, as to the effects of the reverse stock split described in Notes 1 and 14)

We have served as the Company's auditor since 2023.

Table of Contents

Shoulder Innovations, Inc.

Balance Sheets
(in thousands, except share and per share amounts)

December 31,	2024	2023
Assets		
Current Assets		
Cash and cash equivalents	$ 6,123	$ 6,236
Marketable securities	8,921	26,577
Trade accounts receivable, net of allowance for credit losses	5,122	4,693
Inventories, net	13,955	9,923
Prepaid expenses	431	419
Other current assets	573	—
Total Current Assets	35,125	47,848
Property and equipment, net	7,487	5,635
Operating lease right-of-use asset	68	111
Intangible assets, net	400	700
Total Assets	43,080	54,294
Liabilities, Convertible Preferred Stock, and Stockholders' Deficit		
Current Liabilities		
Accounts payable ($813 and $966 to related parties, respectively)	4,860	1,959
Current operating lease obligations	47	43
Accrued liabilities	2,740	2,281
Total Current Liabilities	7,647	4,283
Long-Term Liabilities		
Preferred stock warrant liability	970	851
Long-term debt	14,658	14,351
Long-term operating lease obligations	25	71
Total Long-Term Liabilities	15,653	15,273
Total Liabilities	23,300	19,556
Commitments and contingencies (note 13)		
Convertible Preferred Stock		
Series Seed, $0.001 par value, 16,840,000 shares authorized, and 15,851,401 issued and outstanding as of December 31, 2024 and 2023, respectively	1,337	1,337
Series A, $0.001 par value, 22,399,370 shares authorized, issued, and outstanding as of December 31, 2024 and 2023	5,600	5,600
Series B, $0.001 par value, 6,913,964 shares authorized, and 5,913,964 issued and outstanding as of December 31, 2024 and 2023, respectively	2,200	2,200
Series C, $0.001 par value, 50,116,284 shares authorized, issued, and outstanding as of December 31, 2024 and 2023, respectively	21,550	21,550
Series D, $0.001 par value, 83,403,626 shares authorized, and 80,909,169 shares issued, and outstanding as of December 31, 2024 and 2023, respectively	43,788	43,788
Total Convertible Preferred Stock	74,475	74,475
Stockholders' Deficit		
Common stock, $0.001 par value, 212,366,763 shares authorized, and 83,882 issued and outstanding as of December 31, 2024 and 204,958,600 shares authorized and 57,354 issued and outstanding as of December 31, 2023	1	1
Additional paid-in capital	2,148	1,328

Shoulder Innovations, Inc.

Balance Sheets
(in thousands, except share and per share amounts)

December 31,	2024	2023
Accumulated deficit	(57,041)	(41,422)
Accumulated other comprehensive income	197	356
Total Stockholders' Deficit	(54,695)	(39,737)
Total Liabilities, Convertible Preferred Stock, and Stockholders' Deficit	$ 43,080	$ 54,294

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Statements of Operations and Comprehensive Loss
(in thousands, except share and per share amounts)

Year Ended December 31,	2024	2023
Net Revenue	$ 31,623	$ 19,274
Cost of Goods Sold	7,282	4,004
Gross Profit	24,341	15,270
Selling, General, and Administrative Expenses	34,505	23,113
(Includes $3,325 and $3,161 to related parties, respectively)		
Research and Development	4,489	3,021
(Includes $3,574 and $3,112 to related parties, respectively)		
Operating Loss	(14,653)	(10,864)
Other Expense		
Interest expense, net	1,316	761
Loss on extinguishment of convertible promissory notes	—	1,127
Other (income) expense, net	(350)	(97)
Total Other Expense	966	1,791
Loss before income tax expense	(15,619)	(12,655)
Income Tax Expense		—
Net Loss	$ (15,619)	$ (12,655)
Other Comprehensive (loss) income, net		
Unrealized (loss) gain on marketable securities	(159)	356
Total Other Comprehensive (loss) Income, net	(159)	356
Comprehensive loss	$ (15,778)	$ (12,299)
Net loss per share attributed to common stock – basic and diluted:		
Net loss per share	$ (242.04)	$ (220.65)
Weighted average shares outstanding:		
Weighted average common shares outstanding – basic and diluted	64,530	57,354

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Statements of Changes in Convertible Preferred Stock and Stockholders' Deficit
(in thousands, except shares and per share amounts)

	Preferred Stock	Common Stock	Additional Paid-in Capital	Accumulated Other Comprehensive Income	Accumulated Deficit	Total Stockholders' Deficit
Balance, December 31, 2022	$ 30,687	$ 1	$ 877	$ —	$ (28,767)	$ (27,889)
Issuance of preferred stock in private placement	34,877	—	—	—	—	—
Conversion of convertible notes payable into preferred stock	8,911	—	—	—	—	—
Stock compensation expense	—	—	451	—	—	451
Net loss	—	—	—	—	(12,655)	(12,655)
Comprehensive income (loss)	—	—	—	356	—	356
Balance, December 31, 2023	$ 74,475	$ 1	$ 1,328	$ 356	$ (41,422)	$ (39,737)
Issuance of common stock	—	—	66	—	—	66
Stock compensation expense	—	—	754	—	—	754
Net Loss	—	—	—	—	(15,619)	(15,619)
Comprehensive (loss) income	—	—	—	(159)	—	(159)
Balance, December 31, 2024	$ 74,475	$ 1	$ 2,148	$ 197	$ (57,041)	$ (54,695)

See accompanying notes to financial statements.

Shoulder Innovations, Inc.

Statements of Cash Flows
(amounts in thousands)

Year Ended December 31,	2024	2023
Cash Flows from Operating Activities		
Net loss	$ (15,619)	$ (12,655)
Adjustments to reconcile net loss to net cash used in operating activities		
Depreciation and amortization	2,196	1,641
Amortization of debt discounts	307	249
Credit losses, net of recoveries	146	67
Loss on disposal of long-lived assets	44	129
Stock compensation expense	754	451
Realized gain on marketable securities	(482)	—
Change in derivative liability	—	78
Change in fair value of warrant liabilities	119	(80)
Loss on extinguishment of convertible promissory notes	—	1,127
Other	(571)	614
Change in Operating Assets and Liabilities		
Trade accounts receivable	(575)	(2,693)
Inventory	(4,032)	(2,413)
Prepaid expenses	(12)	(314)
Accounts payable	3,124	(737)
Other current liabilities	458	1,392
Net Cash Used in Operating Activities	(14,143)	(13,144)
Cash Flows from Investing Activities		
Fixed asset purchases	(4,015)	(2,748)
Purchases of marketable securities	(5,281)	(29,250)
Sales of marketable securities	23,260	3,025
Net Cash Provided (Used) in Investing Activities	13,964	(28,973)
Cash Flows from Financing Activities		
Proceeds from exercise of common stock options	66	—
Proceeds from series D convertible preferred stock	—	34,877
Proceeds from borrowings on debt	—	15,000
Debt issuance costs, net	—	(149)
Payments on debt	—	(5,000)
Net Cash Provided in Financing Activities	66	44,728
Net Change in Cash for Period	(113)	2,611
Cash and Cash Equivalents, beginning of year	6,236	3,625
Cash and Cash Equivalents, end of year	6,123	$ 6,236
Supplemental Cash Flows information		
Cash paid for interest	$ 1,785	$ 901
Non-Cash Investing and Financing Activities		
Capital expenditures included in accounts payable	$ 455	$ 678
Conversion of convertible notes payable into preferred stock	$ —	$ 8,911

See accompanying notes to financial statements.

1. Summary of Significant Accounting Policies

Business Activity

Shoulder Innovations, Inc. (the Company) is principally involved in developing next generation shoulder replacement implants, utilizing contract manufacturing partners, and distributing them nationwide for surgeries through a network of employed and contracted sales representatives. The Company is headquartered in Grand Rapids, Michigan and markets and sells its products throughout the United States.

Emerging Growth Company

The Company is an "emerging growth company," as defined in Section 2(a) of the Securities Act, as modified by the Jumpstart Our Business Startups Act of 2012 (the "JOBS Act"), and it may take advantage of certain exemptions from various reporting requirements that are applicable to other public companies that are not emerging growth companies including, but not limited to, not being required to comply with the independent registered public accounting firm attestation requirements of Section 404 of the Sarbanes-Oxley Act, reduced disclosure obligations regarding executive compensation in its periodic reports and proxy statements, and exemptions from the requirements of holding a nonbinding advisory vote on executive compensation and stockholder approval of any golden parachute payments not previously approved.

Further, Section 102(b)(1) of the JOBS Act exempts emerging growth companies from being required to comply with new or revised financial accounting standards until private companies (that is, those that have not had a Securities Act registration statement declared effective or do not have a class of securities registered under the Exchange Act) are required to comply with the new or revised financial accounting standards. The JOBS Act provides that an emerging growth company can elect to opt out of the extended transition period and comply with the requirements that apply to non-emerging growth companies but any such an election to opt out is irrevocable. The Company has elected not to opt out of such extended transition period, which means that when a standard is issued or revised and it has different application dates for public or private companies, the Company, as an emerging growth company, can adopt the new or revised standard at the time private companies adopt the new or revised standard. This may make comparison of the Company's financial statements with another public company that is neither an emerging growth company nor an emerging growth company that has opted out of using the extended transition period difficult or impossible because of the potential differences in accounting standards used.

Reverse Stock Split

On July 23, 2025, the Company amended its amended and restated certificate of incorporation to effect a reverse stock split of shares of the Company's common stock on a 1-for-19.08 basis (the "Reverse Stock Split"). The common stock warrants and options to purchase common stock were subsequently adjusted as a result of the Reverse Stock Split. All impacted share and per-share information included in these financial statements and notes thereto have been retroactively adjusted to give effect to the Reverse Stock Split.

Use of Estimates

The preparation of financial statements in conformity with accounting principles generally accepted in the United States of America ("GAAP") requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results could differ from those estimates.

Concentrations of Credit Risk

Financial instruments, which potentially subject us to concentrations of credit risk, are primarily marketable securities and accounts receivable. Concentrations of credit risk with respect to accounts receivable are limited due to the number of entities comprising our customer base. We perform ongoing credit evaluations of our customers and generally do not require collateral.

There was no customer that accounted for 10% or more of revenue, or accounts receivable for the years ended December 31, 2024 and 2023.

Cash and Cash Equivalents

The Company considers all highly liquid funds with an original contractual maturity on the date of purchase of three months or less to be classified and presented as cash equivalents. The Company maintains cash balances in bank checking accounts. From time to time during the year, cash deposited in these accounts may be in excess of the federally insured limit.

Marketable Securities

The Company's marketable securities consist predominately of investment-grade, U.S. dollar-denominated fixed and floating-rate debt, which are classified as available-for-sale and are valued in accordance with the fair value measurement guidance. Available-for-sale securities are carried at fair value with unrealized gains and losses reported in the statements of operations and comprehensive loss as other comprehensive income, net of tax. Realized gains and losses, if any, are calculated on the specific identification method and are included in the statements of operations and comprehensive loss.

Available-for-sale debt securities are recorded at fair value. When the fair value of the securities declines below the amortized cost basis, impairment is indicated and it must be determined whether it is other than temporary. Impairment is considered to be other than temporary if the Company: (i) intends to sell the security, (ii) will more likely than not be forced to sell the security before recovering its cost, or (iii) does not expect to recover the security's amortized cost basis. If the decline in fair value is considered other than temporary, the cost basis of the security is adjusted to its fair market value and the realized loss is reported in other (income) expense, net. Subsequent increases or decreases in fair value are reported as a component of stockholders' equity in accumulated other comprehensive income.

Trade Accounts Receivable

Trade accounts receivable are stated at the amount management expects to collect from outstanding customer obligations due under normal trade terms. Generally, management provides for probable uncollectible amounts through a charge to the allowance for credit losses on accounts receivable based on its evaluation of the status of individual accounts, past credit history with customers, and the customers' current financial condition. A rollforward of the allowance for credit loss is as follows:

	2024	2023
Beginning Balance	$ 50	$ 10
Provision	146	52
Write-offs	(33)	(12)
Ending Balance	$ 163	$ 50

Inventories

Inventories are stated at the lower of average cost or net realizable value. Inventories consist of finished goods purchased from third-party suppliers. Two suppliers provide substantially all of the Company's finished goods. The Company records inventory reserves for estimated obsolescence or unmarketable inventory equal to the difference between the cost of inventory and the estimated net realizable value based upon the age of specific inventory on hand and assumptions about future demand and market conditions. The Company had an inventory reserve of $287 and $119 as of December 31, 2024 and 2023, respectively.

	2024	2023
Finished Goods	$ 14,242	$ 10,042

Inventory reserve	(287)	(119)
Inventories, net	$ 13,955	$ 9,923

Property and Equipment and Depreciation

Property and equipment are stated at cost, less accumulated depreciation. Property and equipment are depreciated using straight-line methods over the estimated useful lives of the related assets. Leasehold improvements are amortized over the shorter of the remaining lease term or the estimated useful lives of the assets. Maintenance and repairs are charged to expense as incurred.

When items of property or equipment are sold or retired, the related costs and accumulated depreciation are removed from the accounts and any gain or loss is included in the results of operations.

Surgical instruments are provided to surgeons and hospitals by the Company and its distributors to facilitate the use of the Company's products. The Company does not receive additional or separate consideration for the use of its surgical instruments by surgeons and hospital staff. No customer or distributor agreements convey the transfer of ownership of the Company's instrument sets.

The Company has the ability to transfer surgical instruments amongst hospitals, ambulatory surgery centers, or distributors at our discretion. The Company recovers the instruments after each surgery for decontamination and sterilization, and may transfer the instruments based on anticipated demand. The Company performs physical counts of the surgical instruments along with a confirmation process to validate the existence of the surgical instruments. Surgical instruments are owned and controlled by the Company and are used to generate long term economic benefits. The Company has concluded that surgical instruments meet the criteria under ASC 360 - Property, Plant and Equipment and that appropriate expense recognition is through depreciation over the useful life of the surgical instruments. Surgical instruments have been assigned a five-year useful life based on the Company's historical experience and future expected frequency of use.

Estimated useful lives for financial reporting purposes are as follows:

Asset Category	Useful Life (Years)
Machinery and equipment	5
Instruments	5
Computer equipment	5
Furniture and Fixtures	7
Leasehold Improvements	10

Long-Lived Assets

The Company reviews long-lived assets for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset may not be recoverable. Impairment is determined by comparing the carrying value of the assets to their estimated future undiscounted cash flows. If the sum of the estimated future undiscounted cash flows expected to result from the use and eventual disposition of an asset is less than the carrying amount of the asset, an impairment loss is recognized. If it is determined that an impairment has occurred, the asset is written down to its estimated fair value and a charge to income is recognized. The Company has not recorded any impairment losses on long-lived assets during the years ended December 31, 2024 and 2023. As of December 31, 2024 and 2023, all long-lived assets were located in the U.S.

Intangible Assets

The Company's intangible assets relate to an acquired software license agreement. Intangible assets are amortized on a straight-line basis over their estimated useful lives. As of May 1, 2021, upon FDA clearance of the licensed materials, the software license agreement was considered to be in service and amortization began being

recognized. Amortization expense is included in selling, general, and administrative expenses in the statements of operations and comprehensive loss. The Company did not identify any triggering events and determined there was no impairment to recognize as of December 31, 2024 and 2023. The useful life, gross carrying amount, accumulated amortization, and net carrying value of the intangible asset is as follows:

December 31, 2024	Useful Life	Gross Carrying Amount	Accumulated Amortization	Net Carrying Value
Software License	5	$ 1,500	$ 1,100	$ 400

December 31, 2023	Useful Life	Gross Carrying Amount	Accumulated Amortization	Net Carrying Value
Software License	5	$ 1,500	$ 800	$ 700

Amortization expense in each of the next three years is expected to be the following:

Year ending December 31	
2025	$ 300
2026	100
Total	$ 400

Advertising Costs

The Company has elected to expense advertising costs as incurred. Total advertising costs were $865 and $594, for the years ended December 31, 2024 and 2023, respectively.

Research and Development Costs

Research and development costs are expensed in the period in which they are incurred and totaled $4,489 and $3,021 for the years ended December 31, 2024 and 2023, respectively. Research and development expenses may include costs incurred in performing research and development activities, including clinical trial costs, manufacturing costs for both clinical and pre-clinical materials as well as other contracted services, and other external costs. Non-refundable advance payments for goods and services that will be used in future research and development activities are expensed when the activity is performed or when the goods have been received, rather than when payment is made, in accordance with ASC 730, *Research and Development*.

Presentation of Sales Taxes

Various states impose a sales or similar tax on certain of the Company's sales to non-exempt customers. The Company collects that tax from customers and remits the entire amount to the state. The Company's accounting policy is to exclude the tax collected and remitted to the state from sales revenue and cost of goods sold.

Income Taxes

Provision for income taxes is based on amounts reported in the statements of operations and comprehensive loss and include deferred taxes on temporary differences for tax and financial statement purposes. Temporary differences arise from differences between methods used for tax purposes and book purposes. Deferred taxes are computed using the liability method as prescribed by ASC 740, *Income Taxes*. The Company recorded a valuation allowance to reduce deferred tax assets to the amount that is believed more likely than not to be realized.

The Company follows the provisions of ASC 740-10-25, *Accounting for Uncertainty in Income Taxes*, which seeks to reduce the diversity in practice associated with certain aspects of the recognition and measurement related to accounting for income taxes. Under ASC 740-10-25, an organization must recognize the tax benefit associated with tax positions taken for tax return purposes when it is more likely than not that the position will be sustained. The Company recognizes any corresponding interest and penalties associated with its income tax positions in

income tax expense. The Company records liabilities in which it believes the position is not more likely than not sustainable.

There are no uncertain tax positions to recognize as of December 31, 2024 and 2023. The net operating losses for prior years are subject to adjustment under examination to the extent they remain unutilized in an open year.

Convertible Preferred Stock Warrants

The Company accounts for its freestanding warrants on its convertible preferred stock as liabilities, as the instruments underlying the warrants are classified outside of permanent equity. The underlying shares of convertible preferred stock are classified as mezzanine equity due to contingent redemption provisions outside of the Company's control. Such warrants are measured and recognized at fair value and subject to re-measurement at each balance sheet date. At the end of each reporting period, changes in fair value during the period are recognized as a component of other expense in the accompanying statements of operations and comprehensive loss until the warrants are exercised or expire.

Revenue Recognition

Revenue is recognized as the performance obligation to deliver products is satisfied and are recorded based on the amount of consideration the Company expects to receive in exchange for satisfying the performance obligations. Our sales are recognized primarily when the Company transfers control to the customer, which is generally when the Company has received a purchase order and appropriate notification that the product has been used or implanted. Products are primarily transferred to customers at a point in time.

Revenue represents the amount of consideration the Company expects to receive from customers in exchange for transferring products. Net revenue exclude sales taxes the Company collects from customers. Other costs to obtain and fulfill contracts are generally expensed as incurred due to the short-term nature of most of our sales. The Company extends terms of payment to our customers based on commercially reasonable terms for the markets of our customers, while also considering their credit quality. Shipping and handling costs charged to customers are included in net sales. Total shipping and handling costs for the years ended December 31, 2024 and 2023 were $456 and $275, respectively.

The Company's payment terms with customers are customary and vary by customer but typically range from 30 to 60 days. The Company has evaluated the terms of its arrangements and determined that they do not contain significant financing components.

The Company does not provide any warranty on their products other than for implied use. The Company has not experienced any warranty claims and does not carry a reserve.

Employee Benefit Plan

The Company maintains a defined-contribution plan, which covers all employees, and was established during 2021. All employees are eligible as of the first of the month following their hire date. The Company's policy is to provide a discretionary match, which for the year ended December 31, 2024 was 100% of the first 3% contributed, and a 50% match of the next 2% contributed, which results in an overall 4% Company contribution that is vested immediately. The amount of expense recognized for the years ended December 31, 2024, and 2023 was $273 and $166, respectively.

New Accounting Pronouncements

In June 2016, the FASB issued ASU 2016-13: *Financial Instruments – Credit Losses: Measurement of Credit Losses on Financial Instruments.* ASU 2016-13 was issued to provide financial statement users with more useful information regarding the expected credit losses on financial instruments held as assets. Under current GAAP, financial statement recognition for credit losses on financial instruments is generally delayed until the occurrence of the loss was probable. The amendments of ASU 2016-13 eliminate the probable initial recognition threshold and instead reflect an entity's current estimate of all expected credit losses. The amendments also broaden the

information that an entity must consider in developing its expected credit loss estimates for those assets measured at amortized cost by using forecasted information instead of the current methodology which only considered past events and current conditions. Under ASU 2016-13, credit losses on certain types of financial instruments will be measured in a manner similar to current GAAP; however, the amendments require that credit losses be presented as an allowance against the financial instrument, rather than as a write-down. The amendments also allow the entity to record reversals of credit losses in current period net income, which is prohibited under current GAAP. The Company adopted the new standard on January 1, 2023. There was not a significant impact to the financial statements as a result of this pronouncement.

In August 2020, the FASB issued ASU 2020-06, *Debt — Debt with Conversion and Other Options (Subtopic 470-20) and Derivatives and Hedging — Contracts in Entity's Own Equity (Subtopic 815-40)*, which simplifies the accounting for certain financial instruments with characteristics of liabilities and equity, including convertible instruments and contracts on an entity's own equity. ASU 2020-06 is effective for private companies for fiscal years beginning after December 15, 2023, and interim periods within those fiscal years, with early adoption permitted in years beginning after December 15, 2020. ASU 2020-06 permits adoption on a retrospective basis to financial instruments outstanding as of the beginning of the first comparative reporting period presented. The Company adopted the new standard on January 1, 2024. There was not a significant impact to the financial statements as a result of this pronouncement.

In November 2023, the FASB issued ASU 2023-07—Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures was designed to improve reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses. This ASU is effective for fiscal years beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024. The Company adopted the new standard on January 1, 2024. The impact to the Company's financial statements is disclosed in Note 12.

In December 2023, the FASB issued ASU 2023-09*, Income Taxes (Topic 740) ("ASC 740")*. The update requires all public business entities on an annual basis to (1) disclose specific categories in the rate reconciliation and (2) provide additional information for reconciling items that meet a quantitative threshold and an explanation, if not otherwise evident, of the individual reconciling items disclosed, such as the nature, effect, and underlying causes of the reconciling items and the judgment used in categorizing the reconciling items. In addition, the update requires certain new disclosures of the amount of income taxes paid (net of refunds received) disaggregated by federal, state, and foreign taxes and the amount of income taxes paid (net of refunds received) disaggregated by individual jurisdictions in which income taxes paid is equal to or greater than five percent of total income taxes paid (net of refunds received). Other new disclosures required include income (or loss) from continuing operations before income tax expense (or benefit) disaggregated between domestic and foreign and income tax expense (or benefit) from continuing operations disaggregated by federal, state, and foreign. The new guidance is effective for annual periods beginning after December 15, 2024. Early adoption is permitted. The amendments are to be applied on a prospective basis, with retrospective application permitted. The Company is currently evaluating the impact of the new standard on its financial statements and related disclosures.

In November 2024, the FASB issued ASU 2024-03, Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses requires additional, disaggregated disclosure around certain income statement expense line items. This ASU mandates that entities, at each interim and annual period, disclose the amounts of (a) inventory purchases, (b) employee compensation, (c) depreciation, (d) intangible asset amortization, and (e) depletion, depreciation, and amortization for oil and gas activities included within each relevant expense caption presented on the income statement within continuing operations. Entities are also required to (1) combine certain disclosures already mandated under GAAP with these new requirements, (2) provide qualitative descriptions of expenses that are not disaggregated quantitatively, and (3) disclose total selling expenses and, annually, the definition of selling expenses. The guidance is effective for annual periods beginning after December 15, 2026, and interim periods beginning after December 15, 2027. Early adoption is permitted. The Company is currently evaluating the potential impact of adopting this new guidance on the Company's financial statements and related disclosures.

Change in Accounting Policies and Reclassifications

The Company has made certain reclassifications to the previously issued financial statement to conform to current period presentation, including reclassification of depreciation expense on surgical instruments on the statement of operations and comprehensive loss, resulting in a decrease of $1,311 of cost of goods sold, and an increase in gross margin along with selling, general, and administrative expense by the same amount for the year ended December 31, 2023. Operating and Net loss for the year ended December 31, 2023 were not impacted by the change in presentation.

2. Fair Value Measurements

The Company utilizes fair value measurement guidance prescribed by accounting standards to value its financial instruments. The guidance establishes a fair value hierarchy for financial instruments measured at fair value that distinguishes between assumptions based on market data (observable inputs) and the Company's own assumptions (unobservable inputs). Observable inputs are inputs that market participants would use in pricing the asset or liability based on market data obtained from sources independent of the Company. Unobservable inputs are inputs that reflect the Company's assumptions about the inputs that market participants would use in pricing the asset or liability and are developed based on the best information available in the circumstances.

Fair value is defined as the exchange price, or exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants on the measurement date. The fair value hierarchy contains three levels of inputs that may be used to measure fair value, in accordance with ASC 820, *Fair Value Measurement*, of which the first two are considered observable and the last is considered unobservable. These levels are as follows:

Level 1 – Inputs at this level include unadjusted quoted prices in active markets for identical assets or liabilities that the Company has the ability to access.

Level 2 – Inputs at this level include observable inputs other than Level 1 inputs, such as quoted prices for similar assets or liabilities in active markets, quoted prices for identical or similar assets or liabilities in markets that are not active, or other inputs that are observable or can be corroborated by observable market data for substantially the full term of the asset or liability.

Level 3 – Inputs at this level include unobservable inputs that are supported by little or no market activity and that are significant to the fair value of the underlying asset or liability. Level 3 assets and liabilities include those whose fair value measurements are determined using pricing models, discounted cash flow methodologies, or similar valuation techniques, as well as significant management judgment or estimation.

To the extent the valuation is based on models or inputs that are less observable or unobservable in the market, the determination of fair value requires more judgement. Accordingly, the degree of judgement exercised by the Company in determining fair value is greatest for instruments categorized in Level 3. A financial instrument's level within the fair value hierarchy is based on the lowest level of any input that is significant to the fair value measurement.

The recorded amounts of certain financial instruments, including cash, cash equivalents, accounts receivable, accounts payable, and accrued liabilities approximate fair value due to their relatively short maturities.

The fair value of marketable securities as of December 31, 2024 and 2023 are summarized below:

	December 31, 2024		
	Level 1	Level 2	Level 3
Assets:			
Cash Equivalents			
Money market funds	$ 85	$ —	$ —

Shoulder Innovations, Inc.

Notes to the Financial Statements
(in thousands, except share, per share data and percentages)

	December 31, 2024		
Short-term marketable securities at fair value			
U.S. treasury and government agencies	6,053	—	—
Corporate and international bonds	—	2,868	—
Total assets	$ 6,138	$ 2,868	$ —
Liabilities:			
Preferred stock warrant liability			$ 970

	December 31, 2023		
	Level 1	Level 2	Level 3
Assets:			
Cash Equivalents			
Money market funds	$ 151	$ —	$ —
Short-term marketable securities at fair value			
U.S. treasury and government agencies	18,933	—	—
Corporate and international bonds	—	7,493	—
Total assets	$ 19,084	$ 7,493	$ —
Liabilities:			
Preferred stock warrant liability			$ 851

Warrants

Liabilities related to stock warrants are remeasured at fair value on a recurring basis using the Black-Scholes option pricing model. The following table presents the change in fair value of the stock warrants which are classified in Level 3 of the fair value hierarchy for the years ended December 31, 2024 and 2023:

	2024	2023
Balance, January 1	$ 851	$ 484
Fair value of warrants issued	—	447
Change in fair value	119	(80)
Balance, December 31	$ 970	$ 851

The stock warrants were valued under the option pricing model, which considers the estimated volatility of the Company's common stock at the date of measurement based on selected metrics of applicable volatility calculations from guideline public companies. The remeasurement of the convertible preferred stock warrant liability resulted in $119 and $80 recognized as other (income) expense, net for the years ended December 31, 2024 and 2023, respectively.

The fair value of the warrants were estimated as of December 31, 2024 and 2023 using an option pricing model with the following weighted average assumptions:

December 31, Description	2024	2023
Weighted average volatility	55.0 %	50.0 %
Weighted average risk-free rate	4.26 %	4.94 %
Expected dividend yield	0.00 %	0.00 %

Derivative Liability

The derivative liability associated with the Convertible Promissory Notes is remeasured at fair value on a recurring basis using a scenario-based analysis by performing a probability-weighted present value assessment of the Convertible Promissory Notes payoff under possible outcomes available to noteholders and certain bifurcated features within the Convertible Promissory Notes. The derivative liability was extinguished in 2023 as a result of the conversion of the convertible promissory notes. The following table presents the change in fair value of the derivative liability which is classified in Level 3 of the fair value hierarchy:

	2024	2023
Balance, January 1	**$ —**	$ 1,890
Fair value at issuance	—	—
Change in fair value	—	78
Fair value at extinguishment of derivative	—	(1,968)
Balance, December 31	$ —	$ —

3. Accrued Liabilities and Other Current Liabilities

Accrued liabilities and other current liabilities consist of the following:

December 31,	2024	2023
Commissions payable	$ 836	$ 619
Accrued payroll	738	977
Accrued royalties	612	457
Other	554	229
Accrued liabilities and other liabilities	$ 2,740	$ 2,282

4. Property and Equipment, Net

Net property and equipment consist of the following:

December 31,	2024	2023
Instruments	$ 11,223	$ 8,011
Machinery and equipment	94	57
Computer equipment	190	47
Furniture and fixtures	70	54
Leasehold improvements	150	146
Total property and equipment	11,727	8,315
Less: Accumulated depreciation	(4,240)	(2,680)
Property and Equipment, Net	$ 7,487	$ 5,635

Depreciation expense for the years ended December 31, 2024 and 2023 totaled $ 1,896 and $1,341, respectively. Depreciation expense of $1,842 and $1,311 for instruments is included in selling, general, and administrative expense in the statements of operations and comprehensive loss for the years ended December 31, 2024 and 2023 respectively. Depreciation expense of $37 and $30 for computer equipment, furniture and fixtures, and leasehold improvements are included in selling, general, and administrative expenses in the statements of operations and comprehensive loss for the years ended December 31, 2024 and 2023, respectively.

5. Debt

On August 7, 2023, the Company entered into a loan and security agreement (the "Trinity Loan Agreement") with Trinity Capital Inc. as lender, administrative agent, and collateral agent.

The Trinity Loan Agreement provides a term loan commitment of $45,000 in three potential tranches: (i) a $15,000 term loan facility funded on August 7, 2023, (ii) a $15,000 term loan facility available at the Company's request on or before December 31, 2024, if the Company achieves certain revenue milestones, and (iii) a $15,000 term loan facility available at the Company's request on or before December 31, 2025, if the Company achieves certain additional revenue milestones. All three of these term loans have a maturity date of September 1, 2028. Borrowings under all three term loan facilities bear interest at a variable annual rate equal to the greater of (i) 11.50% and (ii) the Prime Rate plus 3.50%. The second tranche expired on December 31, 2024, prior to Company drawing on the tranche.

In connection with the closing of the Trinity Loan Agreement, the Company issued a warrant to the Lender to purchase shares of the Company's Series D convertible preferred stock ("Trinity Warrant"). The Trinity Warrant has an exercise price of $0.5412 per share and expires ten years from the issue date of the Trinity Loan Agreement; refer to Note 9. Upon the advance of the tranche A loan, 831,486 shares vested, and an additional 831,486 shares will vest upon the funding of the second and third term tranche loans, adjusted for the Reverse Stock Split as described in Note 1.

The Trinity Loan Agreement provides that the Company can at any time prepay the term loans, in whole or in part, subject to a prepayment premium equal to: (a) 2.50% of the then-outstanding principal amount of the term loans, if such prepayment occurs on or prior to the first anniversary of the Trinity Loan Agreement; (b) 1.50% of the then-outstanding principal amount of the advance, if such prepayment occurs after the first anniversary of the Trinity Loan Agreement and on or prior to the second anniversary of the Trinity Loan Agreement; and (c) 1.00% of the then-outstanding principal amount of the advance, if such prepayment occurs after the second anniversary of the Trinity Loan Agreement and prior to the Maturity Date. We are required to make an end of term payment equal to 3.00% of the aggregate principal amount of the term loans funded on the earlier of (i) the Maturity Date, (ii) the date that the Company prepays all of the outstanding principal in full or (iii) the date of acceleration of the balance of the outstanding term loans by the Agent. The term loans are secured by substantially all our assets, including intellectual property.

Proceeds related to the Trinity Loan Agreement were $14,850, net of issuance costs of $250 and return of deposit of $100. As of December 31, 2023, the Company recorded long-term debt related to the Trinity Loan Agreement of $14,351, which includes a principal amount of $15,000 and unamortized discount and debt issuance costs of $649, based on an imputed interest rate of 13%. As of December 31, 2024, the Company recorded long-term debt related to the Trinity Loan Agreement of $14,658, which included principal amount of $15,000 and unamortized debt discount, issuance cost and deferred interest of $623, based on an imputed interest rate of 14%.

6. Leases

Month-to-Month Lease

The Company leases a facility in Michigan from a related party. The lease is on a month-to-month basis. Total rent expense under this agreement for the years ended December 31, 2024 and 2023, was $33 and $33, respectively. There are no future minimum lease payments associated with the operating lease. The Company has elected to apply the short-term lease exception guidance and therefore has not recorded a right-of-use asset or liability related to this lease.

Lease Obligations

The Company leases office space in Michigan under a five-year lease agreement, which commenced in July 2021. The Company has determined this lease agreement qualifies as an operating lease arrangement. The Company's remaining operating lease term is approximately 1.5 years at December 31, 2024.

As the lease contract does not provide an implicit discount rate, the Company used 5.25% representing its incremental borrowing rate based on information available at the commencement date, to determine the present value of lease payments. The incremental borrowing rate is based on a borrowing with a term similar to that of the associated lease. The Company does not include renewal, termination, or purchase options that are not reasonably certain of exercise when determining the term of the borrowing.

At December 31, 2024, the right-of-use asset and lease liability relating to the operating lease were $68 and $72, respectively. The lease expense for the year ended December 31, 2024 was $65.

At December 31, 2024, the commitment for operation lease liabilities for future annual periods was as follows:

Year Ending December 31	
2025	$ 47
2026	24
Total Lease Payments	71
Less imputed interest	(6)
Present Value of Lease Liabilities	$ 65

7. Income Taxes

Provision Expense for Income Taxes

The Company had no current or deferred tax expense recorded as of December 31, 2024 and 2023.

Income tax expense differs from the statutory rate due primarily to the impact of the valuation allowance against the Company's deferred tax assets.

Deferred Income Tax Assets and Liabilities

The components of deferred income tax assets and liabilities are as follows:

December 31,	**2024**	**2023**
Deferred Tax Assets		
Net operating loss carryforwards	$ 9,512	$ 6,960
Tax credits and other carryforwards	464	466
Section 174 expenditures	1,993	1,454
Accrued liabilities and other items	332	219
Deferred Tax Assets	12,301	9,099
Deferred Tax Liability		
Depreciation and amortization	(1,338)	(1,213)
Deferred Tax Liability	(1,338)	(1,213)
Deferred Tax Assets, Net	10,963	7,886
Valuation Allowance	(10,963)	(7,886)
Net Deferred Tax Asset	$ —	$ —

A reconciliation of the Company’s provision for income taxes at the federal statutory rate of 21% to the reported income tax provision for the years ended December 31, 2024 and 2023 is as follows:

December 31,	**2024**	**2023**
Computed “expected” tax benefit	$ (3,267)	$ (2,658)
Permanent differences	33	24

December 31,	2024	2023
General Business tax credits	—	145
Other	157	677
Change in valuation allowance	3,077	1,812
Income Tax expense (benefit)	$ —	$ —

The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income during the periods in which those temporary differences become deductible. Management considers the scheduled reversal of deferred tax liabilities (including the impact of available carryback and carryforward periods), projected future taxable income, and tax-planning strategies in making this assessment. The Company has determined for the years ended December 31, 2024 and 2023, based on all available evidence analyzed on a more likely than not basis, a valuation allowance should be established on all of the net deferred tax asset in its U.S. jurisdiction since there is not enough positive evidence to support the more likely than not position for the future realization in the U.S. jurisdiction.

On December 22, 2017, the U.S. government enacted comprehensive tax legislation commonly referred to as the Tax Cuts and Jobs Act (the Tax Act). The Tax Act allows for the indefinite carryforward of net operating losses (NOLs) arising in a taxable year ending after December 31, 2017, which would be considered an indefinite lived asset. The amendments limit the future usage of such NOLs to 80% of taxable income in a single year and disallow the carryback of NOLs to prior years with taxable income.

As of December 31, 2024 and 2023, the Company has federal net operating loss carryforwards of $45,295 and $33,144, respectively. The Company has federal tax credit carryforwards of $464 and $464 for the years ended December 31, 2024 and 2023, respectively. The NOLs and tax credit carryforwards recorded are subject to limitations under Section 382 and 383. However, the Section 382 limitation did not limit the usage of the net operating loss carryforwards in 2024 or 2023 for federal tax purposes. In addition, the Company’s ability to utilize the current NOL carryforwards might be further limited by future equity issuances. The federal net operating loss carryforwards originated after 2016 and have an indefinite life which may be used to offset 80% of a future year's taxable income. The Company has various state net operating loss carryforwards with carryforward periods ranging from 15 -20 years and will begin to expire in 2034.

The Company operates in various states within the United States and files income tax returns in these various jurisdictions and has generated various net operating loss carryforwards in these jurisdictions to date. The Company does not believe a material uncertain tax position exists as of December 31, 2024 and 2023. Based on the Company’s assessment of many factors, including past experience and complex judgements about future events, the Company does not currently anticipate significant changes in its uncertain tax positions over the next 12 months. In connection with the adoption of the referenced provision, the Company recognizes interest and penalties accrued related to unrecognized tax benefits in income tax expense. As of December 31, 2023 and 2024, the Company had no accrued interest and penalties.

The Company’s federal and state tax returns are open for review going back to the 2019 tax year.

8. Related Party Transactions

On October 22, 2020, the Company entered into a software license agreement with Genesis Software which was amended and restated on January 1, 2023. Robert Ball, the Company’s Chief Executive Officer and Executive Chairman, is a co-founder and director of Genesis Software. Mr. Ball and Matthew Ahearn, the Company’s Chief Operating Officer and a director, are directors of Genesis Investment Holdings, LLC, which has an ownership interest in Genesis Software. In addition, cultivate(MD) Capital Accelerator Fund L.P. and Genesis Investment Holdings, LLC, each beneficial owners of more than 5% of our capital stock, are investors in Genesis Software. The software licensing agreement is a 5 year term. The agreement required an upfront payment of $1,000, an incremental $500 payment when FDA clearance was obtained, and quarterly payments of royalties equal to 4% of the net sales

price of each licensed product sold. For each of the years ended December 31, 2024 and 2023, the Company paid Genesis Software $1,169 and $634, respectively, pursuant to the license agreement.

The Company has entered into a consulting agreement with an entity under common ownership. The consulting agreement is currently on a year-to-year basis. The agreement requires compensation for services performed. If services performed are on an hourly basis, the Company shall be responsible to pay for hours actually worked by the consultant's employees. The Company will reimburse the consultant for all reasonable expenses incurred in connection with performing services for the Company. For the years ended December 31, 2024 and 2023, the Company paid $3,589 and $2,761, respectively. Amounts payable of $537 and $552 are included in accounts payable on the balance sheets at December 31, 2024, and 2023, respectively.

In 2024 and 2023, the Company paid $1,925 and $2,585, respectively for software development, to parties related through common ownership and board member ownership. Amounts payable of $276 and $134 are included in accounts payable on the balance sheets at December 31, 2024 and 2023.

On June 3, 2024, the Company entered into a new supply agreement (the "RMD Supply Agreement") with RMD, pursuant to which RMD manufactures substantially all the surgical instruments used during procedures involving the Company's implant systems. During such periods, Robert Ball, the Company's Chief Executive Officer and Executive Chairman, was an investor in RMD. Mr. Ball is no longer an investor in RMD. For each of the years ended December 31, 2024 and 2023, the Company paid RMD $859 and $2,262, respectively, pursuant to the RMD Supply Agreement.

9. Convertible Preferred Stock

On February 10, 2023, the Company held the initial closing of the sale and issuance of shares of its newly authorized Series D Preferred Stock, par value $0.001 per share, at which the Company sold and issued 46,706,316 Series D Preferred shares at a per share price of $0.5412 resulting in aggregate new money proceeds to the Company of $25,277 and, in connection therewith, converted its outstanding convertible promissory notes into 16,464,493 shares of Series D Preferred Stock at a conversion price of $8,911. The conversion resulted in a loss of $1,127, which is recorded in other expense. On March 3, 2023, the Company held a subsequent closing of the sale and issuance of shares of its Series D Preferred Stock, at which the Company sold and issued 17,738,360 Series D Preferred shares at a per share price of $0.5412, resulting in additional aggregate new money proceeds to the Company of $9,600.

A summary of preferred stock is as follows:

	Series Seed Convertible Preferred Stock		Series A Convertible Preferred Stock		Series B Convertible Preferred Stock		Series C Convertible Preferred Stock		Series D Convertible Preferred Stock		Total	
	Number of Shares	Amount	Number of Shares	Amount	Number of Shares	Amount	Number of Shares	Amount	Number of Shares	Amount	Number of Shares	Amount
Balance at December 31, 2022	15,851,401	$ 1,337	22,399,370	$ 5,600	5,913,964	$ 2,200	50,116,284	$ 21,550	—	$ —	94,281,019	$ 30,687
Issuance of Series D	—	—	—	—	—	—	—	—	80,909,169	43,788	80,909,169	43,788
Balance at December 31, 2023	15,851,401	1,337	22,399,370	5,600	5,913,964	2,200	50,116,284	21,550	80,909,169	43,788	175,190,188	74,475
Balance at December 31, 2024	15,851,401	$ 1,337	22,399,370	$ 5,600	5,913,964	$ 2,200	50,116,284	$ 21,550	80,909,169	$ 43,788	175,190,188	$ 74,475

Issued and Outstanding Preferred Shares

Authorized, issued, and outstanding Preferred Stock were as follows:

December 31, 2024

Series	Par Value	Authorized	Issued and Outstanding	Issuance Price Per Share	Liquidation Value
Seed	$ 0.001	16,840,400	15,851,401	$ 0.0843	$ 1,337
A	0.001	22,399,370	22,399,370	0.2500	5,600
B	0.001	6,913,964	5,913,964	0.3720	2,200

December 31, 2024

Series	Par Value	Authorized	Issued and Outstanding	Issuance Price Per Share	Liquidation Value
C	0.001	50,116,284	50,116,284	0.4300	21,550
D	0.001	83,403,626	80,909,169	0.5412	43,788
Total		179,673,644	175,190,188		$ 74,475

December 31, 2023

Series	Par Value	Authorized	Issued and Outstanding	Issuance Price Per Share	Liquidation Value
Seed	$ 0.001	16,840,400	15,851,401	$ 0.0843	$ 1,337
A	0.001	22,399,370	22,399,370	0.2500	5,600
B	0.001	6,913,964	5,913,964	0.3720	2,200
C	0.001	50,116,284	50,116,284	0.4300	21,550
D	0.001	83,403,626	80,909,169	0.5412	43,788
Total		179,673,644	175,190,188		$ 74,475

Preferred Stock Provisions

Conversion of Convertible Preferred Stock to Equity

Preferred Stock contains a provision at the option of the holder, at any time and from time to time, and without the payment of additional consideration by the holder thereof, to be converted into Common Stock. The number of shares of Common Stock to which a holder of Preferred Stock shall be entitled upon conversion shall be the product obtained by multiplying the applicable conversion prices then in effect by the number of shares of Preferred Stock being converted. The conversion price of Preferred Stock shall be subject to adjustment in the event of any stock split, stock dividend, combination, subdivision, recapitalization, or the like. All Preferred Stock shall be automatically converted into Common Stock at their then respective conversion prices: (i) in the event of an underwritten public offering of shares of the Common Stock at an aggregate offering price (prior to underwriting discounts, commissions, and expenses) of at least $50,000 or (ii) the date upon which the Company obtains the vote or consent of the Preferred Majority and the vote or consent of the holders of at least 65% of the Series D Preferred Stock. Convertible preferred stock is presented as temporary equity in the mezzanine section of the balance sheets, as conversion of the preferred stock is not within the control of the Company. The purpose of this classification is to convey that such a security may not be permanently a part of equity and could result in a demand for cash, securities, or other assets of the entity in the future. For all series of preferred stock, the initial conversion price equals their original issuance price per share, at December 31, 2024 and 2023, adjusted for the Reverse Stock Split as described in Note 1.

Dividends

The Company's Preferred Stock Series Seed, A, B, C and D pay non-cumulative dividends, at the rate of 8% per annum, whenever funds are legally available and when, as and if declared by the Company's Board of Directors (Board) prior and in preference to dividends on any other equity of the Company. Dividends, when, as, and if declared, will first be paid in full to the holders of Series D Preferred Stock, then Series C Preferred Stock, then Series B Preferred Stock, then Series A Preferred Stock, then Series Seed Preferred Stock, then to holders of Common Stock. In the event that any dividends are declared to the then-outstanding shares of Common Stock (other than dividends payable in Common Stock for which adjustment to the conversion price of the Preferred Stock is made) an additional dividend shall be declared with respect to the then-outstanding shares of Preferred Stock, in an amount equal to the amount of dividends per share that would have been payable on the number of shares of the Common Stock that the Preferred Stock would have been converted into on an as-converted basis. No dividends were declared by the Board of Directors in 2024 or 2023.

Voting Rights

The holders of Preferred Stock shall be entitled to the number of votes equal to the whole number of shares of the Common Stock into which such shares of the Preferred Stock are then convertible to. Additionally, separate vote of all series of preferred stock requires, in addition to any other vote or consent, the vote or consent of at least a majority of each series of preferred stock then-outstanding votes for certain significant actions. Further, as long as at least 20% of the authorized shares of the Series C Preferred Stock or Series D Preferred Stock remain outstanding, certain significant actions require the vote or consent of at least 60% of the then-outstanding Series C Preferred Stock or Series D Preferred Stock, respectively.

Liquidation Rights

In the event of a deemed liquidation event of the Company, voluntary or involuntary liquidation, dissolution or winding up of the Company, all holders of Preferred Stock are entitled to full payment under their liquidation preferences out of the assets of the Company legally available for distribution to its stockholders (or the consideration received in such transaction), before any payment shall be made to the holders of the Common Stock. After payment in full to the Series D liquidation preference, the remaining proceeds shall be pursuantly distributed to the holders of existing Series C, followed by Series B, Series A, and Series Seed Preferred Stock and common stock, in an amount equal to their original issue price plus any declared and unpaid dividends.

All remaining legally available assets of the Company that are not payable to the holders of shares of Preferred Stock shall be distributed among the holders of shares of common stock, pro rata based on the number of shares held by each such holder.

Anti-Dilution and Downround Protections

The conversion price is subject to adjustment for certain dilutive events, including certain types of (i) stock splits and combinations, (ii) common stock dividends and distributions, (iii) reclassification, exchange, and substitutions, (iv) reorganizations, mergers, or consolidations, or (v) sale of shares below conversion price.

Stock Warrants

In 2020, in connection with its issuance of Series B Preferred Stock, the Company issued 1,000,000 of Series B Stock Warrants. The Series B Preferred Stock issuable upon exercise of the Series B Stock Warrants is convertible into shares of common stock in the same manner as each respective underlying series of outstanding Series B Preferred Stock and will be entitled to the same dividend rights as each respective series.

In August 2023, in connection with the closing of the Trinity Loan Agreement, the Company issued a warrant to the Lender to purchase 2,494,457 shares of the Company's Series D convertible preferred stock ("Trinity Warrant"). The Trinity Warrant has an exercise price of $0.5412 per share, and expires ten years from the issue date of the Trinity Loan Agreement. Upon the advance of the tranche A loan, 831,486 shares vested, and an additional 831,486 shares will vest upon the funding of the second and third term tranche loans.

The following tables summarize the Company's outstanding warrants:

December 31, 2024

Description	Outstanding	Exercise Price per share	Expiration Date	Fair Value
Common	17,827	$ 2.10000	April 1, 2031	$ 41
Series Seed	988,999	0.08434	April 24, 2025	205
Series B	1,000,000	0.37200	January 28, 2030	207
Series D	2,494,457	0.54120	August 7, 2033	517
Total	4,501,283			$ 970

December 31, 2023

Description	Outstanding	Exercise Price per share	Expiration Date	Fair Value
Common	17,827	$ 2.10000	April 1, 2031	$ 48
Series Seed	988,999	0.08434	April 24, 2025	177
Series B	1,000,000	0.37200	January 28, 2030	179
Series D	2,494,457	0.54120	August 7, 2033	447
Total	4,501,283			$ 851

10. Stock Compensation

Common Stock

The Company increased the authorized capital from 114,000,000 to 204,958,600 shares of common stock in 2023, and from 204,958,600 to 212,366,763 shares of common stock in 2024.

Stock Options

In 2017, the Company adopted a stock compensation plan (the "Plan") pursuant to which the Company's Board of Directors may grant stock options or nonvested shares to its full-time employees, directors, and advisors to purchase common stock of the Company. The Plan was amended in 2023 to authorize 1,281,282 shares issuable. The Plan was subsequently amended in 2024 to authorize grants to purchase up to 1,669,550 shares. Stock options can be granted with an exercise price equal to or greater than the stock's fair value at the date of grant. All awards have 10-year terms and vest based on terms defined in each individual grant agreement, and the expense is recognized on a straight-line basis over the vesting period, which is generally the service period.

Because the Company's shares are not publicly traded and its shares are rarely traded privately, expected volatility is estimated based on the average historical volatility of similar entities with publicly traded shares. The risk-free rate for the expected term of the option is based on the U.S. Treasury yield curve at the date of grant.

The Company uses a simplified method to determine the expected term for the valuation of employee options. This method effectively assumes that exercise occurs over the period from vesting until expiration, and therefore the expected term is the midpoint between the service period and the contractual term of the award. The simplified method is applicable to options with service conditions.

The Company utilizes a third-party valuation firm to assist in the estimation of the fair value of the underlying shares of the Company utilizing option pricing models (OPM). Under the OPM, shares are valued by creating a series of call options with exercise prices based on the liquidation preferences and conversion terms of each equity class. The estimated fair values of the convertible preferred stock and common stock are inferred by analyzing these options. This method is appropriate to use when the range of possible future outcomes is so difficult to predict that estimates would be highly speculative, and dissolution or liquidation is not imminent. The Company accounts for forfeitures as they occur. Shares of the Company's common stock granted under the stock option plan in the form of stock options are counted against the share reserve on a one for one basis.

The following tables summarize the Company's outstanding stock options:

	Options	Weighted Average Exercise Price	Weighted Average Remaining Contractual Life (in years)
Outstanding as of December 31, 2022	534,224	$ 1.91	7.54
Granted	670,889	$ 2.48	—
Exercised	—	—	—
Forfeited	—	—	—
Options Outstanding December 31, 2023	1,205,113	$ 2.10	8.03
Granted	210,585	$ 2.11	—
Exercised	(26,528)	2.48	—
Forfeited	(54,367)	0.76	—
Options Outstanding December 31, 2024	1,334,803	$ 2.29	7.47
Options Exercisable, December 31, 2024	791,658	$ 2.10	$ 6.50

The aggregate intrinsic value of stock options is calculated as the pre-tax difference between the weighted-average exercise price of the stock options and the valuation of the Company's common shares of $2.86 and $2.48 as of December 31, 2024 and 2023 respectively. The calculation excludes any stock options with an exercise price higher than the valuation of shares of the Company's common stock, if any. The stock options were valued under an option pricing model, which considers the estimated volatility of the Company's common stock at the date of measurement based on selected metrics of applicable volatility calculations from guideline public companies. Stock compensation expense related to the stock options issued totaled $754 in 2024, and $451 in 2023 and is recognized as selling, general, and administrative expenses in the statements of operations and comprehensive loss. The total intrinsic value of options exercised was $15 and $0 for the years ended December 31, 2024 and 2023, respectively.

As of December 31, 2024, the Company had unrecognized stock-based compensation of $1,381 that is to be recognized over a weighted average period of 2.55 years.

The fair value of the stock options granted were estimated as of the grant date using an option pricing model with the following weighted average assumptions:

	2024	2023
Weighted average volatility	101.50%	50.00%
Weighted average risk-free rate	3.90%	4.94%
Expected dividend yield	0.00	0.00
Expected term (in years)	6.0	2.8

11. Loss Per Share

Basic loss per share is computed by dividing the net loss after tax attributable to common stockholders by the weighted average shares outstanding during the period. Diluted loss per share is computed by including potentially dilutive securities outstanding during the period in the calculation of weighted average shares outstanding. The Company did not have any dilutive securities during the periods presented; therefore, diluted loss per share is equal to basic loss per share.

Presented in the table below is a reconciliation of the numerator and denominator for the basic and diluted loss per share ("LPS") calculations for the years ended December 31:

December 31,

	2024	2023
Net loss	$ (15,619)	$ (12,655)
Basic and diluted weighted average shares outstanding	64,530	57,354
Net loss attributable to common shareholders, basic and diluted	$ (242.04)	$ (220.65)

The following securities were not included in the computation of diluted shares outstanding because the effect would be anti-dilutive (convertible preferred stock outstanding is presented on an as-converted basis):

	2024	2023
Series Seed	830,786	830,786
Series A	1,173,971	1,173,971
Series B	309,956	309,956
Series C	2,626,639	2,626,639
Series D	4,240,522	4,240,522
Common Options	1,334,803	1,205,113
Common Warrants	17,827	17,827
Series Seed Warrants	51,834	51,834
Series B Warrants	52,410	52,410
Series D Warrants	130,736	130,736
Total	10,769,484	10,639,794

12. Segment Information

The Company reports segment information based on how the Company's chief operating decision maker ("CODM") is the Chief Executive Officer, regularly reviews operating results, allocates resources and makes decisions regarding business operations. For segment reporting purposes the Company's business structure is comprised of one operating and reportable segment. The CODM uses segment gross margin and net loss for determining the allocation of resources (including employees, financial, or capital resources) to the segment to achieve the Company's strategic plan and to assess the performance of the segment by monitoring actual results against performance targets established in the Company's annual budget and forecasting process.

All revenue for the years ended December 31, 2024 and 2023 was generated from customers located in the United States. No customers represent 10% or more of the Company's total revenue for the years ended December 31, 2024 and 2023. The measure of segment assets is reported on the balance sheets as total assets.

The table below is a summary of the segment net loss, including significant segment expenses:

December 31,	2024	2023
Total Revenue	$ 31,623	$ 19,274
Cost of goods Sold	7,282	4,004
Gross margin	24,341	15,270
Operating expenses:		
General and administrative	13,975	10,705
Sales and marketing	17,140	12,288
Medical education	3,391	120
Research and development	4,489	3,021
Interest/other expense	966	1,791
Segment net loss and total net loss	$ (15,619)	(12,655)

Depreciation expense for the years ended December 31, 2024 and 2023 totaled $ 1,896 and $1,341, respectively. Depreciation expense of $1,842 and $1,311 for instruments is in sales and marketing and $37 and $30 for computer equipment, furniture and fixtures, and leasehold improvements are included in general, and administrative expenses in the statements of operations and comprehensive loss for the years ended December 31, 2024 and 2023, respectively.

Amortization expense related to the Company's software license for the years ended December 31 2024 and 2023 totaled $300 for both years, and is included in general and administrative expense.

13. Commitments and Contingencies

Litigation

The Company from time to time is involved in legal matters incidental to the conduct of its business. Legal fees and other costs associated with such actions are expensed as incurred. The Company assesses the need to record a liability for litigation and contingencies. Accrual estimates are recorded when and if it is determinable that such a liability for litigation and contingencies are both probable and reasonably estimable.

In February 2024, the Company filed a complaint against Catalyst Orthoscience Inc. ("Catalyst") claiming patent infringement through Catalyst's making, using, selling, offering for sale in the United States, and/or importing into the United States, reverse shoulder systems. In response to the Company's suit, Catalyst filed a counterclaim claiming patent infringement of certain of the Company's products. The Company believes that it has substantial and meritorious defenses to Catalyst's claims and intend to vigorously defend its position, including through the trial and appellate stages if necessary. As the counterclaim is ongoing, the Company is unable to determine the likelihood of an outcome or estimate a range of reasonably possible settlement, if any. Accordingly, the Company has not made an accrual for any possible loss. The outcome of any litigation, however, is inherently uncertain, and an adverse judgment or settlement in the counterclaim proceeding, if any, could have a material and adverse effect the Company's business, financial position, results of operations or cash flows.

14. Subsequent Events

On March 6, 2025, the Company entered into a Series E Preferred Stock Purchase Agreement pursuant to the Series E Preferred Stock Financing, whereby it received a total commitment amount of $40,130 for the issuance and sale of 58,774,332 shares of Series E Preferred Stock pursuant to two separate closing trances, the first of which closed on March 6, 2025 whereby the Company issued 29,455,169 shares of series E preferred stock resulting in total net proceeds to the Company of $20,111 and the second tranche can occur any time prior to September 15, 2025, for the issuance and sale of 29,319,143 shares of Series E Preferred Stock for an aggregate purchase price of $20,019.

In conjunction with entering into the Series E Preferred Stock Purchase Agreement, the Company increased the number of shares reserved under the Plan by 516,010 to a total of 2,185,560.

The Company evaluated subsequent events from December 31, 2024, the date of these financial statements, through April 17, 2025, which represents the date the financial statements were issued, for events requiring recording or disclosure in the financial statements for the year ended December 31, 2024, and with respect to the Reverse Stock Split, described in Note 1, through July 24, 2025.

On April 23, 2025, the Company granted 453,898 common stock options to certain employees and officers of the Company with an exercise price of $2.86. Common stock options vest over a four year period. The Company estimates it will recognize $0.8 million in stock compensation expense over the vesting period.

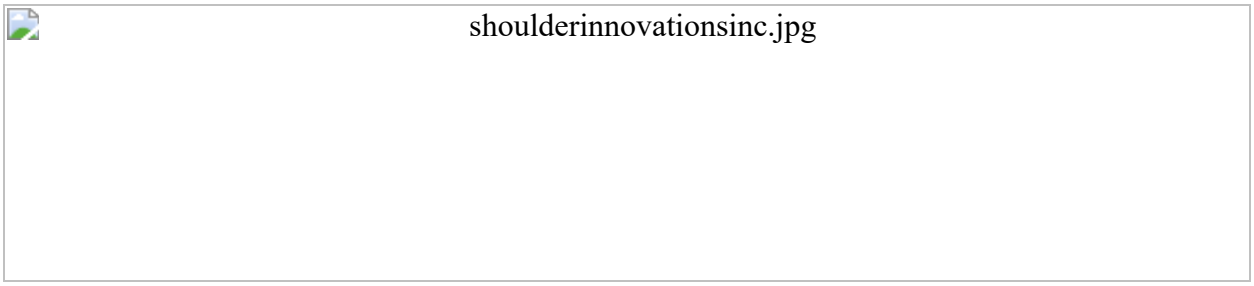
In June 2025 the Company called the second tranche of its Series E Preferred Stock for the issuance of 29,319,143 shares of Series E Preferred Stock resulting in total net proceeds to the Company of $20.0 million.

On July 18, 2025, the Company issued $40.0 million aggregate principal amount of convertible notes. The convertible notes mature on September 1, 2028 and accrue interest at a rate of 5.0% per annum through June 30, 2026, at which time the interest rate shall increase to 10.0% per annum. Upon the consummation of an initial public offering by the Company prior to the maturity date, the convertible notes will automatically convert into a number of shares of common stock equal to the outstanding principal amount of the notes and any accrued interest divided by the applicable conversion price. The conversion price is the lower of (i) a 20% discount to the initial public offering price and (ii) $280 million divided by the number of fully diluted shares of capital stock (on an as-converted basis) outstanding immediately prior to the initial public offering but excluding the convertible notes (the “IPO Capitalization”); provided that, in no event shall such conversion price be less than the quotient obtained by dividing $210 million by the IPO Capitalization. Upon a liquidation event, the holders of the convertible notes shall receive a cash payment equal to the sum of 1.3 times the outstanding principal amount of the convertible notes plus any accrued interest.

On July 21, 2025, the Company amended the Trinity Loan Agreement to extend the commitment date of the tranche (ii) $15,000 draw from before December 31, 2024 to before December 31, 2025 and to extend the commitment date of the tranche (iii) $15,000 draw from before December 31, 2025 to before December 31, 2026. The interest rate on the instrument was amended from a floor of 11.50% to 11.00%, along with increasing the interest only period from an initial term of 48-months to 60-months. Additionally, warrants originally vesting upon the funding of the tranche (iii) $15,000 draw were cancelled and are no longer outstanding.

On July 23, 2025, the Company amended its amended and restated certificate of incorporation to effect the Reverse Stock Split of shares of the Company’s common stock on a 1-for-19.08 basis. The number of authorized shares was increased to 730,000,000 shares of common stock. The par values of the common stock and redeemable convertible preferred stock were not adjusted as a result of the Reverse Stock Split. In connection with the Reverse Stock Split, the conversion ratio for the Company’s outstanding redeemable convertible preferred stock was adjusted. All references to common stock, common stock warrants, and options to purchase common stock share data, per share data, and related information contained in the financial statements have been retroactively adjusted to reflect the effect of the Reverse Stock Split for all periods presented.

5,000,000 Shares



Common Stock

Prospectus

Morgan Stanley **Goldman Sachs & Co. LLC** **Piper Sandler**

Jefferies

BTIG

July 30, 2025

Through and including August 24, 2025 (the 25th day after the date of this prospectus), all dealers that effect transactions in these securities, whether or not participating in this offering, may be required to deliver a prospectus. This is in addition to the dealers' obligation to deliver a prospectus when acting as underwriters and with respect to their unsold allotments or subscriptions.